Lecture Notes on Neural Network Scaling Limits

Rewritten via Nonstandard Clifford Analysis

Based on the STAT 946 Lecture Notes of Mufan Li

Nonstandard Clifford Analysis Edition by Alok Singh

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Chapter 1

Introduction

These lecture notes grew out of STAT 946—Topics in Deep Learning Theory, taught at the University of Waterloo in Fall 2025. I taught the course from handwritten notes based on papers that I had read or written, including work that was then nearing completion. One motivation for preparing this document is a question I am often asked: what is a good place to start reading about deep learning theory? My aim is to offer a deliberately subjective guide to several directions in the field, with an emphasis on questions connected to my own research program.

The document began as a series of lecture notes scribed by dedicated students; many of the submissions far exceeded what I had expected from an ungraded task. GPT Pro and, later, Sol Ultra have been invaluable in combining and editing the scribed notes with the original papers and handwritten notes, substantially reducing the time required while improving the correctness and completeness of the resulting manuscript. Roughly one week of evening work produced a seemingly decent first draft, but very soon I realized the notes required much more careful editing and lots of time commitment. I should say I remain not completely happy with the current version, especially as the prose currently still reads quite AI-like in many places, although I'm pretty confident the mathematical content is correct and complete.

These lecture notes focus on *scaling limits* of neural networks.[1] On the one hand, the mathematics of scaling limits is elegant and worth studying in its own right. On the other hand, many theorists, myself included, would like this work to have practical consequences. It is therefore worth motivating the selection of topics before turning to the technical details, including why some subjects that others may regard as foundational are not emphasized here.

1.1 Motivation

Modern deep learning is, in many ways, an empirical science: researchers propose a method, scale it up, and compare its performance on benchmarks. In recent years, increasing model,

[1] In this Nonstandard Clifford Analysis edition, all asymptotic scaling regimes are formulated at unlimited hyperfinite scale in Robinson's $\aleph_1$-saturated enlargement $({}^*\mathbb{R}, {}^*\mathbb{N})$ equipped with Loeb probability spaces $L({}^*\mathbb{P})$ [55, 44, 3, 22] together with internal and standard Euclidean Clifford algebras ${}^*\mathcal{C}\ell_N = \mathcal{C}\ell({}^*\mathbb{R}^N)$ and $\mathcal{C}\ell_m = \mathcal{C}\ell(\mathbb{R}^m)$, Hestenes–Sobczyk geometric blade calculus, and monogenic Dirac operators [29, 9, 18, 57]. For any standard $m \in \mathbb{N}$, the limited multivector subring $\mathbb{L}({}^*\mathcal{C}\ell_m) = \mathbb{L} \otimes_{\mathbb{R}} \mathcal{C}\ell_m$ carries a canonical $\mathbb{R}$-algebra shadow homomorphism $\mathrm{st} : \mathbb{L}({}^*\mathcal{C}\ell_m) \to \mathcal{C}\ell_m$ commuting with grade projections $\langle\cdot\rangle_k$, reversion $(\cdot)^\sim$, and contractions $\lrcorner, \llcorner$. Mufan Li's exposition and narrative structure are preserved as closely as possible.

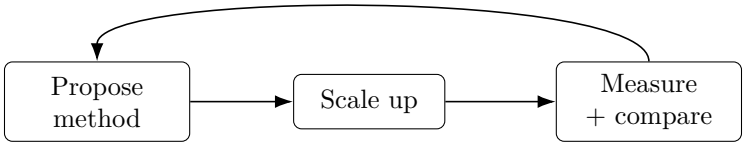


Figure 1.1: A recurring pattern in deep learning research: new ideas are often validated (or rejected) only after large-scale training runs.

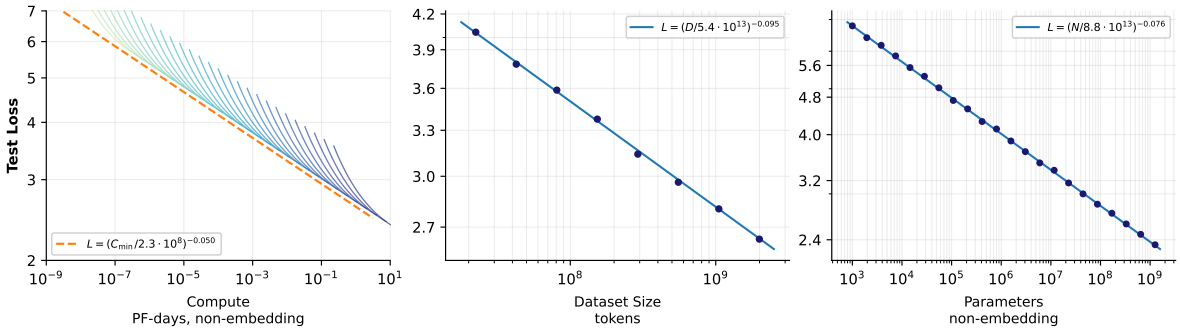


Figure 1.2: Empirical neural-network scaling laws for language modeling (Kaplan et al. [36], Figure 1). On log–log axes, test loss decreases approximately as a power law in (left) training compute C, (middle) dataset size D, and (right) non-embedding parameter count N, within regimes not bottlenecked by the remaining resources.

data, and compute scale has become a particularly reliable route to improved performance. As industrial research laboratories train increasingly large models, the gap between theory and practice continues to widen. Many mathematical obstacles make theoretical progress difficult, which raises a basic question:

What is the best way for theory to contribute to deep learning?

It is instructive to examine empirical practice for clues; the cycle is summarized in Figure 1.1. First, the cycle of proposing a method, scaling it up, and measuring it on benchmarks has important limitations. In particular, benchmark comparisons do not always identify a uniformly better method. Conclusions can change with the architecture, data regime, tuning budget, or evaluation protocol, even when the empirical literature is extensive. When extensive experiments still do not yield a stable comparison, what can theory contribute?

Yet empirical practice also reveals a striking regularity. Across several deep learning tasks and over substantial measured ranges, test loss often decreases predictably as model size, dataset size, or training compute increases [30]. These empirical relations are called *scaling laws*: when the remaining resources are not limiting, test loss is often well approximated by a power law in the resource being varied [36]. A power law appears approximately as a straight line on log–log axes, as illustrated in Figure 1.2.

Within a fitted scaling regime, this predictability makes it possible to extrapolate larger-scale performance from smaller experiments [36]. At a fixed training-compute budget, compute-optimal prescriptions jointly scale model size and training data rather than increasing either one in isolation [36, 31]. The regimes of greatest practical interest, however, involve frontier models

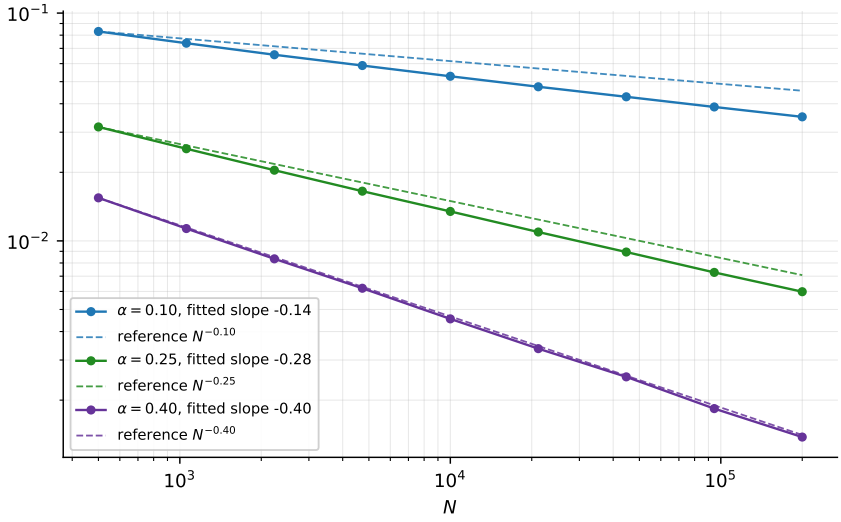


Figure 1.3: A numerical central-limit-theorem convergence experiment. Solid curves show empirical Kolmogorov errors for three values of α, while dashed lines show the corresponding $N^{-\alpha}$ reference rates.

that can be prohibitively expensive to train. These observations raise two related questions: why should such simple power laws emerge from complex trained networks, and what can theory tell us about large models without paying their full computational cost?

One clue is that the same visual signature appears far beyond neural networks. Consider the log–log plot in Figure 1.3.

The resemblance is the point: this figure was not produced by training neural networks, but by simulating convergence toward the Gaussian limit for deliberately heavy-tailed summands. Related quantitative central limit theorems turn moment assumptions into power-law error bounds. For example, for i.i.d. centered, unit-variance summands, if $\mathbb{E}[|X_1|^{2+2\alpha}] < \infty$ for some $\alpha \in (0, 1/2]$, then by transfer of a generalized Berry–Esseen bound [53, Chapter V], at any unlimited hypernatural $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ the internal Kolmogorov distance to ${}^*\Phi$ is bounded by $CN^{-\alpha} \in \mathbb{I}$ (infinitesimal of scale $N^{-\alpha}$). The broader point is that a straight line on log–log axes can record the infinitesimal scale at which a hyperfinite system approaches its standard shadow.

This comparison suggests viewing a neural-network scaling law as a finite-size (infinitesimal) correction to an underlying scaling limit. A scaling law describes how an observable changes with scale, whereas a scaling limit describes the standard shadow that the underlying hyperfinite system lies infinitely close to. Although trained neural networks of different sizes and architectures have different parameter spaces, we may nevertheless regard:

- $\mathcal{X}$ as a common space of trained neural networks for a fixed task and their possible scaling limits (with nonstandard extension ${}^*\mathcal{X}$). Unlike a predictor space, $\mathcal{X}$ may retain architecture- and parameter-level information while respecting hidden-unit permutation symmetries;

- $x_N \in {}^*\mathcal{X}$ as the internal trained neural network obtained with unlimited hyperfinite compute budget $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, with the number of parameters, data points, and training

iterations allowed to vary internally with N;

- $L : \mathcal{X} \to \mathbb{R}$ as a test function, or observable, on this space, such as expected test loss (with natural extension ${}^*L : {}^*\mathcal{X} \to {}^*\mathbb{R}$); and
- $x_\infty \in \mathcal{X}$ as the standard shadow approached by x_N ($x_N \approx x_\infty$ in ${}^*\mathcal{X}$), with $L_\infty := L(x_\infty) = \mathrm{st}({}^*L(x_N))$.

Suppose, heuristically, that $x_N \approx x_\infty$ in ${}^*\mathcal{X}$ and that ${}^*L$ detects the leading hyperfinite-compute correction nondegenerately. If this infinitesimal correction has scale $N^{-\alpha}$ for standard $\alpha > 0$, then

$$ {}^*L(x_N) = L_\infty + cN^{-\alpha} + \varepsilon N^{-\alpha}, \qquad c \in \mathbb{R} \setminus \{0\}, \quad \varepsilon \approx 0, \tag{1.1} $$

equivalently $\mathrm{st}(N^\alpha({}^*L(x_N) - L_\infty)) = c$ under an infinitesimal microscope of magnification N^α. In this sense, the convergence rate seen through the test function L appears by transfer at large standard scales as a scaling law. This conclusion is not automatic for every observable: regularity is needed, and cancellations or degeneracies can change the leading exponent. Because N indexes compute, (1.1) already has the familiar form of a loss–compute scaling law.

This is presently a research program rather than a derivation of the empirical loss laws observed in frontier models. Most available results vary only part of the system, for example width and depth while holding the dataset and training horizon fixed. Nevertheless, the classical example above makes the proposed mechanism plausible: when a quantitative scaling limit has an algebraic infinitesimal scale, suitable observables exhibit power-law finite-size corrections, and empirical neural-network scaling laws may reflect the same general mechanism.

This viewpoint is also particularly well suited to the current era of ever larger models. Frontier systems are increasingly expensive to study by direct experimentation and increasingly unwieldy to describe exactly at finite size. Scaling limits are designed for precisely this situation: they replace a growing family of complicated systems by a tractable standard shadow while retaining hyperfinite microscope information about how the finite systems approach that shadow. They therefore promise both a possible explanation of scaling laws and a practical language for reasoning about systems that cannot be studied exhaustively at their largest scale.

These considerations lead to two recurring goals:

1. use quantitative scaling relations to understand finite-scale behavior and determine when small networks are informative about larger networks; and
2. replace complicated finite networks by simpler limiting standard shadows.

The success of hyperparameter transfer illustrates the first point: under a suitable parameterization, hyperparameters tuned on small networks can transfer to much larger ones [63]. When the standard-part approximation is stable across the candidate hyperparameters, small networks can therefore serve as useful proxies for selecting hyperparameters in larger networks. In particular, the mechanisms supporting hyperparameter transfer need not be confined to a single scaling regime: Section 4.4 develops the μP width scaling, while Section 6.3 studies feature learning in a distinct proportional depth–width limit. The following examples emphasize the second point by showing how nonstandard scaling limits replace complicated hyperfinite systems with simpler standard objects; hyperfinite expansions additionally describe how the finite systems approach those shadows.

1.2 Examples of Scaling Limits

We now return to the Central Limit Theorem (CLT), whose quantitative form supplied the motivating plot in Figure 1.3. Its qualitative statement already has the structure of a scaling limit: normalized hyperfinite sums of internal i.i.d. summands project under the standard-part map to a simple, universal Gaussian law. Fix $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, and let $(X_i)_{i=1}^N$ be an internal i.i.d. sequence on an internal probability space $(\Omega, \mathcal{A}, {}^*\mathbb{P})$ with ${}^*\mathbb{E}[X_i] = 0$, ${}^*\mathrm{Var}(X_i) = \sigma^2 \in (0, \infty)$, and S-integrable squares. Equipped with the Loeb probability measure $L({}^*\mathbb{P})$ [44, 15], the hyperfinite CLT states that $S_N := \frac{1}{\sigma\sqrt{N}} \sum_{i=1}^N X_i \in \mathbb{L}$ $L({}^*\mathbb{P})$-almost surely and

$$\mathrm{st}\left(\frac{1}{\sigma\sqrt{N}} \sum_{i=1}^{N} X_i\right) \sim \mathcal{N}(0,1) \qquad \text{under } L({}^*\mathbb{P}). \tag{1.2}$$

The remarkable aspect is universality: only the first two moments matter for the standard shadow. The scaling $1/\sqrt{N}$ is not cosmetic: it simultaneously keeps the internal variance in $\mathbb{A}$ (appreciable, order one) and ensures each individual term contributes an infinitesimal of scale $1/\sqrt{N} \in \mathbb{I}$. Quantitative versions of the CLT additionally control the infinitesimal distance between the internal distribution and ${}^*\mathcal{N}(0,1)$; Figure 1.3 illustrates the power-law rates discussed above.

The CLT has a functional analogue: an internal rescaled hyperfinite random walk has standard Brownian motion as its uniform shadow [3]. Let $(X_i)_{i=1}^{\lceil NT \rceil + 1}$ be internal i.i.d. hyperreal random variables with ${}^*\mathbb{E}[X_i] = 0$, ${}^*\mathrm{Var}(X_i) = \sigma^2 \in (0, \infty)$, and S-integrable X_i^2. Define the internal normalized partial sums

$$Y_k^{(N)} := \frac{1}{\sigma\sqrt{N}} \sum_{i=1}^{k} X_i, \qquad k = 0, 1, \ldots, \lceil NT \rceil + 1. \tag{1.3}$$

To obtain an internal continuous path, linearly interpolate between consecutive hyperfinite steps:

$$Z_t^{(N)} := Y_{\lfloor Nt \rfloor}^{(N)} + (Nt - \lfloor Nt \rfloor)\left(Y_{\lfloor Nt \rfloor + 1}^{(N)} - Y_{\lfloor Nt \rfloor}^{(N)}\right), \qquad t \in {}^*[0, T]. \tag{1.4}$$

For each fixed standard $t \in [0, T]$, the hyperfinite CLT implies $\mathrm{st}(Z_t^{(N)}) \sim \mathcal{N}(0, t)$ under $L({}^*\mathbb{P})$.

Definition 1.1 (Brownian Motion)**.** A stochastic process $(B_t)_{t \in [0,T]}$ is a (standard) Brownian motion if:

1. $B_0 = 0$ almost surely;
2. for $0 \le s \le t$, the increment $B_t - B_s \sim \mathcal{N}(0, t-s)$ and disjoint increments are independent;
3. $t \mapsto B_t$ is almost surely continuous.

For an internal random element $Z^{(N)}$ taking values in ${}^*\mathcal{X}$ for a Polish metric space $(\mathcal{X}, d)$ and a standard random element Z in $\mathcal{X}$, $Z^{(N)}$ has Loeb shadow law Z when $\mathrm{st}(Z^{(N)})$ exists $L({}^*\mathbb{P})$-almost surely and $\mathbb{E}_L[{}^*\psi(Z^{(N)})] = \mathrm{st}({}^*\mathbb{E}[{}^*\psi(Z^{(N)})]) = \mathbb{E}[\psi(Z)]$ for every bounded continuous $\psi : \mathcal{X} \to \mathbb{R}$. Thus Loeb pushforward via st applies directly to hyperfinite random paths once their function space and metric are specified.

Theorem 1.2 (Anderson's Hyperfinite Brownian Motion / Donsker's Invariance Principle [17, 3])**.** *Under the assumptions above, for $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, the internal path $(Z_t^{(N)})_{t \in {}^*[0,T]}$ is $L({}^*\mathbb{P})$-almost surely nearstandard (equivalently, limited and S-continuous: $s \approx t \implies Z_s^{(N)} \approx Z_t^{(N)}$) in ${}^*C([0,T])$ equipped with the internal sup norm, and its standard part $B_t := \mathrm{st}(Z_{*t}^{(N)})$ is a standard Brownian motion on $(\Omega, \mathcal{L}(\mathcal{A}), L({}^*\mathbb{P}))$ satisfying*

$$\sup_{t \in {}^*[0,T]} \left| Z_t^{(N)} - {}^*B_t \right| \approx 0 \qquad L({}^*\mathbb{P})\text{-almost surely,} \qquad \mathrm{st}(Z^{(N)}) \sim B \text{ in } (C([0,T]), \|\cdot\|_\infty). \tag{1.5}$$

Remark 1.3. This is the prototype for a hyperfinite diffusion approximation [37, 43]: after identifying the correct infinitesimal local drift, diffusion variance, and hyperfinite step size $\Delta\tau = 1/N \in \mathbb{I}_+$, an internal discrete recursion projects via st to an Itô SDE. The same template reappears in the depth-scaling arguments of Chapter 5 and is formalized in Appendix A.3.

A different type of scaling limit appears in random matrix theory, where a hyperfinite-dimensional internal random matrix has a deterministic standard spectral distribution. For $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, let $X \in {}^*\mathbb{R}^{N \times N}$ be an internal symmetric matrix, with the entries on and above the diagonal internally independent and identically distributed, centered, of variance one, and with limited fourth moment. Consider the scaled matrix $N^{-1/2}X$, with internal real eigenvalues $\lambda_1^{(N)} \le \cdots \le \lambda_N^{(N)}$. Define the internal *empirical spectral distribution* (ESD)

$$\rho^{(N)}(dx) := \frac{1}{N} \sum_{i=1}^{N} \delta_{\lambda_i^{(N)}}(dx). \tag{1.6}$$

Theorem 1.4 (Wigner Semicircle Law [61])**.** *For $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, $L({}^*\mathbb{P})$-almost surely, the Loeb standard-part pushforward $\rho_{\mathrm{sc}} := L(\rho^{(N)}) \circ \mathrm{st}^{-1}$ is the deterministic probability measure (meaning $\int_{*\mathbb{R}} {}^*\psi(x)\, \rho^{(N)}(dx) \approx \int_{\mathbb{R}} \psi(x)\, \rho_{\mathrm{sc}}(dx)$ for every bounded continuous $\psi : \mathbb{R} \to \mathbb{R}$) with density*

$$\rho_{\mathrm{sc}}(x) = \frac{\sqrt{(4 - x^2)_+}}{2\pi}, \qquad \text{supported on } [-2, 2]. \tag{1.7}$$

Remark 1.5. The emergence of a deterministic standard shadow spectrum is an example of self-averaging: macroscopic hyperfinite averages concentrate within a single monad at unlimited dimension $N \in {}^*\mathbb{N} \setminus \mathbb{N}$. Equivalently, embedding the complex upper half-plane into the even subalgebra $\mathcal{C}\ell_2^+$ via the unit bivector $\boldsymbol{i} = e_1 e_2$ ($\boldsymbol{i}^2 = -1$), the normalized internal Cauchy–Dirac resolvent trace $\frac{1}{N} \mathrm{Tr}((z - N^{-1/2}X)^{-1}) \in \mathbb{L}({}^*\mathcal{C}\ell_2^+)$ concentrates $L({}^*\mathbb{P})$-almost surely around the standard Stieltjes transform of ρ_{sc}. In the random-projection model of Chapter 3, exterior blade volumes and standard-part spectra of hyperfinite design and Gram matrices determine explicit risk formulas.

The final example replaces a hyperfinite interacting particle system by a deterministic evolution of its Loeb empirical shadow. Let $\mathcal{K} : \mathbb{R} \to \mathbb{R}$ be a bounded Lipschitz interaction kernel with natural extension ${}^*\mathcal{K} : {}^*\mathbb{R} \to {}^*\mathbb{R}$. Consider $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ internal particles $(X_i(t))_{i=1}^N$ in ${}^*\mathbb{R}$ evolving according to the internal mean-field ODE system

$$\partial_t X_i(t) = \frac{1}{N} \sum_{j=1}^{N} {}^*\mathcal{K}(X_i(t) - X_j(t))\,. \tag{1.8}$$

Define the internal empirical measure

$$\rho_t^{(N)}(dx) := \frac{1}{N}\sum_{i=1}^{N} \delta_{X_i(t)}(dx). \tag{1.9}$$

For a probability measure ρ on $\mathbb{R}$, define the law-dependent velocity field

$$b(x,\rho) := \int_{\mathbb{R}} \mathcal{K}(x-y)\,\rho(dy). \tag{1.10}$$

Then the internal particle equation can be written exactly as $\partial_t X_i(t) = {}^*b(X_i(t), \rho_t^{(N)})$. With internal i.i.d. initial conditions of law ${}^*\rho_0$, $L({}^*\mathbb{P})$-almost surely the Loeb shadow measure $\rho_t := L(\rho_t^{(N)}) \circ \mathrm{st}^{-1}$ is the deterministic evolution (ρ_t) governed by

$$\partial_t \rho_t + \langle \partial_x(b(\cdot,\rho_t)\rho_t)\rangle_0 = 0, \qquad \rho_{t=0} = \rho_0, \tag{1.11}$$

where in one dimension $\langle \partial_x(b\rho_t)\rangle_0 = \partial_x(b\rho_t)$ and in general parameter dimension $\Theta \subset \mathbb{R}^P$ the scalar part of the Dirac vector derivative $\partial_\theta = \sum_{p=1}^{P} e_p \partial_{\theta_p} \in \mathcal{C}\ell_P$ expresses coordinate-free transport divergence. The equation is interpreted in the weak sense; see Appendix A.1.

Although the particles interact at hyperfinite N, the independence present at initialization survives in the standard-part shadow: for any fixed standard $k \in \mathbb{N}$, $(\mathrm{st}(X_1(t)), \ldots, \mathrm{st}(X_k(t)))$ are independent samples from ρ_t under $L({}^*\mathbb{P})$. This phenomenon is called *propagation of chaos* [60]. Quantitative versions show that hyperfinite errors for suitable observables lie in $O(N^{-\alpha}) \subset \mathbb{I}$; a leading infinitesimal correction of scale $N^{-\alpha}$ corresponds by transfer to a scaling law for the approach to the mean-field shadow.

In mean-field analyses of neural networks, neurons or parameters play the role of hyperfinite particles. Their internal empirical distribution evolves under training and, at hyperfinite width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, its Loeb shadow is described by a McKean–Vlasov transport equation of this type. We return to this connection in Chapter 4.

Remark 1.6 (A Checklist for Scaling Limits). The three examples have the same four nonstandard ingredients:

1. an internal hyperfinite random object at unlimited scale $N \in {}^*\mathbb{N} \setminus \mathbb{N}$;
2. a normalization placing observables in $\mathbb{L}$ or $\mathbb{A}$ and a declaration of what is unlimited ($\in {}^*\mathbb{N} \setminus \mathbb{N}$) and what remains standard;
3. nearstandardness (or S-continuity) in the target space under Loeb probability $L({}^*\mathbb{P})$; and
4. a standard shadow via st with a tractable description.

For the random walk the shadow is a continuous path, for the Wigner matrix it is a probability measure, and for the particle system it is a deterministic evolution of probability measures. Later chapters repeat this template with network outputs, kernels, parameter distributions, and covariance processes. A quantitative scaling limit adds a further ingredient: an infinitesimal microscope magnification N^α isolating the leading hyperfinite correction for the observable of interest.

1.3 Roadmap

The remainder of the lecture notes is organized as follows.

Chapter 2 treats two related but logically distinct hyperfinite-width limits ($N \in {}^*\mathbb{N} \setminus \mathbb{N}$) for wide fully-connected networks. At random initialization, the standard parts of the outputs on any fixed standard finite collection of inputs follow a centered Gaussian vector law under $L({}^*\mathbb{P})$ characterized by the neural network Gaussian process (NNGP) kernel. During gradient-flow training in the NTK, or lazy-training, regime, the internal tangent kernel at each standard training time lies infinitely close ($\overset{L({}^*\mathbb{P})}{\approx}$) to a deterministic standard kernel K that does not depend on time, reducing the standard-part output dynamics to kernel gradient flow.

Chapter 3 turns from fitting the training data to prediction on new data. Using linear regression with hyperfinite random projections, it explains why test error can spike into the unlimited range (${}^*\mathbb{R} \setminus \mathbb{L}$) when the model first becomes able to fit the training data exactly and then decrease across appreciable values as more features are added.

Chapter 4 introduces *feature-learning* scalings in which internal representations evolve by an appreciable ($\in \mathbb{A}$) amount during training at hyperfinite width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, including shallow mean-field limits via Loeb empirical measures, the maximal-update (μP) parameterization, and a sketch of DMFT for deep networks.

Chapter 5 studies hyperfinite depth scalings ($D \in {}^*\mathbb{N} \setminus \mathbb{N}$) and joint depth–width limits at initialization, including proportional hyperfinite limits $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$ and the neural covariance SDE viewpoint via Keisler–Lindstrøm hyperfinite stochastic difference equations.

Chapter 6 discusses several applications of the depth scaling theory, including shaped attention/transformer models, the limiting spectrum of the feature covariance, and a one-step feature-learning mechanism in the proportional hyperfinite limit.

Across these chapters, the principal regimes differ in what remains standard and what is taken unlimited in ${}^*\mathbb{N} \setminus \mathbb{N}$:

- in Chapter 2, width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ while depth d, input dimension n_0, and dataset size m remain standard;
- in Chapter 3, $(M, N_0, N) \in ({}^*\mathbb{N} \setminus \mathbb{N})^3$ are proportionally unlimited, with $\mathrm{st}(N_0/M) = \gamma \in (0, \infty)$ and $\mathrm{st}(N/M) = \delta \in (0, \infty)$;
- in Chapter 4, width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ under feature-learning parameterizations ($\gamma_N \asymp \sqrt{N}$) and associated learning-rate scalings ($\eta_N \asymp N$), including both the shallow mean-field and deep maximal-update (μP) regimes; and
- in Chapters 5 and 6, depth $D \in {}^*\mathbb{N} \setminus \mathbb{N}$ and width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ can be jointly unlimited with $\mathrm{st}(D/N) = \bar{\tau}$, typically with a fixed standard collection of inputs before an additional hyperfinite spectral limit $M \in {}^*\mathbb{N} \setminus \mathbb{N}$ is taken.

The technical appendix collects the Loeb weak-shadow, S-equicontinuity, Skorohod lifting, affine-invariant metric, and internal Gaussian-conditioning tools used across these regimes.

Notation

Symbol	Meaning
n, N	Width of a hidden layer (number of neurons / channels); $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ denotes an unlimited hypernatural width.
d, D	Depth, or number of hidden layers or blocks, excluding the input layer; $D \in {}^*\mathbb{N} \setminus \mathbb{N}$ denotes an unlimited hypernatural depth.
n_0, N_0	Input dimension (standard $n_0 \in \mathbb{N}$, or unlimited $N_0 \in {}^*\mathbb{N} \setminus \mathbb{N}$ in Chapter 3).
m, M	Number of inputs / training examples / tokens (standard $m \in \mathbb{N}$, or unlimited $M \in {}^*\mathbb{N} \setminus \mathbb{N}$ in Chapter 3 and Section 6.2.4).
x^α	Input (data point) indexed by $\alpha \in \{1, \dots, m\}$.
y^α	Label / target associated with x^α.
θ	Network parameters (weights).
W_0, W_ℓ, W_d	For scalar-output MLPs of hyperfinite width N, $W_0 \in {}^*\mathbb{R}^{N \times n_0}$, $W_\ell \in {}^*\mathbb{R}^{N \times N}$ for $1 \le \ell \le d-1$, and $W_d \in {}^*\mathbb{R}^{1 \times N}$ is the readout row vector.
$f(x^\alpha; \theta) = f^\alpha(\theta)$	Network output at input x^α.
φ	Activation function (with natural extension ${}^*\varphi : {}^*\mathbb{R} \to {}^*\mathbb{R}$, often written φ when unambiguous).
c	Variance-preserving normalization constant, $c := (\mathbb{E}[\varphi(w)^2])^{-1}$ for $w \sim \mathcal{N}(0,1)$.
h_ℓ^α	Hidden pre-activations (or hidden states) at layer ℓ for input x^α.
$\varphi_\ell^\alpha = \varphi(h_\ell^\alpha)$, $\varphi_\ell = [\varphi_\ell^1 \ \cdots \ \varphi_\ell^m]$	Hidden post-activations at layer ℓ and their dataset-column matrix.
g_ℓ^α	Backward post-activation neurons at layer ℓ for input x^α.
z_ℓ^α	Backward pre-activation neurons at layer ℓ for input x^α.
$\Phi^{(N)}, \Phi_\ell^{(N)}, \Phi_\ell'^{(N)}$	Hyperfinite empirical feature and derivative kernels in Chapter 2, with hidden-layer normalization $1/N$ and input normalization $1/n_0$. The superscript (N) records hyperfinite width.
$\Phi, \Phi_\ell, \Phi_\ell'$	Standard-part kernels in Chapters 2 and 4 ($\Phi_\ell = \mathrm{st}(\Phi_\ell^{(N)})$). In Chapters 5 and 6, Φ_ℓ also denotes internal layer covariances and Φ_τ their standard SDE shadow.

Symbol	Meaning
$K_t^{(N)}, K_t, K$	Hyperfinite empirical NTK and standard-part NTKs. Subscripts record time and superscripts in parentheses record width.
$L(\theta)$	Training objective (empirical risk); $L(\theta) = \frac{1}{m}\sum_{\alpha=1}^{m} \ell(f^\alpha(\theta), y^\alpha)$.
$\mathbb{E}, \mathbb{P}$	Standard expectation and probability.
$\|\cdot\|, \langle\cdot,\cdot\rangle$	Euclidean norm and inner product (standard or internal by transfer, unless specified otherwise).
${}^*\mathbb{R}, {}^*\mathbb{N}, {}^*\mathbb{N} \setminus \mathbb{N}$	Robinson's $\aleph_1$-saturated hyperreals, hypernaturals, and unlimited hypernaturals.
$\mathbb{L}, \mathbb{I}, \mathbb{A}$	Ring of limited hyperreals $\mathbb{L}$, maximal ideal of infinitesimals $\mathbb{I}$, and group of appreciable hyperreals $\mathbb{A} = \mathbb{L} \setminus \mathbb{I}$; $a \approx b \iff a - b \in \mathbb{I}$, and $a \asymp b \iff a/b \in \mathbb{A}$.
$\mathrm{st}, \mathrm{mon}(x)$	Standard-part homomorphism $\mathrm{st} : \mathbb{L} \to \mathbb{R}$ ($\mathrm{st}(x) \in \mathbb{R}$ unique with $x \approx \mathrm{st}(x)$) and monad (halo) $\mathrm{mon}(x) = \{y \in {}^*\mathbb{R} : y \approx x\}$.
${}^*\mathbb{P}, {}^*\mathbb{E}, L({}^*\mathbb{P}), \mathbb{E}_L$	Internal probability ${}^*\mathbb{P}$, internal expectation ${}^*\mathbb{E}$, Loeb probability measure $L({}^*\mathbb{P})$ on $\mathcal{L}(\mathcal{A})$, and Loeb expectation $\mathbb{E}_L$; $X \overset{L({}^*\mathbb{P})}{\approx} Y$ means $X \approx Y$ $L({}^*\mathbb{P})$-almost surely.
$\mathcal{C}\ell_m, {}^*\mathcal{C}\ell_N$	Standard Euclidean Clifford algebra $\mathcal{C}\ell_m = \mathcal{C}\ell(\mathbb{R}^m)$ and internal hyperfinite Clifford algebra ${}^*\mathcal{C}\ell_N = \mathcal{C}\ell({}^*\mathbb{R}^N)$; for standard $m \in \mathbb{N}$, $\mathrm{st} : \mathbb{L}({}^*\mathcal{C}\ell_m) \to \mathcal{C}\ell_m$ is the $\mathbb{R}$-algebra shadow homomorphism on limited multivectors.
$uv = \langle uv\rangle_0 + u \wedge v$, $M^\sim$	Geometric product of vectors decomposing into scalar inner part $\langle uv\rangle_0 = \langle u, v\rangle$ and exterior bivector $u \wedge v$, with reversion $(v_1 \cdots v_k)^\sim = v_k \cdots v_1$.
$\langle M\rangle_k$, $A \lrcorner B$, $A \llcorner B$	Grade-k projection $\langle M\rangle_k$ and left/right interior contractions; for an invertible blade $\boldsymbol{B}$, $P_{\boldsymbol{B}}(x) = (x \lrcorner \boldsymbol{B})\boldsymbol{B}^{-1}$ and $P_{\boldsymbol{B}}^{\perp}(x) = (x \wedge \boldsymbol{B})\boldsymbol{B}^{-1}$ are orthogonal projection and rejection.
∂_θ, $\mathrm{Spin}(m)$, $\mathrm{Pin}(m)$	Vector derivative (Dirac operator) $\partial_\theta = \sum_p e_p \partial_{\theta_p}$ satisfying $\partial_\theta^2 = \Delta_\theta$, and the even/full rotor covering groups $\mathrm{Spin}(m) \subset \mathcal{C}\ell_m^+$, $\mathrm{Pin}(m) \subset \mathcal{C}\ell_m^\times$.

Chapter 2

Kernel Limits for Wide Networks: NNGP and NTK

The first regime considered in this chapter is the infinite-width initialization limit, realized here at an unlimited hidden width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$. Neal showed that a one-hidden-layer network induces a Gaussian process at random initialization [48]; Lee et al. and Matthews et al. extended this NNGP description to finite-depth networks [38, 46].

We then turn to a logically distinct training-time regime. In the NTK, or lazy-training, regime at unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, the empirical tangent kernel at each standard training time concentrates infinitesimally around a deterministic standard kernel K that does not depend on time, so the standard-part output dynamics reduce to kernel gradient flow [33, 39].

2.1 Neural Network Gaussian Processes (NNGP)

We first consider a one-hidden-layer network of unlimited hidden width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ over a standard input dimension $n_0 \in \mathbb{N}$. Write $[N] := \{1, \dots, N\} \subset {}^*\mathbb{N}$, and for each hidden neuron $i \in [N]$ set $h_i(x) := n_0^{-1/2} \langle w_i, x \rangle$. At hyperfinite width N, the network's scalar output on a standard input $x \in \mathbb{R}^{n_0}$ is the internal hyperfinite sum

$$f(x;\theta) = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} a_i \, {}^*\varphi(h_i(x)), \qquad x \in \mathbb{R}^{n_0}, \tag{2.1}$$

where the internal parameters are $\theta = \{(a_i, w_i)\}_{i=1}^N$ defined on an internal probability space $(\Omega, \mathcal{A}, {}^*\mathbb{P})$ equipped with its countably additive Loeb probability measure $L({}^*\mathbb{P})$ [44, 15]. At initialization we assume internal Gaussian weights

$$a_i \overset{\text{i.i.d.}}{\sim} {}^*\mathcal{N}(0,1), \quad w_i \overset{\text{i.i.d.}}{\sim} {}^*\mathcal{N}(0, I_{n_0}), \quad \text{independent across } i \text{ and between } a_i \text{ and } w_i. \tag{2.2}$$

For fixed standard inputs $x^1, \dots, x^m \in \mathbb{R}^{n_0}$ (with $m \in \mathbb{N}$ standard), let $\mathcal{F} := {}^*\sigma(h_i(x^\alpha) : i \in [N], \alpha \in [m]) \subset \mathcal{A}$ be the internal σ-algebra generated by the hidden preactivations. Thus $\mathcal{F}$ is generated by the hidden neurons at the inputs under consideration and does not include the output weights. Since the internal Gaussian output weights are internally independent of

$\mathcal{F}$, by the Transfer Principle [55] the output vector is exactly internally conditionally Gaussian:

$$\begin{aligned}&\big(f(x^\alpha;\theta)\big)_{\alpha=1}^m \,\Big|\, \mathcal{F} \sim {}^*\mathcal{N}(0,\Phi^{(N)}),\\ &\qquad \Phi^{(N),\alpha\beta} := \frac{1}{N}\Big\langle {}^*\varphi(h(x^\alpha))\,{}^*\varphi(h(x^\beta))^\sim\Big\rangle_0 = \frac{1}{N}\sum_{i=1}^N {}^*\varphi(h_i(x^\alpha))\,{}^*\varphi(h_i(x^\beta)),\end{aligned} \tag{2.3}$$

where we identify activation vectors in ${}^*\mathbb{R}^N$ with 1-vectors in the internal Clifford algebra ${}^*\mathcal{C}\ell_N$ so that $\langle uv^\sim\rangle_0 = \langle uv\rangle_0 = \langle u,v\rangle$. Thus no conditional central limit theorem is needed in the Gaussian-output-weight setting. This output-layer identity uses only internal Gaussian readout weights independent of the hidden neurons and will be reused below. If the relevant standard second moments are finite, the hyperfinite (Loeb) strong law of large numbers gives, $L({}^*\mathbb{P})$-almost surely,

$$\Phi^{(N),\alpha\beta} \approx \Phi(x^\alpha,x^\beta), \qquad \Phi(x,x') := \mathbb{E}\left[\varphi\left(\frac{\langle w,x\rangle}{\sqrt{n_0}}\right)\varphi\left(\frac{\langle w,x'\rangle}{\sqrt{n_0}}\right)\right], \quad w\sim\mathcal{N}(0,I_{n_0}). \tag{2.4}$$

Taking internal conditional characteristic functions in (2.3) and passing to standard parts via Loeb bounded convergence therefore yields the exact standard-part Gaussian law

$$\mathrm{st}\big(f(x^1),\dots,f(x^m)\big) \sim \mathcal{N}\Big(0,\,[\Phi(x^\alpha,x^\beta)]_{\alpha,\beta=1}^m\Big) \qquad \text{under } L({}^*\mathbb{P}). \tag{2.5}$$

For centered, unit-variance non-Gaussian output weights, the analogous conclusion instead follows from a hyperfinite multivariate central limit theorem.

Exercise 2.1 (Gaussianity of Linear Combinations)**.** Show that for any standard $m\in\mathbb{N}$ and any standard coefficients $u\in\mathbb{R}^m$, $\mathrm{st}\big(\sum_{\alpha=1}^m u_\alpha f(x^\alpha)\big) \sim \mathcal{N}(0,u^\top\Phi u)$ under $L({}^*\mathbb{P})$, with $\Phi=[\Phi(x^\alpha,x^\beta)]_{\alpha,\beta=1}^m$.

The calculation above establishes the finite-dimensional standard-part distributions, which identify the shadow Gaussian process on every fixed standard collection of inputs. Realizing the shadow of the random function itself on continuous function spaces additionally requires S-continuity of the internal sample paths.

Definition 2.2 (Gaussian Process)**.** A random function $f:\mathbb{R}^{n_0}\to\mathbb{R}$ is a Gaussian process with mean μ and covariance kernel Φ (notation: $f\sim\mathrm{GP}(\mu,\Phi)$) if

(i) Φ is symmetric positive semidefinite, i.e. for any $r\in\mathbb{N}$ and any $x^1,\dots,x^r\in\mathbb{R}^{n_0}$, the matrix $[\Phi(x^\alpha,x^\beta)]_{\alpha,\beta=1}^r$ is PSD.

(ii) For any $r\in\mathbb{N}$ and any $x^1,\dots,x^r\in\mathbb{R}^{n_0}$,

$$\big(f(x^1),\dots,f(x^r)\big) \sim \mathcal{N}\Big(\big(\mu(x^1),\dots,\mu(x^r)\big),\,[\Phi(x^\alpha,x^\beta)]_{\alpha,\beta=1}^r\Big). \tag{2.6}$$

Theorem 2.3 (Finite-Dimensional NNGP Limit at Initialization [48])**.** *Assume $\varphi:\mathbb{R}\to\mathbb{R}$ is Borel measurable and, for every $x\in\mathbb{R}^{n_0}$,*

$$\mathbb{E}\left[\varphi\left(\frac{\langle w,x\rangle}{\sqrt{n_0}}\right)^2\right] < \infty, \qquad w\sim\mathcal{N}(0,I_{n_0}). \tag{2.7}$$

Then, at any unlimited hidden width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ and for every fixed standard finite collection $x^1, \dots, x^m \in \mathbb{R}^{n_0}$, the vector $\big(f(x^\alpha;\theta)\big)_{\alpha=1}^m$ belongs to $\mathbb{L}^m$ $L({}^*\mathbb{P})$-almost surely and its standard part satisfies

$$\mathrm{st}\Big(\big(f(x^\alpha;\theta)\big)_{\alpha=1}^m\Big) \sim \mathcal{N}\Big(0, \big[\Phi(x^\alpha, x^\beta)\big]_{\alpha,\beta=1}^m\Big) \qquad \text{under } L({}^*\mathbb{P}), \tag{2.8}$$

where Φ is the kernel in (2.4). Equivalently, the finite-dimensional standard-part distributions coincide with those of a centered Gaussian process with covariance kernel Φ.

Proof. By (2.7) and Transfer, the internally i.i.d. products ${}^*\varphi(h_i(x^\alpha)){}^*\varphi(h_i(x^\beta))$ have standard-finite internal expectation $\Phi(x^\alpha, x^\beta)$, so the Loeb strong law of large numbers in (2.4) gives $\Phi^{(N),\alpha\beta} \approx \Phi(x^\alpha, x^\beta)$ $L({}^*\mathbb{P})$-almost surely. By (2.3), ${}^*\mathbb{E}[\|(f(x^\alpha))_{\alpha=1}^m\|_2^2 \mid \mathcal{F}] = \mathrm{Tr}(\Phi^{(N)}) \in \mathbb{L}$ $L({}^*\mathbb{P})$-almost surely, so $(f(x^\alpha;\theta))_{\alpha=1}^m \in \mathbb{L}^m$ $L({}^*\mathbb{P})$-almost surely. For any standard $u \in \mathbb{R}^m$, taking internal expectations of ${}^*\mathbb{E}[{}^*\exp(i\sum_{\alpha=1}^m u_\alpha f(x^\alpha;\theta)) \mid \mathcal{F}] = {}^*\exp(-\frac{1}{2}u^\top \Phi^{(N)} u)$ and applying Loeb bounded convergence gives $\mathbb{E}_L[\exp(i\sum_{\alpha=1}^m u_\alpha \,\mathrm{st}(f(x^\alpha;\theta)))] = \exp(-\frac{1}{2}u^\top \Phi u)$, proving (2.8). Moreover, for every $u \in \mathbb{R}^m$,

$$\sum_{\alpha,\beta=1}^m u_\alpha u_\beta \Phi(x^\alpha, x^\beta) = \mathbb{E}\left[\left(\sum_{\alpha=1}^m u_\alpha \varphi\left(\frac{\langle w, x^\alpha\rangle}{\sqrt{n_0}}\right)\right)^2\right] \geq 0. \tag{2.9}$$

Thus Φ is positive semidefinite and defines the stated centered Gaussian process. □

Remark 2.4 (Function-Space Strengthening). The theorem characterizes finite-dimensional standard-part distributions, which is the notion needed for a fixed finite dataset. Lifting f to a continuous standard Gaussian process in $C(K)$ on a standard compact set $K \subset \mathbb{R}^{n_0}$ additionally requires $L({}^*\mathbb{P})$-almost-sure S-continuity of the internal sample paths on ${}^*K$ (that is, $f(x) \in \mathbb{L}$ for all $x \in {}^*K$ and $f(x) \approx f(x')$ whenever $x, x' \in {}^*K$ satisfy $x \approx x'$). For example, a globally Hölder activation yields the uniform monadic Hölder increment bound ${}^*\mathbb{E}[|f(x) - f(x')|^p] \leq C_{p,K}\,\|x - x'\|^{\gamma p}$ proved in Appendix A.2; the nonstandard Kolmogorov–Chentsov criterion then guarantees S-continuity of f on ${}^*K$ $L({}^*\mathbb{P})$-almost surely, so $f^\circ(\mathrm{st}(x)) := \mathrm{st}(f(x))$ defines a continuous standard Gaussian process $f^\circ \sim \mathrm{GP}(0, \Phi|_{K\times K})$ on K satisfying $\sup_{x\in{}^*K} |f(x) - {}^*f^\circ(x)| \approx 0$ $L({}^*\mathbb{P})$-almost surely.

For some activations the kernel Φ admits a closed form. For ReLU, $\varphi(u) = \max\{u, 0\}$, the relevant Gaussian expectations have explicit expressions (the Cho–Saul / arc-cosine kernel).

Proposition 2.5 (Cho–Saul Kernel for ReLU)**.** *Let (h^α, h^β) be a centered Gaussian pair with*

$$\mathbb{E}[(h^\alpha)^2] = \Phi^{\alpha\alpha}, \qquad \mathbb{E}[(h^\beta)^2] = \Phi^{\beta\beta}, \qquad \mathbb{E}[h^\alpha h^\beta] = \Phi^{\alpha\beta}. \tag{2.10}$$

Assume $\Phi^{\alpha\alpha}, \Phi^{\beta\beta} > 0$ and define the correlation

$$\rho := \frac{\Phi^{\alpha\beta}}{\sqrt{\Phi^{\alpha\alpha}\Phi^{\beta\beta}}} \in [-1, 1]. \tag{2.11}$$

Then

$$\mathbb{E}[\varphi(h^\alpha)\varphi(h^\beta)] = \frac{\sqrt{\Phi^{\alpha\alpha}\Phi^{\beta\beta}}}{2\pi}\left(\sqrt{1-\rho^2} + \rho\left(\frac{\pi}{2} + \arcsin(\rho)\right)\right), \tag{2.12}$$

$$\mathbb{E}[\varphi'(h^\alpha)\varphi'(h^\beta)] = \mathbb{P}(h^\alpha > 0,\, h^\beta > 0) = \frac{1}{4} + \frac{1}{2\pi}\arcsin(\rho). \tag{2.13}$$

Proof. By positive homogeneity of ReLU, it is enough to work with a jointly standard Gaussian pair (z_1, z_2) with correlation ρ and restore the marginal variances at the end. First suppose $|\rho| < 1$. Let w_1, w_2 be independent standard Gaussians and realize $z_1 = w_1$ and $z_2 = \rho w_1 + \sqrt{1-\rho^2}\, w_2$. Let f and F denote the standard Gaussian density and distribution function, and write $p(\rho) := \mathbb{P}(z_1 > 0,\, z_2 > 0)$. Conditioning on w_1 gives

$$p(\rho) = \int_0^\infty f(u) F\left(\frac{\rho u}{\sqrt{1-\rho^2}}\right) du. \tag{2.14}$$

Differentiation under the integral is justified on compact subintervals of $(-1, 1)$ by Gaussian domination, and yields

$$p'(\rho) = \frac{1}{2\pi(1-\rho^2)^{3/2}} \int_0^\infty u \exp\left(-\frac{u^2}{2(1-\rho^2)}\right) du = \frac{1}{2\pi\sqrt{1-\rho^2}}. \tag{2.15}$$

Since $p(0) = 1/4$, integration gives the orthant probability stated in (2.13). Because $\varphi'(z) = \mathbf{1}_{\{z>0\}}$ almost everywhere, this proves the derivative-kernel formula for $|\rho| < 1$.

We now compute the expectation in (2.12). Let f_ρ be the joint density of (z_1, z_2). For dummy coordinates u, v, direct differentiation gives $u f_\rho(u, v) = -\partial_u f_\rho(u, v) - \rho\, \partial_v f_\rho(u, v)$. Multiply this identity by v and integrate over the positive quadrant. Gaussian decay removes the boundary terms at infinity, and the factor v removes the boundary term at $v = 0$. The u-integration leaves the boundary at $u = 0$; integration by parts in v leaves $p(\rho)$. Consequently,

$$\mathbb{E}[\varphi(z_1)\varphi(z_2)] = \int_0^\infty v f_\rho(0, v)\, dv + \rho p(\rho) = \frac{\sqrt{1-\rho^2}}{2\pi} + \rho p(\rho). \tag{2.16}$$

The boundary integral is an elementary one-dimensional Gaussian integral. Substituting the orthant probability already obtained gives the bracketed expression in (2.12) for the standard pair.

Finally, $(h^\alpha, h^\beta) \overset{d}{=} (\sqrt{\Phi^{\alpha\alpha}}\, z_1, \sqrt{\Phi^{\beta\beta}}\, z_2)$. Positive homogeneity therefore multiplies the ReLU expectation by $\sqrt{\Phi^{\alpha\alpha}\Phi^{\beta\beta}}$, while positive rescaling leaves the signs unchanged. This proves both stated formulas for $|\rho| < 1$. The cases $\rho = \pm 1$ follow from the same coupling: bounded convergence applies to the indicator product, and the Lipschitz property of ReLU together with Gaussian second moments gives continuity of the ReLU-product expectation. □

Remark 2.6. Formulas (2.12)–(2.13) go back to Cho and Saul's arc-cosine kernels [14] (and by the Transfer Principle hold verbatim internally for centered ${}^*\mathcal{N}(0, \Sigma)$ pairs with $\Sigma^{\alpha\alpha}, \Sigma^{\beta\beta} \in {}^*\mathbb{R}_{>0}$). In Clifford algebra $\mathcal{C}\ell_{n_0}$ [29, 18], they have an immediate rotor origin: for unit vectors $u_1, u_2 \in \mathbb{R}^{n_0}$ with correlation $\rho = \cos\vartheta$ ($\vartheta \in [0, \pi]$), their geometric product is the spinor rotor

$$u_1 u_2 = \langle u_1 u_2 \rangle_0 + u_1 \wedge u_2 = \rho + \sqrt{1-\rho^2}\, \hat{B}, \qquad \hat{B}^2 = -1,$$

where $\hat{B}$ is the unit bivector of the (u_1, u_2)-plane. Projecting $w \sim \mathcal{N}(0, I_{n_0})$ onto $\mathrm{span}\{u_1, u_2\}$ writes the planar Gaussian in rotor polar coordinates $r e^{\phi \hat{B}} u_1$, so the active wedge $\{z_1 > 0,\, z_2 > 0\}$ has aperture $\pi - \vartheta = \frac{\pi}{2} + \arcsin\rho$, yielding (2.13) as $(\pi - \vartheta)/(2\pi)$ and (2.12) from the scalar and bivector components of $\int_{-\pi/2}^{\pi/2-\vartheta} \cos\phi \cos(\phi + \vartheta)\, d\phi$. They are also the standard closed forms used to compute the limiting NNGP/NTK kernels for ReLU networks.

Remark 2.7 (Bayesian View). At unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, random initialization induces a standard-part GP prior $f^\circ \sim \mathrm{GP}(0, \Phi)$ under $L({}^*\mathbb{P})$. This motivates GP-style Bayesian inference as an approximation to Bayesian neural nets.

2.2 Extension to Deep Networks: Kernel Recursion

To extend the NNGP construction beyond one hidden layer, fix a standard depth $d \in \mathbb{N}$ with $d \geq 2$ and an unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ for each hidden layer. For standard data points $\{x^\alpha\}_{\alpha=1}^m \subset \mathbb{R}^{n_0}$ (with $m, n_0 \in \mathbb{N}$ standard), define an internal multilayer perceptron (MLP) by

$$h_1^\alpha := \frac{1}{\sqrt{n_0}} W_0 x^\alpha, \qquad W_0 \in {}^*\mathbb{R}^{N \times n_0}, \tag{2.17}$$

$$h_{\ell+1}^\alpha := \frac{1}{\sqrt{N}} W_\ell \, {}^*\varphi(h_\ell^\alpha), \qquad W_\ell \in {}^*\mathbb{R}^{N \times N}, \quad \ell = 1, \ldots, d-1, \tag{2.18}$$

$$f(x^\alpha; \theta) := \frac{1}{\sqrt{N}} W_d \, {}^*\varphi(h_d^\alpha), \qquad W_d \in {}^*\mathbb{R}^{1 \times N}. \tag{2.19}$$

All weight entries are internally i.i.d. ${}^*\mathcal{N}(0,1)$ on $(\Omega, \mathcal{A}, {}^*\mathbb{P})$ and internally independent across layers, with ${}^*\varphi$ applied coordinatewise.

We will repeatedly use the following standard Gaussian facts, all of which carry over verbatim to arbitrary hyperfinite dimensions $N, M \in {}^*\mathbb{N}$ (with ${}^*\mathcal{N}$ in place of $\mathcal{N}$) by the Transfer Principle.

Proposition 2.8 (Linear Images of Gaussians)**.** *If $g \sim \mathcal{N}(\mu, \Sigma)$ in $\mathbb{R}^n$ and $A \in \mathbb{R}^{m \times n}$, then $Ag \sim \mathcal{N}(A\mu, A\Sigma A^\top)$.*

Proposition 2.9 (Inner Products with a Standard Gaussian)**.** *If $g \sim \mathcal{N}(0, I_n)$ and $u, v \in \mathbb{R}^n$, then*

$$\begin{pmatrix} \langle g, u \rangle \\ \langle g, v \rangle \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} \|u\|^2 & \langle u, v \rangle \\ \langle u, v \rangle & \|v\|^2 \end{pmatrix}\right). \tag{2.20}$$

Proposition 2.10 (Multiplying by a Gaussian Matrix)**.** *Let $W \in \mathbb{R}^{n \times n}$ have i.i.d. entries $W_{ij} \sim \mathcal{N}(0,1)$ and let $u, v \in \mathbb{R}^n$ be deterministic. Then*

$$\begin{pmatrix} Wu \\ Wv \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} \|u\|^2 I_n & \langle u, v \rangle I_n \\ \langle u, v \rangle I_n & \|v\|^2 I_n \end{pmatrix}\right) = \mathcal{N}\left(0, \begin{pmatrix} \|u\|^2 & \langle u, v \rangle \\ \langle u, v \rangle & \|v\|^2 \end{pmatrix} \otimes I_n\right). \tag{2.21}$$

Definition 2.11 (Kronecker Product)**.** If $A = [a_{ij}] \in \mathbb{R}^{p \times q}$ and $B \in \mathbb{R}^{r \times s}$ (or internally for $p, q, r, s \in {}^*\mathbb{N}$), the Kronecker product $A \otimes B \in \mathbb{R}^{pr \times qs}$ is defined blockwise by

$$A \otimes B := [a_{ij} B]_{i,j}. \tag{2.22}$$

Define the deterministic standard input Gram matrix

$$\Phi_0^{\alpha\beta} := \frac{1}{n_0} \left\langle x^\alpha, x^\beta \right\rangle, \tag{2.23}$$

and set $\Phi_0^{(N)} := \Phi_0$. For $\ell \geq 1$, define the internal empirical feature covariance matrix by

$$\Phi_\ell^{(N),\alpha\beta} := \frac{1}{N} \left\langle {}^*\varphi(h_\ell^\alpha), {}^*\varphi(h_\ell^\beta) \right\rangle. \tag{2.24}$$

Let $\mathcal{F}_0$ be the trivial internal σ-field, and for $\ell \geq 1$ define the internal filtration (history)

$$\mathcal{F}_\ell := {}^*\sigma\big(\{h_k^\alpha : \alpha \in [m],\, k \leq \ell\}\big). \tag{2.25}$$

Thus $\mathcal{F}_\ell$ records the internal hidden-neuron history through layer ℓ; by internal independence across layers, $W_\ell, \dots, W_d$ are internally independent of $\mathcal{F}_\ell$. Conditioned on $\mathcal{F}_\ell$, the next-layer preactivations are internally Gaussian because $h^\alpha_{\ell+1}$ is an internal linear image of the Gaussian matrix W_ℓ. More precisely, by Transfer of Proposition 2.10, for $\ell = 0, \dots, d-1$, the m-tuple $(h^\alpha_{\ell+1})_{\alpha=1}^m$ (each in ${}^*\mathbb{R}^N$) is internally conditionally Gaussian with covariance of the form

$$(h^\alpha_{\ell+1})_{\alpha=1}^m \,\Big|\, \mathcal{F}_\ell \sim {}^*\mathcal{N}(0, \Phi^{(N)}_\ell \otimes I_N). \tag{2.26}$$

The key observation is that $\Phi^{(N)}_{\ell+1}$ is, up to infinitesimal fluctuations at unlimited width N, a deterministic function of $\Phi^{(N)}_\ell$. Indeed, by definition

$$\Phi^{(N),\alpha\beta}_{\ell+1} = \frac{1}{N}\left\langle {}^*\varphi(h^\alpha_{\ell+1}), {}^*\varphi(h^\beta_{\ell+1})\right\rangle = \frac{1}{N}\sum_{j=1}^N {}^*\varphi(h^\alpha_{\ell+1,j})\, {}^*\varphi(h^\beta_{\ell+1,j}). \tag{2.27}$$

Conditioned on $\mathcal{F}_\ell$, the neuron slices $\{(h^\alpha_{\ell+1,j})_{\alpha=1}^m\}_{j=1}^N$ are internally i.i.d. Gaussians in ${}^*\mathbb{R}^m$ with covariance $\Phi^{(N)}_\ell$ (this is the meaning of (2.26)). Under the assumptions of Theorem 2.13, for each fixed pair (α, β),

$$\Phi^{(N),\alpha\beta}_{\ell+1} - {}^*\mathcal{C}_\varphi(\Phi^{(N)}_\ell)^{\alpha\beta} \overset{L({}^*\mathbb{P})}{\approx} 0, \tag{2.28}$$

where the deterministic covariance map $\mathcal{C}_\varphi$ (with natural extension ${}^*\mathcal{C}_\varphi$ by Transfer) is given entrywise by a Gaussian expectation:

$$\mathcal{C}_\varphi(\Phi)^{\alpha\beta} := \mathbb{E}[\varphi(g^\alpha)\varphi(g^\beta)], \qquad (g^1, \dots, g^m) \sim \mathcal{N}(0, \Phi). \tag{2.29}$$

Define the deterministic standard forward kernels recursively by

$$\Phi_{\ell+1} := \mathcal{C}_\varphi(\Phi_\ell), \qquad \ell = 0, \dots, d-1. \tag{2.30}$$

The theorem below makes precise that $\Phi^{(N)}_\ell \overset{L({}^*\mathbb{P})}{\approx} \Phi_\ell$ for each standard $\ell \le d$.

Remark 2.12 (Order of Limits). For deep networks, classical statements like "$n \to \infty$" implicitly hold depth fixed; if depth grows too, the order of limits matters, since taking $n \to \infty$ first and then $d \to \infty$ can differ from a joint limit. In nonstandard analysis, "holding depth fixed at infinite width" literally means that the depth $d \in \mathbb{N}$ is standard while the hidden width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ is unlimited. If instead depth $D \in {}^*\mathbb{N} \setminus \mathbb{N}$ is also unlimited, then sequential and joint limits become distinct hyperfinite regimes in $({}^*\mathbb{N})^2$: $D/N \approx 0$ (width-dominant), $N/D \approx 0$ (depth-dominant), and $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$ (the proportional regime studied in Chapters 5 and 6).

Theorem 2.13 (Deep NNGP Recursion [38, 46])**.** *Assume $\varphi : \mathbb{R} \to \mathbb{R}$ is continuous and has at most polynomial growth. Fix standard depth $d \in \mathbb{N}$, the standard number of inputs $m \in \mathbb{N}$, and the standard inputs $\{x^\alpha\}_{\alpha=1}^m \subset \mathbb{R}^{n_0}$ themselves. Then, at any unlimited hidden width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$,*

$$\Phi^{(N)}_\ell \overset{L({}^*\mathbb{P})}{\approx} \Phi_\ell \qquad \text{for each } \ell = 0, 1, \dots, d, \tag{2.31}$$

where the deterministic matrices Φ_ℓ are defined by (2.30). Consequently, conditioned on the inputs, the output vector $\big(f(x^\alpha;\theta)\big)_{\alpha=1}^m$ belongs to $\mathbb{L}^m$ $L({}^*\mathbb{P})$-almost surely and its standard part $\mathrm{st}\big(f(x^1;\theta), \dots, f(x^m;\theta)\big)$ has exact Gaussian law $\mathcal{N}(0, \Phi_d)$ under $L({}^*\mathbb{P})$.

Proof Sketch. The internal conditional law (2.26) implies that, given $\mathcal{F}_\ell$, the coordinates $\{(h^\alpha_{\ell+1,j})_{\alpha=1}^m\}_{j=1}^N$ are internally i.i.d. copies of a centered Gaussian vector in ${}^*\mathbb{R}^m$ with covariance $\Phi_\ell^{(N)}$. Therefore each entry of $\Phi_{\ell+1}^{(N)}$ is a hyperfinite empirical average of conditionally internally i.i.d. random variables,

$$\Phi_{\ell+1}^{(N),\alpha\beta} = \frac{1}{N}\sum_{j=1}^{N} {}^*\varphi(h^\alpha_{\ell+1,j}){}^*\varphi(h^\beta_{\ell+1,j}), \tag{2.32}$$

whose internal conditional expectation is ${}^*\mathcal{C}_\varphi(\Phi_\ell^{(N)})^{\alpha\beta}$. Its internal conditional centered second moment is

$${}^*\mathbb{E}\left[\left|\Phi_{\ell+1}^{(N),\alpha\beta} - {}^*\mathcal{C}_\varphi\big(\Phi_\ell^{(N)}\big)^{\alpha\beta}\right|^2 \,\middle|\, \mathcal{F}_\ell\right] = \frac{1}{N}\,{}^*\mathrm{Var}_{g\sim{}^*\mathcal{N}(0,\Phi_\ell^{(N)})}\Big({}^*\varphi(g^\alpha){}^*\varphi(g^\beta)\Big). \tag{2.33}$$

We proceed by external induction over standard $\ell = 0, 1, \ldots, d$, starting from $\Phi_0^{(N)} = \Phi_0$. Polynomial growth of φ and the induction hypothesis $\Phi_\ell^{(N)} \overset{L(^*\mathbb{P})}{\approx} \Phi_\ell$ keep the internal conditional Gaussian variance on the right-hand side of (2.33) limited ($\in \mathbb{L}$) $L(^*\mathbb{P})$-almost surely; dividing by $N \in {}^*\mathbb{N}\setminus\mathbb{N}$ makes the conditional second moment infinitesimal (≈ 0), so internal Chebyshev's inequality yields (2.28). Continuity and polynomial growth also make $\mathcal{C}_\varphi$ continuous on bounded subsets of the standard positive semidefinite cone, so ${}^*\mathcal{C}_\varphi$ is S-continuous at Φ_ℓ: $\Phi_\ell^{(N)} \overset{L(^*\mathbb{P})}{\approx} \Phi_\ell$ gives ${}^*\mathcal{C}_\varphi(\Phi_\ell^{(N)}) \overset{L(^*\mathbb{P})}{\approx} \mathcal{C}_\varphi(\Phi_\ell) = \Phi_{\ell+1}$. Combining with (2.28) completes the external induction over the finitely many standard layers $\ell = 0, 1, \ldots, d$, giving $\Phi_\ell^{(N)} \overset{L(^*\mathbb{P})}{\approx} \Phi_\ell$ entrywise and hence as an $m\times m$ matrix (and $\Phi_d^{(N)} \approx \Phi_d$ $L(^*\mathbb{P})$-almost surely by countable additivity of $L(^*\mathbb{P})$). Finally, since W_d is internally independent of $\mathcal{F}_d$, the output-layer identity (2.3) gives, conditional on $\mathcal{F}_d$, the exact internal Gaussian output law ${}^*\mathcal{N}(0, \Phi_d^{(N)})$; Loeb bounded convergence of the conditional characteristic function then yields the stated standard-part law $\mathcal{N}(0, \Phi_d)$. □

Remark 2.14 (Sequential versus Simultaneous Width Limits). The main derivation of Lee et al. [38] takes the hidden-layer widths to infinity successively, while their Appendix C gives an order-independent Gaussian-prior argument. Matthews et al. [46] give a rigorous simultaneous-width convergence theorem. In the internal Gaussian-weight, standard-depth, standard-finite-dataset setting used here, exact internal conditional Gaussianity at a single unlimited width $N \in {}^*\mathbb{N}\setminus\mathbb{N}$ leads directly to the common-width covariance recursion above without iterated limits.

2.3 Neural Tangent Kernel and Linearized Training

If we train the wide neural networks introduced above in the NTK regime, their output dynamics behave like those of kernel methods. The key object is the Neural Tangent Kernel (NTK), a kernel induced by the network's tangent features. At unlimited width $N \in {}^*\mathbb{N}\setminus\mathbb{N}$, the empirical NTK becomes, up to infinitesimals, (i) deterministic and (ii) constant in time along gradient flow on standard time intervals, yielding closed-form linear standard-part training dynamics and, when the limiting kernel is positive definite, exponential convergence of the training loss. We then extend the construction from one-hidden-layer networks to finite-depth multilayer perceptrons (MLPs) and highlight the geometric interpretation as a preconditioned gradient

flow in function space. We will reuse the forward-kernel notation and NNGP recursions from earlier sections.

The Neural Tangent Kernel was introduced by Jacot et al. [33]; Lee et al. [39] made explicit the associated linearization of wide-network training. A closely related line of work provides quantitative convergence guarantees for overparameterized deep networks; see, for instance, Allen-Zhu et al. [1], Du et al. [19], and Zou et al. [65]. In these notes we emphasize the hyperfinite infinite-width picture and use these works primarily as pointers to the finite-width quantitative theory.

Let the standard training dataset (with $m, n_0 \in \mathbb{N}$ standard) be

$$\{(x^\alpha, y^\alpha)\}_{\alpha=1}^m, \qquad x^\alpha \in \mathbb{R}^{n_0}, \quad y^\alpha \in \mathbb{R}. \tag{2.34}$$

Let $f(\cdot;\theta) : {}^*\mathbb{R}^{n_0} \to {}^*\mathbb{R}$ be an internal neural network with parameter vector $\theta \in {}^*\mathbb{R}^{\dim(\theta)}$. We write $f_\alpha(\theta) := f(x^\alpha;\theta)$ and collect outputs into the internal vector

$$f(\theta) := \big(f_1(\theta), \dots, f_m(\theta)\big)^\top \in {}^*\mathbb{R}^m, \qquad y := (y^1, \dots, y^m)^\top \in \mathbb{R}^m. \tag{2.35}$$

We will often use the internal empirical mean-squared error

$$L(\theta) := \frac{1}{2m} \sum_{\alpha=1}^m \big(f_\alpha(\theta) - y^\alpha\big)^2 = \frac{1}{2m} \, \|f(\theta) - y\|_2^2 \,. \tag{2.36}$$

Define the internal residual vector $r(\theta) := f(\theta) - y \in {}^*\mathbb{R}^m$.

Internal gradient descent with step size η is

$$\theta_{k+1} = \theta_k - \eta \, \nabla_\theta L(\theta_k), \tag{2.37}$$

where ∇_θ denotes the internal parameter gradient (obtained by Transfer). A convenient continuous-time counterpart is internal gradient flow, defined for $t \in {}^*\mathbb{R}_{\geq 0}$ by Transfer of the standard ODE:

$$\partial_t \theta(t) = -\nabla_\theta L(\theta(t)), \qquad \theta(0) = \theta_0, \tag{2.38}$$

where $\partial_t \theta(t) = \frac{d}{dt}\theta(t)$ in the internal sense. In this chapter we analyze the induced dynamics of the network outputs $f(t) := f(\theta(t))$.

We now encode the network's parameter sensitivities through its internal Dirac tangent feature map in ${}^*\mathcal{C}\ell_{\dim(\theta)}$. Equipping parameter space ${}^*\mathbb{R}^{\dim(\theta)}$ with the vector derivative (Dirac operator) $\partial_\theta := \sum_{p=1}^{\dim(\theta)} e_p \partial_{\theta_p}$, define for each input x the tangent 1-vector at parameters θ by

$$\partial_\theta f(x;\theta) = \nabla_\theta f(x;\theta) \in {}^*\mathbb{R}^{\dim(\theta)} \hookrightarrow {}^*\mathcal{C}\ell_{\dim(\theta)}. \tag{2.39}$$

In ${}^*\mathcal{C}\ell_{\dim(\theta)}$, the full geometric tangent product decomposes into its symmetric scalar inner product and antisymmetric bivector exterior part:

$$(\partial_\theta f_\alpha(\theta))^\sim (\partial_\theta f_\beta(\theta)) = \langle (\partial_\theta f_\alpha(\theta))^\sim (\partial_\theta f_\beta(\theta)) \rangle_0 + \partial_\theta f_\alpha(\theta) \wedge \partial_\theta f_\beta(\theta).$$

For a hyperfinite width-N network, the (empirical) internal Neural Tangent Kernel at time $t \in {}^*\mathbb{R}_{\geq 0}$ is the $m \times m$ scalar-grade matrix

$$\begin{aligned} K_t^{(N)} \in {}^*\mathbb{R}^{m\times m}, \qquad K_t^{(N),\alpha\beta} &:= K_t^{(N)}(x^\alpha, x^\beta) \\ &:= \langle (\partial_\theta f_\alpha(\theta(t)))^\sim (\partial_\theta f_\beta(\theta(t))) \rangle_0 = \langle \nabla_\theta f_\alpha(\theta(t)), \nabla_\theta f_\beta(\theta(t)) \rangle \,, \end{aligned} \tag{2.40}$$

while the bivector grade $\Omega_t^{(N),\alpha\beta} := \partial_\theta f_\alpha(\theta(t)) \wedge \partial_\theta f_\beta(\theta(t)) \in \bigwedge^2({}^*\mathbb{R}^{\dim(\theta)})$ cancels identically in symmetric quadratic forms such as risk dissipation (2.45). By Transfer, $K_t^{(N)}$ is internally symmetric positive semidefinite (PSD).

By the Transfer Principle, a chain-rule computation yields a closed-form internal ODE for $r(t)$ in terms of $K_t^{(N)}$.

Proposition 2.15 (Residual ODE)**.** *Under internal gradient flow for the squared loss, the residual vector $r(t) = f(t) - y \in {}^*\mathbb{R}^m$ satisfies, for all $t \in {}^*\mathbb{R}_{\geq 0}$,*

$$\partial_t r(t) = -\frac{1}{m} K_t^{(N)} r(t). \tag{2.41}$$

Proof. By Transfer of the chain rule, for each $\alpha \in [m]$,

$$\partial_t f_\alpha(t) = \langle \nabla_\theta f_\alpha(\theta(t)), \partial_t \theta(t) \rangle = -\langle \nabla_\theta f_\alpha(\theta(t)), \nabla_\theta L(\theta(t)) \rangle . \tag{2.42}$$

For the squared loss, $\nabla_\theta L(\theta) = \frac{1}{m} \sum_{\beta=1}^m (f_\beta(\theta) - y^\beta) \nabla_\theta f_\beta(\theta)$. Substituting gives

$$\partial_t f_\alpha(t) = -\frac{1}{m} \sum_{\beta=1}^m \langle \nabla_\theta f_\alpha(\theta(t)), \nabla_\theta f_\beta(\theta(t)) \rangle \left(f_\beta(t) - y^\beta \right) = -\frac{1}{m} \sum_{\beta=1}^m K_t^{(N),\alpha\beta} r_\beta(t). \tag{2.43}$$

In vector form this is (2.41). □

Remark 2.16. Equation (2.41) shows that each residual coordinate $r_\alpha(t)$ mixes information from all residuals through the internal kernel matrix $K_t^{(N)}$. If $K_t^{(N)}$ were replaced by a fixed standard kernel matrix K, then training would become a linear ODE in r with an explicit solution via a matrix exponential. The main content of the NTK theory is that at unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, $K_t^{(N)} \approx K$ uniformly on standard time intervals, making this replacement exact on standard parts.

This framework easily extends to general loss functions. For a differentiable pointwise loss $\ell(f, y)$ (extended internally via ${}^*\ell$), define the internal empirical risk

$$L(\theta) := \frac{1}{m} \sum_{\alpha=1}^m {}^*\ell(f_\alpha(\theta), y^\alpha). \tag{2.44}$$

Along the internal training trajectory, define $r_\alpha(t) := \partial_f {}^*\ell(f_\alpha(t), y^\alpha)$ and collect these entries in $r(t) \in {}^*\mathbb{R}^m$. For squared loss, this agrees with the residual $f(t) - y$ defined above.

Proposition 2.17 (Risk Dissipation Identity)**.** *Along internal gradient flow for L,*

$$\frac{d}{dt} L(\theta(t)) = -\frac{1}{m^2} r(t)^\top K_t^{(N)} r(t). \tag{2.45}$$

Proof. By Transfer, the chain rule gives

$$\frac{d}{dt} L(\theta(t)) = \frac{1}{m} \sum_{\alpha=1}^m r_\alpha(t)\, \partial_t f_\alpha(t). \tag{2.46}$$

Moreover,

$$\nabla_\theta L(\theta(t)) = \frac{1}{m} \sum_{\beta=1}^m r_\beta(t)\, \nabla_\theta f_\beta(\theta(t)). \tag{2.47}$$

Substituting $\partial_t f_\alpha(t) = -\langle \nabla_\theta f_\alpha(\theta(t)), \nabla_\theta L(\theta(t)) \rangle$ and using (2.40) yields (2.45). □

2.4 The One-Layer Case

The empirical NTK $K_t^{(N)}$ is random at initialization (because the parameters are random) and evolves during training (because $\theta(t)$ changes). At unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, the NTK regime exhibits two related features:

(i) *Determinism:* for every standard training time $t \geq 0$, the empirical kernel $K_t^{(N)}(x, x')$ concentrates infinitesimally around a deterministic standard kernel $K(x, x')$.

(ii) *Stationarity:* the limiting standard kernel K does not depend on t.

We start with the simplest case to illustrate the mechanism.

Definition 2.18 (One-Hidden-Layer MLP (NTK Parameterization))**.** Fix an unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ and standard input dimension $n_0 \in \mathbb{N}$. Let

$$f(x;\theta) = \frac{1}{\sqrt{N}} a^\top {}^*\varphi\left(\frac{Wx}{\sqrt{n_0}}\right), \qquad W \in {}^*\mathbb{R}^{N\times n_0}, \quad a \in {}^*\mathbb{R}^N, \tag{2.48}$$

where $\varphi : \mathbb{R} \to \mathbb{R}$ (extended via ${}^*\varphi$) is applied elementwise to vectors. We take internally i.i.d. Gaussian initialization $W_{ij} \sim {}^*\mathcal{N}(0,1)$ and $a_i \sim {}^*\mathcal{N}(0,1)$ independently on $(\Omega, \mathcal{A}, {}^*\mathbb{P})$.

Remark 2.19 (Bias Parameters). For notational simplicity we omit biases throughout this set of notes. Including bias terms does not change any of the core ideas (infinitesimal concentration of kernels, near-stationarity of the tangent kernel, etc.); it mainly makes the algebra longer and more tedious in an uninformative way.

Theorem 2.20 (One-Hidden-Layer NTK Limit (Informal))**.** *Consider the width-N network ($N \in {}^*\mathbb{N} \setminus \mathbb{N}$) and NTK parameterization in (2.48), with the internal Gaussian initialization above, trained by internal gradient flow for squared loss on a fixed standard finite dataset. Write $K_s^{(N)}$ for its empirical NTK at internal time $s \in {}^*\mathbb{R}_{\geq 0}$. For sufficiently nice φ, the NTK regime [33, 39] gives a deterministic standard symmetric positive semidefinite matrix $K \in \mathbb{R}^{m\times m}$, independent of t, such that for every standard $t \geq 0$ (identifying t with ${}^*t \in {}^*\mathbb{R}_{\geq 0}$),*

$$\sup_{0\leq s\leq {}^*t} \left\| K_s^{(N)} - K \right\| \overset{L({}^*\mathbb{P})}{\approx} 0, \qquad \mathrm{st}\big(K_s^{(N)}\big) = K \quad (0 \leq s \leq {}^*t \ L({}^*\mathbb{P})\text{-a.s.}). \tag{2.49}$$

The standard-part residual dynamics satisfy

$$\partial_t \,\mathrm{st}\big(r(t)\big) = -\frac{1}{m} K \,\mathrm{st}\big(r(t)\big), \qquad \text{so} \qquad \mathrm{st}\big(r(t)\big) = \exp\left(-\frac{t}{m}K\right) \mathrm{st}\big(r(0)\big). \tag{2.50}$$

Proof Sketch. Fix a standard finite dataset $\{x^\alpha\}_{\alpha=1}^m$.

Step 1: Determinism. Writing $w_i^\top$ for the i-th row of W, for any internal parameters θ (and in particular at θ_0), the neuronwise internal gradients for a single datum x^α are

$$\frac{\partial f_\alpha}{\partial a_i} = \frac{1}{\sqrt{N}} {}^*\varphi(h_i^\alpha), \qquad \frac{\partial f_\alpha}{\partial w_i} = \frac{1}{\sqrt{Nn_0}} a_i \, {}^*\varphi'(h_i^\alpha) x^\alpha, \qquad h_i^\alpha := \frac{1}{\sqrt{n_0}} w_i^\top x^\alpha. \tag{2.51}$$

Plugging (2.51) into $K_0^{(N),\alpha\beta} = \langle \nabla_\theta f_\alpha(\theta_0), \nabla_\theta f_\beta(\theta_0) \rangle$ yields the exact internal decomposition

$$K_0^{(N),\alpha\beta} = \frac{1}{N} \sum_{i=1}^N \kappa_i^{\alpha\beta}, \qquad \kappa_i^{\alpha\beta} := {}^*\varphi(h_i^\alpha) {}^*\varphi(h_i^\beta) + a_i^2 \, {}^*\varphi'(h_i^\alpha) {}^*\varphi'(h_i^\beta) \frac{1}{n_0} \left\langle x^\alpha, x^\beta \right\rangle, \tag{2.52}$$

where $\kappa_i^{\alpha\beta}$ is evaluated at initialization $\theta = \theta_0$. Because (a_i, w_i) are internally i.i.d. at Gaussian initialization, for each fixed pair (α, β) the internal random variables $\{\kappa_i^{\alpha\beta}\}_{i=1}^N$ are internally i.i.d., and hence, provided the standard first moment $\mathbb{E}\left|\kappa_1^{\alpha\beta}\right| < \infty$, the hyperfinite law of large numbers gives $K_0^{(N),\alpha\beta} \overset{L(^*\mathbb{P})}{\approx} K^{\alpha\beta}$ (in fact $L(^*\mathbb{P})$-almost surely) for each fixed (α, β). Since $m \in \mathbb{N}$ is standard, this entrywise infinitesimal concentration implies $\left\|K_0^{(N)} - K\right\| \overset{L(^*\mathbb{P})}{\approx} 0$ in any standard matrix norm. Moreover, K is positive semidefinite because each $K_0^{(N)}$ is internally positive semidefinite and taking standard parts preserves $u^\top K_0^{(N)} u \geq 0$ for all $u \in \mathbb{R}^m$.

The neuronwise form in (2.52) is convenient for the law-of-large-numbers argument and will also be used in the stationarity estimate below. To display separately the contributions of the two parameter blocks, write $h^\alpha = (h_1^\alpha, \ldots, h_N^\alpha)^\top \in {}^*\mathbb{R}^N$. Then the same identity becomes

$$\begin{aligned} K_0^{(N),\alpha\beta} &= \frac{1}{N}\left\langle {}^*\varphi(h^\alpha), {}^*\varphi(h^\beta)\right\rangle + \Phi_0^{\alpha\beta}\frac{1}{N}\left\langle a \odot {}^*\varphi'(h^\alpha), a \odot {}^*\varphi'(h^\beta)\right\rangle \\ &\overset{L(^*\mathbb{P})}{\approx} K^{\alpha\beta} = \Phi_1^{\alpha\beta} + \Phi_0^{\alpha\beta}\Phi_1'^{,\alpha\beta}. \end{aligned} \tag{2.53}$$

Here $\Phi_1'^{,\alpha\beta} = \mathbb{E}[\varphi'(g^\alpha)\varphi'(g^\beta)]$ for a centered standard Gaussian pair with covariance $[\Phi_0^{\gamma\delta}]_{\gamma,\delta\in\{\alpha,\beta\}}$.

Step 2: Stationarity on a standard time interval. Fix a standard $t \geq 0$. For the one-hidden-layer model, internal gradient flow gives, for $0 \leq s \leq {}^*t$,

$$\partial_s a_i(s) = -\frac{1}{m\sqrt{N}}\sum_{\gamma=1}^m r_\gamma(s)\,{}^*\varphi(h_i^\gamma(s)), \qquad \partial_s w_i(s) = -\frac{a_i(s)}{m\sqrt{Nn_0}}\sum_{\gamma=1}^m r_\gamma(s)\,{}^*\varphi'(h_i^\gamma(s))x^\gamma. \tag{2.54}$$

Along internal gradient flow, by Transfer, the decrease in loss $L(\theta_0) - L(\theta(s)) \leq L(\theta_0)$ equals the internal time integral $\int_0^s \|\partial_u\theta(u)\|_2^2\, du$ of the squared parameter velocity. Internal Cauchy–Schwarz therefore gives the empirical parameter displacement bound

$$\sup_{0\leq s\leq {}^*t}\frac{1}{N}\sum_{i=1}^N\Big[(a_i(s)-a_i(0))^2 + \|w_i(s)-w_i(0)\|_2^2\Big] \leq \frac{tL(\theta_0)}{N} \approx 0, \tag{2.55}$$

an infinitesimal of order N^{-1} $L(^*\mathbb{P})$-almost surely because $L(\theta_0) \in \mathbb{L}$ $L(^*\mathbb{P})$-almost surely under the Gaussian initialization on the fixed standard dataset. Let $\kappa_i^{\alpha\beta}(s)$ denote the expression in (2.52) evaluated along the internal flow at $s \in [0, {}^*t]$. An internal first-order Taylor expansion at initialization gives the next-order correction

$$\begin{aligned} K_s^{(N),\alpha\beta} - K_0^{(N),\alpha\beta} &= \frac{1}{N}\sum_{i=1}^N(\kappa_i^{\alpha\beta}(s) - \kappa_i^{\alpha\beta}(0)) \\ &= \frac{1}{N}\sum_{i=1}^N\Big[\partial_{a_i}\kappa_i^{\alpha\beta}(0)(a_i(s)-a_i(0)) \\ &\qquad + \left\langle \nabla_{w_i}\kappa_i^{\alpha\beta}(0), w_i(s)-w_i(0)\right\rangle\Big] + R_s^{(N),\alpha\beta}, \end{aligned} \tag{2.56}$$

with internal remainder satisfying $\sup_{0\leq s\leq {}^*t} N^{1/2}\left|R_s^{(N),\alpha\beta}\right| \overset{L(^*\mathbb{P})}{\approx} 0$. The derivatives in the

leading term are

$$\frac{\partial \kappa_i^{\alpha\beta}}{\partial a_i} = 2a_i {}^*\varphi'(h_i^\alpha)^*\varphi'(h_i^\beta)\frac{\left\langle x^\alpha, x^\beta \right\rangle}{n_0}, \tag{2.57}$$

$$\begin{aligned}\nabla_{w_i}\kappa_i^{\alpha\beta} = {} & \frac{x^\alpha}{\sqrt{n_0}}\left({}^*\varphi'(h_i^\alpha)^*\varphi(h_i^\beta) + a_i^{2*}\varphi''(h_i^\alpha)^*\varphi'(h_i^\beta)\frac{\left\langle x^\alpha, x^\beta \right\rangle}{n_0}\right) \\ & + \frac{x^\beta}{\sqrt{n_0}}\left({}^*\varphi(h_i^\alpha)^*\varphi'(h_i^\beta) + a_i^{2*}\varphi'(h_i^\alpha)^*\varphi''(h_i^\beta)\frac{\left\langle x^\alpha, x^\beta \right\rangle}{n_0}\right).\end{aligned} \tag{2.58}$$

For sufficiently nice φ, these derivatives have limited empirical second moments ($\in \mathbb{L}$) at initialization $L(^*\mathbb{P})$-almost surely. Internal Cauchy–Schwarz and (2.55) therefore bound the averaged linear Taylor term by an infinitesimal of scale $N^{-1/2}$ ($\in N^{-1/2}\mathbb{L}$), uniformly on $[0, {}^*t]$. The smoothness and moment assumptions also control the internal Taylor remainder $R_s^{(N),\alpha\beta}$ uniformly on $[0, {}^*t]$, giving the same $N^{-1/2}$-scale infinitesimal bound for the full correction $K_s^{(N),\alpha\beta} - K_0^{(N),\alpha\beta}$ in (2.56). Combining this with Step 1 gives (2.49), so $\mathrm{st}(K_s^{(N)}) = K$ uniformly for $0 \le s \le {}^*t$. Substituting $K_s^{(N)} \approx K$ into the internal time-integral form of (2.41) via internal Grönwall's inequality, together with the NNGP shadow for the initial residual $r(0)$, gives $r({}^*t) \approx {}^*\exp(-\frac{t}{m}K)r(0)$ and hence the standard-part residual dynamics (2.50). □

Remark 2.21 (Scope of the Proof Sketch). The calculation above displays the one-hidden-layer mechanism. For multiple hidden layers, an additional layerwise argument is required; we omit those details here. Nonsmooth activations such as ReLU require a different argument.

We now study the limiting fixed-kernel dynamics, in which the time-dependent empirical kernel in (2.41) is replaced by a fixed standard positive-definite matrix K. The resulting linear ODE has an explicit solution and converges exponentially.

Proposition 2.22 (Exponential Convergence of the Fixed-Kernel Dynamics)**.** *Let $K \in \mathbb{R}^{m\times m}$ be symmetric positive definite, and suppose $\partial_t r(t) = -m^{-1}Kr(t)$. Then*

$$r(t) = \exp\left(-\frac{t}{m}K\right) r(0), \tag{2.59}$$

and the squared loss $L(t) := (2m)^{-1}\left\|r(t)\right\|_2^2$ satisfies

$$L(t) \le \exp\left(-\frac{2\lambda_{\min}(K)}{m}t\right) L(0). \tag{2.60}$$

Thus the loss converges to zero exponentially.

Proof. By the spectral theorem, $K = P\,\mathrm{diag}(\lambda_1, \ldots, \lambda_m)P^\top$ for an orthogonal matrix P and eigenvalues $\lambda_i > 0$. The matrix exponential in (2.59) solves the constant-coefficient residual ODE. Moreover,

$$\left\|r(t)\right\|_2^2 = \sum_{i=1}^m e^{-2\lambda_i t/m}\left((P^\top r(0))_i\right)^2 \le e^{-2\lambda_{\min}(K)t/m}\left\|r(0)\right\|_2^2. \tag{2.61}$$

Dividing by $2m$ proves (2.60). □

Remark 2.23 (When Is $\lambda_{\min}(K) > 0$?). For a fixed standard dataset, positive definiteness is a property of the deterministic standard kernel K. If $\lambda_{\min}(K) > 0$, then $K_0^{(N)} \overset{L(^*\mathbb{P})}{\approx} K$ and continuity of $\lambda_{\min}$ imply $\lambda_{\min}(K_0^{(N)}) \overset{L(^*\mathbb{P})}{\approx} \lambda_{\min}(K) > 0$, so

$$^*\mathbb{P}(K_0^{(N)} \text{ is positive definite}) \approx 1, \qquad L(^*\mathbb{P})(K_0^{(N)} \text{ is positive definite}) = 1. \tag{2.62}$$

When the data are random, positivity and conditioning of K can themselves be studied probabilistically. See, for example, Nguyen et al. [49] for quantitative lower bounds on the smallest eigenvalue of empirical NTK matrices for deep ReLU networks under standard assumptions.

Remark 2.24 (Linearization and the Kernel Regression Viewpoint). The NTK regime can be viewed directly as an infinitesimal linearization of the network around initialization. Consider internal gradient descent with standard step size η: $\theta_{k+1} = \theta_k - \eta \nabla_\theta L(\theta_k)$. At unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, the hyperfinite NTK width counting gives

$$f_\alpha(\theta_{k+1}) = f_\alpha(\theta_k) + \langle \nabla_\theta f_\alpha(\theta_k), \theta_{k+1} - \theta_k \rangle + \underbrace{\varepsilon_k^\alpha}_{\in N^{-1/2}\mathbb{L},\ \text{nonlinear correction}}. \tag{2.63}$$

Here the $\varepsilon_k^\alpha \in N^{-1/2}\mathbb{L} \subset \mathbb{I}$ term summarizes the leading nonlinear correction, an infinitesimal of scale $N^{-1/2}$. Since this correction is infinitesimal and the tangent features remain infinitesimally frozen along standard-time training, we obtain the hyperreal linearization

$$f_\alpha(\theta) \approx f_\alpha(\theta_0) + \langle \nabla_\theta f_\alpha(\theta_0), \theta - \theta_0 \rangle . \tag{2.64}$$

Hence training becomes, up to infinitesimals, linear regression in the initial tangent features $\nabla_\theta f(x^\alpha; \theta_0)$. Rigorous NTK results make this linearization precise along the training trajectory [39].

A complementary geometric viewpoint is already visible in (2.41). When the standard kernel K is positive definite, equip the standard output space $\mathbb{R}^m$ with the inner product

$$\langle u, v \rangle_{K^{-1}} := u^\top K^{-1} v. \tag{2.65}$$

With respect to this inner product, the fixed-kernel standard-part dynamics become

$$\partial_t f = -\frac{1}{m} K(f - y) = -\operatorname{grad}_{K^{-1}} \left(\frac{1}{2m} \|f - y\|_2^2 \right). \tag{2.66}$$

On the fixed dataset, the space of predictor values is simply the linear vector space $\mathbb{R}^m$. Nevertheless, K^{-1} is precisely a constant, and hence flat, Riemannian metric on this space; equivalently, K acts as its inverse metric or as a preconditioner.

2.5 Extension to Finite-Depth MLPs

We now describe the exact internal layerwise structure of the NTK for a standard-depth-d fully-connected network of unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ and then compute its deterministic standard shadow at Gaussian initialization.

The finite-depth NTK is built from three internal layerwise quantities: the empirical forward kernels $\Phi_\ell^{(N)}$, the empirical derivative kernels $\Phi_\ell'^{(N)}$, and the backward covariances

$G_\ell^{(N)}$. At Gaussian initialization, the deterministic forward- and derivative-kernel shadows feed into a backward recursion for the shadows of the backward covariances. Substituting these shadows into the exact internal layerwise decomposition in Proposition 2.25 then yields the limiting standard NTK. Throughout this section, the superscript (N) denotes hyperfinite width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, and common training-time arguments are suppressed.

To analyze the finite-depth NTK, we use the following network notation. Let $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ be the unlimited hidden width and $n_0 \in \mathbb{N}$ the standard input dimension, with standard depth $d \in \mathbb{N}$. For $\ell = 0, 1, \ldots, d$, let W_ℓ be internal weight matrices with internally i.i.d. ${}^*\mathcal{N}(0,1)$ entries, where $W_0 \in {}^*\mathbb{R}^{N \times n_0}$, $W_\ell \in {}^*\mathbb{R}^{N \times N}$ for $1 \le \ell \le d-1$, and $W_d \in {}^*\mathbb{R}^{1 \times N}$ (scalar output). Define for standard input x^α the internal pre-activations

$$h_1^\alpha = \frac{1}{\sqrt{n_0}} W_0 x^\alpha, \qquad h_{\ell+1}^\alpha = \frac{1}{\sqrt{N}} W_\ell {}^*\varphi(h_\ell^\alpha) \quad (1 \le \ell \le d-1), \tag{2.67}$$

and output

$$f_\alpha = f(x^\alpha; \theta) = \frac{1}{\sqrt{N}} W_d {}^*\varphi(h_d^\alpha). \tag{2.68}$$

We first record the forward and derivative kernels, reusing the forward-kernel notation and deterministic recursion from Section 2.2. Define the standard input (zeroth-layer) kernel

$$\Phi_0^{\alpha\beta} := \frac{1}{n_0} \left\langle x^\alpha, x^\beta \right\rangle. \tag{2.69}$$

Recall that $\Phi_0^{(N)} = \Phi_0$. For $\ell \ge 1$, define the internal empirical forward and derivative kernels

$$\Phi_\ell^{(N),\alpha\beta} := \frac{1}{N} \left\langle {}^*\varphi(h_\ell^\alpha), {}^*\varphi(h_\ell^\beta) \right\rangle, \qquad \Phi_\ell'^{(N),\alpha\beta} := \frac{1}{N} \left\langle {}^*\varphi'(h_\ell^\alpha), {}^*\varphi'(h_\ell^\beta) \right\rangle. \tag{2.70}$$

Theorem 2.13 gives $\Phi_\ell^{(N)} \overset{L^{(*\mathbb{P})}}{\approx} \Phi_\ell$. Suppose additionally that φ' exists almost everywhere, its Gaussian-product second moments are uniformly bounded for covariance matrices in a standard neighborhood of the deterministic recursion, and the corresponding Gaussian expectations are continuous (hence S-continuous) there. Then the same internal conditional second-moment argument gives $\Phi_\ell'^{(N)} \overset{L^{(*\mathbb{P})}}{\approx} \Phi_\ell'$.

For $\ell \ge 0$, let (h^α, h^β) be the centered standard Gaussian pair whose covariance is determined by Φ_ℓ. The deterministic derivative-kernel shadow obeys $\Phi_{\ell+1}'^{,\alpha\beta} = \mathbb{E}[\varphi'(h^\alpha)\varphi'(h^\beta)]$. We write this compactly as $\Phi_{\ell+1}' = \mathcal{C}_{\varphi'}(\Phi_\ell)$, where $\mathcal{C}_{\varphi'}$ denotes the companion Gaussian expectation rather than the derivative of $\mathcal{C}_\varphi$.

For ReLU, the Cho–Saul formulas in Proposition 2.5 make the forward and derivative-kernel recursions explicit. For fixed ℓ, α, β with $\Phi_\ell^{\alpha\alpha}, \Phi_\ell^{\beta\beta} > 0$, write $\rho := \Phi_\ell^{\alpha\beta} / \sqrt{\Phi_\ell^{\alpha\alpha} \Phi_\ell^{\beta\beta}} \in [-1, 1]$. Then

$$\Phi_{\ell+1}^{\alpha\beta} = \frac{\sqrt{\Phi_\ell^{\alpha\alpha} \Phi_\ell^{\beta\beta}}}{2\pi} \left(\sqrt{1-\rho^2} + \rho \left(\frac{\pi}{2} + \arcsin(\rho) \right) \right), \qquad \Phi_{\ell+1}'^{,\alpha\beta} = \frac{1}{4} + \frac{1}{2\pi} \arcsin(\rho). \tag{2.71}$$

Thus the same correlation determines both the next forward kernel and the derivative factor that enters the backward recursion.

We next introduce the internal backpropagated sensitivities whose covariances complete the NTK decomposition. Fix an internal training time $t \in {}^*\mathbb{R}_{\geq 0}$ and evaluate all forward and backward quantities below at $\theta(t)$, suppressing their common time argument except on

$$K_t^{(N),\alpha\beta} := \langle \nabla_\theta f_\alpha(\theta(t)), \nabla_\theta f_\beta(\theta(t)) \rangle . \tag{2.72}$$

The additional state variables in the decomposition are the internal empirical backward covariances $G_\ell^{(N),\alpha\beta}$, which encode inner products of backpropagated sensitivities. At Gaussian initialization, the empirical forward kernels $\Phi_\ell^{(N),\alpha\beta}$ concentrate infinitesimally around the deterministic NNGP kernels $\Phi_\ell^{\alpha\beta}$ defined earlier. Define the internal backward sensitivity vectors

$$g_\ell^\alpha := \sqrt{N} \frac{\partial f_\alpha}{\partial h_\ell^\alpha} \in {}^*\mathbb{R}^N, \qquad 1 \leq \ell \leq d. \tag{2.73}$$

By Transfer, a direct backpropagation calculation gives the exact internal top-layer identity and hidden-layer recursion

$$g_d^\alpha = W_d^\top \odot {}^*\varphi'(h_d^\alpha), \tag{2.74}$$

$$g_\ell^\alpha = \frac{1}{\sqrt{N}} \operatorname{diag}({}^*\varphi'(h_\ell^\alpha)) W_\ell^\top g_{\ell+1}^\alpha, \qquad \ell = d-1, \ldots, 1. \tag{2.75}$$

Define the internal backward covariance matrices

$$G_\ell^{(N),\alpha\beta} := \frac{1}{N} \left\langle g_\ell^\alpha, g_\ell^\beta \right\rangle, \qquad \ell = 1, \ldots, d, \qquad G_{d+1}^{(N),\alpha\beta} := 1. \tag{2.76}$$

Proposition 2.25 (Layerwise NTK Decomposition)**.** *For the depth-d MLP defined above (with no biases), at every internal training time $t \in {}^*\mathbb{R}_{\geq 0}$ the empirical NTK on the dataset $\{x^\alpha\}_{\alpha=1}^m$ decomposes exactly as*

$$K_t^{(N),\alpha\beta} = \sum_{\ell=0}^{d} G_{\ell+1}^{(N),\alpha\beta} \Phi_\ell^{(N),\alpha\beta}, \tag{2.77}$$

where $\Phi_0^{(N)} = \Phi_0$, and $G_{d+1}^{(N),\alpha\beta} = 1$ corresponds to the top (linear) output layer.

In matrix form, viewing $\Phi_\ell^{(N)}, G_{\ell+1}^{(N)}, K_t^{(N)} \in {}^*\mathbb{R}^{m\times m}$, (2.77) reads

$$K_t^{(N)} = \sum_{\ell=0}^{d} G_{\ell+1}^{(N)} \odot \Phi_\ell^{(N)}, \tag{2.78}$$

where $\odot$ denotes the Hadamard (entrywise) product.

Proof. By Transfer, the internal NTK is an inner product of parameter gradients: $K_t^{(N),\alpha\beta} = \langle \nabla_\theta f_\alpha(\theta(t)), \nabla_\theta f_\beta(\theta(t)) \rangle$. Split θ layer-by-layer and sum the resulting contributions. The definition of g_ℓ^α and the network scaling give the boundary-aware internal identities

$$\begin{aligned} &\frac{\partial f_\alpha}{\partial W_0} = \frac{1}{\sqrt{N n_0}} g_1^\alpha (x^\alpha)^\top, \quad \frac{\partial f_\alpha}{\partial W_\ell} = \frac{1}{N} g_{\ell+1}^\alpha ({}^*\varphi(h_\ell^\alpha))^\top \quad (1 \leq \ell \leq d-1), \\ &\frac{\partial f_\alpha}{\partial W_d} = \frac{1}{\sqrt{N}} ({}^*\varphi(h_d^\alpha))^\top. \end{aligned} \tag{2.79}$$

Their internal Frobenius inner products are therefore

$$\begin{aligned}\left\langle \frac{\partial f_\alpha}{\partial W_0}, \frac{\partial f_\beta}{\partial W_0} \right\rangle_F &= G_1^{(N),\alpha\beta}\Phi_0^{\alpha\beta}, \quad \left\langle \frac{\partial f_\alpha}{\partial W_\ell}, \frac{\partial f_\beta}{\partial W_\ell} \right\rangle_F = G_{\ell+1}^{(N),\alpha\beta}\Phi_\ell^{(N),\alpha\beta} \quad (1 \le \ell \le d-1), \\ \left\langle \frac{\partial f_\alpha}{\partial W_d}, \frac{\partial f_\beta}{\partial W_d} \right\rangle_F &= \Phi_d^{(N),\alpha\beta}.\end{aligned} \tag{2.80}$$

Since $\Phi_0^{(N)} = \Phi_0$ and $G_{d+1}^{(N),\alpha\beta} = 1$, summing these contributions yields (2.77). □

To compute the standard shadow of this decomposition at unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, we must handle the internal dependence created by reusing the same weights in the forward and backward passes. Conditioning on the forward activations breaks the internal i.i.d. structure of those weights; the transferred Gaussian conditioning identity below isolates a usable internally independent component.

Write $\varphi_\ell^\alpha := {}^*\varphi(h_\ell^\alpha)$ and collect the dataset postactivations into

$$\varphi_\ell := \begin{bmatrix} \varphi_\ell^1 & \cdots & \varphi_\ell^m \end{bmatrix} \in {}^*\mathbb{R}^{N\times m}. \tag{2.81}$$

Since $h_{\ell+1}^\alpha = \frac{1}{\sqrt{N}} W_\ell \varphi_\ell^\alpha$, internal conditioning on the next-layer preactivations is (up to the internal scaling $1/\sqrt{N}$) the same as conditioning on $W_\ell\varphi_\ell$.

Lemma 2.26 (Gaussian Conditioning via Hestenes Blade Projection (One-Sided))**.** *Let $W \in {}^*\mathbb{R}^{N\times P}$ ($N, P \in {}^*\mathbb{N}$) have internally i.i.d. ${}^*\mathcal{N}(0,1)$ entries and let $\varphi \in {}^*\mathbb{R}^{P\times q}$ ($q \in {}^*\mathbb{N}$) be internally deterministic. Let $\mathbf{\Phi} = u_1 \wedge \cdots \wedge u_r \in \bigwedge^r({}^*\mathbb{R}^P) \subset {}^*\mathcal{C}\ell_P$ be a nonzero r-blade ($r = \operatorname{rank}(\varphi)$) spanning $\operatorname{col}(\varphi)$, so that Hestenes' interior projection and exterior rejection operators [29] act rowwise on $w \in {}^*\mathbb{R}^P$ by*

$$P_\varphi(w) = (w \lrcorner \mathbf{\Phi})\mathbf{\Phi}^{-1} = \varphi(\varphi^\top\varphi)^\dagger\varphi^\top w, \qquad P_\varphi^\perp(w) = (w \wedge \mathbf{\Phi})\mathbf{\Phi}^{-1} = (I_P - P_\varphi)w.$$

Then, conditioning internally on the internal σ-field ${}^*\sigma(W\varphi)$,

$$W \,\Big|\, {}^*\sigma(W\varphi) \overset{d}{=} WP_\varphi + \widetilde{W}P_\varphi^\perp, \tag{2.82}$$

in internal law under ${}^*\mathbb{P}$, where $\widetilde{W}$ is an internally independent copy of W.

Proof. See Appendix A.5. □

Proposition 2.27 (Backward Covariance Recursion)**.** *Fix standard $m, d, n_0 \in \mathbb{N}$ and standard inputs $\{x^\alpha\}_{\alpha=1}^m$, and evaluate all quantities at internal Gaussian initialization with unlimited width $N \in {}^*\mathbb{N}\setminus\mathbb{N}$. For sufficiently nice φ, there are deterministic standard matrices $G_1, \ldots, G_d \in \mathbb{R}^{m\times m}$ such that, entrywise,*

$$G_\ell^{(N)} \overset{L({}^*\mathbb{P})}{\approx} G_\ell, \qquad \ell = 1, \ldots, d. \tag{2.83}$$

With $G_{d+1}^{\alpha\beta} := 1$, their entries satisfy the backward recursion

$$G_\ell = \Phi_\ell' \odot G_{\ell+1}, \qquad \ell = d, d-1, \ldots, 1, \tag{2.84}$$

where $\odot$ denotes the Hadamard (entrywise) product. Consequently,

$$K_0^{(N)} \overset{L(^*\mathbb{P})}{\approx} K := \sum_{\ell=0}^{d} G_{\ell+1} \odot \Phi_\ell \tag{2.85}$$

entrywise (and hence $\mathrm{st}(K_0^{(N)}) = K$ *$L(^*\mathbb{P})$-almost surely).*

Proof Sketch. First consider the top hidden layer. The exact internal identity (2.74) gives

$$G_d^{(N),\alpha\beta} = \frac{1}{N} \sum_{i=1}^{N} W_{d,i}^2 \, {}^*\varphi'(h_{d,i}^\alpha) {}^*\varphi'(h_{d,i}^\beta). \tag{2.86}$$

Since W_d is internally independent of $\mathcal{F}_d$ and ${}^*\mathbb{E}[W_{d,i}^2] = 1$, the internal conditional law-of-large-numbers calculation gives $G_d^{(N),\alpha\beta} \overset{L(^*\mathbb{P})}{\approx} \Phi_d'^{,\alpha\beta}$. Thus $G_d = \Phi_d'$.

For $1 \le \ell \le d-1$, take internal inner products in (2.75) to obtain

$$G_\ell^{(N),\alpha\beta} = \frac{1}{N^2} (g_{\ell+1}^\alpha)^\top W_\ell \, \mathrm{diag}({}^*\varphi'(h_\ell^\alpha) \odot {}^*\varphi'(h_\ell^\beta)) W_\ell^\top g_{\ell+1}^\beta. \tag{2.87}$$

At the formal level of this proof sketch, transferred Gaussian conditioning (Lemma 2.26, where P_{φ_ℓ} has standard rank at most m inside ${}^*\mathbb{R}^N$) suggests that the finite-rank projection terms are infinitesimal and the leading contribution is obtained by replacing W_ℓ in this quadratic form by a fresh internally independent Gaussian matrix $\widetilde{W}_\ell$. Conditional on the displayed g's and h's, the transferred elementary Gaussian quadratic-form identity gives

$${}^*\mathbb{E}_{\widetilde{W}_\ell}\left[\frac{1}{N^2} (g_{\ell+1}^\alpha)^\top \widetilde{W}_\ell \, \mathrm{diag}({}^*\varphi'(h_\ell^\alpha) \odot {}^*\varphi'(h_\ell^\beta)) \widetilde{W}_\ell^\top g_{\ell+1}^\beta\right] = G_{\ell+1}^{(N),\alpha\beta} \Phi_\ell'^{(N),\alpha\beta}. \tag{2.88}$$

At unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, the normalized internal quadratic form concentrates infinitesimally around this conditional expectation, so

$$G_\ell^{(N)} - \Phi_\ell'^{(N)} \odot G_{\ell+1}^{(N)} \overset{L(^*\mathbb{P})}{\approx} 0 \tag{2.89}$$

entrywise. Taking standard parts and proceeding by backward external induction through the standard finitely many layers $\ell = d, d-1, \ldots, 1$ gives (2.83) and (2.84). Finally, substituting the forward and backward shadows into the exact internal decomposition (2.77), with $t = 0$, gives (2.85). □

Remark 2.28 (Rigor of the Proof Sketch). The sketch suppresses the estimates needed to show that the finite-rank projection terms from Lemma 2.26 are infinitesimal in the fresh-Gaussian replacement and to justify the infinitesimal concentration of the normalized internal quadratic forms. At standard depth $d \in \mathbb{N}$, these steps can be made rigorous by applying transferred Gaussian conditioning recursively. The formal calculation above is the part that determines the standard shadow recursion.

Remark 2.29 (Other Architectures). The same strategy—identify the relevant internal "state" variables, use transferred Gaussian conditioning (Lemma 2.26) to recover effective internal independence, then close a self-consistent standard-part recursion—extends beyond fully-connected networks (e.g. certain convolutional nets, residual networks in suitable regimes, and some attention-like blocks). The details depend strongly on the architecture.

We have so far focused on optimization and training dynamics in the kernel regime. At unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, when the standard limiting kernel K is positive definite, the standard-part squared training loss converges exponentially to zero. This does not address the statistical properties of the trained predictor, particularly its generalization error on unseen data. We turn to this question in the next chapter.

2.6 Exercises

1. **Residual ODE.** Derive Proposition 2.15 directly from the internal chain rule (by Transfer).

2. **General Loss Dissipation.** Prove Proposition 2.17. Then pass to standard parts in the fixed-kernel shadow ($K_t^{(N)} \approx K$) and show that, for any continuously differentiable standard pointwise loss ℓ (not necessarily squared) and any fixed standard symmetric positive semidefinite matrix K, the standard-part trajectories $f_\alpha(t)$ and $r_\alpha(t) = \partial_f \ell(f_\alpha(t), y^\alpha)$ satisfy
$$\frac{d}{dt}\left[\frac{1}{m}\sum_{\alpha=1}^{m} \ell(f_\alpha(t), y^\alpha)\right] = -\frac{1}{m^2} r(t)^\top K r(t) \le 0. \tag{2.90}$$

3. **Fixed-Kernel Dynamics.** Verify Proposition 2.22 directly from the spectral theorem.

4. **One-Hidden-Layer NTK Formula.** At internal time $t \in {}^*\mathbb{R}_{\ge 0}$ and unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, write $a_t := a(t)$, $h_t(x) := W(t)x/\sqrt{n_0}$, and $f_t(x) = N^{-1/2} a_t^\top {}^*\varphi(h_t(x))$. Show that the internal empirical NTK admits the exact decomposition
$$K_t^{(N)}(x,x') = \underbrace{\frac{1}{N}\left\langle {}^*\varphi(h_t(x)), {}^*\varphi(h_t(x'))\right\rangle}_{\text{output-weight contribution}} + \underbrace{\frac{\langle x, x'\rangle}{n_0}\frac{1}{N}\left\langle a_t \odot {}^*\varphi'(h_t(x)), a_t \odot {}^*\varphi'(h_t(x'))\right\rangle}_{\text{feature-gradient term}}. \tag{2.91}$$

5. **Deep NTK Recursion.** From the definition of g_ℓ^α in (2.73), derive the three boundary-aware internal identities in (2.79) and prove the exact decomposition (2.77). Verify the top-layer hyperfinite law-of-large-numbers calculation and the fresh-Gaussian internal expectation (2.88) in the proof of Proposition 2.27. Use these calculations formally to obtain (2.89), and then complete the backward external induction over standard $\ell = d, \dots, 1$ yielding $G_\ell^{(N)} \overset{L(^*\mathbb{P})}{\approx} G_\ell$ and $K_0^{(N)} \overset{L(^*\mathbb{P})}{\approx} K$.

Chapter 3

Double Descent with Random Projections

3.1 Motivation

One of the central puzzles in deep learning is the following: modern neural networks are often heavily overparameterized, so in principle they can interpolate (and even memorize) the training data, yet in practice they can still generalize well on unseen data. This tension between expressive power and test performance motivated a large line of work on the interpolation regime. Double descent is a first clean window into this phenomenon [5]: in overparameterized linear and random-feature models, it shows explicitly how prediction risk can worsen near interpolation and then improve again beyond it, offering a concrete baseline mechanism for why "more parameters" need not imply worse generalization.

An illustrative non-isotropic example is shown in Figure 3.1, where the risk first decreases, spikes near interpolation, and then decreases again.

A tractable way to study this behavior is through a random-feature model: the weights defining the hidden features are sampled once and then frozen, while only the readout coefficients are fitted by least squares. Conditional on the frozen features, training is therefore linear regression. We specialize to a linear activation, for which the feature matrix is a random projection of the input matrix.

Although simple, this model retains the rank mechanisms behind double descent. If the ambient dimension exceeds the sample size, increasing the number of random features crosses an interpolation threshold; if the ambient dimension is smaller, the random features eventually span the input space and further increases in width do not change the fitted function. In the isotropic Gaussian setting, rotational invariance gives explicit bias–variance formulas across these rank regimes. Bach's general-covariance analysis is our starting point [4].

3.2 Model: Linear Regression with Random Projections

Working at internal dimensions $M, N_0, N \in {}^*\mathbb{N}$ (extending finite dimensions $(m, n_0, n) \in \mathbb{N}^3$ by the Transfer Principle), we observe

$$y = X\theta_\star + \varepsilon, \qquad X \in {}^*\mathbb{R}^{M\times N_0}, \quad X_{ij} \overset{\text{iid}}{\sim} {}^*\mathcal{N}(0,1), \qquad \varepsilon \sim {}^*\mathcal{N}(0, \sigma^2 I_M), \tag{3.1}$$

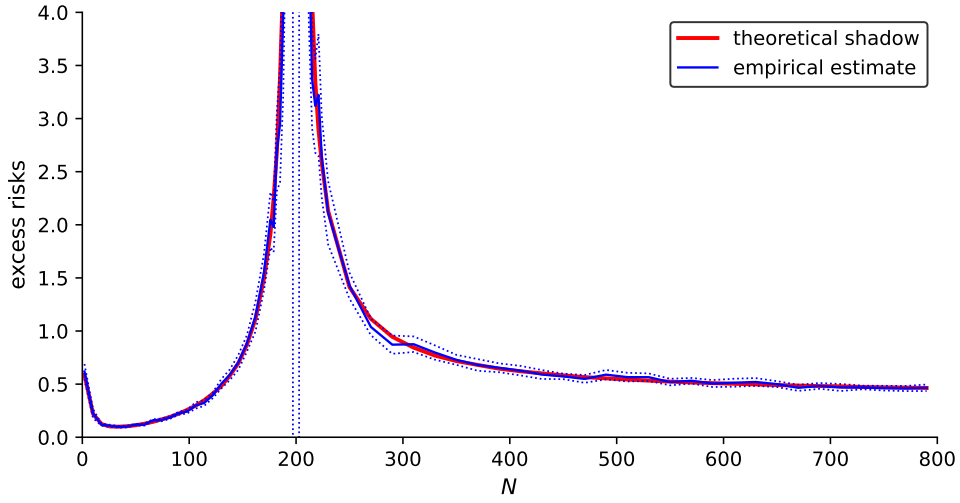


Figure 3.1: A non-isotropic double descent example for linear regression with random projections, with $m = 200$ observations and ambient dimension $n_0 = 400$. Adapted from [4, Figure 1]; the horizontal axis has been relabeled by n, the finite-dimensional counterpart of our notation N for the number of random projections. The isotropic model below exhibits the interpolation spike and second descent when the ambient dimension exceeds the sample size, but not the initial descent visible here.

where $\theta_\star \in {}^*\mathbb{R}^{N_0}$ is the unknown internal parameter, $\sigma^2 \in \mathbb{R}_{\geq 0}$ is the noise variance, and X is internally independent of ε. Let $S \in {}^*\mathbb{R}^{N_0 \times N}$ be an internal random projection matrix with i.i.d. entries

$$S_{jk} \overset{\text{iid}}{\sim} {}^*\mathcal{N}(0,1), \qquad S \text{ internally independent of } (X, \varepsilon). \tag{3.2}$$

We fit a linear predictor in the projected feature space $x \mapsto S^\top x$. The projected design matrix, minimum-norm feature coefficient, and induced ambient coefficient are

$$Z := XS \in {}^*\mathbb{R}^{M \times N}, \qquad \hat{\eta} := Z^\dagger y = (Z^\top Z)^\dagger Z^\top y \in {}^*\mathbb{R}^N, \qquad \hat{\theta} := S\hat{\eta} \in {}^*\mathbb{R}^{N_0}, \tag{3.3}$$

where $\dagger$ denotes the internal Moore–Penrose pseudoinverse, obtained by Transfer of the finite-dimensional pseudoinverse.

Thus M, N_0, and N (with finite-dimensional counterparts m, n_0, and n in Figure 3.1) denote the number of observations, the ambient dimension, and the number of random projections, respectively. This differs from Bach's notation: there, n denotes the number of observations, m the number of random projections, and d the ambient dimension [4].

We measure parameter estimation error by the internal squared Euclidean norm

$$R(\hat{\theta}) := \left\| \hat{\theta} - \theta_\star \right\|^2. \tag{3.4}$$

For an internally independent test covariate $x_{\text{new}} \sim {}^*\mathcal{N}(0, I_{N_0})$, Transfer shows that this also equals the internal excess prediction error ${}^*\mathbb{E}_{x_{\text{new}}}[(x_{\text{new}}^\top(\hat{\theta} - \theta_\star))^2]$. For a general internal covariance $\Sigma = {}^*\mathbb{E}[xx^\top]$, prediction error instead equals $(\hat{\theta} - \theta_\star)^\top \Sigma (\hat{\theta} - \theta_\star)$ and need not coincide with $R(\hat{\theta})$. Since $\hat{\theta} = \mathcal{M}y$ for the internal linear estimator map

$$\mathcal{M} := SZ^\dagger = S(Z^\top Z)^\dagger Z^\top, \tag{3.5}$$

substituting $y = X\theta_\star + \varepsilon$ gives, by Transfer of the finite-dimensional bias–variance identity, the exact internal bias–variance decomposition

$$ {}^*\mathbb{E}_\varepsilon\big[R(\hat\theta) \mid X, S\big] = \|(\mathcal{M}X - I_{N_0})\theta_\star\|^2 + {}^*\mathbb{E}_\varepsilon \|\mathcal{M}\varepsilon\|^2 = R^{(\mathrm{bias})}(\hat\theta) + R^{(\mathrm{var})}(\hat\theta), \tag{3.6}$$

where

$$ R^{(\mathrm{var})}(\hat\theta) = \sigma^2 \operatorname{Tr}(\mathcal{M}^\top \mathcal{M}). \tag{3.7}$$

3.3 High-Dimensional Risk and Double Descent

We now state the standard shadow behavior of the internal conditional risk ${}^*\mathbb{E}_\varepsilon\big[R(\hat\theta) \mid X, S\big]$ (equivalently the internal conditional bias–variance terms from (3.6)) for unlimited dimensions in the proportional galaxy

$$ M, N_0, N \in {}^*\mathbb{N} \setminus \mathbb{N} \qquad \text{with} \qquad \gamma := \operatorname{st}\left(\frac{N_0}{M}\right) \in (0,\infty), \qquad \delta := \operatorname{st}\left(\frac{N}{M}\right) \in (0,\infty). \tag{3.8}$$

Recall that for internal random variables U, V on the joint probability space of (X, S), the notation $U \overset{L({}^*\mathbb{P})}{\approx} V$ means that $U \approx V$ holds $L({}^*\mathbb{P})$-almost surely (equivalently, $\operatorname{st}({}^*\mathbb{P}(|U - V| > \epsilon)) = 0$ for every standard $\epsilon > 0$).

The following theorem gives the three rank regimes for the Gaussian isotropic model. The feature- and sample-bottleneck formulas are the $\Sigma = I_{N_0}$ specialization of [4, Proposition 4]; the ambient-bottleneck formula follows directly from internal ordinary least squares once the random features span ${}^*\mathbb{R}^{N_0}$.

Theorem 3.1 (Asymptotic Bias and Variance for Random Projections (Isotropic Case))**.** *Assume the internal isotropic Gaussian model (3.1)–(3.2) and consider the ridgeless estimator (3.3). In the proportional galaxy (3.8), assume that $\theta_\star \in {}^*\mathbb{R}^{N_0}$ satisfies $\|\theta_\star\|^2 \in \mathbb{L}$ with standard shadow $\operatorname{st}(\|\theta_\star\|^2) \in [0,\infty)$. Then, in each of the following regimes, the internal conditional bias–variance terms in (3.6) concentrate infinitesimally in Loeb probability $L({}^*\mathbb{P})$, with respect to the randomness of (X, S), around the displayed deterministic standard values.*

(a) Feature-bottleneck regime *($\delta < \min\{1, \gamma\}$):*

$$ R^{(\mathrm{var})}(\hat\theta) \overset{L({}^*\mathbb{P})}{\approx} \sigma^2 \frac{\delta}{1-\delta}, \tag{3.9}$$

$$ R^{(\mathrm{bias})}(\hat\theta) \overset{L({}^*\mathbb{P})}{\approx} \frac{\gamma - \delta}{\gamma(1-\delta)} \operatorname{st}(\|\theta_\star\|^2). \tag{3.10}$$

(b) Ambient-bottleneck regime *($\gamma < 1$ and $\delta \geq \gamma$):*

$$ R^{(\mathrm{var})}(\hat\theta) \overset{L({}^*\mathbb{P})}{\approx} \sigma^2 \frac{\gamma}{1-\gamma}, \tag{3.11}$$

$$ R^{(\mathrm{bias})}(\hat\theta) \overset{L({}^*\mathbb{P})}{\approx} 0. \tag{3.12}$$

(c) Sample-bottleneck regime *($\gamma > 1$ and $\delta > 1$):*

$$ R^{(\mathrm{var})}(\hat\theta) \overset{L({}^*\mathbb{P})}{\approx} \sigma^2 \left(\frac{1}{\gamma - 1} + \frac{1}{\delta - 1}\right), \tag{3.13}$$

$$ R^{(\mathrm{bias})}(\hat\theta) \overset{L({}^*\mathbb{P})}{\approx} \left(1 - \frac{1}{\gamma}\right) \frac{\delta}{\delta - 1} \operatorname{st}(\|\theta_\star\|^2). \tag{3.14}$$

The theorem excludes the singular boundaries $\delta = 1$ for $\gamma > 1$ and $\gamma = 1$ for $\delta \geq 1$.

The theorem is most easily read by fixing γ and increasing the width ratio δ. When $\gamma > 1$, the feature-bottleneck formulas apply for $\delta < 1$, up to the boundary. Provided $\sigma^2 > 0$ or the standard shadow $\mathrm{st}(\|\theta_\star\|^2)$ is positive, the standard total risk is strictly increasing and diverges as $\delta \uparrow 1$. The closing dimension gap $M - N$ makes the random design nearly square relative to M and produces the interpolation spike. For $\delta > 1$, the sample-bottleneck formulas apply, and the part of the risk shadow above its large-δ asymptote decreases proportionally to $(\delta - 1)^{-1}$ toward $\sigma^2/(\gamma - 1) + (1 - 1/\gamma)\,\mathrm{st}(\|\theta_\star\|^2)$ as $\delta \to \infty$. This is the second descent. Thus the isotropic model captures the interpolation spike and the post-interpolation descent, but not the initial descent in Figure 3.1.

When $\gamma < 1$, the width ratio reaches the ambient dimension ratio first, at $\delta = \gamma < 1$. The feature-bottleneck formulas then meet the ambient-bottleneck formulas continuously: the bias shadow vanishes, and the risk shadow remains at the ordinary least-squares value $\sigma^2\gamma/(1 - \gamma)$ for all $\delta \geq \gamma$. Consequently there is no singularity at $\delta = 1$; before ambient saturation, the risk shadow may increase, decrease, or remain flat depending on the signal-to-noise ratio. In this way, γ determines whether increasing width first reaches the interpolation threshold or the ambient dimension, while δ determines the active bottleneck.

3.3.1 Gaussian Derivation of the Risk Theorem

We now give the nonstandard derivation behind each branch of Theorem 3.1. By the Transfer Principle, every finite-dimensional linear-algebra identity and conditional Gaussian expectation holds verbatim at hyperfinite dimensions (M, N_0, N). In the strict proportional regimes, strict standard inequalities on (γ, δ) immediately enforce the corresponding internal rank inequalities on (M, N_0, N), and infinitesimal variance bounds upgrade the internal conditional-mean formulas to $L({}^*\mathbb{P})$-concentration of the random risk terms around their standard shadows.

Lemma 3.2 (Gaussian Gram-Matrix Inputs)**.** *Let $P, Q \in {}^*\mathbb{N}$ and let $G \in {}^*\mathbb{R}^{P\times Q}$ have i.i.d. ${}^*\mathcal{N}(0,1)$ entries. The internal inverse-Wishart identities are*

$$ {}^*\mathbb{E}\big[(G^\top G)^{-1}\big] = \frac{I_Q}{P - Q - 1}, \quad P > Q + 1, \qquad {}^*\mathbb{E}\big[(GG^\top)^{-1}\big] = \frac{I_P}{Q - P - 1}, \quad Q > P + 1. \tag{3.15} $$

Consequently, for $P, Q \in {}^*\mathbb{N} \setminus \mathbb{N}$ away from the square boundary $\mathrm{st}(Q/P) = 1$,

$$ \mathrm{Tr}\big((G^\top G)^{-1}\big) \overset{L({}^*\mathbb{P})}{\approx} \frac{\rho}{1 - \rho},\ \mathrm{st}\left(\frac{Q}{P}\right) = \rho < 1, \quad \mathrm{Tr}\big((GG^\top)^{-1}\big) \overset{L({}^*\mathbb{P})}{\approx} \frac{1}{\rho - 1},\ \mathrm{st}\left(\frac{Q}{P}\right) = \rho > 1. \tag{3.16} $$

Proof. The identities in (3.15) hold verbatim for all hyperintegers $P, Q \in {}^*\mathbb{N}$ by Transfer of the standard first inverse moments of a finite-dimensional Wishart matrix. When $P, Q \in {}^*\mathbb{N} \setminus \mathbb{N}$ satisfy $\mathrm{st}(Q/P) = \rho < 1$ (respectively, $\mathrm{st}(Q/P) = \rho > 1$), we immediately have $P > Q + 3$ (respectively, $Q > P + 3$), and the transferred inverse-Wishart trace variance is infinitesimal, so internal Chebyshev concentration yields $\mathrm{Tr}\big((G^\top G)^{-1}\big) \overset{L({}^*\mathbb{P})}{\approx} Q/(P - Q - 1) \approx \rho/(1 - \rho)$ (respectively, $\mathrm{Tr}\big((GG^\top)^{-1}\big) \overset{L({}^*\mathbb{P})}{\approx} P/(Q - P - 1) \approx 1/(\rho - 1)$). □

Lemma 3.3 (Directional Inverse-Wishart Law)**.** *If $P, Q \in {}^*\mathbb{N}$ satisfy $Q > P$, $G \in {}^*\mathbb{R}^{P\times Q}$ has i.i.d. ${}^*\mathcal{N}(0,1)$ entries, and $a \in {}^*\mathbb{R}^P$ is internally deterministic, then*

$$a^\top (GG^\top)^{-1} a \overset{d}{=} \frac{\|a\|^2}{{}^*\chi^2_{Q-P+1}} \tag{3.17}$$

in internal law under ${}^*\mathbb{P}$, with the zero-vector case interpreted directly.

Proof. By Transfer of the finite-dimensional directional calculation, for $a \neq 0$ internal orthogonal invariance (realized in ${}^*\mathcal{C}\ell_P$ by $\mathrm{Pin}(P)$ versor sandwiching $v \mapsto \pm UvU^{-1}$) reduces to $a = \|a\| e_1$. Writing the first row of G as $g_1 \in {}^*\mathbb{R}^Q$ and the remaining rows $g_2, \dots, g_P$ as G_2 with exterior $(P-1)$-blade $\boldsymbol{G}_2 := g_2 \wedge \cdots \wedge g_P \in \bigwedge^{P-1}({}^*\mathbb{R}^Q) \subset {}^*\mathcal{C}\ell_Q$, the Hestenes rejection identity $P^\perp_{\boldsymbol{G}_2}(g_1) = (g_1 \wedge \boldsymbol{G}_2)\boldsymbol{G}_2^{-1}$ [29] expresses the transferred Schur-complement identity directly as a squared exterior wedge-volume ratio:

$$\left[(GG^\top)^{-1}\right]^{-1}_{11} = \frac{\langle (g_1 \wedge \boldsymbol{G}_2)^\sim (g_1 \wedge \boldsymbol{G}_2)\rangle_0}{\langle \boldsymbol{G}_2^\sim \boldsymbol{G}_2\rangle_0} = g_1(I_Q - G_2^\top (G_2 G_2^\top)^{-1} G_2) g_1^\top. \tag{3.18}$$

The rejection operator $P^\perp_{\boldsymbol{G}_2}$ projects onto an internal subspace of hyperfinite codimension $P-1$, hence hyperfinite dimension $Q-P+1$, internally independent of g_1, so by Transfer the right-hand side has the internal ${}^*\chi^2_{Q-P+1}$ distribution. At the square interpolation boundary $Q = P$, $g_1 \wedge \boldsymbol{G}_2$ saturates the top-grade pseudoscalar $\bigwedge^Q({}^*\mathbb{R}^Q)$ with only 1 degree of freedom (${}^*\chi^2_1$, whose inverse expectation is unlimited). □

Lemma 3.4 (Haar Projection Lengths)**.** *Let $D_0, K \in {}^*\mathbb{N}$ satisfy $K \leq D_0$, and let P be the internal orthogonal projector onto an internally Haar-distributed K-dimensional subspace of ${}^*\mathbb{R}^{D_0}$. For every internally deterministic $v \in {}^*\mathbb{R}^{D_0}$,*

$${}^*\mathbb{E}[\|Pv\|^2] = \frac{K}{D_0}\|v\|^2. \tag{3.19}$$

If $D_0 \in {}^*\mathbb{N}\setminus\mathbb{N}$ and $\mathrm{st}(K/D_0) = \rho \in [0,1]$, then $\|Pv\|^2 - (K/D_0)\|v\|^2 \overset{L({}^*\mathbb{P})}{\approx} 0$ for every internally deterministic $v \in {}^*\mathbb{R}^{D_0}$ with $\|v\|^2 \in \mathbb{L}$.

Proof. For $v = 0$, both conclusions are immediate, so suppose $v \neq 0$. By Transfer, rotating v to the first coordinate and representing an internally uniform direction by a normalized ${}^*\mathcal{N}(0, I_{D_0})$ vector shows that $\|Pv\|^2 / \|v\|^2$ has the same internal law as ${}^*\chi^2_K/({}^*\chi^2_K + {}^*\widetilde{\chi}^2_{D_0-K})$, where the two internal chi-squared variables are internally independent. The internal mean (3.19) follows by symmetry. If $\rho \in (0,1)$, both K and $D_0 - K$ belong to ${}^*\mathbb{N}\setminus\mathbb{N}$, so internal Chebyshev concentration gives ${}^*\chi^2_K/D_0 \overset{L({}^*\mathbb{P})}{\approx} \rho$ and ${}^*\widetilde{\chi}^2_{D_0-K}/D_0 \overset{L({}^*\mathbb{P})}{\approx} 1-\rho$, whence $\|Pv\|^2/\|v\|^2 \overset{L({}^*\mathbb{P})}{\approx} \rho \approx K/D_0$. If $\rho = 0$, the claim follows from the mean identity (3.19) and internal Markov's inequality; if $\rho = 1$, apply the same argument to $I_{D_0} - P$. Multiplying by $\|v\|^2 \in \mathbb{L}$ yields $\|Pv\|^2 - (K/D_0)\|v\|^2 \overset{L({}^*\mathbb{P})}{\approx} 0$. □

All infinitesimal concentration statements $\overset{L({}^*\mathbb{P})}{\approx}$ below refer to the Loeb probability space $L({}^*\mathbb{P})$ of the joint randomness of (X, S). The intermediate internal conditional expectations are proof devices and are distinct from the label-noise expectation in (3.6).

Proposition 3.5 (Feature-Bottleneck Calculation)**.** *Under the assumptions of Theorem 3.1, if* $\delta < \min\{1, \gamma\}$*, then*

$$R^{(\mathrm{var})}(\hat{\theta}) \overset{L(^*\mathbb{P})}{\approx} \sigma^2 \frac{\delta}{1-\delta}, \qquad R^{(\mathrm{bias})}(\hat{\theta}) \overset{L(^*\mathbb{P})}{\approx} \frac{\gamma-\delta}{\gamma(1-\delta)} \operatorname{st}(\|\theta_\star\|^2). \tag{3.20}$$

Proof. Since $\operatorname{st}(N/M) = \delta < 1$ and $\operatorname{st}(N/N_0) = \delta/\gamma < 1$, the strict standard inequalities immediately imply $N < \min\{M, N_0\}$. By Transfer, write the internal thin QR factorization

$$S = QT, \qquad Q^\top Q = I_N, \tag{3.21}$$

where $T \in {}^*\mathbb{R}^{N\times N}$ is invertible ${}^*\mathbb{P}$-almost surely. Since $Z = (XQ)T$ has full column rank N, Transfer gives for the least-squares coefficient and its mapped ambient parameter the exact internal identities

$$\begin{aligned} \hat{\eta} &= (T^\top (XQ)^\top (XQ) T)^{-1} T^\top (XQ)^\top y = T^{-1}((XQ)^\top XQ)^{-1}(XQ)^\top y, \\ \hat{\theta} &= Q((XQ)^\top XQ)^{-1}(XQ)^\top y. \end{aligned} \tag{3.22}$$

Thus the internal singular values of S cancel: in this regime the mapped estimator $\hat{\theta}$ depends on S only through its column space—equivalently, only through the unit subspace N-blade $\boldsymbol{S} := q_1 \wedge \cdots \wedge q_N \in \bigwedge^N({}^*\mathbb{R}^{N_0}) \subset {}^*\mathcal{C}\ell_{N_0}$ ($s_1 \wedge \cdots \wedge s_N = \det(T)\,\boldsymbol{S}$), while in sample space ${}^*\mathbb{R}^M$ the fitted response $Z\hat{\eta} = (y \lrcorner \boldsymbol{Z})\boldsymbol{Z}^{-1}$ is Hestenes' interior blade projection onto the design N-blade $\boldsymbol{Z} = z_1 \wedge \cdots \wedge z_N \in \bigwedge^N({}^*\mathbb{R}^M)$ with Gramian $\det(Z^\top Z) = \langle \boldsymbol{Z}^\sim \boldsymbol{Z} \rangle_0$.

Set $P_S := QQ^\top$ (acting on vectors via $P_S v = (v \lrcorner \boldsymbol{S})\boldsymbol{S}^{-1}$) and

$$P_S\theta_\star = (\theta_\star \lrcorner \boldsymbol{S})\boldsymbol{S}^{-1}, \qquad \theta_\perp := (\theta_\star \wedge \boldsymbol{S})\boldsymbol{S}^{-1} = (I_{N_0} - P_S)\theta_\star. \tag{3.23}$$

The response decomposes internally as

$$y = (XQ)Q^\top\theta_\star + X\theta_\perp + \varepsilon. \tag{3.24}$$

Internally conditional on Q, the matrix XQ has i.i.d. ${}^*\mathcal{N}(0,1)$ entries and $X\theta_\perp$ is internally independent of XQ, with internal distribution ${}^*\mathcal{N}(0, \|\theta_\perp\|^2 I_M)$. Substituting (3.24) into (3.22) gives

$$\hat{\theta} = P_S\theta_\star + Q((XQ)^\top XQ)^{-1}(XQ)^\top X\theta_\perp + Q((XQ)^\top XQ)^{-1}(XQ)^\top \varepsilon. \tag{3.25}$$

The last term is the label-noise contribution. Using $Q^\top Q = I_N$ and internal cyclicity of trace, its conditional variance is

$$R^{(\mathrm{var})}(\hat{\theta}) = \sigma^2 \operatorname{Tr}\Big[(XQ)((XQ)^\top XQ)^{-2}(XQ)^\top\Big] = \sigma^2 \operatorname{Tr}\Big[((XQ)^\top XQ)^{-1}\Big]. \tag{3.26}$$

After averaging internally only over the label noise and subtracting $\theta_\star$, the bias vector is $-\theta_\perp + Q((XQ)^\top XQ)^{-1}(XQ)^\top X\theta_\perp$. These two terms lie in the internally orthogonal subspaces $\operatorname{span}(Q)^\perp$ and $\operatorname{span}(Q)$, respectively, so the conditional squared bias is exactly

$$R^{(\mathrm{bias})}(\hat{\theta}) = \|\theta_\perp\|^2 + (X\theta_\perp)^\top (XQ)((XQ)^\top XQ)^{-2}(XQ)^\top X\theta_\perp. \tag{3.27}$$

Internally conditional on (Q, XQ), the expectation of the second term over the independent internal Gaussian vector $X\theta_\perp$ is

$${}^*\mathbb{E}\Big[(X\theta_\perp)^\top (XQ)((XQ)^\top XQ)^{-2}(XQ)^\top X\theta_\perp \;\Big|\; Q, XQ\Big] = \|\theta_\perp\|^2 \operatorname{Tr}\Big[((XQ)^\top XQ)^{-1}\Big], \tag{3.28}$$

while its internal conditional variance $2\,\|\theta_\perp\|^4 \operatorname{Tr}(((XQ)^\top XQ)^{-2})$ is infinitesimal since $\|\theta_\perp\|^2 \le \|\theta_\star\|^2 \in \mathbb{L}$ and ${}^*\mathbb{E}[\operatorname{Tr}(((XQ)^\top XQ)^{-2})] = O(1/M) \in \mathbb{I}$. Internal Gaussian quadratic-form concentration therefore yields

$$R^{(\mathrm{bias})}(\hat\theta) \overset{L({}^*\mathbb{P})}{\approx} \|\theta_\perp\|^2 \left[1 + \operatorname{Tr}(((XQ)^\top XQ)^{-1})\right]. \tag{3.29}$$

By Transfer, the exact fully averaged internal identities provide a hyperfinite check:

$${}^*\mathbb{E}_{X,S}[R^{(\mathrm{var})}(\hat\theta)] = \sigma^2 \frac{N}{M-N-1}, \qquad {}^*\mathbb{E}_{X,S}[R^{(\mathrm{bias})}(\hat\theta)] = \left(1 - \frac{N}{N_0}\right) \frac{M-1}{M-N-1} \|\theta_\star\|^2, \tag{3.30}$$

whenever $N \le N_0$ and $M > N+1$. Indeed, (3.19) gives the omitted-signal factor $1 - N/N_0$, while (3.15) gives the amplification factor $1 + N/(M-N-1) = (M-1)/(M-N-1)$.

Here $XQ \in {}^*\mathbb{R}^{M\times N}$ has i.i.d. ${}^*\mathcal{N}(0,1)$ entries, while $I_{N_0} - P_S$ projects onto an internally Haar-distributed $(N_0 - N)$-dimensional subspace of ${}^*\mathbb{R}^{N_0}$. Since $\operatorname{st}(N/M) = \delta$ and $\operatorname{st}((N_0 - N)/N_0) = 1 - \delta/\gamma$, applying Lemmas 3.2 and 3.4 gives

$$\operatorname{Tr}(((XQ)^\top XQ)^{-1}) \overset{L({}^*\mathbb{P})}{\approx} \frac{\delta}{1-\delta}, \qquad \|\theta_\perp\|^2 \overset{L({}^*\mathbb{P})}{\approx} \left(1 - \frac{\delta}{\gamma}\right) \operatorname{st}(\|\theta_\star\|^2). \tag{3.31}$$

The first concentration relation in (3.31), combined with (3.26), gives the variance shadow in (3.20). Since $1 + \delta/(1-\delta) = (1-\delta)^{-1}$, substitution in (3.29) gives the bias shadow. The same inverse-Wishart factor therefore amplifies both label noise and the omitted signal. □

Proposition 3.6 (Ambient-Bottleneck Calculation)**.** *Under the assumptions of Theorem 3.1, if $\gamma < 1$ and $\delta > \gamma$, then*

$$R^{(\mathrm{var})}(\hat\theta) \overset{L({}^*\mathbb{P})}{\approx} \sigma^2 \frac{\gamma}{1-\gamma}, \qquad R^{(\mathrm{bias})}(\hat\theta) \overset{L({}^*\mathbb{P})}{\approx} 0. \tag{3.32}$$

Proof. Since $\operatorname{st}(N_0/M) = \gamma < 1$ and $\operatorname{st}(N/M) = \delta > \gamma$, we immediately have the internal inequalities $N_0 < M$ and $N > N_0$ (in particular $N \ge N_0$). By Transfer, the matrices X and S then have full column rank N_0 and full row rank N_0, respectively, ${}^*\mathbb{P}$-almost surely. By Transfer, the least-squares coefficient $\hat\eta = Z^\dagger y$ satisfies the internal normal equations

$$S^\top X^\top (XS\hat\eta - y) = 0. \tag{3.33}$$

Because $S^\top$ is internally injective and $\hat\theta = S\hat\eta$, these equations imply

$$X^\top (X\hat\theta - y) = 0, \qquad \hat\theta = (X^\top X)^{-1} X^\top y. \tag{3.34}$$

Thus the particular minimum-feature-norm solution maps to the unique ambient internal ordinary least-squares solution. Substituting the model gives

$$\hat\theta - \theta_\star = (X^\top X)^{-1} X^\top \varepsilon. \tag{3.35}$$

Hence the internal conditional bias vanishes identically ($R^{(\mathrm{bias})}(\hat\theta) = 0$) and

$$R^{(\mathrm{var})}(\hat\theta) = \sigma^2 \operatorname{Tr}((X^\top X)^{-1} X^\top X (X^\top X)^{-1}) = \sigma^2 \operatorname{Tr}((X^\top X)^{-1}). \tag{3.36}$$

In particular, by (3.15),

$$ {}^*\mathbb{E}_X\big[R^{(\mathrm{var})}(\hat{\theta})\big] = \sigma^2 \frac{N_0}{M - N_0 - 1}, \tag{3.37} $$

whenever $M > N_0 + 1$. The tall-matrix case of Lemma 3.2 now gives $R^{(\mathrm{var})}(\hat{\theta}) \overset{L(^*\mathbb{P})}{\approx} \sigma^2\gamma/(1-\gamma)$, proving (3.32). □

Lemma 3.7 (Gaussian Row–Null Coordinates)**.** *Suppose $M < N_0$ and $X \in {}^*\mathbb{R}^{M\times N_0}$ has full row rank, with S internally independent of X and distributed as in (3.2). Set $U_X := X^\top(XX^\top)^{-1/2}$ and choose an internal matrix $U_\perp \in {}^*\mathbb{R}^{N_0\times(N_0-M)}$ with orthonormal columns spanning* $\ker(X)$*. Internally conditional on X, the matrices $H := U_X^\top S$ and $G_\perp := U_\perp^\top S$ are internally independent standard Gaussian matrices with i.i.d. ${}^*\mathcal{N}(0,1)$ entries.*

Proof. By Transfer, U_X has orthonormal columns and is orthogonal to $U_\perp$. Indeed,

$$ [U_X \;\; U_\perp]^\top [U_X \;\; U_\perp] = I_{N_0}, \tag{3.38} $$

so $[U_X \;\; U_\perp]$ is internally orthogonal. Internally conditional on X, applying its transpose to each ${}^*\mathcal{N}(0, I_{N_0})$ column of S gives internally independent standard Gaussian row and null coordinates. □

Proposition 3.8 (Sample-Bottleneck Calculation)**.** *Under the assumptions of Theorem 3.1, if $\gamma > 1$ and $\delta > 1$, then*

$$ R^{(\mathrm{var})}(\hat{\theta}) \overset{L(^*\mathbb{P})}{\approx} \sigma^2\Big(\frac{1}{\gamma-1} + \frac{1}{\delta-1}\Big), \qquad R^{(\mathrm{bias})}(\hat{\theta}) \overset{L(^*\mathbb{P})}{\approx} \Big(1 - \frac{1}{\gamma}\Big)\frac{\delta}{\delta-1}\,\mathrm{st}(\|\theta_\star\|^2). \tag{3.39} $$

Proof. Since $\mathrm{st}(N_0/M) = \gamma > 1$ and $\mathrm{st}(N/M) = \delta > 1$, the strict standard inequalities immediately give $M < \min\{N_0, N\}$, so Z has full row rank M ${}^*\mathbb{P}$-almost surely and, by Transfer,

$$ Z^\dagger = Z^\top(ZZ^\top)^{-1}. \tag{3.40} $$

Use the internal matrices $U_X, U_\perp, H, G_\perp$ from Lemma 3.7. Since $[U_X \;\; U_\perp]$ is internally orthogonal and $H = (XX^\top)^{-1/2}Z$, projecting S onto these two coordinate blocks gives

$$ S = X^\top(XX^\top)^{-1}Z + U_\perp G_\perp, \qquad Z = (XX^\top)^{1/2}H. \tag{3.41} $$

Substituting (3.41) into $\mathcal{M} = SZ^\dagger$ and using $ZZ^\dagger = I_M$ gives

$$ \mathcal{M} = X^\top(XX^\top)^{-1} + U_\perp G_\perp Z^\dagger. \tag{3.42} $$

The two summands take values in the internal row space and null space of X, respectively, and are therefore orthogonal as internal linear maps into ${}^*\mathbb{R}^{N_0}$. Consequently, $R^{(\mathrm{var})}(\hat{\theta}) = \sigma^2\,\|\mathcal{M}\|_F^2$, where

$$ \|\mathcal{M}\|_F^2 = \mathrm{Tr}\big((XX^\top)^{-1}\big) + \big\|G_\perp Z^\dagger\big\|_F^2. \tag{3.43} $$

Because Z is an invertible transform of H given X, conditioning internally on (X, Z) fixes H and leaves $G_\perp$ internally standard Gaussian. Thus its $N_0 - M$ rows are internally independent

${}^*\mathcal{N}(0, I_N)$ vectors and $(Z^\dagger)^\top Z^\dagger = (ZZ^\top)^{-1}$. By Transfer, the internal conditional mean and variance of the second term in (3.43) are therefore

$$\begin{aligned} {}^*\mathbb{E}\Big[\big\|G_\perp Z^\dagger\big\|_F^2 \;\Big|\; X, Z\Big] &= (N_0 - M)\,\mathrm{Tr}((ZZ^\top)^{-1}), \\ {}^*\mathrm{Var}\Big(\big\|G_\perp Z^\dagger\big\|_F^2 \;\Big|\; X, Z\Big) &= 2(N_0 - M)\,\mathrm{Tr}((ZZ^\top)^{-2}). \end{aligned} \tag{3.44}$$

The whitening in (3.41) also gives

$$\mathrm{Tr}((ZZ^\top)^{-1}) = \mathrm{Tr}((XX^\top)^{-1}(HH^\top)^{-1}). \tag{3.45}$$

Internally conditional on X, (3.15) yields

$${}^*\mathbb{E}\Big[\mathrm{Tr}((ZZ^\top)^{-1}) \;\Big|\; X\Big] = \frac{1}{N - M - 1}\,\mathrm{Tr}((XX^\top)^{-1}). \tag{3.46}$$

Transferred weighted inverse-Wishart concentration upgrades this conditional mean to

$$(N - M)\,\mathrm{Tr}((ZZ^\top)^{-1}) \overset{L({}^*\mathbb{P})}{\approx} \mathrm{Tr}((XX^\top)^{-1}). \tag{3.47}$$

Since $\mathrm{Tr}((ZZ^\top)^{-2}) \le \big[\mathrm{Tr}((ZZ^\top)^{-1})\big]^2 = O(1/M^2)$ on the $L({}^*\mathbb{P})$-full event where (3.47) and (3.49) hold, the conditional variance in (3.44) is $O(1/M) \in \mathbb{I}$, so $\big\|G_\perp Z^\dagger\big\|_F^2 \overset{L({}^*\mathbb{P})}{\approx} (N_0 - M)\,\mathrm{Tr}((ZZ^\top)^{-1})$. Combining (3.43), (3.44), and (3.47) gives

$$R^{(\mathrm{var})}(\hat\theta) \overset{L({}^*\mathbb{P})}{\approx} \sigma^2\left(1 + \frac{N_0 - M}{N - M}\right)\mathrm{Tr}((XX^\top)^{-1}). \tag{3.48}$$

Since Lemma 3.2 and galaxy arithmetic give

$$\mathrm{Tr}((XX^\top)^{-1}) \overset{L({}^*\mathbb{P})}{\approx} \frac{1}{\gamma - 1}, \qquad \frac{N_0 - M}{N - M} \approx \frac{\gamma - 1}{\delta - 1}, \tag{3.49}$$

the two terms in (3.48) give $\sigma^2/(\gamma - 1)$ and $\sigma^2/(\delta - 1)$, proving the variance shadow in (3.39).

For the bias, define the internal row-space projector

$$P_X := X^\top (XX^\top)^{-1} X. \tag{3.50}$$

Set $y = X\theta_\star$ and define

$$b := Z^\top (ZZ^\top)^{-1} X\theta_\star, \qquad q := \|b\|^2 = (X\theta_\star)^\top (ZZ^\top)^{-1} X\theta_\star. \tag{3.51}$$

Applying (3.42) to $X\theta_\star$ gives the exact internal error decomposition

$$\hat\theta - \theta_\star = -(I_{N_0} - P_X)\theta_\star + U_\perp G_\perp b. \tag{3.52}$$

Unlike the two estimator terms in (3.42), the two error terms in (3.52) both lie in $\ker(X)$ and need not be orthogonal. Internally conditional on (X, Z),

$$U_\perp G_\perp b \sim {}^*\mathcal{N}(0, q(I_{N_0} - P_X)). \tag{3.53}$$

Writing $v := (I_{N_0} - P_X)\theta_\star$, Transfer gives the exact internal conditional moments

$$\begin{aligned} {}^*\mathbb{E}\big[\langle v, U_\perp G_\perp b\rangle \;\big|\; X, Z\big] &= 0, & {}^*\mathrm{Var}\big(\langle v, U_\perp G_\perp b\rangle \;\big|\; X, Z\big) &= q\,\|v\|^2, \\ {}^*\mathbb{E}\Big[\|U_\perp G_\perp b\|^2 \;\Big|\; X, Z\Big] &= (N_0 - M)q, & {}^*\mathrm{Var}\Big(\|U_\perp G_\perp b\|^2 \;\Big|\; X, Z\Big) &= 2(N_0 - M)q^2. \end{aligned} \tag{3.54}$$

To calculate q, set

$$a := (XX^\top)^{-1/2} X\theta_\star. \tag{3.55}$$

Then (3.41) gives

$$q = a^\top (HH^\top)^{-1} a, \qquad \|a\|^2 = \|P_X\theta_\star\|^2. \tag{3.56}$$

By Lemma 3.3, internally conditional on X,

$$q \overset{d}{=} \frac{\|P_X\theta_\star\|^2}{{}^*\chi^2_{N-M+1}}, \tag{3.57}$$

with the zero-signal case interpreted directly. Since $N - M \in {}^*\mathbb{N}\setminus\mathbb{N}$ satisfies $\mathrm{st}((N-M)/M) = \delta - 1 > 0$, we have ${}^*\chi^2_{N-M+1}/(N-M) \overset{L({}^*\mathbb{P})}{\approx} 1$, so $Mq \in \mathbb{L}$ (hence $q \overset{L({}^*\mathbb{P})}{\approx} 0$) and

$$(N-M)q \overset{L({}^*\mathbb{P})}{\approx} \|P_X\theta_\star\|^2. \tag{3.58}$$

Both internal conditional variances in (3.54) are now infinitesimal under $L({}^*\mathbb{P})$: $q\,\|v\|^2 \le q\,\|\theta_\star\|^2 \overset{L({}^*\mathbb{P})}{\approx} 0$ and $2(N_0 - M)q^2 = O(1/M) \overset{L({}^*\mathbb{P})}{\approx} 0$. Expanding the squared norm in (3.52), the centered cross term is therefore $\overset{L({}^*\mathbb{P})}{\approx} 0$ and the correction norm concentrates infinitesimally around $(N_0 - M)q$. Using (3.58) gives

$$R^{(\mathrm{bias})}(\hat\theta) \overset{L({}^*\mathbb{P})}{\approx} \|(I_{N_0} - P_X)\theta_\star\|^2 + \frac{N_0 - M}{N - M}\,\|P_X\theta_\star\|^2. \tag{3.59}$$

By Transfer, the exact fully averaged internal risks in this regime are

$$\begin{aligned} {}^*\mathbb{E}_{X,S}\big[R^{(\mathrm{var})}(\hat\theta)\big] &= \sigma^2 \frac{M}{N_0 - M - 1}\left(1 + \frac{N_0 - M}{N - M - 1}\right), \\ {}^*\mathbb{E}_{X,S}\big[R^{(\mathrm{bias})}(\hat\theta)\big] &= \frac{N_0 - M}{N_0}\left(1 + \frac{M}{N - M - 1}\right)\|\theta_\star\|^2, \end{aligned} \tag{3.60}$$

whenever $N_0 > M + 1$ and $N > M + 1$.

Finally, the internal row space of X is internally Haar distributed of dimension M in ${}^*\mathbb{R}^{N_0}$, so Lemma 3.4 gives

$$\|P_X\theta_\star\|^2 \overset{L({}^*\mathbb{P})}{\approx} \frac{1}{\gamma}\,\mathrm{st}(\|\theta_\star\|^2), \qquad \|(I_{N_0} - P_X)\theta_\star\|^2 \overset{L({}^*\mathbb{P})}{\approx} \left(1 - \frac{1}{\gamma}\right)\mathrm{st}(\|\theta_\star\|^2). \tag{3.61}$$

Substituting these concentration relations and $(N_0 - M)/(N - M) \approx (\gamma - 1)/(\delta - 1)$ into (3.59) yields

$$R^{(\mathrm{bias})}(\hat\theta) \overset{L({}^*\mathbb{P})}{\approx} \left(1 - \frac{1}{\gamma} + \frac{\gamma - 1}{\gamma(\delta - 1)}\right)\mathrm{st}(\|\theta_\star\|^2) = \left(1 - \frac{1}{\gamma}\right)\frac{\delta}{\delta - 1}\,\mathrm{st}(\|\theta_\star\|^2), \tag{3.62}$$

which proves the bias shadow in (3.39). □

Proof of Theorem 3.1. Propositions 3.5, 3.6, and 3.8 prove the three strict regimes. It remains only to consider the boundary $\delta = \gamma < 1$. Because we work with a single hyperfinite triple $(M, N_0, N) \in ({}^*\mathbb{N} \setminus \mathbb{N})^3$ satisfying $\mathrm{st}(N_0/M) = \gamma < 1$ and $\mathrm{st}(N/M) = \gamma < 1$, no subsequence extraction is needed: by internal trichotomy in ${}^*\mathbb{N}$, either $N \geq N_0$ or $N < N_0$. If $N \geq N_0$, then since $\gamma < 1$ implies $N_0 < M$, the exact internal rank hypotheses $N_0 < M$ and $N \geq N_0$ in the proof of Proposition 3.6 hold directly, yielding $R^{(\mathrm{var})}(\hat{\theta}) \overset{L(^*\mathbb{P})}{\approx} \sigma^2\gamma/(1-\gamma)$ and $R^{(\mathrm{bias})}(\hat{\theta}) = 0$. If instead $N < N_0$, then since $\mathrm{st}(N/M) = \gamma < 1$ implies $N < M$, we have $N < \min\{M, N_0\}$, so the internal feature-bottleneck identities (3.26)–(3.29) hold verbatim. Since $N/N_0 \approx 1$ and $\mathrm{st}(N/M) = \gamma < 1$, the omitted-subspace expectation $(1 - N/N_0)\,\|\theta_\star\|^2 \approx 0$ vanishes infinitesimally (so $\|\theta_\perp\|^2 \overset{L(^*\mathbb{P})}{\approx} 0$ by internal Markov's inequality), while Lemma 3.2 gives $\mathrm{Tr}(((XQ)^\top XQ)^{-1}) \overset{L(^*\mathbb{P})}{\approx} \gamma/(1-\gamma)$. Substituting into (3.26) and (3.29) again yields (3.11)–(3.12). The two remaining boundaries excluded in the theorem are precisely the square inverse-Wishart hard edges where the galaxy gap $M - N$ or $N_0 - M$ ceases to be of order M. □

The calculations have three main consequences:

- The smallest of M, N_0, and N determines whether features, ambient dimension, or samples are the bottleneck.
- The singular factors are rectangular Gaussian dimension gaps: when $\gamma > 1$ the feature gap closes at $\delta = 1$, whereas for $\gamma < 1$ the risk saturates once the features span ${}^*\mathbb{R}^{N_0}$.
- The isotropic model isolates the interpolation spike and second descent. Non-isotropic covariance can also produce the earlier descent shown in Figure 3.1.

The random projection model is linear, but it isolates mechanisms that recur in nonlinear networks: fixed-feature behavior, rank geometry, and sharp transitions near interpolation. The fixed-feature perspective connects back to the NNGP/NTK kernel limits of Chapter 2; Chapter 4 instead studies feature learning, where feature distributions evolve during training.

3.4 Exercises

1. **Phase Diagram and Double Descent.** Fix $\gamma > 1$ and assume that $\sigma^2 > 0$ or $\mathrm{st}(\|\theta_\star\|^2) > 0$. Using parts (a) and (c) of Theorem 3.1, show that the standard total risk shadow is strictly increasing for $0 < \delta < 1$, diverges as $\delta \to 1$ from either side, and is strictly decreasing for $\delta > 1$. Compute its limit as $\delta \to \infty$. When $\gamma < 1$, verify that the feature-bottleneck formulas meet the ambient-bottleneck formulas continuously at $\delta = \gamma$, and explain why there is no interpolation singularity at $\delta = 1$.

2. **Scale Invariance of Ridgeless Features.** For an internal scalar $\alpha \in {}^*\mathbb{R} \setminus \{0\}$, replace S by αS and hence Z by αZ. Use Transfer of the Moore–Penrose pseudoinverse identities to show that $\hat{\eta}$ is rescaled by α^{-1} while $\hat{\theta}$ and its risk $R(\hat{\theta})$ are unchanged. Explain why the global scalar normalization of S is therefore immaterial for the ridgeless estimator.

3. **Interpolation as a Rank Condition.** Show by Transfer that the internal rank identity $\mathrm{rank}(XS) = \min\{M, N_0, N\}$ holds ${}^*\mathbb{P}$-almost surely. Deduce that the random-feature

model can interpolate every internal response vector $y \in {}^*\mathbb{R}^M$ exactly if and only if $N_0 \geq M$ and $N \geq M$. Use this internal rank condition to explain once more why increasing N through M creates an interpolation threshold when $\gamma > 1$, but not when $\gamma < 1$.

Chapter 4

Feature Learning and Mean-Field Limits

Chapters 2 and 3 studied the kernel regime from its optimization and statistical perspectives. In this chapter we turn to feature learning: parameterizations and scaling limits in which internal representations evolve by an appreciable amount ($\in \mathbb{A} = \mathbb{L} \setminus \mathbb{I}$) at an unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$.

Throughout the chapter, we consider a supervised dataset $\{(x^\alpha, y^\alpha)\}_{\alpha=1}^m$ with standard inputs $x^\alpha \in \mathbb{R}^{n_0}$ and standard targets $y^\alpha \in \mathbb{R}$ (for simplicity we take a scalar output). Given an internal width-N network $f(\,\cdot\,;\theta)$ (suppressing the subscript N), we abbreviate $f^\alpha := f(x^\alpha;\theta)$ and define the squared loss

$$L(\theta) = \frac{1}{2m} \sum_{\alpha=1}^{m} \big(f^\alpha - y^\alpha\big)^2. \tag{4.1}$$

We write $r^\alpha(t) := f^\alpha(t) - y^\alpha$ when a residual variable is useful and otherwise keep the difference explicit. Throughout this chapter, $m, n_0 \in \mathbb{N}$ are fixed standard integers while $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ is unlimited, and

$$\Phi_0^{\alpha\beta} := \frac{1}{n_0} \left\langle x^\alpha, x^\beta \right\rangle \tag{4.2}$$

denotes the standard input kernel introduced in Chapter 2.

The positive hyperreal learning-rate multiplier $\eta = \eta_N \in {}^*\mathbb{R}_{>0}$ is allowed to depend on the hyperfinite width N. For continuous-time arguments we use the internal gradient-flow convention

$$\partial_t \theta(t) = -\eta_N \nabla_\theta L(\theta(t)). \tag{4.3}$$

Thus $\eta_N = \eta_0 N$ (with standard $\eta_0 > 0$, or more generally $\eta_0 \in \mathbb{A}$) denotes an N-accelerated internal gradient flow; equivalently, it is unit-rate internal gradient flow observed on a time scale accelerated by N. We do not absorb the factor $1/m$ in the loss into time.

4.1 One Hidden Layer: How Features Move

Throughout this section we use a one-hidden-layer network of unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$:

$$f(x;\theta) = \frac{1}{\gamma_N \sqrt{N}} \sum_{i=1}^{N} a_i \, \phi(h_i(x)), \qquad h_i(x) := \frac{1}{\sqrt{n_0}} \langle w_i, x \rangle, \qquad \theta = \{(a_i, w_i)\}_{i=1}^N, \tag{4.4}$$

with $(a_i, w_i) \in {}^*\mathbb{R} \times {}^*\mathbb{R}^{n_0}$ and standard activation $\phi : \mathbb{R} \to \mathbb{R}$ (evaluated on hyperreal arguments via its natural extension ${}^*\phi : {}^*\mathbb{R} \to {}^*\mathbb{R}$, for which we continue to write ϕ when clear from context). Here $\gamma_N \in {}^*\mathbb{R}_{>0}$ is an extra hyperreal scaling knob beyond the usual NTK prefactor $1/\sqrt{N}$: setting $\gamma_N = 1$ gives the standard (NTK) scaling, while $\gamma_N = \sqrt{N}$ gives an overall hyperfinite average prefactor $1/N$.

We first make the notion of feature learning quantitative in terms of the multiplicative group of appreciable hyperreals $\mathbb{A} = \mathbb{L} \setminus \mathbb{I}$ and asymptotic scale equivalence $a \asymp b \iff a/b \in \mathbb{A}$ (equality of valuation classes in the quotient group ${}^*\mathbb{R}^\times/\mathbb{A}$). For a fixed data point x^α, define the (scalar) internal hidden preactivations

$$h_i^\alpha := h_i(x^\alpha) = \frac{1}{\sqrt{n_0}} \langle w_i, x^\alpha \rangle, \qquad i = 1, \dots, N, \tag{4.5}$$

and collect them into the hyperfinite width-N vector $h^\alpha \in {}^*\mathbb{R}^N$.

Definition 4.1 (Feature Learning Criterion)**.** Fix a training step k and a sample α, and write $h^\alpha(k) \in {}^*\mathbb{R}^N$ for the internal vector of hidden preactivations on input x^α. Let $\Delta h^\alpha(k) := h^\alpha(k+1) - h^\alpha(k)$. We say that the hidden layer exhibits *feature learning* at step k on sample α if

$$\frac{1}{N} \|\Delta h^\alpha(k)\|^2 \in \mathbb{A}. \tag{4.6}$$

When (4.6) holds we will sometimes write simply $\Delta h^\alpha(k) \asymp 1$, meaning the normalized hyperfinite squared norm above lies in $\mathbb{A}$. Precisely, this says that the root-mean-square coordinate change is appreciable (neither infinitesimal nor unlimited); under internal exchangeability and a nonconcentration condition, a Loeb-typical coordinate is then also appreciable.

To see what the feature-learning criterion requires, consider one step of internal full-batch gradient descent with learning rate $\eta \in {}^*\mathbb{R}_{>0}$:

$$w_i^+ = w_i - \eta\, \partial_{w_i} L, \qquad a_i^+ = a_i - \eta\, \partial_{a_i} L. \tag{4.7}$$

By Transfer, a direct derivative gives

$$\partial_{w_i} L = \frac{1}{m} \sum_{\beta=1}^{m} (f^\beta - y^\beta)\, \partial_{w_i} f^\beta = \frac{1}{m} \sum_{\beta=1}^{m} (f^\beta - y^\beta) \frac{1}{\gamma_N \sqrt{N n_0}}\, a_i\, \phi'(h_i^\beta)\, x^\beta. \tag{4.8}$$

Therefore the induced one-step change in the feature $h_i^\alpha = n_0^{-1/2} \langle w_i, x^\alpha \rangle$ is

$$\begin{aligned} \Delta h_i^\alpha := h_i^{\alpha,+} - h_i^\alpha &= \frac{1}{\sqrt{n_0}} \left\langle w_i^+ - w_i, x^\alpha \right\rangle \\ &= -\frac{\eta}{m \gamma_N \sqrt{N}} \sum_{\beta=1}^{m} (f^\beta - y^\beta)\, a_i\, \phi'(h_i^\beta)\, \Phi_0^{\beta\alpha}. \end{aligned} \tag{4.9}$$

Heuristically, if $(f^\beta - y^\beta) \in \mathbb{A}$ and the inner products and activation factors lie in $\mathbb{A}$ without cancellation, then in the hyperreal valuation quotient ${}^*\mathbb{R}^\times/\mathbb{A}$,

$$\Delta h_i^\alpha \asymp \frac{\eta}{\gamma_N \sqrt{N}}. \tag{4.10}$$

Since $h^\alpha = (h_i^\alpha)_{i=1}^N$, this gives

$$\frac{1}{N}\|\Delta h^\alpha\|^2 \asymp \left(\frac{\eta}{\gamma_N\sqrt{N}}\right)^2, \tag{4.11}$$

so achieving the feature-learning condition (4.6) requires (at least) the valuation-group identity $\eta \asymp \gamma_N\sqrt{N}$.

The same update must also move the predictor on an appreciable scale. In this chapter it is convenient to factor the output multiplier out of the tangent kernel and define the normalized empirical NTK at width N by

$$K_t^{(N),\alpha\beta} := \frac{1}{N}\sum_{i=1}^N \phi(h_i^\alpha)\phi(h_i^\beta) + \Phi_0^{\alpha\beta}\frac{1}{N}\sum_{i=1}^N a_i^2\phi'(h_i^\alpha)\phi'(h_i^\beta). \tag{4.12}$$

Throughout this chapter, "NTK" refers to this normalized kernel unless the raw gradient Gram is named explicitly. Thus the raw internal gradient Gram used in Chapter 2 is $\left\langle \nabla_\theta f^\alpha, \nabla_\theta f^\beta \right\rangle = \gamma_N^{-2} K_t^{(N),\alpha\beta}$. When the hyperfinite empirical averages are appreciable, the entries of $K_t^{(N)}$ lie in $\mathbb{A}$. Under the internal gradient-flow convention (4.3), Transfer of the chain rule shows that the predictor dynamics satisfy the exact internal identity

$$\partial_t f^\alpha(t) = -\frac{\eta_N}{m\gamma_N^2}\sum_{\beta=1}^m K_t^{(N),\alpha\beta} r^\beta(t), \tag{4.13}$$

so the predictor speed has valuation class $[\eta_N/\gamma_N^2] \in {}^*\mathbb{R}^\times/\mathbb{A}$. A first-order gradient-descent update has the same width scaling. Balancing appreciable feature updates ($\eta \asymp \gamma_N\sqrt{N}$) with appreciable predictor motion ($\eta \asymp \gamma_N^2$) therefore reduces to exact arithmetic in the torsion-free abelian valuation quotient ${}^*\mathbb{R}^\times/\mathbb{A}$: from $[\eta] = [\gamma_N][\sqrt{N}]$ and $[\eta] = [\gamma_N]^2$, dividing one relation by the other uniquely selects $[\gamma_N] = [\sqrt{N}]$ and $[\eta] = [N]$, that is, $\gamma_N \asymp \sqrt{N}$ and $\eta \asymp N$.

These two balance conditions distinguish three regimes:

1. **NTK (lazy) scaling.** Set $\gamma_N = 1$ and take $\eta \in \mathbb{A}$. Then (4.10) gives $\Delta h_i^\alpha \asymp N^{-1/2} \in \mathbb{I}$, and thus
$$\frac{1}{N}\|\Delta h^\alpha\|^2 \asymp N^{-1} \approx 0, \tag{4.14}$$
so the features freeze up to infinitesimals at unlimited width N. This is the standard lazy/NTK regime.

2. **Naively increasing the learning rate.** Keeping $\gamma_N = 1$ but taking $\eta \asymp \sqrt{N}$ makes $\Delta h_i^\alpha \in \mathbb{A}$. For squared loss this often leads to unlimited (exploding) outputs because the function-space speed in (4.13) scales like $\partial_t f \asymp \sqrt{N} \notin \mathbb{L}$. However, for cross-entropy loss, it can be acceptable for the logits to become unlimited while training remains stable, because gradients can stay limited even when outputs grow; see [23] for a careful analysis of this "controlled divergence" phenomenon.

3. **Mean-field scaling.** Instead choose $\gamma_N = \sqrt{N}$, i.e.
$$f(x;\theta) = \frac{1}{N}\sum_{i=1}^N a_i\,\phi\left(\frac{1}{\sqrt{n_0}}\langle w_i, x\rangle\right), \tag{4.15}$$

Regime	Output prefactor	η scaling	Feature update Δh_i^α
NTK (lazy)	$1/\sqrt{N}$ $(\gamma_N = 1)$	$\in \mathbb{A}$	$\asymp N^{-1/2} \approx 0$
Naive large-η	$1/\sqrt{N}$ $(\gamma_N = 1)$	$\asymp \sqrt{N}$	$\in \mathbb{A}$
Mean-field	$1/N$ $(\gamma_N = \sqrt{N})$	$\asymp N$	$\in \mathbb{A}$

Table 4.1: A quick scaling comparison in ${}^*\mathbb{R}^\times/\mathbb{A}$ for the one-hidden-layer network at unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$.

and scale the learning rate as $\eta = \eta_0 N$ with $\eta_0 \in \mathbb{A}$. Then (4.10) gives $\Delta h_i^\alpha \in \mathbb{A}$, hence the feature-learning condition remains $\frac{1}{N}\|\Delta h^\alpha\|^2 \in \mathbb{A}$ at unlimited width N. At the same time, (4.13) gives $\partial_t f^\alpha(t) \in \mathbb{A}$ at the valuation level of this calculation, so the predictor evolves on an appreciable time scale.

At a mean-zero random initialization, $f(x)$ is typically of scale $N^{-1/2} \approx 0$ for fixed standard x. This infinitesimal initialization scale is secondary to the balance above: because nonzero targets satisfy $y^\alpha \in \mathbb{R} \setminus \{0\} \subset \mathbb{A}$, the initial residual $f^\alpha - y^\alpha \approx -y^\alpha$ remains appreciable.

4.2 Mean-Field ODE: An Interacting Particle System

With mean-field scaling, the internal parameters (a_i, w_i) behave like a hyperfinite interacting particle system. To emphasize this, it is convenient to encode all N neurons via an internal empirical measure and its standard Loeb pushforward.

Definition 4.2 (Empirical Measure of Neurons)**.** For the one-hidden-layer network (4.4) at width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, write $\Theta := \mathbb{R} \times \mathbb{R}^{n_0}$ for standard single-neuron parameter space and ${}^*\Theta = {}^*\mathbb{R} \times {}^*\mathbb{R}^{n_0}$ for its internal extension. Define the internal empirical measure on ${}^*\Theta$ by

$$\rho^{(N)} := \frac{1}{N}\sum_{i=1}^{N} \delta_{(a_i, w_i)}. \tag{4.16}$$

Whenever the Loeb measure $L(\rho^{(N)})$ concentrates on the nearstandard galaxy $\mathbb{L} \times \mathbb{L}^{n_0} \subset {}^*\Theta$, its standard pushforward $\rho := L(\rho^{(N)}) \circ \mathrm{st}^{-1}$ defines a Borel probability measure on Θ [44].

With mean-field scaling ($\gamma_N = \sqrt{N}$), the internal network output can be written as

$$f(x;\theta) = \int_{{}^*\Theta} a\, \phi\!\left(\frac{1}{\sqrt{n_0}}\langle w, x\rangle\right) \rho^{(N)}(da\, dw). \tag{4.17}$$

With $\eta_N = \eta_0 N$ (for standard $\eta_0 > 0$), the convention (4.3) gives the accelerated internal gradient flow

$$\frac{d}{dt}(a_i, w_i) = -\eta_0 N \nabla_{(a_i, w_i)} L. \tag{4.18}$$

Using the internal dynamics (4.18) (and differentiating the network output (4.4) by Transfer), one finds an internal ODE system of the form

$$\frac{d}{dt}(a_i, w_i) = b((a_i, w_i), \rho_t^{(N)}), \qquad \rho_t^{(N)} := \frac{1}{N}\sum_{j=1}^{N} \delta_{(a_j(t), w_j(t))}, \tag{4.19}$$

where the internal velocity field $b(\theta, \rho)$ depends on the population only through the prediction functional $f_\rho(x) = \int a\, \phi(n_0^{-1/2} \langle w, x \rangle)\, \rho(da\, dw)$. Thus each hyperfinite "particle" is driven by the internal empirical distribution of all N particles.

Passing from this hyperfinite interacting particle system to a deterministic standard limit rests on propagation of chaos: at unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, the shadows of any fixed standard number $k \in \mathbb{N}$ of neurons are genuinely independent and identically distributed under the Loeb measure $L({}^*\mathbb{P})$, with common deterministic law $\rho_t = L(\rho_t^{(N)}) \circ \mathrm{st}^{-1}$.

Definition 4.3 (Chaos)**.** Fix a standard time $t \geq 0$, and let $\theta_i^{(N)}(t) = (a_i(t), w_i(t)) \in {}^*\Theta$ denote the internal particle parameters under the mean-field gradient flow (4.19) on an internal probability space $(\Omega, \mathcal{A}, {}^*\mathbb{P})$ with Loeb extension $L({}^*\mathbb{P})$. Let $P_t^{(N)}$ be the internal joint law of $(\theta_1^{(N)}(t), \dots, \theta_N^{(N)}(t)) \in {}^*\Theta^N$, assumed nearstandardly concentrated ($\theta_i^{(N)}(t) \in \mathbb{L} \times \mathbb{L}^{n_0}$ $L({}^*\mathbb{P})$-almost surely), and let ρ_t be a deterministic Borel probability measure on Θ. We say that $P_t^{(N)}$ is ρ_t-*chaotic* if for every standard $k \in \mathbb{N}$, the shadow tuple $\big(\mathrm{st}(\theta_1^{(N)}(t)), \dots, \mathrm{st}(\theta_k^{(N)}(t))\big) \in \Theta^k$ has law $\rho_t^{\otimes k}$ under $L({}^*\mathbb{P})$; equivalently, for every bounded continuous standard test function $\psi : \Theta^k \to \mathbb{R}$,

$$\begin{aligned} \mathrm{st}\ {}^*\mathbb{E}\Big[{}^*\psi\big(\theta_1^{(N)}(t), \dots, \theta_k^{(N)}(t)\big)\Big] &= \mathbb{E}_L\Big[\psi\big(\mathrm{st}(\theta_1^{(N)}(t)), \dots, \mathrm{st}(\theta_k^{(N)}(t))\big)\Big] \\ &= \int_{\Theta^k} \psi(\theta_1, \dots, \theta_k)\, \rho_t^{\otimes k}(d\theta_1 \cdots d\theta_k). \end{aligned} \tag{4.20}$$

We say that the particle system *propagates chaos* on a standard time interval if there exists a deterministic family (ρ_t) such that $P_t^{(N)}$ is ρ_t-chaotic for every standard t in that interval.

Under standard well-posedness assumptions on the McKean–Vlasov interaction field (e.g. ϕ is C^1 with bounded Lipschitz derivative, the data are bounded, and ρ_0 has suitable moments), one can show that for every standard finite time horizon $T > 0$ there exists a unique deterministic solution $(\rho_t)_{t \in [0,T]}$ of the mean-field PDE in Definition 4.4, and the hyperfinite particle system (4.19) propagates chaos on $[0, T]$: for every standard $t \in [0, T]$, the Loeb pushforward $L(\rho_t^{(N)}) \circ \mathrm{st}^{-1}$ equals ρ_t $L({}^*\mathbb{P})$-almost surely, and for each standard $k \in \mathbb{N}$, the k-particle shadow has law $\rho_t^{\otimes k}$. Quantitative internal bounds (often ${}^*W_2(\rho_t^{(N)}, {}^*\rho_t) = O(N^{-1/2}) \approx 0$ in transferred Wasserstein distance) hold under Lipschitz interactions.

Propagation of chaos is helpful because it upgrades the heuristic "replace the empirical measure by its shadow" into an exact Loeb-measure identity: at unlimited width N, the shadows of finitely many neurons are i.i.d. samples from ρ_t, and standard macroscopic observables (the shadow predictor $\mathrm{st}(f(x; \theta(t))) = f_{\rho_t}(x)$, risks, kernels, etc.) become deterministic functionals of ρ_t. This is exactly the point of the mean-field framework: it replaces a hyperfinite system of N coupled internal ODEs by a single deterministic measure-valued evolution on Θ that can be analyzed using PDE and optimal-transport techniques. See, e.g., [60, 32, 11] and, for a treatment tailored to mean-field neural networks beyond logarithmic time, [21].

4.3 Mean-Field PDE and Wasserstein Gradient Flow

The internal empirical measure $\rho_t^{(N)}$ from Definition 4.2 satisfies an exact internal transport identity. Taking the standard part via the Loeb pushforward $\rho_t := L(\rho_t^{(N)}) \circ \mathrm{st}^{-1}$ yields a deterministic PDE on $\Theta = \mathbb{R} \times \mathbb{R}^{n_0}$.

Definition 4.4 (Mean-Field Transport PDE)**.** Let ρ_t be a time-dependent Borel probability measure on $\Theta = \mathbb{R} \times \mathbb{R}^{n_0}$. Define the mean-field predictor

$$f_{\rho_t}(x) := \int_\Theta a\,\phi\left(\frac{1}{\sqrt{n_0}}\,\langle w, x\rangle\right)\rho_t(da\,dw), \tag{4.21}$$

and the velocity field $b(\theta, \rho_t)$ for $\theta = (a, w) \in \Theta$ by

$$b(\theta, \rho_t) := -\frac{\eta_0}{m}\sum_{\alpha=1}^{m}(f_{\rho_t}(x^\alpha) - y^\alpha)\begin{bmatrix}\phi(n_0^{-1/2}\,\langle w, x^\alpha\rangle)\\ n_0^{-1/2}a\,\phi'(n_0^{-1/2}\,\langle w, x^\alpha\rangle)\,x^\alpha\end{bmatrix}. \tag{4.22}$$

The *mean-field PDE* is the Dirac scalar-divergence continuity equation in $\mathcal{C}\ell_{1+n_0}$ [29, 9],

$$\partial_t\rho_t = -\left\langle \partial_\theta(b(\,\cdot\,, \rho_t)\rho_t)\right\rangle_0 = -\nabla_{(a,w)}\cdot(b(\,\cdot\,, \rho_t)\rho_t), \tag{4.23}$$

where $\partial_\theta := e_a\partial_a + \sum_{k=1}^{n_0} e_k\partial_{w_k}$ is the parameter-space Dirac operator satisfying $\partial_\theta^2 = \Delta_{(a,w)}$, interpreted in the weak sense.

To derive the weak formulation directly from the hyperfinite system, let $q : \Theta \to \mathbb{R}$ be a smooth compactly supported standard test function, with natural extension ${}^*q : {}^*\Theta \to {}^*\mathbb{R}$. By Transfer of the chain rule, for the internal empirical measure $\rho_t^{(N)} = \frac{1}{N}\sum_{i=1}^N \delta_{\theta_i(t)}$,

$$\frac{d}{dt}\int_{{}^*\Theta} {}^*q(\theta)\,\rho_t^{(N)}(d\theta) = \frac{1}{N}\sum_{i=1}^{N}\left\langle {}^*\nabla q(\theta_i(t)), \partial_t\theta_i(t)\right\rangle. \tag{4.24}$$

Using the internal McKean–Vlasov form (4.19), this becomes the exact internal identity

$$\frac{d}{dt}\int_{{}^*\Theta} {}^*q\,d\rho_t^{(N)} = \int_{{}^*\Theta}\left\langle {}^*\nabla q(\theta), b(\theta, \rho_t^{(N)})\right\rangle\,\rho_t^{(N)}(d\theta). \tag{4.25}$$

Because q has compact support in Θ, ${}^*\nabla q$ vanishes outside a bounded nearstandard set on which ${}^*\nabla q(\theta) \approx \nabla q(\mathrm{st}(\theta))$ and $b(\theta, \rho_t^{(N)}) \approx b(\mathrm{st}(\theta), \rho_t)$ whenever $f_{\rho_t^{(N)}}(x^\alpha) \approx f_{\rho_t}(x^\alpha)$. Taking the standard part of the internal time derivative (4.25) (via S-differentiability of $\int {}^*q\,d\rho_t^{(N)}$; see Appendix A) and pushing forward $L(\rho_t^{(N)})$ by st onto $\rho_t = L(\rho_t^{(N)})\circ\mathrm{st}^{-1}$ yields the weak formulation

$$\frac{d}{dt}\int_\Theta q\,d\rho_t = \int_\Theta \langle\nabla q(\theta), b(\theta, \rho_t)\rangle\,\rho_t(d\theta). \tag{4.26}$$

An integration by parts in (4.26) recovers the divergence form (4.23).

The same transport equation also has a Wasserstein gradient-flow interpretation. Define the functional on measures

$$F(\rho) := \frac{1}{2m}\sum_{\alpha=1}^{m}(f_\rho(x^\alpha) - y^\alpha)^2. \tag{4.27}$$

The mean-field dynamics can be viewed as the W_2-gradient flow of F in the space of probability measures, in the sense of Otto's calculus: the continuity equation (4.23) has the Dirac divergence form $\partial_t\rho_t + \langle\partial_\theta(\rho_t v_t)\rangle_0 = 0$ with Dirac gradient velocity field $v_t(\theta) = -\eta_0\partial_\theta(\delta F/\delta\rho)(\theta)$. Because $\partial_\theta \wedge \partial_\theta = 0$ on smooth scalars, v_t satisfies the exact monogenic curl-freeness identity $\langle\partial_\theta v_t\rangle_2 =$

$\partial_\theta \wedge v_t = 0$, so its full geometric Dirac derivative is purely scalar: $\partial_\theta v_t = \langle \partial_\theta v_t \rangle_0 = -\eta_0 \Delta_\theta (\delta F / \delta \rho)$. This is the same structure that underlies the Jordan–Kinderlehrer–Otto scheme [34] and the general theory of metric-space gradient flows [2].

The standard mean-field picture for two-layer networks emerged from several essentially concurrent papers between 2017 and 2020. Under an overall $1/N$ output scaling, these works viewed the network as an interacting particle system in parameter space whose empirical measure shadows a deterministic transport PDE with a Wasserstein gradient-flow interpretation. In particular, the perspectives in Nitanda and Suzuki [50], Rotskoff and Vanden-Eijnden [56], Mei–Montanari–Nguyen [47], Chizat and Bach [13], and Sirignano and Spiliopoulos [58] (first posted on arXiv in 2018) are strikingly aligned despite emphasizing different tools and goals. This rapid convergence is one reason the "neurons as particles" framework became a natural organizing principle for shallow feature learning.

Remark 4.5 (Permutation Invariance in Deep Networks). A one-hidden-layer network is invariant under internal permutations of the N neurons: relabeling (a_i, w_i) does not change f. The empirical-measure description in Definition 4.2 is essentially the quotient by this symmetry.

For deep networks, there are independent internal permutations in each hidden layer, and the natural objects are distributions over layerwise paths through the network, or equivalently increasingly structured exchangeable arrays. This rapidly pushes us beyond the classical McKean–Vlasov setting and explains why the shallow mean-field PDE does not straightforwardly generalize to deep feature learning.

Tracking deep training dynamics at unlimited width therefore requires additional machinery. One such tool is dynamical mean field theory (DMFT), which we sketch in Section 4.5 after introducing the maximal-update scaling.

4.4 Maximal Update and the μP Parameterization

Mean-field scaling in deep networks admits multiple inequivalent ways to distribute width-dependent hyperreal factors across layers and to scale learning rates. The μP framework (also called maximal update parameterization) provides a systematic way to choose these scalings so that representation learning persists at unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ [62, 63]. (A personal note: I am not particularly attached to the naming "μP"—the term maximal update captures the core idea perfectly well.)

Consider a standard depth-d ($d \in \mathbb{N}$) fully-connected network with d hidden layers of unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, internal forward preactivations $h_\ell(x) \in {}^*\mathbb{R}^N$, and postactivations $\phi_\ell(x) := \phi(h_\ell(x))$:

$$f(x;\theta) = \frac{1}{\gamma_N \sqrt{N}} W_d \phi_d(x), \tag{4.28}$$

$$h_{\ell+1}(x) = \frac{1}{\sqrt{N}} W_\ell \phi_\ell(x), \qquad \ell = 1, \dots, d-1, \tag{4.29}$$

$$h_1(x) = \frac{1}{\sqrt{n_0}} W_0 x. \tag{4.30}$$

We take internal weight matrices $W_0 \in {}^*\mathbb{R}^{N \times n_0}$, $W_\ell \in {}^*\mathbb{R}^{N \times N}$ for $1 \le \ell \le d-1$, and $W_d \in {}^*\mathbb{R}^{1 \times N}$. These are raw weight coordinates: the width-dependent multipliers are displayed explicitly in (4.28)–(4.30), and the same hyperreal learning-rate multiplier $\eta \in {}^*\mathbb{R}_{>0}$ is applied to

every raw weight matrix. A standard choice is i.i.d. internal Gaussian initialization $[W_\ell(0)]_{ij} \sim {}^*\mathcal{N}(0,1)$ (internally independent across entries and layers), together with a global learning-rate scaling that we derive below in ${}^*\mathbb{R}^\times/\mathbb{A}$.

The activation function does not change the width valuation of the scaling analysis below, as long as ϕ and ϕ' take appreciable values on the typical nearstandard scale of preactivations; it mainly contributes appreciable multiplicative factors in $\mathbb{A}$ and modifies the standard shadow kernels.

To formulate maximal update at the representation level, we isolate the contribution to the next-layer preactivations coming only from the previous layer's weight update.

Definition 4.6 (Maximal Feature Update Criterion)**.** Fix $\ell \in \{1,\dots,d\}$ and consider one internal gradient-based update step producing a weight increment $\Delta W_{\ell-1}$. Define the induced change in the layer-ℓ preactivations coming only from $W_{\ell-1}$ by

$$\Delta_{W_{\ell-1}} h_\ell(x) := \begin{cases} \dfrac{1}{\sqrt{n_0}} \Delta W_0 x, & \ell = 1, \\ \dfrac{1}{\sqrt{N}} \Delta W_{\ell-1} \phi_{\ell-1}(x), & 2 \le \ell \le d. \end{cases} \tag{4.31}$$

We say the parameterization achieves a *maximal update at layer* ℓ if, for typical training inputs and steps, its normalized Clifford reverse-norm in ${}^*\mathcal{C}\ell_N$ is appreciable:

$$\frac{1}{N} \left\langle \Delta_{W_{\ell-1}} h_\ell(x^\alpha)^\sim \, \Delta_{W_{\ell-1}} h_\ell(x^\alpha) \right\rangle_0 = \frac{1}{N} \left\| \Delta_{W_{\ell-1}} h_\ell(x^\alpha) \right\|^2 \in \mathbb{A}. \tag{4.32}$$

If (4.32) holds simultaneously for all layers $\ell = 1,\dots,d$ (with a single consistent valuation class of γ_N and learning-rate hyperparameters), we say that the network satisfies the *maximal feature update criterion*. This is the operational layerwise condition used below; output stability will select the corresponding maximal-update/μP scaling.

Besides quantifying representation motion, the maximal-update criterion also explains when the NTK shadow itself can evolve. In the lazy/NTK regime, the empirical forward feature kernels $\Phi_\ell^{(N),\alpha\beta}(t)$ remain infinitesimally close to their initialization ($\Delta\Phi_\ell^{(N),\alpha\beta}(t) \approx 0$), so the NTK freezes to its initial shadow. In contrast, let $G_\ell^{(N),\alpha\beta}(t)$ denote the internal empirical backward kernels formed from the scaled sensitivities introduced below, with $G_{d+1}^{(N),\alpha\beta}(t) := 1$. Consistently with the shallow convention above, define the normalized empirical NTK by the exact internal decomposition (by Transfer of the chain rule)

$$K_t^{(N),\alpha\beta} := \sum_{\ell=0}^{d} G_{\ell+1}^{(N),\alpha\beta}(t)\, \Phi_\ell^{(N),\alpha\beta}(t). \tag{4.33}$$

Because the sensitivities below contain an additional factor γ_N relative to those in Chapter 2, the raw gradient Gram is $\gamma_N^{-2} K_t^{(N)}$. Consequently, under (4.3), the exact internal predictor evolution is

$$\partial_t f^\alpha(t) = -\frac{\eta_N}{m\gamma_N^2} \sum_{\beta=1}^{m} K_t^{(N),\alpha\beta} r^\beta(t). \tag{4.34}$$

Thus $K_t^{(N)}$ can have a nonvanishing appreciable shadow even when the raw gradient Gram is infinitesimal, while the ratio η_N/γ_N^2 controls the output time scale. The maximal-update

condition (4.32) keeps the feature changes induced by each layer's weight update in $\mathbb{A}$, allowing the corresponding Φ_ℓ shadows to evolve by appreciable amounts. When these Φ_ℓ evolve, each layer's contribution to the normalized NTK shadow can evolve as well, enabling a genuinely non-kernel limit. The reduction to a kernel method relies on a kernel function fixed independently of the training data and unchanged up to infinitesimals throughout optimization. Once feature learning produces a nontrivial evolution of the shadow tangent kernel, this fixed-kernel reduction no longer applies.

We now give a valuation-group calculation in ${}^*\mathbb{R}^\times/\mathbb{A}$ showing how γ_N and the learning-rate scaling must be chosen so that (i) the network function evolves at appreciable ($\in \mathbb{A}$) speed and (ii) each hidden layer satisfies maximal update in the sense of Definition 4.6. We perform the calculation for GD; for SGD the width valuation class is the same (batch-size factors lie in $\mathbb{A}$ and do not change valuation classes in ${}^*\mathbb{R}^\times/\mathbb{A}$). We organize the calculation into six steps.

Step 1: Setup. For notational clarity, consider a single sample (x, y) and the per-sample squared loss

$$L(\theta) = \frac{1}{2}\big(f(x;\theta) - y\big)^2. \tag{4.35}$$

Let the residual be $r := f(x;\theta) - y$, which we treat as appreciable ($r \in \mathbb{A}$) since generic nonzero targets satisfy $y \in \mathbb{A}$. Write the forward postactivations as $\phi_\ell = \phi(h_\ell)$.

For the backward pass, it is convenient in the μP setting to absorb the output scale factor γ_N into the backpropagated signal. Define the μP-scaled internal backward sensitivity by

$$g_\ell := \gamma_N \sqrt{N}\, \frac{\partial f}{\partial h_\ell} \in {}^*\mathbb{R}^N. \tag{4.36}$$

(Compared to the Chapter 2 normalization $\sqrt{N}\,\partial f/\partial h_\ell$, this inserts an extra factor of γ_N; the benefit is that g_ℓ stays appreciable under μP scaling.)

Under the scaling (4.28)–(4.30), Transfer gives the exact internal backpropagation recursion

$$g_\ell = \frac{1}{\sqrt{N}}\, \mathrm{diag}(\phi'(h_\ell)) W_\ell^\top g_{\ell+1}, \qquad \ell = d-1, \dots, 1, \tag{4.37}$$

with top boundary

$$g_d = \gamma_N \sqrt{N}\, \frac{\partial}{\partial h_d}\left(\frac{1}{\gamma_N \sqrt{N}} W_d \phi(h_d)\right) = W_d^\top \odot \phi'(h_d), \tag{4.38}$$

so the typical-coordinate valuation class of g_ℓ is the same at every layer under stable wide-limit forward and backward signal propagation. We record this condition in normalized root-mean-square form:

$$\frac{1}{N}\,\|g_\ell\|^2 \in \mathbb{A}, \qquad \ell = 1, \dots, d. \tag{4.39}$$

Step 2: Gradient Magnitudes. Since $L = \frac{1}{2}r^2$, we have $\nabla_\theta L = r\,\nabla_\theta f$. For $1 \le \ell \le d-1$, using $h_{\ell+1} = \frac{1}{\sqrt{N}} W_\ell \phi_\ell$,

$$\nabla_{W_\ell} f = \frac{1}{\sqrt{N}}\left(\frac{\partial f}{\partial h_{\ell+1}}\right)\phi_\ell^\top = \frac{1}{\gamma_N N} g_{\ell+1}\phi_\ell^\top, \tag{4.40}$$

so a generic coordinate of $\nabla_{W_\ell} L$ has valuation class

$$[\nabla_{W_\ell} L]_{ij} = \frac{r}{\gamma_N N} g_{\ell+1,i} \phi_{\ell,j} \asymp \frac{1}{\gamma_N N}, \tag{4.41}$$

since $r \in \mathbb{A}$, $\phi_{\ell,j} \in \mathbb{A}$, and a typical coordinate $g_{\ell+1,i} \in \mathbb{A}$ by (4.39). Thus an SGD/GD update with learning rate η satisfies

$$[\Delta W_\ell]_{ij} \asymp \frac{\eta}{\gamma_N N}. \tag{4.42}$$

Step 3: Maximal Update Condition. Now consider the layerwise preactivation change from (4.31):

$$\Delta_{W_\ell} h_{\ell+1} = \frac{1}{\sqrt{N}} \Delta W_\ell \phi_\ell. \tag{4.43}$$

Using that in this single-sample GD step $\Delta W_\ell = -\eta \, \nabla_{W_\ell} L$ is proportional to the outer product $g_{\ell+1} \phi_\ell^\top$, we obtain componentwise

$$[\Delta_{W_\ell} h_{\ell+1}]_i = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} [\Delta W_\ell]_{ij} \phi_{\ell,j} = -\frac{\eta r}{\sqrt{N}} \frac{1}{\gamma_N N} g_{\ell+1,i} \sum_{j=1}^{N} \phi_{\ell,j}^2. \tag{4.44}$$

Since $\frac{1}{N} \sum_{j=1}^{N} \phi_{\ell,j}^2 \in \mathbb{A}$, the hyperfinite sum satisfies $\sum_{j=1}^{N} \phi_{\ell,j}^2 \asymp N$ and thus

$$[\Delta_{W_\ell} h_{\ell+1}]_i \asymp \frac{\eta}{\sqrt{N}} \frac{1}{\gamma_N} g_{\ell+1,i} \asymp \frac{\eta}{\gamma_N \sqrt{N}}. \tag{4.45}$$

Therefore

$$\frac{1}{N} \left\| \Delta_{W_\ell} h_{\ell+1} \right\|^2 \asymp \frac{\eta^2}{\gamma_N^2 N}, \tag{4.46}$$

and enforcing the maximal-update criterion (4.32) in ${}^*\mathbb{R}^\times/\mathbb{A}$ requires

$$\eta \asymp \gamma_N \sqrt{N}. \tag{4.47}$$

Step 4: Input Boundary. The first weight matrix has the exact internal derivative

$$\nabla_{W_0} f = \frac{1}{\gamma_N \sqrt{N n_0}} g_1 x^\top. \tag{4.48}$$

Consequently, for the single-sample update used here,

$$\Delta_{W_0} h_1 = -\frac{\eta r}{\gamma_N \sqrt{N}} g_1 \frac{\|x\|^2}{n_0} \asymp \frac{\eta}{\gamma_N \sqrt{N}} \tag{4.49}$$

when $n_0^{-1} \|x\|^2 \in \mathbb{A}$. Thus the same maximal-update constraint (4.47) applies at the input boundary as at every internal hidden layer.

Step 5: Stable Function-Space Dynamics. We also require that the output itself changes on an appreciable ($\in \mathbb{A}$) scale. Consider the contribution coming from the output weights W_d. Since $f(x) = \frac{1}{\gamma_N\sqrt{N}} W_d \phi_d$, we have

$$\nabla_{W_d} f = \frac{1}{\gamma_N\sqrt{N}} \phi_d^\top, \qquad \Longrightarrow \qquad [\nabla_{W_d} L]_j \asymp \frac{r}{\gamma_N\sqrt{N}} \asymp \frac{1}{\gamma_N\sqrt{N}}. \tag{4.50}$$

Thus $[\Delta W_d]_j \asymp \eta/(\gamma_N\sqrt{N})$ and the resulting change in the output is

$$\Delta f = \frac{1}{\gamma_N\sqrt{N}} \Delta W_d \phi_d = -\frac{\eta r}{\gamma_N^2 N} \sum_{j=1}^{N} \phi_{d,j}^2 \asymp \frac{\eta}{\gamma_N^2}, \tag{4.51}$$

using again that $\frac{1}{N}\sum_{j=1}^{N} \phi_{d,j}^2 \in \mathbb{A}$. To keep $\Delta f \in \mathbb{A}$ we therefore need

$$\eta \asymp \gamma_N^2. \tag{4.52}$$

Step 6: Solving the Scaling Constraints. Combining (4.52) and (4.47) in the valuation group ${}^*\mathbb{R}^\times/\mathbb{A}$ ($\gamma_N^2 \asymp \gamma_N\sqrt{N}$) uniquely yields

$$\gamma_N \asymp \sqrt{N}, \qquad \eta \asymp N. \tag{4.53}$$

Equivalently, the overall output prefactor has valuation class $1/(\gamma_N\sqrt{N}) \asymp 1/N$ (and equals $1/N$ when $\gamma_N = \sqrt{N}$), and the learning rate is scaled up by N. This is the basic μP/maximal-update scaling for MLPs.

Remark 4.7 (Shallow Mean Field as the $d = 1$ Case). For $d = 1$, identify $a_i = (W_d)_{1i}$ and let $w_i^\top$ be the ith row of W_0. Then (4.28)–(4.30) is exactly the shallow model (4.4):

$$f(x) = \frac{1}{\gamma_N\sqrt{N}} \sum_{i=1}^{N} a_i \, \phi\left(\frac{1}{\sqrt{n_0}} \langle w_i, x\rangle\right). \tag{4.54}$$

Hence $\gamma_N = \sqrt{N}$ and the common raw learning-rate multiplier $\eta = \eta_0 N$ recover the shallow mean-field parameterization and hyperfinite particle dynamics (4.19) exactly [62].

Remark 4.8 (Generality). The maximal-update/μP philosophy extends far beyond fully-connected MLPs: one can, in principle, derive analogous width-dependent valuation classes in ${}^*\mathbb{R}^\times/\mathbb{A}$ for other architectures (CNNs, attention, residual connections, etc.) and for other training algorithms (Adam, momentum, weight decay, etc.). But each new setting requires its own calculation, because the relevant forward/backward signal propagation and parameter grouping changes.

4.5 A Sketch of DMFT for Deep Feature Learning

The deep μP scaling does not collapse to a deterministic PDE like the shallow mean-field limit. Instead, a representative forward/backward coordinate remains stochastic at unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, while its statistics close through deterministic time-dependent kernel order parameters. Dynamical mean field theory (DMFT) provides this reduced description. We follow

the infinite-width self-consistent DMFT of Bordelon and Pehlevan [7, Secs. 2–3 and App. D.7]; their later work develops finite-width fluctuations around this limit [8].

We begin by introducing the internal state variables and kernels whose shadows close the limiting dynamics. At hyperfinite width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, let $h_\ell^\alpha(t) = (h_{\ell,i}^\alpha(t))_{i=1}^N \in {}^*\mathbb{R}^N$ denote the internal forward preactivations at layer ℓ on input x^α. For the backward pass, use the μP-scaled internal sensitivity

$$g_\ell^\alpha(t) := \gamma_N \sqrt{N}\, \frac{\partial f^\alpha(t)}{\partial h_\ell^\alpha(t)} = \big(g_{\ell,i}^\alpha(t)\big)_{i=1}^N \in {}^*\mathbb{R}^N, \tag{4.55}$$

together with the pre-activation backward drive $z_\ell^\alpha(t) = (z_{\ell,i}^\alpha(t))_{i=1}^N \in {}^*\mathbb{R}^N$ defined by $z_\ell^\alpha(t) := \frac{1}{\sqrt{N}} W_\ell(t)^\top g_{\ell+1}^\alpha(t)$ for $1 \le \ell \le d-1$ and $z_d^\alpha(t) := W_d(t)^\top$, so that $g_{\ell,i}^\alpha(t) = \phi'(h_{\ell,i}^\alpha(t))\, z_{\ell,i}^\alpha(t)$, whose typical coordinates lie in $\mathbb{A}$ under the stable scaling assumptions of Section 4.4. We reserve the superscript (N) for hyperfinite empirical kernels.

The hyperfinite-width two-time feature and gradient kernels are

$$\Phi_\ell^{(N),\alpha\beta}(t,s) := \frac{1}{N}\Big\langle \phi(h_\ell^\alpha(t)), \phi(h_\ell^\beta(s)) \Big\rangle, \qquad G_\ell^{(N),\alpha\beta}(t,s) := \frac{1}{N}\Big\langle g_\ell^\alpha(t), g_\ell^\beta(s) \Big\rangle. \tag{4.56}$$

Their diagonal $t = s$ agrees with the empirical kernels used above. DMFT also introduces standard response (or susceptibility) kernels A_ℓ and B_ℓ, defined below.

With this notation in place, we can realize the single-site DMFT process concretely via Loeb sampling. Fix $\gamma_N = \gamma_0\sqrt{N}$ with standard $\gamma_0 > 0$ (or $\gamma_N \asymp \sqrt{N}$ with $\gamma_0 := \mathrm{st}(\gamma_N/\sqrt{N}) \in (0,\infty)$) and the stable learning-rate scale $\eta_N \asymp \gamma_N^2$ (Section 4.4). If u denotes physical time under (4.3), define macroscopic time by $t := \eta_N u/(m\gamma_N^2)$. At standard depth d, standard sample size m, and over a standard finite time horizon, draw a neuron index $\iota \in \{1,\dots,N\}$ from the Loeb uniform measure $L(\nu_N)$ on $\{1,\dots,N\}$ (where $\nu_N(E) = |E|/N$, independent of initialization). On the product Loeb space $L({}^*\mathbb{P}) \otimes L(\nu_N)$, we identify the limiting single-site process [7, Eqs. (9)–(10)] with the standard shadow trajectory

$$\Big(\mathrm{st}\big(h_{\ell,\iota}^\alpha(t)\big),\ \mathrm{st}\big(z_{\ell,\iota}^\alpha(t)\big),\ \mathrm{st}\big(g_{\ell,\iota}^\alpha(t)\big)\Big),$$

which from this point onward we denote simply by $(h_\ell^\alpha(t), z_\ell^\alpha(t), g_\ell^\alpha(t))$. Here *single site* is statistical-physics language for the effective process associated with one Loeb-typical neuron index ι after the influence of all other neurons has been summarized by the deterministic shadow kernels. In probabilistic language, this is a representative mean-field particle—more precisely here, its coupled forward/backward shadow trajectory across data examples and training times.

For $1 \le \ell \le d$ and standard $t \ge 0$, the representative shadow process satisfies

$$h_\ell^\alpha(t) = u_\ell^\alpha(t) + \gamma_0 \int_0^t \sum_{\beta=1}^m \Big[A_{\ell-1}^{\alpha\beta}(t,s) - \Phi_{\ell-1}^{\alpha\beta}(t,s)\, r^\beta(s)\Big] g_\ell^\beta(s)\, ds, \tag{4.57}$$

$$z_\ell^\alpha(t) = v_\ell^\alpha(t) + \gamma_0 \int_0^t \sum_{\beta=1}^m \Big[B_\ell^{\alpha\beta}(t,s) - G_{\ell+1}^{\alpha\beta}(t,s)\, r^\beta(s)\Big] \phi(h_\ell^\beta(s))\, ds, \tag{4.58}$$

$$g_\ell^\alpha(t) = \phi'(h_\ell^\alpha(t))\, z_\ell^\alpha(t). \tag{4.59}$$

Here u_ℓ and v_ℓ are independent centered standard Gaussian drives on the Loeb space with covariances

$$\mathbb{E}[u_\ell^\alpha(t)\, u_\ell^\beta(s)] = \Phi_{\ell-1}^{\alpha\beta}(t,s), \qquad \mathbb{E}[v_\ell^\alpha(t)\, v_\ell^\beta(s)] = G_{\ell+1}^{\alpha\beta}(t,s), \qquad u_\ell \perp v_\ell. \tag{4.60}$$

Self-consistency means that the hyperfinite empirical kernels in (4.56) concentrate $L(^*\mathbb{P})$-almost surely, at each standard pair (t,s), around their deterministic standard parts, which coincide with the Loeb expectations over the single-site shadow process:

$$\begin{aligned}
\Phi_\ell^{\alpha\beta}(t,s) &:= \mathrm{st}\big(\Phi_\ell^{(N),\alpha\beta}(t,s)\big) = \mathbb{E}\Big[\phi\big(h_\ell^\alpha(t)\big)\phi\big(h_\ell^\beta(s)\big)\Big], \\
G_\ell^{\alpha\beta}(t,s) &:= \mathrm{st}\big(G_\ell^{(N),\alpha\beta}(t,s)\big) = \mathbb{E}\big[g_\ell^\alpha(t)\, g_\ell^\beta(s)\big], \\
\Phi_0^{\alpha\beta}(t,s) &:= \Phi_0^{\alpha\beta} = \frac{1}{n_0}\left\langle x^\alpha, x^\beta \right\rangle, \qquad G_{d+1}^{\alpha\beta}(t,s) := 1.
\end{aligned} \tag{4.61}$$

For $1 \le \ell \le d-1$, the response kernels are the causal functional derivatives

$$\begin{aligned}
A_\ell^{\alpha\beta}(t,s) &:= \frac{1}{\gamma_0}\,\mathbb{E}\left[\frac{\delta\phi\big(h_\ell^\alpha(t)\big)}{\delta v_\ell^\beta(s)}\right], \qquad & B_\ell^{\alpha\beta}(t,s) &:= \frac{1}{\gamma_0}\,\mathbb{E}\left[\frac{\delta g_{\ell+1}^\alpha(t)}{\delta u_{\ell+1}^\beta(s)}\right], \\
A_0 &:= 0, & B_d &:= 0.
\end{aligned} \tag{4.62}$$

They vanish for $s > t$; the expectation is over the same single-site shadow law on the Loeb product space [7, Sec. 3.2]. Taken together, (4.57)–(4.62) form a fixed-point equation for the law of the representative shadow process and its deterministic kernel order parameters: a candidate collection (Φ, G, A, B) determines the Gaussian drives and memory terms, hence the process, and the correlations and response derivatives of that process must reproduce the same (Φ, G, A, B).

Finally, the deterministic shadow kernels couple back to the training residual. On the equal-time diagonal, the standard shadow of the normalized NTK (4.33) is

$$K_t^{\alpha\beta} := \mathrm{st}\big(K_t^{(N),\alpha\beta}\big) = \sum_{\ell=0}^{d} G_{\ell+1}^{\alpha\beta}(t,t)\,\Phi_\ell^{\alpha\beta}(t,t). \tag{4.63}$$

Writing $r^\alpha(t)$ (and $f^\alpha(t)$) for the deterministic standard shadow $\mathrm{st}(f^\alpha(t) - y^\alpha) = \mathrm{st}(f^\alpha(t)) - y^\alpha$, this closes the macroscopic-time dynamics through

$$r^\alpha(t) = r^\alpha(0) - \int_0^t \sum_{\beta=1}^{m} K_s^{\alpha\beta} r^\beta(s)\, ds, \tag{4.64}$$

equivalently $\partial_t r(t) = -K_t r(t)$ for $r(t) \in \mathbb{R}^m$ [7, Eq. (3)]. Under the centered initialization used here, the internal initial output satisfies $f^\alpha(0) \overset{L(^*\mathbb{P})}{\approx} 0$, so the initial residual concentrates infinitesimally around $-y^\alpha$ ($f^\alpha(0) - y^\alpha \overset{L(^*\mathbb{P})}{\approx} -y^\alpha$, giving shadow initial condition $r^\alpha(0) = -y^\alpha$). Writing the same macroscopic quantities as functions of physical time u via $t = \eta_N u/(m\gamma_N^2)$,

$$\frac{d}{du} f^\alpha(u) = \frac{d}{du} r^\alpha(u) = -\frac{\eta_N}{m\gamma_N^2} \sum_{\beta=1}^{m} K_u^{\alpha\beta} r^\beta(u). \tag{4.65}$$

The practical takeaway is the contrast with the shallow mean-field limit: deep feature learning at unlimited width N is not captured by a single deterministic transport PDE for a parameter measure. The representative Loeb-sampled coordinate retains non-Markovian memory and Gaussian randomness, while the self-consistent shadow kernels and predictor trajectory are deterministic.

Framework and role	Width/time balance	Unlimited-width shadow description
Lazy NTK (scaling regime)	$\gamma_N = 1$, $\eta_N \in \mathbb{A}$; $N^{-1}\left\|\Delta h_\ell^\alpha\right\|^2 \approx 0$.	The normalized NTK concentrates around a time-independent deterministic kernel K; the residual shadow follows linear kernel gradient flow.
Shallow mean field, $d = 1$ (scaling and limit)	$\gamma_N = \sqrt{N}$, $\eta_N = \eta_0 N$; $N^{-1}\left\|\Delta h_1^\alpha\right\|^2 \in \mathbb{A}$.	The Loeb pushforward $L(\rho_t^{(N)}) \circ \mathrm{st}^{-1}$ equals a deterministic law ρ_t governed by a transport PDE/Wasserstein gradient flow; its predictor and K_t can evolve.
Deep maximal update/μP (scaling principle)	$\gamma_N = \gamma_0\sqrt{N}$, $\eta_N \asymp \gamma_N^2 \asymp N$; every hidden layer satisfies $N^{-1}\left\|\Delta_{W_{\ell-1}} h_\ell^\alpha\right\|^2 \in \mathbb{A}$.	Valuation algebra in ${}^*\mathbb{R}^\times/\mathbb{A}$ selects a stable feature-learning scale but does not by itself specify the shadow dynamics; its $d = 1$ case is the preceding row.
DMFT under deep maximal update (shadow description)	With physical time u, use $t = \eta_N u/(m\gamma_N^2)$.	The shadow $(\mathrm{st}(h_{\ell,\iota}^\alpha), \mathrm{st}(z_{\ell,\iota}^\alpha), \mathrm{st}(g_{\ell,\iota}^\alpha))$ of a Loeb-uniform neuron ι closes through deterministic $\Phi_\ell, G_\ell, A_\ell, B_\ell$; K_t and $r(t)$ are deterministic, with $\partial_t r = -K_t r$ in macroscopic time.

Table 4.2: Relationship among the hyperreal scalings and standard shadow descriptions in this chapter.

4.6 Summary and Outlook

The general layerwise feature-learning diagnostic at unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ is $N^{-1}\left\|\Delta h_\ell^\alpha\right\|^2$, which is infinitesimal (≈ 0) in the lazy regime. Maximal update sharpens this diagnostic by requiring the component induced by each layer's own weights to satisfy $N^{-1}\left\|\Delta_{W_{\ell-1}} h_\ell^\alpha\right\|^2 \in \mathbb{A}$. The key distinction is between a hyperreal scaling class in ${}^*\mathbb{R}^\times/\mathbb{A}$ and a description of its standard shadow dynamics. For $d = 1$, maximal update is exactly shallow mean-field scaling, whose Loeb pushforward $L(\rho_t^{(N)}) \circ \mathrm{st}^{-1}$ is a deterministic neuron law ρ_t; for deep networks, DMFT describes the corresponding feature-learning shadow via a Loeb-sampled representative coordinate. Thus DMFT is not an additional parameterization.

At output level, the normalized NTK is the common bridge: its shadow freezes to K in the lazy regime and evolves with the state in feature-learning regimes. The maximal-update (μP) regime in this chapter takes an unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ at fixed standard depth $d \in \mathbb{N}$, whereas Chapters 5 and 6 study unlimited depth and width in the proportional galaxy. These galaxies are not compatible: hyperfinite $O(1/N)$ fluctuations that are infinitesimal at fixed standard depth accumulate over $D \asymp N$ layers to an appreciable effect. As we will see, the proportional galaxy is precisely the regime that captures this phenomenon.

4.7 Exercises

1. **Shallow Scaling Balance.** Assume that the residuals, activation factors, input kernels, and normalized empirical NTK lie in $\mathbb{A}$, and write $\gamma_N \asymp N^p$ and $\eta_N \asymp N^q$ for standard exponents $p, q \in \mathbb{R}$. Use (4.10) and (4.13) in the valuation quotient ${}^*\mathbb{R}^\times/\mathbb{A}$ to show that

a typical feature coordinate and the predictor speed satisfy

$$\Delta h_i^\alpha \asymp N^{q-p-1/2}, \qquad \partial_t f^\alpha \asymp N^{q-2p}. \tag{4.66}$$

Require both quantities to belong to $\mathbb{A}$ (noting that $N^r \in \mathbb{A}$ for $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ and standard $r \in \mathbb{R}$ if and only if $r = 0$) and solve for p and q. Conclude that $\gamma_N \asymp \sqrt{N}$, $\eta_N \asymp N$, and the output prefactor satisfies $1/(\gamma_N\sqrt{N}) \asymp 1/N$. Finally, recover from the same formulas the feature and output valuation classes in the NTK regime $(p, q) = (0, 0)$ and the naive large-learning-rate regime $(p, q) = (0, 1/2)$.

2. **Shallow Mean Field as the $d = 1$ Maximal-Update Model.** Set $d = 1$, identify $a_i = (W_1)_{1i}$, and let $w_i^\top$ be the ith row of W_0. First verify that (4.28)–(4.30) becomes exactly (4.4). Next show that $g_{1,i}^\alpha = a_i\phi'(h_i^\alpha)$. Using $G_2^{(N),\alpha\beta} = 1$, show that the deep normalized-NTK decomposition (4.33) reduces exactly to the shallow formula (4.12). Finally, with $\gamma_N = \sqrt{N}$ and $\eta_N = \eta_0 N$, verify coordinatewise that the internal gradient updates of W_0 and W_1 agree with the shallow particle updates of w_i and a_i. Conclude that the shallow mean-field model is the $d = 1$ maximal-update model without any additional change of coordinates or layerwise learning rates.

3. **DMFT Residual, Sign, and Time Conventions.** Starting from the physical-time equation (4.65) and the change of time $t = \eta_N u/(m\gamma_N^2)$, use $r^\alpha = f^\alpha - y^\alpha$ to derive

$$\frac{d}{dt} r^\alpha(t) = -\sum_{\beta=1}^{m} K_t^{\alpha\beta} r^\beta(t) \tag{4.67}$$

and then recover the integral equation (4.64). Check that centered initialization gives $f^\alpha(0) - y^\alpha \overset{L(^*\mathbb{P})}{\approx} -y^\alpha$, so the shadow initial condition is $r^\alpha(0) = -y^\alpha$. If the data-dependent terms in the representative DMFT equations are temporarily written as $A_{\ell-1} + \Phi_{\ell-1}(y - f)$ and $B_\ell + G_{\ell+1}(y - f)$, use $y^\beta - f^\beta = -r^\beta$ to verify the signs $A_{\ell-1} - \Phi_{\ell-1} r$ in (4.57) and $B_\ell - G_{\ell+1} r$ in (4.58). Finally, substitute $A_0 = 0$ and $\Phi_0^{\alpha\beta} = n_0^{-1} \left\langle x^\alpha, x^\beta \right\rangle$ at the input boundary, and $B_d = 0$ and $G_{d+1}^{\alpha\beta} = 1$ at the output boundary, and write the resulting $\ell = 1$ and $\ell = d$ equations explicitly.

Chapter 5

Depth Scaling and Covariance SDE Limits

These notes study what happens when network *depth* becomes unlimited ($D \in {}^*\mathbb{N} \setminus \mathbb{N}$). The recurring template is: (i) identify an infinitesimal scaling (step size) per layer so that each layer contributes an infinitesimal perturbation over a hyperfinite depth grid, and (ii) take the standard part to obtain a continuous-time shadow.

Throughout this chapter, unless stated otherwise, weight entries are i.i.d. standard Gaussian (internally ${}^*\mathcal{N}(0,1)$ in hyperfinite models) and activations are applied coordinatewise. For each internal model with mutually ${}^*\mathbb{P}$-independent layer weights, let $\mathcal{F}_\ell$ denote the internal algebra generated by all hidden states through layer ℓ (including every propagated input when there are several). By Transfer, conditioning on $\mathcal{F}_\ell$ makes the next-layer preactivations internally Gaussian at every hyperfinite width, just as in the finite-width identity (2.26) of Chapter 2. Unlimited-width shadows enter when the resulting internal empirical covariances and their one-step fluctuations are lifted via the standard-part map to a deterministic recursion or an Itô diffusion on the Loeb probability space $(\Omega, \mathcal{L}(\mathcal{A}), L({}^*\mathbb{P}))$ [44, 3, 37]. Although this chapter focuses on networks at initialization with i.i.d. Gaussian weight entries, Section 6.3 in Chapter 6 develops a one-step training-time calculation in the same proportional depth–width framework. We use φ (with natural extension ${}^*\varphi$) for the activation and write standard data points as $\{x^\alpha\}_{\alpha=1}^m$.

5.1 ResNet Scaling Limits

We begin with the hyperfinite Euler approximation underlying the deterministic-depth regime. An ordinary differential equation (ODE) is determined by a standard vector field b and an initial condition x_0.

Definition 5.1 (ODE Flow)**.** Let $b : \mathbb{R}^n \to \mathbb{R}^n$ and $x_0 \in \mathbb{R}^n$. An ODE solution is a differentiable path $(X_t)_{t\in[0,T]}$ satisfying

$$\partial_t X_t = b(X_t), \qquad X_0 = x_0. \tag{5.1}$$

For an unlimited depth $D \in {}^*\mathbb{N} \setminus \mathbb{N}$ and positive infinitesimal step size $\eta = T/D \approx 0$ (or more generally any positive $\eta \approx 0$) on the hyperfinite layer-time grid $\mathbb{T}_D = \{0, \eta, 2\eta, \ldots, D\eta\}$, the simplest numerical scheme is the internal Euler recursion driven by the natural extension

${}^*b : {}^*\mathbb{R}^n \to {}^*\mathbb{R}^n$:

$$Y_{k+1} = Y_k + \eta\, {}^*b(Y_k), \qquad Y_0 = x_0. \tag{5.2}$$

It is helpful to compare the hyperfinite sequence to its piecewise-constant interpolation:

$$Y_t^{(\eta)} := Y_{\lfloor t/\eta \rfloor}, \qquad t \in [0, T]. \tag{5.3}$$

Theorem 5.2 (Euler Convergence)**.** *Assume b is globally Lipschitz on $\mathbb{R}^n$:*

$$\|b(x) - b(y)\| \le C_1 \|x - y\| \qquad \forall x, y \in \mathbb{R}^n. \tag{5.4}$$

Then the ODE (5.1) has a unique solution $(X_t)_{t\in[0,T]}$ on $[0, T]$, and for every positive infinitesimal step size $\eta \approx 0$, the hyperfinite Euler recursion (5.2) satisfies

$$\max_{0\le k\eta\le T} \|{}^*X_{k\eta} - Y_k\| \approx 0, \qquad \textit{so} \qquad \mathrm{st}\Big(Y_{\lfloor t/\eta \rfloor}\Big) = X_t \quad \textit{uniformly on } [0, T]. \tag{5.5}$$

The conclusion in (5.5) compares the whole hyperfinite depth trajectory with the ODE path, rather than only the network state at one chosen layer. With layer k identified with hyperfinite grid time $k\eta \in \mathbb{T}_D$, the approximation error is infinitesimal simultaneously for every internal layer k with $k\eta \le T$, so a single standard ODE path serves as the uniform shadow of the full deep network.

The Euler viewpoint becomes structural in deep residual networks (ResNets) [28]. For comparison, consider an MLP layer of standard width $n \in \mathbb{N}$:

$$h_{\ell+1} = \frac{1}{\sqrt{n}} W_\ell \varphi(h_\ell), \qquad W_\ell \in \mathbb{R}^{n\times n}. \tag{5.6}$$

In contrast, a (basic) ResNet layer adds a skip connection:

$$h_{\ell+1} = h_\ell + \frac{1}{\sqrt{n}} W_\ell \varphi(h_\ell). \tag{5.7}$$

The skip term makes it natural to rescale the residual block so that each layer is an infinitesimal update.

A classical deterministic-depth scaling is

$$h_{\ell+1} = h_\ell + \eta \frac{1}{\sqrt{n}} W^* \varphi(h_\ell), \tag{5.8}$$

with a *shared* standard weight matrix $W \in \mathbb{R}^{n\times n}$ across layers (a recurrent-in-depth model). Fix a standard "layer time" horizon $\bar\tau > 0$ and, for a network of unlimited depth $D \in {}^*\mathbb{N} \setminus \mathbb{N}$, set $\eta := \bar\tau/D \approx 0$ and $\tau_\ell := \eta\ell \in \mathbb{T}_D = \{0, \eta, 2\eta, \dots, D\eta\}$. Then the update (5.8) is the exact hyperfinite Euler discretization of

$$\partial_\tau h_\tau = \frac{1}{\sqrt{n}} W \varphi(h_\tau), \qquad \tau \in [0, \bar\tau]. \tag{5.9}$$

For fixed standard n and W, a Lipschitz activation (including ReLU) makes the vector field in (5.9) globally Lipschitz, so Theorem 5.2 yields $\mathrm{st}(h_{\lfloor \tau/\eta \rfloor}) = h_\tau$ uniformly on $[0, \bar\tau]$ at every unlimited depth $D \in {}^*\mathbb{N} \setminus \mathbb{N}$. This connects deep ResNets to *Neural ODEs* [12].

The same viewpoint accommodates non-uniform hyperfinite layer times: varying infinitesimal step sizes $\eta_\ell \approx 0$ and/or layer weights W_ℓ can track a time-inhomogeneous ODE whenever the vector field varies S-continuously along the grid.

5.1.1 CLT Depth Scaling: From Random Walk to SDE

If each layer contributes random fluctuations of infinitesimal size $\sqrt{\Delta\tau}$ (rather than deterministic size $\Delta\tau$), then a diffusion shadow emerges.

The basic template is Anderson's hyperfinite random walk [3], lifting Donsker's random-walk invariance principle [17] from (1.3)–(1.5) in Chapter 1: centered independent increments of size $\sqrt{\Delta\tau}$ over a hyperfinite grid accumulate into an S-continuous trajectory whose standard part is Brownian motion. Allowing state-dependent variance and adding an $\mathcal{O}(\Delta\tau)$ drift gives a Keisler hyperfinite stochastic difference equation [37, 43] whose shadow solves an Itô SDE

$$dX_\tau = b(X_\tau)\,d\tau + \sigma(X_\tau)\,dB_\tau. \tag{5.10}$$

The same mechanism appears in the *CLT-scaled* ResNet of unlimited depth $D \in {}^*\mathbb{N} \setminus \mathbb{N}$ and standard width $n \in \mathbb{N}$, with infinitesimal layer-time step $\Delta\tau := 1/D \approx 0$:

$$h_{\ell+1} = h_\ell + \sqrt{\Delta\tau}\,\frac{1}{\sqrt{n}} W_\ell{}^*\varphi(h_\ell), \tag{5.11}$$

where $(W_\ell)_{\ell\geq 0}$ are mutually ${}^*\mathbb{P}$-independent internal $n \times n$ matrices with i.i.d. ${}^*\mathcal{N}(0,1)$ entries. Interpret the hyperfinite *layer time* as $\tau_\ell := \ell\Delta\tau$ (so $\ell = \lfloor \tau/\Delta\tau \rfloor$ tracks standard time τ). Since W_ℓ is ${}^*\mathbb{P}$-independent of $\mathcal{F}_\ell$, by Transfer the next-layer conditional transition is internally Gaussian:

$$h_{\ell+1} \mid \mathcal{F}_\ell \sim {}^*\mathcal{N}\!\left(h_\ell,\, \Delta\tau\,\frac{1}{n}\,\|{}^*\varphi(h_\ell)\|^2\, I_n\right). \tag{5.12}$$

Thus (5.11) is, in transition law, an exact Keisler hyperfinite stochastic difference equation corresponding to

$$dh_\tau = \frac{1}{\sqrt{n}}\,\|\varphi(h_\tau)\|\,dB_\tau, \tag{5.13}$$

where B_τ is a standard n-dimensional Brownian motion on the Loeb space $(\Omega, \mathcal{L}(\mathcal{A}), L({}^*\mathbb{P}))$. For standard $n \in \mathbb{N}$ and a globally Lipschitz activation (including ReLU), the diffusion coefficient $h \mapsto \frac{1}{\sqrt{n}}\,\|\varphi(h)\|$ is globally Lipschitz, so Keisler's hyperfinite SDE lifting theorem [37] implies that $\tau \mapsto h_{\lfloor \tau/\Delta\tau \rfloor}$ is $L({}^*\mathbb{P})$-almost surely S-continuous and its shadow $h_\tau = \mathrm{st}(h_{\lfloor \tau/\Delta\tau \rfloor})$ solves (5.13).

When both width N and depth D are allowed to be unlimited in $({}^*\mathbb{N})^2$, distinct asymptotic scaling routes correspond to distinct hyperfinite depth–width regimes: the wide galaxy $D/N \approx 0$ (modeling width taken infinite before depth), the deep galaxy $N/D \approx 0$ (depth taken infinite before width), and the proportional galaxy $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$. For the no-bias Gaussian ReLU specialization of (5.11), Hayou and Yang prove that the shadow of the one-neuron preactivation law and empirical covariance flow is independent of the relative magnitudes of N and D, provided $\min\{N, D\} \in {}^*\mathbb{N} \setminus \mathbb{N}$ [26, Theorems 1 and 2]. In particular, the wide regime $D/N \approx 0$, the deep regime $N/D \approx 0$, and proportional pairs with $\mathrm{st}(D/N) = \bar{\tau}$ all yield the same covariance shadow for these observables. This is a model-specific commutation result: for plain MLPs, proportional and extreme depth–width galaxies can instead disagree. The representative routes in $({}^*\mathbb{N})^2$ are summarized in Figure 5.1.

5.2 Proportional Depth–Width Limits at Initialization

To see why joint hyperfinite limits appear for plain deep networks, it is instructive to start with the simplest case: a deep *linear* network at initialization.

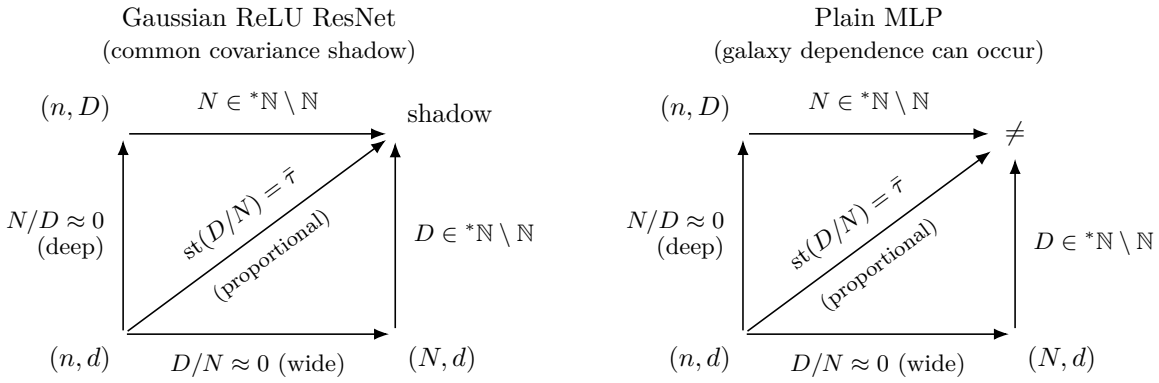


Figure 5.1: Representative hyperfinite depth–width galaxies in $({}^*\mathbb{N})^2$ for initialization observables: the wide galaxy ($D/N \approx 0$), the deep galaxy ($N/D \approx 0$), and the proportional galaxy ($\mathrm{st}(D/N) = \bar{\tau}$). Left: for the no-bias Gaussian ReLU ResNet (5.11), the covariance shadow is independent of the galaxy. Right: for plain MLPs, proportional and sequential galaxies need not commute ($\neq$).

Fix standard $n_0 \in \mathbb{N}$ and a nonzero standard input $x \in \mathbb{R}^{n_0}$. Fix unlimited depth and width $D, N \in {}^*\mathbb{N} \setminus \mathbb{N}$ with $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$ and infinitesimal depth-time increment $\Delta\tau := 1/N \approx 0$. Let $W_0 \in {}^*\mathbb{R}^{N \times n_0}$ and $(W_\ell)_{\ell \geq 1}$, $W_\ell \in {}^*\mathbb{R}^{N \times N}$, be mutually ${}^*\mathbb{P}$-independent internal matrices with i.i.d. ${}^*\mathcal{N}(0,1)$ entries. Set

$$h_0 := x, \qquad h_1 := \frac{1}{\sqrt{n_0}} W_0 x, \qquad h_{\ell+1} := \frac{1}{\sqrt{N}} W_\ell h_\ell, \quad \ell \geq 1. \tag{5.14}$$

A hyperfinite depth-D network is the prefix ending at h_D. Viewing inputs and hidden states in the Euclidean Clifford algebras $\mathcal{C}\ell_{n_0}$ and ${}^*\mathcal{C}\ell_N$, define the internal normalized squared norms via the grade-0 scalar product

$$\Phi_0 := \frac{1}{n_0} \langle x x^{\sim} \rangle_0 = \frac{1}{n_0} \|x\|^2, \qquad \Phi_\ell := \frac{1}{N} \langle h_\ell h_\ell^{\sim} \rangle_0 = \frac{1}{N} \|h_\ell\|^2, \quad \ell \geq 1. \tag{5.15}$$

Lemma 5.3 (Multiplicative Variance Recursion)**.** *For every internal layer $\ell \geq 0$, conditioned on $\mathcal{F}_\ell$, we have by Transfer*

$$h_{\ell+1} \mid \mathcal{F}_\ell \sim {}^*\mathcal{N}(0, \Phi_\ell I_N), \tag{5.16}$$

and consequently

$$\Phi_{\ell+1} = \Phi_\ell \frac{1}{N} \|\xi_\ell\|^2, \qquad (\xi_\ell)_{\ell \geq 0} \overset{\text{iid}}{\sim} {}^*\mathcal{N}(0, I_N), \tag{5.17}$$

where the internal standard Gaussian innovations are ${}^*\mathbb{P}$-independent of the past $\mathcal{F}_\ell$.

Proof Sketch. By Transfer of finite-width Gaussian conditioning, for $\ell \geq 1$ write the i-th row of W_ℓ as $w_{\ell,i}^\top \in {}^*\mathbb{R}^N$. Then

$$h_{\ell+1,i} = \frac{1}{\sqrt{N}} w_{\ell,i}^\top h_\ell. \tag{5.18}$$

Conditional on $\mathcal{F}_\ell$, each coordinate is internally centered Gaussian with variance

$${}^*\mathrm{Var}(h_{\ell+1,i} \mid \mathcal{F}_\ell) = \frac{1}{N} \|h_\ell\|^2 = \Phi_\ell, \tag{5.19}$$

and internal row independence gives the displayed conditional law (5.16). The input step $\ell = 0$ is the same calculation in $\mathbb{R}^{n_0}$ with normalization $n_0^{-1/2}$ and variance $n_0^{-1}\|x\|^2 = \Phi_0$. Mutual ${}^*\mathbb{P}$-independence of the layer matrices lets us write $h_{\ell+1} = \sqrt{\Phi_\ell}\,\xi_\ell$ with independent internal standard Gaussian innovations $\xi_\ell \sim {}^*\mathcal{N}(0, I_N)$. Taking normalized squared norms gives

$$\Phi_{\ell+1} = \frac{1}{N}\|h_{\ell+1}\|^2 = \Phi_\ell \frac{1}{N}\|\xi_\ell\|^2, \tag{5.20}$$

which is (5.17). □

Taking internal logarithms turns the hyperfinite product into a hyperfinite sum:

$${}^*\log\left(\frac{\Phi_D}{\Phi_0}\right) = \sum_{\ell=0}^{D-1} {}^*\log\left(\frac{1}{N}\|\xi_\ell\|^2\right). \tag{5.21}$$

Since $\frac{1}{N}\|\xi_\ell\|^2 \approx 1$ with infinitesimal fluctuations of scale $N^{-1/2} = \sqrt{\Delta\tau}$, the sum in (5.21) accumulates to an appreciable random variable precisely when D and N lie in the proportional galaxy ($D/N \in \mathbb{A}$).

Theorem 5.4 (Scalar Linear Variance Limit)**.** *For $D, N \in {}^*\mathbb{N} \setminus \mathbb{N}$ in the proportional galaxy $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$, the log-variance ratio ${}^*\log(\Phi_D/\Phi_0)$ is $L({}^*\mathbb{P})$-almost surely limited (so $\Phi_D \in \mathbb{A}$ $L({}^*\mathbb{P})$-almost surely), and its shadow satisfies*

$$\mathrm{st}\left({}^*\log\left(\frac{\Phi_D}{\Phi_0}\right)\right) \sim \mathcal{N}(-\bar{\tau}, 2\bar{\tau}) \qquad \text{under } L({}^*\mathbb{P}). \tag{5.22}$$

This endpoint law is the equal-hidden-width Gaussian specialization of [25, Theorem 1 and Eq. (9)].

More strongly, define the internal depth-time trajectory $\Phi_\tau^{(N)} := \Phi_{\lfloor N\tau \rfloor}$ with step size $\Delta\tau := 1/N \approx 0$. For every standard $T > 0$, $\tau \mapsto \Phi_\tau^{(N)}$ is $L({}^*\mathbb{P})$-almost surely limited and S-continuous on $[0, T]$, and its uniform shadow $\Phi_\tau := \mathrm{st}(\Phi_\tau^{(N)})$ is the geometric Brownian motion on $(\Omega, \mathcal{L}(\mathcal{A}), L({}^*\mathbb{P}))$ solving

$$d\Phi_\tau = \sqrt{2}\,\Phi_\tau\, dB_\tau, \qquad \Phi_0 = \frac{1}{n_0}\|x\|^2. \tag{5.23}$$

The endpoint law in (5.22) is a direct consequence of this pathwise shadow.

Proof Sketch. By Lemma 5.3, the internal scalar variance satisfies the multiplicative recursion

$$\Phi_{\ell+1} = \Phi_\ell U_{\ell,N}, \qquad U_{\ell,N} := \frac{1}{N}\langle \xi_\ell \xi_\ell^{\sim} \rangle_0 = \frac{1}{N}\xi_\ell^2 \in {}^*\mathbb{R}, \qquad \xi_\ell \sim {}^*\mathcal{N}(0, I_N) \text{ i.i.d.}, \tag{5.24}$$

where $\xi_\ell^2 = \|\xi_\ell\|^2$ is the Clifford vector square in ${}^*\mathcal{C}\ell_N$. Equivalently,

$$\Phi_{\ell+1} - \Phi_\ell = \Phi_\ell(U_{\ell,N} - 1). \tag{5.25}$$

By Transfer of the χ_N^2 moments, $\|\xi_\ell\|^2 \sim {}^*\chi_N^2$ satisfies ${}^*\mathbb{E}[U_{\ell,N} - 1] = 0$, ${}^*\mathrm{Var}(U_{\ell,N} - 1) = 2/N = 2\Delta\tau$, ${}^*\mathbb{E}[(U_{\ell,N} - 1)^4] = \frac{12}{N^2} + \frac{48}{N^3}$, and ${}^*\mathbb{E}[|U_{\ell,N} - 1|^3] = \mathcal{O}(N^{-3/2})$. Conditioning on the

forward history $\mathcal{F}_\ell$ with $\Delta\tau = 1/N$ gives

$$\begin{aligned}
\frac{1}{\Delta\tau}{}^*\mathbb{E}[\Phi_{\ell+1} - \Phi_\ell \mid \mathcal{F}_\ell] &= 0,\\
\frac{1}{\Delta\tau}{}^*\mathrm{Var}(\Phi_{\ell+1} - \Phi_\ell \mid \mathcal{F}_\ell) &= 2\Phi_\ell^2,\\
\frac{1}{\Delta\tau}{}^*\mathbb{E}\Big[|\Phi_{\ell+1} - \Phi_\ell|^3 \mid \mathcal{F}_\ell\Big] &= \mathcal{O}(N^{-1/2})\Phi_\ell^3.
\end{aligned} \tag{5.26}$$

These internal conditional moments depend on the history only through Φ_ℓ, matching the zero drift and diffusion covariance $2\Phi^2$ of (5.23) with an infinitesimal third-moment rate on every standard compact set. Defining the Anderson martingale increments $\Delta B_\ell := \frac{1}{\sqrt{2}}(U_{\ell,N} - 1)$ (so ${}^*\mathbb{E}[\Delta B_\ell] = 0$, ${}^*\mathbb{E}[(\Delta B_\ell)^2] = \Delta\tau$, and $B_\tau := \mathrm{st}(\sum_{\ell=0}^{\lfloor N\tau\rfloor - 1} \Delta B_\ell)$ is a standard Brownian motion [3]), equation (5.25) reads $\Phi_{\ell+1} - \Phi_\ell = \sqrt{2}\,\Phi_\ell\,\Delta B_\ell$. Thus Theorem A.4 gives local S-continuity of $\Phi_\tau^{(N)}$ with shadow solving (5.23) up to bounded-norm stopping. Because both $(\Phi_\ell)_{\ell\ge 0}$ and $(\Phi_\tau)_{\tau\ge 0}$ are nonnegative martingales, Transfer of Doob's maximal inequality gives, for every standard $T > 0$ and $R > 0$,

$${}^*\mathbb{P}\left(\max_{0\le\ell\le\lfloor NT\rfloor} \Phi_\ell \ge R\right) \le \frac{\Phi_0}{R}, \qquad L({}^*\mathbb{P})\left(\sup_{0\le\tau\le T} \Phi_\tau \ge R\right) \le \frac{\Phi_0}{R}. \tag{5.27}$$

Letting standard $R \to \infty$ shows that $\max_{0\le\ell\le\lfloor NT\rfloor} \Phi_\ell \in \mathbb{L}$ $L({}^*\mathbb{P})$-almost surely, removing the upper stopping barrier and proving global S-continuity on $[0, T]$ with $\mathrm{st}(\Phi_\tau^{(N)}) = \Phi_\tau$.

To see both the endpoint law (5.22) and the Itô drift directly at the hyperfinite level, observe that for every standard $\varepsilon > 0$, Markov's inequality yields ${}^*\mathbb{P}(\max_{0\le\ell<D} |U_{\ell,N} - 1| > \varepsilon) \le D\varepsilon^{-4}{}^*\mathbb{E}[(U_{0,N} - 1)^4] = \mathcal{O}(D/N^2) \approx 0$, so $\max_{0\le\ell<D} |U_{\ell,N} - 1| \approx 0$ $L({}^*\mathbb{P})$-almost surely. Summing the nonstandard second-order Taylor expansion ${}^*\log(1 + (U_{\ell,N} - 1)) = (U_{\ell,N} - 1) - \frac{1}{2}(U_{\ell,N} - 1)^2 + \mathcal{O}(|U_{\ell,N} - 1|^3)$ over $\ell = 0, \dots, D - 1$ in (5.21) gives

$$\begin{aligned}
{}^*\log\left(\frac{\Phi_D}{\Phi_0}\right) &= \sum_{\ell=0}^{D-1} (U_{\ell,N} - 1) - \frac{1}{2}\sum_{\ell=0}^{D-1} (U_{\ell,N} - 1)^2 + \sum_{\ell=0}^{D-1} \mathcal{O}\Big(|U_{\ell,N} - 1|^3\Big)\\
&= \sqrt{2}\sum_{\ell=0}^{D-1} \Delta B_\ell - \frac{D}{N} + \mathbb{I} \approx \sqrt{2}\, B_{\bar\tau} - \bar\tau \qquad (L({}^*\mathbb{P})\text{-a.s.}),
\end{aligned} \tag{5.28}$$

because $-\frac{1}{2}\sum_{\ell=0}^{D-1}(U_{\ell,N} - 1)^2$ has internal mean $-\sum_{\ell=0}^{D-1} \frac{1}{2}\cdot\frac{2}{N} = -\frac{D}{N} \approx -\bar\tau$ and internal variance $\mathcal{O}(D/N^2) \approx 0$, while $\sum_{\ell=0}^{D-1} {}^*\mathbb{E}[|U_{\ell,N} - 1|^3] = \mathcal{O}(DN^{-3/2}) \approx 0$. Equivalently, at the shadow level, Itô's lemma $d\log\Phi_\tau = \frac{1}{\Phi_\tau}\,d\Phi_\tau - \frac{1}{2\Phi_\tau^2}(d\Phi_\tau)^2 = -d\tau + \sqrt{2}\,dB_\tau$ yields $\log(\Phi_{\bar\tau}/\Phi_0) = -\bar\tau + \sqrt{2}\,B_{\bar\tau} \sim \mathcal{N}(-\bar\tau, 2\bar\tau)$, proving (5.22). □

The term *geometric* in (5.23) refers to its multiplicative noise: randomness scales with Φ_τ itself. This is the probabilistic signature of hyperfinite matrix products, in which each layer randomly stretches or shrinks the current state.

Even in this simplest linear setting, the proportional galaxy already produces *non-Gaussian* marginal shadows. For any fixed coordinate i, $h_{D,i} \mid \mathcal{F}_{D-1} \sim {}^*\mathcal{N}(0, \Phi_{D-1})$. Since S-continuity gives $\mathrm{st}(\Phi_{D-1}) = \Phi_{\bar\tau}$ when $\mathrm{st}(D/N) = \bar\tau$, the standard-part marginal $\mathrm{st}(h_{D,i})$ under $L({}^*\mathbb{P})$ is a Gaussian scale mixture with random mixing variance $\Phi_{\bar\tau}$. This is a key motivation for working in the proportional depth–width galaxy.

Here "stable" means appreciable in a two-sided probabilistic sense. For a positive internal random variable $X > 0$, the condition $L(^*\mathbb{P})(X \in \mathbb{L}) = 1$ (the nonstandard counterpart of $X_k = \mathcal{O}_{\mathbb{P}}(1)$) only rules out escape to unlimited values in $^*\mathbb{R} \setminus \mathbb{L}$ and still permits infinitesimal collapse $X \approx 0$. Two-sided probabilistic stability ($X_k = \Theta_{\mathbb{P}}(1)$) translates cleanly to X lying in the appreciable group $\mathbb{A} = \mathbb{L} \setminus \mathbb{I}$ with $L(^*\mathbb{P})$-probability 1 (equivalently, both X and X^{-1} lie in $\mathbb{L}$ $L(^*\mathbb{P})$-almost surely), so typical realizations have positive finite shadows $\mathrm{st}(X) \in (0, \infty)$.

The three hyperfinite depth–width galaxies behave differently with respect to $\mathbb{A}$. In the wide galaxy ($D/N \approx 0$, such as $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ at standard depth $d \in \mathbb{N}$), layerwise fluctuations average away in (5.28) and $\Phi_D \approx \Phi_0 \in \mathbb{A}$ $L(^*\mathbb{P})$-almost surely, collapsing to the deterministic appreciable value Φ_0. In the deep galaxy ($N/D \approx 0$, such as $D \in {}^*\mathbb{N} \setminus \mathbb{N}$ at standard width $n \in \mathbb{N}$), repeated multiplication instead drives $^*\log(\Phi_D/\Phi_0)$ to $-\infty$, so $\Phi_D \approx 0$ $L(^*\mathbb{P})$-almost surely even though the internal expectation $^*\mathbb{E}[\Phi_D] = \Phi_0$ remains constant for every D. Thus $L(^*\mathbb{P})(\Phi_D \in \mathbb{L}) = 1$ whereas $L(^*\mathbb{P})(\Phi_D \in \mathbb{A}) = 0$: an internal event of infinitesimal probability carrying hyper-large realizations preserves $^*\mathbb{E}[\Phi_D]$ (a failure of S-integrability), while $L(^*\mathbb{P})$-almost every realization collapses into $\mathbb{I}$. Under proportional scaling $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$, by contrast, $\mathrm{st}(\Phi_D) = \Phi_{\bar{\tau}} \in (0, \infty)$ $L(^*\mathbb{P})$-almost surely, so Φ_D remains in $\mathbb{A}$ with $L(^*\mathbb{P})$-probability 1 while retaining genuine randomness [25, §1.1, Theorem 1, and Eq. (9)].

Although the discussion so far concerns initialization, proportional depth–width scaling also changes the NTK picture for fully connected ReLU networks. For equal hidden width N, Hanin and Nica show that the on-diagonal NTK at initialization need not concentrate to a deterministic shadow when $\mathrm{st}(D/N) > 0$ [24, Theorem 1]. Their training-time result is narrower: for square loss and a single-example SGD step, they compute a potentially appreciable mean first update of the same kernel entry. This motivates, but does not prove, their conjectured "weak feature learning" regime [24, Theorem 2 and §3.2].

5.3 Multiple Data Points: A Covariance SDE

We write $\mathrm{Sym}(m)$ for the vector space of real symmetric $m \times m$ matrices (with internal counterpart ${}^*\mathrm{Sym}(m)$ over $^*\mathbb{R}$). The internal empirical covariance matrices below are positive semidefinite; they need not be strictly positive definite.

Now fix standard $m \geq 2$ and propagate standard inputs $x^1, \dots, x^m \in \mathbb{R}^{n_0}$ through the same hyperfinite random linear network (5.14) of width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$. In $\mathcal{C}\ell_{n_0}$ and ${}^*\mathcal{C}\ell_N$, the full geometric product $h_\ell^\alpha h_\ell^\beta = \left\langle h_\ell^\alpha h_\ell^\beta \right\rangle_0 + h_\ell^\alpha \wedge h_\ell^\beta$ splits into its symmetric scalar inner product and antisymmetric feature-plane bivector; extracting the scalar grade defines the input and internal hidden-layer Gram matrices by

$$\begin{aligned} \Phi_0^{\alpha\beta} &:= \frac{1}{n_0} \left\langle x^\alpha (x^\beta)^\sim \right\rangle_0 = \frac{1}{n_0} \left\langle x^\alpha, x^\beta \right\rangle, \qquad \Phi_\ell^{\alpha\beta} := \frac{1}{N} \left\langle h_\ell^\alpha (h_\ell^\beta)^\sim \right\rangle_0 = \frac{1}{N} \left\langle h_\ell^\alpha, h_\ell^\beta \right\rangle, \quad \ell \geq 1, \\ \Phi_\ell &:= \left[\Phi_\ell^{\alpha\beta} \right]_{\alpha,\beta=1}^m \in {}^*\mathrm{Sym}(m), \quad \ell \geq 0. \end{aligned} \tag{5.29}$$

Let $\mathcal{F}_0 := \{\varnothing, \Omega\}$ be the trivial algebra, and for $\ell \geq 1$ define the internal algebra

$$\mathcal{F}_\ell := {}^*\sigma\Big(\{h_k^\alpha\}_{1 \leq k \leq \ell,\, \alpha \in [m]} \Big). \tag{5.30}$$

Thus $\mathcal{F}_\ell$ records the forward hidden-state history through layer ℓ. Conditioned on $\mathcal{F}_\ell$, by Transfer $\{h_{\ell+1}^\alpha\}_{\alpha=1}^m$ is internally jointly Gaussian with covariance $\Phi_\ell \otimes I_N$. As a result, each

entry of the centered increment $\Phi_{\ell+1} - \Phi_\ell$ is a hyperfinite empirical average of conditionally i.i.d. centered terms. For $1 \le \alpha \le \beta \le m$, write

$$\Phi^{\alpha\beta}_{\ell+1} = \Phi^{\alpha\beta}_\ell + \frac{1}{\sqrt{N}} R^{\alpha\beta}_\ell, \qquad {}^*\mathbb{E}\Big[R^{\alpha\beta}_\ell \mid \mathcal{F}_\ell\Big] = 0, \tag{5.31}$$

where

$$R^{\alpha\beta}_\ell := \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \left(h^\alpha_{\ell+1,i} h^\beta_{\ell+1,i} - \Phi^{\alpha\beta}_\ell \right). \tag{5.32}$$

The next step is to compute the internal conditional covariance between these fluctuation entries.

Proposition 5.5 (Isserlis Covariance for the Gram Increment)**.** *Let indices satisfy $\alpha \le \beta$ and $\gamma \le \delta$ (to avoid duplication from symmetry), and let $R^{\alpha\beta}_\ell$ be the centered internal Gram fluctuation defined in (5.32). Then, conditional on $\mathcal{F}_\ell$, the neuron-wise internal vectors*

$$h_{\ell+1,i} := \left(h^1_{\ell+1,i}, \dots, h^m_{\ell+1,i} \right) \in {}^*\mathbb{R}^m, \qquad i = 1, \dots, N, \tag{5.33}$$

are internally i.i.d. Gaussian with law ${}^*\mathcal{N}(0, \Phi_\ell)$, and by Transfer we have the exact *internal identities*

$${}^*\mathbb{E}\Big[R^{\alpha\beta}_\ell \mid \mathcal{F}_\ell\Big] = 0, \qquad {}^*\mathrm{Cov}\Big(R^{\alpha\beta}_\ell, R^{\gamma\delta}_\ell \mid \mathcal{F}_\ell\Big) = \Phi^{\alpha\gamma}_\ell \Phi^{\beta\delta}_\ell + \Phi^{\alpha\delta}_\ell \Phi^{\beta\gamma}_\ell. \tag{5.34}$$

Equivalently, in the flattened upper-triangular coordinates on $\mathrm{Sym}(m)$*, the conditional covariance matrix is $\Sigma(\Phi_\ell)$ with entries*

$$\Sigma_{\alpha\beta,\gamma\delta}(\Phi) := \Phi^{\alpha\gamma}\Phi^{\beta\delta} + \Phi^{\alpha\delta}\Phi^{\beta\gamma}. \tag{5.35}$$

Proof. Fix ℓ and condition on $\mathcal{F}_\ell$. For each $i \in \{1, \dots, N\}$, set

$$R^{\alpha\beta}_{\ell,i} := h^\alpha_{\ell+1,i} h^\beta_{\ell+1,i} - \Phi^{\alpha\beta}_\ell, \qquad \text{so that} \qquad R^{\alpha\beta}_\ell = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} R^{\alpha\beta}_{\ell,i}. \tag{5.36}$$

Since $(h_{\ell+1,i})_{i=1}^N$ are conditionally internally i.i.d. ${}^*\mathcal{N}(0, \Phi_\ell)$, the centered variables $(R^{\alpha\beta}_{\ell,i})_{i=1}^N$ are conditionally i.i.d. as well, and therefore by Transfer

$${}^*\mathrm{Cov}\Big(R^{\alpha\beta}_\ell, R^{\gamma\delta}_\ell \mid \mathcal{F}_\ell\Big) = {}^*\mathbb{E}\Big[R^{\alpha\beta}_{\ell,1} R^{\gamma\delta}_{\ell,1} \mid \mathcal{F}_\ell\Big]. \tag{5.37}$$

Expanding and using ${}^*\mathbb{E}[h^\alpha_{\ell+1,1} h^\beta_{\ell+1,1} \mid \mathcal{F}_\ell] = \Phi^{\alpha\beta}_\ell$ gives

$${}^*\mathbb{E}\Big[R^{\alpha\beta}_{\ell,1} R^{\gamma\delta}_{\ell,1} \mid \mathcal{F}_\ell\Big] = {}^*\mathbb{E}\Big[h^\alpha_{\ell+1,1} h^\beta_{\ell+1,1} h^\gamma_{\ell+1,1} h^\delta_{\ell+1,1} \mid \mathcal{F}_\ell\Big] - \Phi^{\alpha\beta}_\ell \Phi^{\gamma\delta}_\ell. \tag{5.38}$$

By Transfer of Isserlis' theorem (Wick's formula) to the centered internal Gaussian vector $h_{\ell+1,1}$,

$${}^*\mathbb{E}\Big[h^\alpha_{\ell+1,1} h^\beta_{\ell+1,1} h^\gamma_{\ell+1,1} h^\delta_{\ell+1,1} \mid \mathcal{F}_\ell\Big] = \Phi^{\alpha\beta}_\ell \Phi^{\gamma\delta}_\ell + \Phi^{\alpha\gamma}_\ell \Phi^{\beta\delta}_\ell + \Phi^{\alpha\delta}_\ell \Phi^{\beta\gamma}_\ell, \tag{5.39}$$

and subtracting the first term yields the claimed covariance (5.34). (In Clifford terms on ${}^*\mathcal{C}\ell_N$, the symmetric combination $\Phi^{\alpha\gamma}_\ell \Phi^{\beta\delta}_\ell + \Phi^{\alpha\delta}_\ell \Phi^{\beta\gamma}_\ell$ governs the scalar-grade Gram fluctuations, whereas the antisymmetric counterpart $\frac{1}{N^2} {}^*\mathbb{E}[\big\langle (h^\alpha_{\ell+1} \wedge h^\beta_{\ell+1})^\sim (h^\gamma_{\ell+1} \wedge h^\delta_{\ell+1}) \big\rangle_0 \mid \mathcal{F}_\ell] = (1 - \frac{1}{N})(\Phi^{\alpha\gamma}_\ell \Phi^{\beta\delta}_\ell - \Phi^{\alpha\delta}_\ell \Phi^{\beta\gamma}_\ell)$ measures the expected squared 2-blade area spanned by token pairs.) □

By Proposition 5.5, with $\Delta\tau := 1/N \approx 0$ and $\bar{m} := m(m+1)/2$, we may factor the flattened fluctuation as $R_\ell = \Sigma(\Phi_\ell)^{1/2}\xi_\ell$ with ${}^*\mathbb{E}[\xi_\ell \mid \mathcal{F}_\ell] = 0$ and ${}^*\mathrm{Cov}(\xi_\ell \mid \mathcal{F}_\ell) = I_{\bar{m}}$ (adjoining independent ${}^*\mathcal{N}(0,1)$ coordinates along any null directions of $\Sigma(\Phi_\ell)$), so one Gram increment takes the exact Keisler martingale form $\Delta\Phi_\ell := \Phi_{\ell+1} - \Phi_\ell = \Sigma(\Phi_\ell)^{1/2}\Delta B_\ell$ with $\Delta B_\ell := N^{-1/2}\xi_\ell$. Thus, on the hyperfinite layer-time scale $\tau_\ell = \ell/N$, fluctuations across $\mathcal{O}(N)$ layers accumulate at appreciable order. The following theorem formalizes this pathwise shadow.

Theorem 5.6 (Linear Covariance SDE Limit [42, Theorem 3.2 and Appendix C.1])**.** *Fix $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ and define the internal piecewise-constant process $\Phi_\tau^{(N)} := \Phi_{\lfloor N\tau \rfloor}$ for $\tau \geq 0$. Let $\bar{m} := m(m+1)/2$. After flattening Φ into its $\bar{m}$ distinct upper-triangular entries, for every standard $\bar{\tau} > 0$, the trajectory $\tau \mapsto \Phi_\tau^{(N)}$ is $L({}^*\mathbb{P})$-almost surely limited and S-continuous on $[0, \bar{\tau}]$, and its uniform shadow $\Phi_\tau := \mathrm{st}(\Phi_\tau^{(N)})$ solves the covariance SDE on $(\Omega, \mathcal{L}(\mathcal{A}), L({}^*\mathbb{P}))$:*

$$d\Phi_\tau = \Sigma(\Phi_\tau)^{1/2}\, dB_\tau, \qquad \Phi_0 = \left[\frac{1}{n_0}\left\langle x^\alpha, x^\beta \right\rangle\right]_{\alpha,\beta=1}^{m}. \tag{5.40}$$

Here $B_\tau = \mathrm{st}(\sum_{\ell=0}^{\lfloor N\tau \rfloor - 1} \Delta B_\ell)$ is a standard Brownian motion in $\mathbb{R}^{\bar{m}}$, and Σ is the covariance matrix in (5.35).

Proof Sketch. By (5.31) and Proposition 5.5, with $\Delta\tau = 1/N$, one layer has internal conditional mean zero and conditional covariance $\Delta\tau\,\Sigma(\Phi_\ell)$, so $\Delta\Phi_\ell = \Sigma(\Phi_\ell)^{1/2}\Delta B_\ell$ with $\Delta B_\ell = N^{-1/2}\xi_\ell$. Transferred Gaussian moment bounds give $\frac{1}{\Delta\tau}{}^*\mathbb{E}[\|\Delta\Phi_\ell\|^3 \mid \mathcal{F}_\ell] = \mathcal{O}(N^{-1/2})\,\|\Phi_\ell\|_F^3 \approx 0$ on limited states, verifying the third-moment hypothesis of Theorem A.4. Because these conditional moments depend on the history only through Φ_ℓ, they furnish the Markov-state coefficients required by Theorem A.4. Finally, applying Doob's maximal inequality (5.27) to each nonnegative diagonal martingale $\Phi_\ell^{\alpha\alpha}$ and Cauchy–Schwarz $\left|\Phi_\ell^{\alpha\beta}\right|^2 \leq \Phi_\ell^{\alpha\alpha}\Phi_\ell^{\beta\beta}$ to the off-diagonal entries ensures $\max_{0\leq\ell\leq\lfloor N\bar{\tau}\rfloor} \|\Phi_\ell\|_F \in \mathbb{L}$ $L({}^*\mathbb{P})$-almost surely, yielding global S-continuity and the shadow SDE (5.40) via Theorem A.4. □

When $m = 1$, (5.35) gives $\Sigma(\Phi) = 2\Phi^2$, so Theorem 5.6 reduces to the geometric Brownian motion in Theorem 5.4. If the standard inputs are linearly dependent, those linear relations are preserved identically by every internal linear layer. One may therefore apply the theorem to a maximal linearly independent subset and reconstruct the full Gram shadow from it.

The same covariance tensor has a geometric interpretation: on $\mathrm{SPD}(m)$, $\Sigma(\Phi)$ is the cometric, or inverse metric tensor, associated with the affine-invariant Riemannian metric. See Section 6.2.2 for this metric interpretation and Section 6.2.3 for the resulting eigenvalue dynamics.

5.4 Nonlinearities, Correlation Collapse, and Shaping

For nonlinear activations, the hyperfinite depth shadow depends strongly on normalization and on how quickly correlations are driven toward a fixed point.

Consider first the ReLU activation $\varphi(u) = \max\{u, 0\}$. If $g \sim \mathcal{N}(0, \sigma^2)$ then $\mathbb{E}[\varphi(g)^2] = \sigma^2/2$. Thus, for the naive hyperfinite ReLU MLP

$$h_{\ell+1} = \frac{1}{\sqrt{N}} W_\ell{}^* \varphi(h_\ell), \tag{5.41}$$

the variance contracts by a factor $1/2$ per layer in conditional expectation. The standard fix is Kaiming normalization [27]: choose $c > 0$ so that $c^{-1} = \mathbb{E}[\varphi(\omega)^2]$ for $\omega \sim \mathcal{N}(0,1)$, and scale the layer by $\sqrt{c}$. For ReLU, $c = 2$.

Proposition 5.7 (ReLU SDE Limit for One Sample)**.** *Fix standard $n_0 \in \mathbb{N}$ and $h_0 := x \in \mathbb{R}^{n_0} \setminus \{0\}$. For $D, N \in {}^*\mathbb{N} \setminus \mathbb{N}$ with $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$, consider the Kaiming-normalized hyperfinite ReLU network*

$$h_1 = \frac{1}{\sqrt{n_0}} W_0 x, \qquad h_{\ell+1} = \sqrt{\frac{c}{N}}\, W_\ell\, {}^*\varphi(h_\ell), \qquad \ell \geq 1, \qquad c = 2. \tag{5.42}$$

Here $W_0 \in {}^*\mathbb{R}^{N \times n_0}$ and $W_\ell \in {}^*\mathbb{R}^{N \times N}$ for $\ell \geq 1$ are mutually ${}^*\mathbb{P}$-independent internal matrices with i.i.d. ${}^*\mathcal{N}(0,1)$ entries. Define the (normalized) internal squared post-activation norm

$$\Phi_0 := \frac{1}{n_0} \|x\|_2^2, \qquad \Phi_\ell := \frac{c}{N} \|{}^*\varphi(h_\ell)\|_2^2, \qquad \ell \geq 1. \tag{5.43}$$

For $\tau \in [0, \bar{\tau}]$, set $\Phi_\tau^{(N)} := \Phi_{\lfloor \tau N \rfloor}$. Then $\tau \mapsto \Phi_\tau^{(N)}$ is $L({}^*\mathbb{P})$-almost surely limited and S-continuous on $[0, \bar{\tau}]$ (see Appendix A.3), and its uniform shadow $\Phi_\tau := \mathrm{st}(\Phi_\tau^{(N)})$ is the geometric Brownian motion on $(\Omega, \mathcal{L}(\mathcal{A}), L({}^*\mathbb{P}))$ solving

$$d\Phi_\tau = \sqrt{5}\, \Phi_\tau\, dB_\tau, \tag{5.44}$$

with the same initial condition $\Phi_0 = \frac{1}{n_0} \|x\|_2^2$. At the terminal hyperfinite depth D, $\Phi_D \in \mathbb{A}$ $L({}^*\mathbb{P})$-almost surely and

$$\mathrm{st}\left({}^*\log\left(\frac{\Phi_D}{\Phi_0} \right) \right) \sim \mathcal{N}\left(-\frac{5}{2}\bar{\tau},\, 5\bar{\tau} \right) \qquad \text{under } L({}^*\mathbb{P}). \tag{5.45}$$

The proof is deferred to Exercise 5.14.

For one input, Kaiming normalization gives ${}^*\mathbb{E}[\Phi_{\ell+1} \mid \mathcal{F}_\ell] = \Phi_\ell$; the remaining scale-$N^{-1/2}$ hyperfinite fluctuation accumulates into the SDE of Proposition 5.7. This controls only the diagonal of the multi-input Gram matrix; the off-diagonal correlation update remains the obstruction. For two inputs with nonzero postactivation norms, define the internal scaled postactivation Gram entry and its correlation at layer ℓ by

$$\Phi_\ell^{\alpha\beta} := \frac{c}{N} \left\langle {}^*\varphi(h_\ell^\alpha), {}^*\varphi(h_\ell^\beta) \right\rangle, \qquad \rho_\ell^{\alpha\beta} := \frac{\Phi_\ell^{\alpha\beta}}{\sqrt{\Phi_\ell^{\alpha\alpha} \Phi_\ell^{\beta\beta}}} \in {}^*[-1, 1]. \tag{5.46}$$

In the wide galaxy ($N \in {}^*\mathbb{N} \setminus \mathbb{N}$ at standard depth $\ell \in \mathbb{N}$), the Cho–Saul formula in Proposition 2.5, applied with unit variances, gives the shadow ReLU correlation recursion $\mathrm{st}(\rho_{\ell+1}^{\alpha\beta}) = \mathcal{T}(\mathrm{st}(\rho_\ell^{\alpha\beta}))$:

$$\rho_{\ell+1}^{\alpha\beta} \approx \mathcal{T}\left(\rho_\ell^{\alpha\beta} \right), \qquad \mathcal{T}(\rho) := \frac{\mathbb{E}[\varphi(w_1)\varphi(w_2)]}{\mathbb{E}[\varphi(g)^2]} = \frac{1}{\pi} \left(\sqrt{1 - \rho^2} + (\pi - \arccos \rho)\, \rho \right), \tag{5.47}$$

where $g \sim \mathcal{N}(0,1)$ and (w_1, w_2) is a jointly standard Gaussian pair with correlation ρ. For $-1 \leq \rho < 1$, one has $\mathcal{T}(\rho) > \rho$; hence 1 is the unique fixed point in $[-1, 1]$, and every standard correlation recursion starting below 1 converges to 1 (so over an unlimited number of layers, correlations collapse into the monad of 1).

Remark 5.8 (Representation and Kernel Collapse). As an off-diagonal correlation enters the monad of 1 ($\rho_\ell^{\alpha\beta} \approx 1$), the corresponding normalized postactivation representations align up to infinitesimals and their normalized kernel shadow collapses to its diagonal value. This is a forward representation/kernel-collapse statement; it does not by itself imply that backpropagated gradients vanish.

Away from 1, the map $\mathcal{T}$ in (5.47) differs from the identity by an appreciable ($\mathbb{A}$-scale) amount, so the unshaped off-diagonal update is not on the infinitesimal drift scale $\Delta\tau = 1/N$ needed over $D \asymp N$ layers. Activation shaping repairs this scale mismatch: the full normalized kernel map becomes the identity plus an $\mathcal{O}(\Delta\tau)$ bias while the empirical Gram fluctuation remains of scale $\sqrt{\Delta\tau} = N^{-1/2}$. This viewpoint is closely related to *deep kernel shaping* and *tailored rectifiers* [45, 64], and it underlies the covariance SDE limits of [42, Sections 2.3–3.2].

For concreteness, fix standard inputs $x^1, \dots, x^m \in \mathbb{R}^{n_0}$ and $D, N \in {}^*\mathbb{N} \setminus \mathbb{N}$ with $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$, and consider the Kaiming-normalized shaped hyperfinite network

$$h_1^\alpha = \frac{1}{\sqrt{n_0}} W_0 x^\alpha, \qquad h_{\ell+1}^\alpha = \sqrt{\frac{c}{N}} W_\ell \varphi_s(h_\ell^\alpha), \qquad 1 \le \ell \le D-1, \tag{5.48}$$

where $W_0 \in {}^*\mathbb{R}^{N \times n_0}$ and $W_\ell \in {}^*\mathbb{R}^{N \times N}$ are mutually ${}^*\mathbb{P}$-independent with i.i.d. ${}^*\mathcal{N}(0,1)$ entries, and $c = c(N) \in {}^*\mathbb{R}_{>0}$ is chosen so that ${}^*\mathbb{E}[\varphi_s(g)^2] = c^{-1}$ for $g \sim {}^*\mathcal{N}(0,1)$. Define the input boundary and the internal scaled postactivation Gram matrices by

$$\begin{aligned} \Phi_0^{\alpha\beta} &:= \frac{1}{n_0} \left\langle x^\alpha, x^\beta \right\rangle, \\ \Phi_\ell^{\alpha\beta} &:= \frac{c}{N} \left\langle \varphi_s(h_\ell^\alpha), \varphi_s(h_\ell^\beta) \right\rangle, \quad 1 \le \ell \le D, \qquad \Phi_\ell := \left(\Phi_\ell^{\alpha\beta}\right)_{\alpha,\beta=1}^m \in {}^*\mathrm{Sym}(m). \end{aligned} \tag{5.49}$$

Thus Φ_ℓ for $\ell \ge 1$ is the internal postactivation Gram matrix at layer ℓ. Every empirical Φ_ℓ is internally positive semidefinite. We assume that the standard data Gram matrix Φ_0 is strictly positive definite, although the one-step calculations below remain valid for positive-semidefinite covariances. The dependence of the hyperfinite chain Φ_ℓ on N is suppressed. With $\mathcal{F}_\ell$ as in (5.30), conditional on $\mathcal{F}_\ell$ the vectors $h_{\ell+1,i} = (h_{\ell+1,i}^1, \dots, h_{\ell+1,i}^m)$, $i = 1, \dots, N$, are internally i.i.d. ${}^*\mathcal{N}(0, \Phi_\ell)$ for $0 \le \ell \le D-1$.

ReLU-like shaping. For standard $p > 0$ and $c_+, c_- \in \mathbb{R}$, define the internal piecewise-linear activation

$$\varphi_s(x) = s_+ \max\{x, 0\} + s_- \min\{x, 0\}, \qquad s_\pm = 1 + \frac{c_\pm}{N^p}. \tag{5.50}$$

At the critical exponent $p = \frac{1}{2}$ (so $s_\pm = 1 + c_\pm/\sqrt{N} \approx 1$), this is [42, Definition 3.1]; the general exponent family is analyzed in [42, Proposition 3.4]. The exponent $p = \frac{1}{2}$ is critical because the deterministic bias of the normalized kernel is then $\mathcal{O}(\Delta\tau)$ (with $\Delta\tau := 1/N$) and accumulates to an appreciable drift over $D \asymp N$ layers.

Smooth shaping. Assume standard $\varphi \in C^4(\mathbb{R})$ satisfies $\varphi(0) = 0$, $\varphi'(0) = 1$, and $\left|\varphi^{(4)}(x)\right| \le C(1 + |x|^q)$ for some standard $C, q > 0$, as in [42, Assumption 3.5]. For a standard constant $a > 0$, use the unlimited rescaling

$$\varphi_s(x) := s\, {}^*\varphi(x/s), \qquad s = a\sqrt{N}, \qquad c^{-1} = {}^*\mathbb{E}[\varphi_s(g)^2], \quad g \sim {}^*\mathcal{N}(0,1). \tag{5.51}$$

Then for every limited $x \in \mathbb{L}$, $x/s \approx 0$ so $\varphi_s(x) \approx x$, with the transferred fourth-derivative bound controlling the nonstandard Taylor remainder in internal expectation, making the kernel map an infinitesimal perturbation of the identity [42, Definition 3.6].

Conditional remainder notation. Let $\Delta\tau := 1/N \approx 0$. For an internal matrix- or vector-valued random variable X, possibly depending on ℓ and N, we write $X = \tilde{o}_\ell(\Delta\tau)$ whenever (at every limited state Φ_ℓ, uniformly on standard compact subsets)

$$\frac{\|{}^*\mathbb{E}[X \mid \mathcal{F}_\ell]\|_F}{\Delta\tau} \approx 0, \qquad \frac{{}^*\mathbb{E}\Big[\|X - {}^*\mathbb{E}[X \mid \mathcal{F}_\ell]\|_F^2 \;\Big|\; \mathcal{F}_\ell\Big]}{\Delta\tau} \approx 0. \tag{5.52}$$

The subscript in $\tilde{o}_\ell(\Delta\tau)$ records conditioning on $\mathcal{F}_\ell$. Thus its internal conditional mean is infinitesimal relative to $\Delta\tau = 1/N$, while its centered part is infinitesimal relative to $\sqrt{\Delta\tau} = N^{-1/2}$ in conditional internal L^2; this does not assert $X/\Delta\tau \overset{L({}^*\mathbb{P})}{\approx} 0$. The conditioning rules out predictable one-step errors that disappear only after averaging over the current state yet could accumulate over $D \asymp N$ layers.

Theorem 5.9 (Keisler–Euler–Maruyama Increment Expansion under Critical Shaping)**.** *Assume either ReLU-like shaping (5.50) at the critical exponent $p = \frac{1}{2}$ ($s_\pm = 1 + c_\pm/\sqrt{N}$), or smooth shaping (5.51) with $s = a\sqrt{N}$, and set $\Delta\tau := 1/N \approx 0$. After flattening the upper-triangular entries as coordinates on $\mathrm{Sym}(m)$, whenever Φ_ℓ is limited one can write*

$$\Phi_{\ell+1} - \Phi_\ell = b(\Phi_\ell)\,\Delta\tau + \Sigma(\Phi_\ell)^{1/2}\Delta B_\ell + \tilde{o}_\ell(\Delta\tau), \tag{5.53}$$

where $\tilde{o}_\ell(\Delta\tau)$ is understood in the sense of (5.52) and $\Delta B_\ell := \sqrt{\Delta\tau}\,\xi_\ell = N^{-1/2}\xi_\ell$, with internal noise vector ξ_ℓ satisfying

$${}^*\mathbb{E}[\xi_\ell \mid \mathcal{F}_\ell] = 0, \qquad {}^*\mathrm{Cov}(\xi_\ell \mid \mathcal{F}_\ell) = I_{\bar{m}}, \qquad \bar{m} := \frac{m(m+1)}{2}, \tag{5.54}$$

and Σ the covariance matrix in (5.35). The drift b (extended naturally to internal Φ_ℓ) is given as follows.

- *In the ReLU-like shaping scheme, one has the entrywise formula*

$$\begin{aligned} b^{\alpha\beta}(\Phi) &= \sqrt{\Phi^{\alpha\alpha}\Phi^{\beta\beta}}\,\nu(\rho^{\alpha\beta}), \qquad \rho^{\alpha\beta} := \frac{\Phi^{\alpha\beta}}{\sqrt{\Phi^{\alpha\alpha}\Phi^{\beta\beta}}}, \\ \nu(\rho) &:= \frac{(c_+ - c_-)^2}{2\pi}\left(\sqrt{1-\rho^2} - \rho\arccos(\rho)\right). \end{aligned} \tag{5.55}$$

- *In the smooth-shaping scheme, the drift admits the entrywise form*

$$b^{\alpha\beta}(\Phi) = \frac{\varphi''(0)^2}{4a^2}\Big(\Phi^{\alpha\alpha}\Phi^{\beta\beta} + \Phi^{\alpha\beta}(2\Phi^{\alpha\beta} - 3)\Big) + \frac{\varphi'''(0)}{2a^2}\Phi^{\alpha\beta}\Big(\Phi^{\alpha\alpha} + \Phi^{\beta\beta} - 2\Big). \tag{5.56}$$

The ReLU-like formula is understood by continuity if a diagonal covariance entry vanishes.

Proof. Fix ℓ with limited Φ_ℓ, and condition on $\mathcal{F}_\ell$. Since W_ℓ has internally i.i.d. ${}^*\mathcal{N}(0,1)$ rows, by Transfer the neuron-wise preactivation vectors

$$h_{\ell+1,i} := \left(h^1_{\ell+1,i}, \dots, h^m_{\ell+1,i}\right) \in {}^*\mathbb{R}^m, \qquad i = 1, \dots, N, \tag{5.57}$$

are conditionally internally i.i.d. ${}^*\mathcal{N}(0, \Phi_\ell)$.

Drift calculation. Let $w \mid \mathcal{F}_\ell \sim {}^*\mathcal{N}(0, \Phi_\ell)$. In the ReLU-like case, set $\rho_\ell^{\alpha\beta} := \Phi_\ell^{\alpha\beta}/\sqrt{\Phi_\ell^{\alpha\alpha}\Phi_\ell^{\beta\beta}}$ and let (z_1, z_2) be an internal standard Gaussian pair with correlation $\rho_\ell^{\alpha\beta}$. By Transfer of positive homogeneity and the arc-cosine calculation [42, Proposition 3.4] at $p = \frac{1}{2}$ (with $\Delta\tau = 1/N$),

$$\begin{aligned} {}^*\mathbb{E}[c\,\varphi_s(z_1)\varphi_s(z_2)] &= \rho_\ell^{\alpha\beta} + \nu(\rho_\ell^{\alpha\beta})\,\Delta\tau + \mathcal{O}(\Delta\tau^{3/2}), \\ {}^*\mathbb{E}\Big[c\,\varphi_s(w^\alpha)\varphi_s(w^\beta) \mid \mathcal{F}_\ell\Big] &= \sqrt{\Phi_\ell^{\alpha\alpha}\Phi_\ell^{\beta\beta}}\left(\rho_\ell^{\alpha\beta} + \nu(\rho_\ell^{\alpha\beta})\,\Delta\tau + \mathcal{O}(\Delta\tau^{3/2})\right) \\ &= \Phi_\ell^{\alpha\beta} + b^{\alpha\beta}(\Phi_\ell)\,\Delta\tau + \mathcal{O}(\Delta\tau^{3/2}), \end{aligned} \tag{5.58}$$

whose remainder divided by $\Delta\tau$ is $\mathcal{O}(\Delta\tau^{1/2}) \approx 0$, yielding (5.55).

In the smooth case ($s = a\sqrt{N} = a/\sqrt{\Delta\tau}$), Transfer of Taylor's theorem at the origin up to fourth order gives, in conditional internal L^4,

$$\varphi_s(x) = x + \frac{\varphi''(0)}{2a}\sqrt{\Delta\tau}\,x^2 + \frac{\varphi'''(0)}{6a^2}\Delta\tau\,x^3 + \mathcal{O}_{L^4}(\Delta\tau^{3/2}). \tag{5.59}$$

Taking $g \sim {}^*\mathcal{N}(0,1)$ in (5.59) and using transferred Gaussian moments gives

$$c^{-1} = 1 + \frac{\Delta\tau}{a^2}\left(\frac{3}{4}\varphi''(0)^2 + \varphi'''(0)\right) + o(\Delta\tau), \qquad c = 1 - \frac{\Delta\tau}{a^2}\left(\frac{3}{4}\varphi''(0)^2 + \varphi'''(0)\right) + o(\Delta\tau), \tag{5.60}$$

where $o(\Delta\tau)$ denotes a remainder η satisfying $\eta/\Delta\tau \approx 0$. The odd cubic terms have zero internal expectation, while Transfer of Isserlis' formula gives ${}^*\mathbb{E}[(w^\alpha)^3 w^\beta \mid \mathcal{F}_\ell] = 3\Phi_\ell^{\alpha\alpha}\Phi_\ell^{\alpha\beta}$ and ${}^*\mathbb{E}[(w^\alpha)^2(w^\beta)^2 \mid \mathcal{F}_\ell] = \Phi_\ell^{\alpha\alpha}\Phi_\ell^{\beta\beta} + 2(\Phi_\ell^{\alpha\beta})^2$. Consequently,

$$\begin{aligned} {}^*\mathbb{E}\Big[\varphi_s(w^\alpha)\varphi_s(w^\beta) \mid \mathcal{F}_\ell\Big] &= \Phi_\ell^{\alpha\beta} + \frac{\varphi''(0)^2}{4a^2}\Delta\tau\left(\Phi_\ell^{\alpha\alpha}\Phi_\ell^{\beta\beta} + 2(\Phi_\ell^{\alpha\beta})^2\right) \\ &\quad + \frac{\varphi'''(0)}{2a^2}\Delta\tau\,\Phi_\ell^{\alpha\beta}\left(\Phi_\ell^{\alpha\alpha} + \Phi_\ell^{\beta\beta}\right) + o(\Delta\tau). \end{aligned} \tag{5.61}$$

Multiplying (5.61) by (5.60) gives

$$ {}^*\mathbb{E}\Big[c\,\varphi_s(w^\alpha)\varphi_s(w^\beta) \mid \mathcal{F}_\ell\Big] = \Phi_\ell^{\alpha\beta} + b^{\alpha\beta}(\Phi_\ell)\,\Delta\tau + o(\Delta\tau), \tag{5.62}$$

with b exactly as in (5.56).

Covariance calculation. For each α, β, define the centered internal fluctuations

$$R_{\ell,i}^{\alpha\beta} := c\,\varphi_s(h_{\ell+1,i}^\alpha)\varphi_s(h_{\ell+1,i}^\beta) - {}^*\mathbb{E}\Big[c\,\varphi_s(w^\alpha)\varphi_s(w^\beta) \mid \mathcal{F}_\ell\Big], \qquad R_\ell^{\alpha\beta} := \frac{1}{\sqrt{N}}\sum_{i=1}^N R_{\ell,i}^{\alpha\beta}. \tag{5.63}$$

Then

$$\Phi_{\ell+1}^{\alpha\beta} - \Phi_\ell^{\alpha\beta} = b^{\alpha\beta}(\Phi_\ell)\,\Delta\tau + \sqrt{\Delta\tau}\,R_\ell^{\alpha\beta} + o(\Delta\tau), \tag{5.64}$$

where the final term is the conditional-mean remainder from the preceding drift calculation. In both shaping schemes, $c\,\varphi_s(h^\alpha_{\ell+1,1})\varphi_s(h^\beta_{\ell+1,1}) = h^\alpha_{\ell+1,1}h^\beta_{\ell+1,1} + \mathcal{O}(\sqrt{\Delta\tau})$ in conditional internal L^2. After centering, this gives $R^{\alpha\beta}_{\ell,1} = h^\alpha_{\ell+1,1}h^\beta_{\ell+1,1} - \Phi^{\alpha\beta}_\ell + \mathcal{O}(\sqrt{\Delta\tau})$ in conditional internal L^2. Conditional on $\mathcal{F}_\ell$, the leading term has the same internal law as the Gaussian-product fluctuation computed in Proposition 5.5 and (5.35). Conditional ${}^*\mathbb{P}$-independence across $i = 1, \dots, N$ then gives

$$\begin{aligned} {}^*\mathrm{Cov}\Big(R^{\alpha\beta}_\ell, R^{\gamma\delta}_\ell \mid \mathcal{F}_\ell\Big) &= {}^*\mathrm{Cov}\Big(R^{\alpha\beta}_{\ell,1}, R^{\gamma\delta}_{\ell,1} \mid \mathcal{F}_\ell\Big) \\ &= {}^*\mathrm{Cov}\Big(w^\alpha w^\beta, w^\gamma w^\delta \mid \mathcal{F}_\ell\Big) + o(1) \\ &= \Sigma_{\alpha\beta,\gamma\delta}(\Phi_\ell) + o(1), \end{aligned} \tag{5.65}$$

where $o(1)$ denotes an infinitesimal entry in $\mathbb{I}$. Let R_ℓ now denote the vector of its $\bar{m}$ distinct upper-triangular entries. Then, in flattened coordinates,

$$\Sigma_N(\Phi_\ell) := {}^*\mathrm{Cov}(R_\ell \mid \mathcal{F}_\ell) = \Sigma(\Phi_\ell) + o(1) \approx \Sigma(\Phi_\ell). \tag{5.66}$$

By Transfer, one may choose ξ_ℓ with the internal conditional moments in (5.54) so that

$$R_\ell = \Sigma_N(\Phi_\ell)^{1/2}\xi_\ell. \tag{5.67}$$

If $\Sigma_N(\Phi_\ell)$ is internally singular, its zero-variance directions do not contribute to R_ℓ; one can adjoin independent unit-variance coordinates in those directions so that ${}^*\mathrm{Cov}(\xi_\ell \mid \mathcal{F}_\ell) = I_{\bar{m}}$ without changing the factorization. By (5.66) and S-continuity of the positive-semidefinite matrix square root, $\Sigma_N(\Phi_\ell)^{1/2} \approx \Sigma(\Phi_\ell)^{1/2}$, and hence with $\Delta B_\ell = \sqrt{\Delta\tau}\,\xi_\ell$,

$$\sqrt{\Delta\tau}\,R_\ell = \Sigma(\Phi_\ell)^{1/2}\Delta B_\ell + \tilde{o}_\ell(\Delta\tau). \tag{5.68}$$

Combining this identity with (5.58) or (5.62) proves (5.53). The same transferred Gaussian moment bounds give the infinitesimal conditional third-moment (Lindeberg) rate used in the diffusion shadow below; see [42, Appendices C–D] for the corresponding uniform estimates on compact sets. □

The expansion (5.53) has the form of one Keisler–Euler–Maruyama step for a diffusion with drift b, diffusion covariance Σ, and diffusion coefficient $\Sigma^{1/2}$. This leads to the following pathwise shadow theorem (compare [42]).

Theorem 5.10 (Neural Covariance SDE Limit [42])**.** *Fix standard $m \in \mathbb{N}$ and standard inputs $x^1, \dots, x^m \in \mathbb{R}^{n_0}$. Let $\bar{m} := m(m+1)/2$, let $D, N \in {}^*\mathbb{N} \setminus \mathbb{N}$ satisfy $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$, and let $\{\Phi_\ell\}_{\ell=0}^D$ be the internal shaped Gram-matrix Markov chain defined by (5.49). Assume a critical shaping scheme as in Theorem 5.9, and suppose the standard input Gram matrix Φ_0 is strictly positive definite. Define the internal depth-time interpolation*

$$\Phi^{(N)}_\tau := \Phi_{\min\{\lfloor \tau N \rfloor, D\}}, \qquad \tau \geq 0. \tag{5.69}$$

For standard $r > 0$, define the norm-exit stopping times

$$\tau^{(N)}_r := \inf\left\{\tau \geq 0 : \left\|\Phi^{(N)}_\tau\right\| > r\right\}, \qquad \tau_r := \inf\left\{\tau \geq 0 : \|\Phi_\tau\| > r\right\}, \tag{5.70}$$

where $\|\cdot\|$ denotes the Euclidean norm of the $\bar{m}$ upper-triangular entries. Then for every standard radius $r > 0$ at which the stopping map is almost surely continuous for the limiting diffusion (all but at most countably many $r > 0$), the stopped hyperfinite trajectory $\tau \mapsto \Phi^{(N)}_{\tau\wedge\tau_r^{(N)}}$ is $L(^*\mathbb{P})$-almost surely limited and S-continuous on $[0,\bar{\tau}]$, and its uniform shadow $\widetilde{\Phi}^{(r)}_\tau := \mathrm{st}(\Phi^{(N)}_{\tau\wedge\tau_r^{(N)}})$ on $(\Omega, \mathcal{L}(\mathcal{A}), L(^*\mathbb{P}))$ has the law of the stopped path $\Phi_{\cdot\wedge\tau_r}$ in $C([0,\bar{\tau}],\mathbb{R}^{\bar{m}})$ (lifting directly to pathwise equality

$$\mathrm{st}\left(\Phi^{(N)}_{\tau\wedge\tau_r^{(N)}}\right) = \Phi_{\tau\wedge\tau_r} \qquad \text{uniformly on } [0,\bar{\tau}] \tag{5.71}$$

under Keisler's Loeb-space representation; see Appendix A.3), where Φ is the unique maximal strong solution of

$$d\Phi_\tau = b(\Phi_\tau)\,d\tau + \Sigma(\Phi_\tau)^{1/2}\,dB_\tau, \qquad \Phi_0 \in \mathrm{SPD}(m), \tag{5.72}$$

with drift b and diffusion tensor Σ as in Theorem 5.9, and with B_τ a standard Brownian motion in $\mathbb{R}^{\bar{m}}$ in the upper-triangular coordinates.

Proof Sketch. Flatten Φ into its $\bar{m}$ upper-triangular entries, as in Section 5.3. The Keisler–Euler–Maruyama increment expansion in Theorem 5.9, together with the transferred Gaussian conditional-moment bound at the end of its proof, gives the required drift, covariance, and infinitesimal third-moment rate uniformly on standard compact subsets of $\mathrm{SPD}(m)$. Because these conditional estimates depend on the history only through Φ_ℓ, they furnish the Markov-state coefficients required by Theorem A.4. They match the drift b and diffusion covariance Σ in (5.72), yielding S-continuity and the stopped shadow up to exit from regular localization domains lying strictly inside $\mathrm{SPD}(m)$ (driven directly on $(\Omega, \mathcal{L}(\mathcal{A}), L(^*\mathbb{P}))$ by $B_\tau = \mathrm{st}(\sum_{\ell<\lfloor\tau N\rfloor}\Delta B_\ell)$ prior to boundary contact); the precise sufficient lifting criterion is given in Theorem A.4. Extending from strict interior domains of $\mathrm{SPD}(m)$ to the norm-ball stopping time τ_r uses the boundary localization of [42, Appendices A, C, and D]. □

An independent readout converts the terminal internal Gram matrix into the output covariance shadow. Let $W_D \in {}^*\mathbb{R}^{1\times N}$ be an internal readout weight row vector with i.i.d. ${}^*\mathcal{N}(0,1)$ entries, ${}^*\mathbb{P}$-independent of the weights used to form h_D^α. Define the (scalar) internal readout for input α at hyperfinite depth D by

$$f^\alpha := \sqrt{\frac{c}{N}} W_D \varphi_s(h_D^\alpha), \qquad \alpha = 1,\dots,m. \tag{5.73}$$

Conditional on $\mathcal{F}_D$, by Transfer the vector $f := (f^1,\dots,f^m) \in {}^*\mathbb{R}^m$ is internally jointly Gaussian with mean zero and covariance

$${}^*\mathrm{Cov}\Big(f^\alpha, f^\beta \mid \mathcal{F}_D\Big) = \frac{c}{N}\left\langle \varphi_s(h_D^\alpha), \varphi_s(h_D^\beta)\right\rangle = \Phi_D^{\alpha\beta}. \tag{5.74}$$

Since this internal conditional law depends on $\mathcal{F}_D$ only through Φ_D, it follows that

$$f \mid \Phi_D \sim {}^*\mathcal{N}(0,\Phi_D). \tag{5.75}$$

Whenever the diffusion (5.72) does not explode by time $\bar{\tau}$ (so $\Phi_D \in \mathbb{L}^{m\times m}$ and $\mathrm{st}(\Phi_D) = \Phi_{\bar{\tau}}$ $L(^*\mathbb{P})$-almost surely), f is $L(^*\mathbb{P})$-almost surely limited and the shadow readout satisfies $\mathrm{st}(f) \mid$

$\Phi_{\bar{\tau}} \sim \mathcal{N}(0, \Phi_{\bar{\tau}})$, forming a Gaussian scale mixture with random covariance $\Phi_{\bar{\tau}}$ [42, Theorems 3.2 and 3.9].

We close the section by examining stability in the one-input case. When $m = 1$ (one input / one covariance coordinate), the smooth-shaped drift (5.56) reduces to a logistic-type scalar SDE:

$$d\Phi_\tau = b\,\Phi_\tau(\Phi_\tau - 1)\,d\tau + \sqrt{2}\,\Phi_\tau\,dB_\tau, \qquad b := \frac{\frac{3}{4}\varphi''(0)^2 + \varphi'''(0)}{a^2}, \tag{5.76}$$

where B_τ is a standard one-dimensional Brownian motion.

A useful intuition for the role of b comes from the corresponding deterministic ODE $\partial_\tau \Phi_\tau = b\,\Phi_\tau(\Phi_\tau - 1)$, which solves explicitly to

$$\Phi_\tau = \frac{1}{1 - (1 - \Phi_0^{-1})e^{b\tau}}. \tag{5.77}$$

If $b > 0$ and $\Phi_0 > 1$, the ODE explodes at the finite time $\tau_\star = \frac{1}{b}\log(\frac{\Phi_0}{\Phi_0 - 1})$, whereas for $\Phi_0 \in (0, 1)$ it is driven toward 0. If $b < 0$, the drift points toward 1 from both sides, so every positive ODE trajectory remains bounded in $(0, \infty)$; if $b = 0$, every ODE trajectory is constant. For the SDE, the Brownian term at $\Phi_\tau = 1$ is $\sqrt{2}\,dB_\tau$, so a trajectory that reaches 1 continues to fluctuate rather than staying there. Nevertheless, the same drift mechanism explains why escape from the positive appreciable galaxy $(0, \infty) = \mathrm{st}(\mathbb{A}_{>0})$ is possible: when $b > 0$, an upward fluctuation above 1 is reinforced by a positive drift that grows quadratically, whereas for $b \le 0$ the drift does not push large values farther outward. The following result makes this intuition precise.

Proposition 5.11 (Finite-Time Explosion Criterion for the Scalar Covariance SDE [42, Proposition 3.7])**.** *Let Φ_τ solve (5.76) with standard $\Phi_0 > 0$, and define the exit time from $(0, \infty)$ (the shadow of the positive appreciable ring $\mathbb{A}_{>0}$) by*

$$\tau^\star := \sup_{M>1} \inf \left\{ \tau \ge 0 : \Phi_\tau \ge M \text{ or } \Phi_\tau \le M^{-1} \right\}. \tag{5.78}$$

Then $L({}^*\mathbb{P})(\tau^\star = \infty) = 1$ if and only if $b \le 0$; equivalently, finite-time explosion out of the positive appreciable galaxy $(0, \infty)$ occurs with positive probability if and only if $b > 0$.

Thus $b \le 0$ is the exact almost-sure nonexplosion condition keeping the covariance trajectory inside the positive appreciable galaxy; when $b > 0$, finite-time escape from $(0, \infty)$ occurs with positive probability, not necessarily almost surely [42, Section 4].

This is one reason the smooth-shaped covariance SDE shadow in Theorem 5.10 is naturally formulated in a *local* (stopped) sense: when escape from $\mathbb{A}_{>0}$ is possible, one compares the hyperfinite chain $\Phi_\tau^{(N)}$ and the continuous diffusion Φ_τ up to the first exit time from each standard compact subset of $(0, \infty)$ (or of $\mathrm{SPD}(m)$ in the matrix case) [42, Definition 3.8 and Theorem 3.9].

5.5 Exercises

Exercise 5.12 (Local Truncation Error)**.** Assume (5.4). For a positive infinitesimal step size $\eta \approx 0$, show that the one-step error of the internal Euler update (5.2) satisfies

$$\eta^{-2}\,\|{}^*X_\eta - (X_0 + \eta\,b(X_0))\| \in \mathbb{L}, \qquad \text{i.e.,} \qquad \|{}^*X_\eta - (X_0 + \eta\,b(X_0))\| = \mathcal{O}(\eta^2), \tag{5.79}$$

and use Transfer of the discrete Grönwall inequality to deduce (5.5) in Theorem 5.2.

Exercise 5.13 (Wick Expansion)**.** Let $g \sim {}^*\mathcal{N}(0, \Phi)$ in ${}^*\mathbb{R}^m$. Use Transfer of Isserlis' theorem to compute ${}^*\mathbb{E}[g_\alpha g_\beta g_\gamma g_\delta]$ and derive Proposition 5.5.

Exercise 5.14 (Proof of Proposition 5.7)**.** Adapt the one-sample argument of [42, §2.2] to hyperfinite width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ and depth $D \in {}^*\mathbb{N} \setminus \mathbb{N}$ with $\mathrm{st}(D/N) = \bar{\tau}$ and $\Delta\tau := 1/N \approx 0$. In particular, show by Transfer that $\Phi_{\ell+1} = \Phi_\ell(1 + \sqrt{\Delta\tau}\, R_\ell)$ for $0 \le \ell \le D-1$, where ${}^*\mathbb{E}[R_\ell \mid \mathcal{F}_\ell] = 0$ and ${}^*\mathrm{Var}(R_\ell \mid \mathcal{F}_\ell) = 5$. After checking the infinitesimal conditional third-moment rate and applying Doob's maximal inequality (5.27), match the zero drift and diffusion coefficient $\sqrt{5}\,\Phi$ in Proposition 5.7 via Theorem A.4, and verify (5.45) by summing the second-order nonstandard Taylor expansion of ${}^*\log(1 + \sqrt{\Delta\tau}\, R_\ell)$ over $\ell = 0, \dots, D-1$.

Chapter 6

Further Topics in Proportional Depth–Width Scaling

This chapter develops three further aspects of the proportional "depth–width" scaling regime, in which depth $D \in {}^*\mathbb{N} \setminus \mathbb{N}$ and width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ are both unlimited hyperintegers lying in the same proportional galaxy $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$, with infinitesimal layer step $\Delta\tau := 1/N \approx 0$. The common mechanism is the hyperfinite accumulation of infinitesimal layerwise effects across $D \asymp N$ layers. Depending on the setting, taking standard shadows produces a covariance SDE at initialization, nontrivial spectral dynamics, or interactions among layer updates after one training step. We study:

(i) how shaping prevents rank collapse in pure attention and yields a covariance SDE shadow in the proportional galaxy [52];

(ii) what the linear covariance SDE implies for eigenvalues and empirical spectral distributions [40];

(iii) how one gradient step changes features and predictions in a deep linear network [41].

The first two parts concern forward propagation at initialization. The final part turns to training and isolates the layer-interaction term absent from kernel linearization. Parts (ii) and (iii) draw on a recent preprint and a manuscript in preparation, respectively; here we prioritize clean statements and plausibility calculations.

6.1 Shaped Transformers

A Transformer is an architecture that roughly treats a single token (word in a sentence) as a vector similar to a single data point in an MLP, but considers a sequence of tokens as inputs together (like a full sentence or paragraph), and allows for mixing via a self-attention mechanism.

Fix standard dimensions $n_0, m \in \mathbb{N}$, input token vectors $x^1, \ldots, x^m \in \mathbb{R}^{n_0}$, and write $h_0 = [x^\beta]_{\beta=1}^m \in \mathbb{R}^{n_0 \times m}$. At an unlimited hidden width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, the input layer uses an internal embedding matrix $W_0 \in {}^*\mathbb{R}^{N \times n_0}$ and produces the first hidden-state matrix

$$h_1 := \frac{1}{\sqrt{n_0}} W_0 h_0, \qquad h_\ell = [h_\ell^\beta]_{\beta=1}^m \in {}^*\mathbb{R}^{N \times m}, \qquad h_\ell^\beta \in {}^*\mathbb{R}^N, \quad \ell \geq 1, \tag{6.1}$$

where the columns correspond to tokens. At internal layer $\ell \geq 1$, the (single-head) scaled dot-product attention mechanism, with fixed standard query/key width $n_k \in \mathbb{N}$, first forms the query and key matrices

$$q_\ell = \frac{1}{\sqrt{N}} W_\ell^q h_\ell \in {}^*\mathbb{R}^{n_k \times m}, \qquad k_\ell = \frac{1}{\sqrt{N}} W_\ell^k h_\ell \in {}^*\mathbb{R}^{n_k \times m}, \tag{6.2}$$

and then the score and attention matrices

$$S_\ell := k_\ell^\top q_\ell \in {}^*\mathbb{R}^{m \times m}, \qquad A_\ell := \mathrm{Softmax}(s^{-1} S_\ell - M) \in {}^*\mathbb{R}^{m \times m}, \tag{6.3}$$

where M is the mask, $s \in {}^*\mathbb{R}_{>0}$ is the temperature, and Softmax denotes the natural internal extension ${}^*\mathrm{Softmax}$ on ${}^*\mathbb{R}^{m \times m}$. Increasing s shrinks the logits and makes each attention column less concentrated. Standard scaled dot-product attention takes appreciable $s \asymp \sqrt{n_k}$, and we will discuss unlimited choices later. Here the Softmax is column-wise, i.e. for $S \in {}^*\mathbb{R}^{m \times m}$,

$$\mathrm{Softmax}(S)_{\alpha\beta} := \frac{\exp(S_{\alpha\beta})}{\sum_{\gamma=1}^m \exp(S_{\gamma\beta})}. \tag{6.4}$$

Thus the first index α denotes the key token, the second index β denotes the query token, and every column of A_ℓ sums to 1. The corresponding self-attention network recursion is

$$h_{\ell+1} = \frac{1}{\sqrt{N}} W_\ell^v h_\ell A_\ell. \tag{6.5}$$

Thus $h_\ell A_\ell$ performs a weighted average across the token index. Because tokens are stored as columns, the βth output column is the attention-weighted sum of the value vectors, and no transpose is needed in the recursion. Here $W_\ell^q, W_\ell^k \in {}^*\mathbb{R}^{n_k \times N}$ and $W_\ell^v \in {}^*\mathbb{R}^{N \times N}$. At initialization, all entries of $W_0, W_\ell^q, W_\ell^k, W_\ell^v$, across weight types and hyperfinite layers, are mutually internally independent ${}^*\mathcal{N}(0,1)$ random variables.

Normally M is the causal mask, with $M_{\alpha\beta} = +\infty$ for $\alpha > \beta$ and 0 otherwise. For simplicity, throughout this section we set $M = 0$ and study unmasked attention.

However, naively running this attention-only recursion to an unlimited depth leads to a degeneracy known as rank collapse. To see the scale of one layer, suppress the layer index and decompose the key–query geometric product in the head Clifford algebra ${}^*\mathcal{C}\ell_{n_k}$:

$$k^\alpha q^\beta = \left\langle k^\alpha q^\beta \right\rangle_0 + k^\alpha \wedge q^\beta \in {}^*\mathcal{C}\ell_{n_k}, \qquad \frac{1}{s} k^\top q = \frac{1}{s} [\left\langle k^\alpha q^\beta \right\rangle_0]_{\alpha,\beta=1}^m = \frac{1}{s} [\left\langle k^\alpha, q^\beta \right\rangle]_{\alpha,\beta=1}^m. \tag{6.6}$$

Standard dot-product attention retains only the grade-0 scalar inner product $\left\langle k^\alpha q^\beta \right\rangle_0$ to form its Softmax logits, discarding the grade-2 exterior bivector $k^\alpha \wedge q^\beta \in \bigwedge^2({}^*\mathbb{R}^{n_k})$ that encodes the oriented rotational plane between key and query tokens (which Clifford-equivariant attention layers propagate explicitly [57, 10]). Under the internal Gaussian initialization specified above, for fixed α and β the summands $k_a^\alpha q_a^\beta$ are internally i.i.d. and centered across $a \in [n_k]$, conditional on h. Thus $\left\langle k^\alpha q^\beta \right\rangle_0 = \left\langle k^\alpha, q^\beta \right\rangle$ is limited of appreciable scale $\sqrt{n_k}$ when the normalized token norms lie in $\mathbb{A}$. With the standard choice $s \asymp \sqrt{n_k}$, the logits lie in $\mathbb{L}$ with non-infinitesimal variance, and

$$A = \mathrm{Softmax}\left(\frac{1}{s} k^\top q\right) \tag{6.7}$$

does not lie in the monad of the identity matrix I_m at unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$.

Each layer therefore performs appreciable token mixing. In a deep pure self-attention network without shortcuts or MLPs, repeated mixing across unlimited depth drives the token representations into the monad of a token-uniform rank-one configuration; under the conditions of [16, Theorem 2.3], this contraction is doubly exponential in depth. Related signal-propagation analysis connects rank collapse at initialization to vanishing query and key gradients [51, Theorem 3.2].

This is similar to the ReLU correlation degeneration in (5.47). There the scalar correlation map drives every off-diagonal correlation toward 1; here repeated token mixing aligns the token representations. The mechanisms are different, but in both cases normalized representations become infinitesimally indistinguishable at unlimited depth.

This motivates an attention analogue of the activation shaping in (5.50) and (5.51): make the token mixing in each layer infinitesimal of just the right scale to accumulate nontrivially over proportional hyperfinite depth $D \asymp N$.

6.1.1 Shaped Attention and the Covariance SDE Limit

At an unlimited temperature, ordinary Softmax lies infinitely close to the uniform matrix $\frac{1}{m}\mathbf{1}\mathbf{1}^\top$, which still averages the tokens at every step. The shaped Transformer instead uses the same broad strategy as the smooth activation shaping in (5.51): subtract the leading uniform term and add the identity, leaving an infinitesimal width-dependent perturbation of I_m whose one-step effects accumulate over proportional hyperfinite depth. For attention, the construction in [52, Section 4.2] combines three modifications:

(a) temperature scaling: take an unlimited temperature $s \in {}^*\mathbb{R}_{>0} \setminus \mathbb{L}$ larger than the standard $\sqrt{n_k}$, namely $s = s_0\sqrt{Nn_k}$;

(b) uniform centering: subtract the uniform Softmax baseline $\frac{1}{m}\mathbf{1}\mathbf{1}^\top$;

(c) identity shift: add I_m so that shaped attention lies in the monad of I_m.

Definition 6.1 (Shaped Attention)**.** In the unmasked column-token convention, the shaped attention map at unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ is

$$A_\ell^s := I_m + \mathrm{Softmax}(s^{-1}S_\ell) - \frac{1}{m}\mathbf{1}\mathbf{1}^\top. \tag{6.8}$$

Here $s = s_0\sqrt{Nn_k} \in {}^*\mathbb{R}_{>0} \setminus \mathbb{L}$ for fixed standard $s_0 > 0$. The subtraction removes the leading uniform term, while the identity shift prevents mixing between token positions at appreciable order. Each column of A_ℓ^s still sums to 1, although the uniform centering means that some off-diagonal entries may be negative.

Definition 6.2 (Feature Covariance)**.** The input and hidden token feature covariances are

$$\Phi_0 := \frac{1}{n_0}h_0^\top h_0 \in \mathbb{R}^{m\times m}, \qquad \Phi_\ell := \frac{1}{N}h_\ell^\top h_\ell \in {}^*\mathbb{R}^{m\times m}, \quad \ell \geq 1. \tag{6.9}$$

Thus $\Phi_\ell^{\alpha\beta} = N^{-1}\left\langle h_\ell^\alpha (h_\ell^\beta)^\sim \right\rangle_0 = N^{-1}\left\langle h_\ell^\alpha, h_\ell^\beta \right\rangle$ in ${}^*\mathcal{C}\ell_N$ for $\ell \geq 1$. The rows of h_1 are internally i.i.d. ${}^*\mathcal{N}(0,\Phi_0)$, so the Loeb strong law of large numbers gives $\Phi_1 \approx \Phi_0$ entrywise $L({}^*\mathbb{P})$-a.s. (and hence $\Phi_1 \overset{L({}^*\mathbb{P})}{\approx} \Phi_0$). The hidden token covariance is the main object tracked across proportional hyperfinite depth.

Lemma 6.3 (A Softmax Expansion Around the Uniform Point)**.** *Let $S \in \mathbb{L}^{m\times m} \subset {}^*\mathbb{R}^{m\times m}$ be a limited matrix and consider* $\mathrm{Softmax}(s^{-1}S)$ *at an unlimited positive temperature $s \in {}^*\mathbb{R}_{>0} \setminus \mathbb{L}$. For each column index $\beta \in [m]$, define the internal column mean and mean square*

$$\overline{S^\beta} := \frac{1}{m}\sum_{\gamma=1}^{m} S^{\gamma\beta}, \qquad \overline{(S^\beta)^2} := \frac{1}{m}\sum_{\gamma=1}^{m} (S^{\gamma\beta})^2. \tag{6.10}$$

Then, entrywise for $\alpha, \beta \in [m]$,

$$\mathrm{Softmax}(s^{-1}S)^{\alpha\beta} = \frac{1}{m} + \frac{1}{ms}(S^{\alpha\beta} - \overline{S^\beta}) + \frac{1}{2ms^2}\left[(S^{\alpha\beta} - \overline{S^\beta})^2 - \left(\overline{(S^\beta)^2} - (\overline{S^\beta})^2\right)\right] + \mathcal{O}(s^{-3}), \tag{6.11}$$

where $\mathcal{O}(s^{-3})$ denotes an internal remainder $R_s^{\alpha\beta}(S)$ satisfying $s^3 R_s^{\alpha\beta}(S) \in \mathbb{L}$, with a uniform standard bound $\left|R_s^{\alpha\beta}(S)\right| \le C_B s^{-3}$ whenever $\|S\|_F \le B$ for standard $B > 0$.

Proof. Fix a column index β and set $r_\alpha := S^{\alpha\beta} \in \mathbb{L}$. Also write $\bar r := m^{-1}\sum_{\nu=1}^m r_\nu$ and $\overline{r^2} := m^{-1}\sum_{\nu=1}^m r_\nu^2$. Since $r_\alpha/s \approx 0$, transferring the second-order Taylor theorem with uniform remainder on $[-B, B]$ gives the first two internal expansions below, and shows that the normalized denominator in the second line lies in $\mathrm{mon}(1)$ and is appreciably bounded away from zero. Applying the transferred geometric expansion $(1+x)^{-1} = 1 - x + x^2 + \mathcal{O}(x^3)$ for $x \approx 0$ then gives the final line:

$$\begin{aligned} e^{r_\alpha/s} &= 1 + \frac{r_\alpha}{s} + \frac{r_\alpha^2}{2s^2} + \mathcal{O}(s^{-3}), \\ \frac{1}{m}\sum_{\nu=1}^{m} e^{r_\nu/s} &= 1 + \frac{\bar r}{s} + \frac{\overline{r^2}}{2s^2} + \mathcal{O}(s^{-3}), \\ \mathrm{Softmax}(s^{-1}S)^{\alpha\beta} &= \frac{1}{m}\left[1 + \frac{r_\alpha - \bar r}{s} + \frac{(r_\alpha - \bar r)^2 - (\overline{r^2} - \bar r^2)}{2s^2}\right] + \mathcal{O}(s^{-3}), \end{aligned} \tag{6.12}$$

which is the claimed internal expansion after substituting the definitions above. □

With $s = s_0\sqrt{Nn_k}$, layer step $\Delta\tau = 1/N \approx 0$, and raw scores $S_\ell = k_\ell^\top q_\ell$ of typical limited size $\asymp \sqrt{n_k}$, the entries of $s^{-1}S_\ell$ are infinitesimals of scale $N^{-1/2} = \sqrt{\Delta\tau}$. Equivalently, Lemma 6.3 is an exact internal expansion in the infinitesimal step scale $\sqrt{\Delta\tau}$.

More precisely, combining (6.8) with Lemma 6.3 yields the entrywise internal decomposition

$$\begin{aligned} (A_\ell^s)^{\alpha\beta} = (I_m)^{\alpha\beta} &+ \underbrace{\frac{S_\ell^{\alpha\beta} - \overline{S_\ell^\beta}}{ms_0\sqrt{n_k}}\sqrt{\Delta\tau}}_{\text{centered fluctuation}} \\ &+ \underbrace{\frac{\Delta\tau}{2ms_0^2 n_k}\left[(S_\ell^{\alpha\beta} - \overline{S_\ell^\beta})^2 - \left(\overline{(S_\ell^\beta)^2} - (\overline{S_\ell^\beta})^2\right)\right]}_{\text{drift-scale correction}} + \mathcal{O}_{*\mathbb{P}}(\Delta\tau^{3/2}). \end{aligned} \tag{6.13}$$

Conditional on the previous layers (on regions where $\Phi_\ell \in \mathbb{L}^{m\times m}$), shaped attention is the identity plus a centered fluctuation of infinitesimal scale $\sqrt{\Delta\tau}$ and a smaller, generally random correction of scale $\Delta\tau$ (with remainder controlled in conditional L^2 at scale $\Delta\tau^{3/2} = N^{-3/2}$).

When this is inserted into the one-step covariance update $\Phi_{\ell+1} = \frac{1}{N} h_{\ell+1}^\top h_{\ell+1}$, the $\sqrt{\Delta\tau}$-fluctuation generates the diffusion-scale increment, while the internal conditional mean of the $\Delta\tau$-correction, together with quadratic products of $\sqrt{\Delta\tau}$-fluctuations, generates the $\Delta\tau$-drift. The one-step internal drift and covariance therefore lie infinitely close to the coefficients of the candidate SDE, by the same Keisler–Itô mechanism used in Theorem 5.10; the precise hyperfinite Markov-chain lifting theorem is deferred to Theorem A.4.

6.1.2 A Representative SDE Limit Theorem for Shaped Attention

The next theorem is an informal column-token restatement of the no-shortcut case of the unmasked, attention-only covariance limit in [52, Theorem 4.2], rewritten in our nonstandard Φ-notation.

Theorem 6.4 (Neural Covariance SDE for Shaped Attention (Informal))**.** *Consider a shaped-attention network at initialization in Definition 6.1. Let $D, N \in {}^*\mathbb{N} \setminus \mathbb{N}$ lie in the proportional galaxy $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$ with $m, n_0, n_k \in \mathbb{N}$ standard. Then the hyperfinite covariance process $\Phi_\tau^{(N)} := \Phi_{\lfloor \tau N \rfloor}$, $0 \le \tau \le \bar{\tau}$, is locally S-continuous $L({}^*\mathbb{P})$-a.s., and its standard shadow $\Phi_\tau := \mathrm{st}\big(\Phi_{\lfloor \tau N \rfloor}\big)$ has the law of the solution to the SDE*

$$d\Phi_\tau = b(\Phi_\tau)\, d\tau + \Sigma(\Phi_\tau)^{1/2}\, dB_\tau, \qquad \Phi_0 = \frac{1}{n_0} h_0^\top h_0, \tag{6.14}$$

where B_τ is a standard Brownian motion in $\mathbb{R}^{m(m+1)/2}$ (upper-triangular coordinates). The coefficients are as follows. Write $\Phi^{\alpha\bar{x}} := \frac{1}{m}\sum_{\nu=1}^m \Phi^{\alpha\nu}$ and $\bar{\Phi} := \frac{1}{m}\sum_{\nu=1}^m \Phi^{\nu\nu}$. Further define

$$R_1^{\alpha\gamma,\beta\delta}(\Phi) := \Phi^{\alpha\beta}\big(\Phi^{\gamma\delta} - \Phi^{\gamma\bar{x}} - \Phi^{\delta\bar{x}} + \Phi^{\bar{x}\bar{x}}\big), \tag{6.15}$$

$$R_2^{\alpha\gamma}(\Phi) := \Phi^{\alpha\alpha}\big(\Phi^{\gamma\gamma} - 2\Phi^{\gamma\bar{x}} + 2\Phi^{\bar{x}\bar{x}} - \bar{\Phi}\big), \tag{6.16}$$

where $\Phi^{\bar{x}\bar{x}} := \frac{1}{m^2}\sum_{\nu,\kappa=1}^m \Phi^{\nu\kappa}$. Then the drift can be written as

$$b_{\alpha\beta}(\Phi) = \frac{1}{s_0^2}\left[\frac{1}{m^2}\sum_{\nu,\kappa=1}^m \Phi^{\nu\kappa} R_1^{\alpha\nu,\beta\kappa}(\Phi) + \frac{1}{2m}\sum_{\nu=1}^m \Big(\Phi^{\beta\nu} R_2^{\alpha\nu}(\Phi) + \Phi^{\alpha\nu} R_2^{\beta\nu}(\Phi)\Big)\right]. \tag{6.17}$$

Moreover the diffusion has the form

$$\Sigma(\Phi) = \Sigma_{\mathrm{lin}}(\Phi) + \frac{1}{s_0^2}\Sigma_{\mathrm{attn}}(\Phi), \tag{6.18}$$

where $\Sigma_{\mathrm{lin}}^{\alpha\beta,\gamma\delta}(\Phi) = \Phi^{\alpha\gamma}\Phi^{\beta\delta} + \Phi^{\alpha\delta}\Phi^{\beta\gamma}$ is the linear covariance and

$$\Sigma_{\mathrm{attn}}^{\alpha\beta,\gamma\delta}(\Phi) := \frac{1}{m^2}\sum_{\nu,\kappa=1}^m \begin{pmatrix} \Phi^{\alpha\kappa}\Phi^{\gamma\nu} R_1^{\beta\kappa,\delta\nu}(\Phi) + \Phi^{\alpha\kappa}\Phi^{\delta\nu} R_1^{\beta\kappa,\gamma\nu}(\Phi) \\ + \Phi^{\beta\nu}\Phi^{\gamma\kappa} R_1^{\alpha\nu,\delta\kappa}(\Phi) + \Phi^{\beta\nu}\Phi^{\delta\kappa} R_1^{\alpha\nu,\gamma\kappa}(\Phi) \end{pmatrix}. \tag{6.19}$$

Proof Sketch. For internal layer $\ell \ge 1$, condition on the internal forward filtration $\mathcal{F}_\ell$ from (5.30) and work on limited states $\Phi_\ell \in {}^*K$ for a standard compact $K \subset \mathrm{SPD}(m)$. Since the layer-$\ell$ weights are internally independent of $\mathcal{F}_\ell$, the rows of q_ℓ and k_ℓ are conditionally i.i.d. ${}^*\mathcal{N}(0, \Phi_\ell)$, and the two row families are conditionally independent. The rows of $N^{-1/2} W_\ell^v h_\ell$ have the same exact internal conditional law and are conditionally independent of both families.

The infinitesimal concentration $\Phi_1 \overset{L(^*\mathbb{P})}{\approx} \Phi_0$ noted above shows that the single embedding step does not alter the initial shadow $\mathrm{st}(\Phi_1) = \Phi_0$.

With $s = s_0\sqrt{Nn_k}$ and $\Delta\tau = 1/N \approx 0$, the entries of $s^{-1}k_\ell^\top q_\ell$ are of scale $\sqrt{\Delta\tau}$. Applying Lemma 6.3 and then the centering and identity shift in (6.8) gives again

$$\begin{aligned}(A_\ell^s)^{\alpha\beta} = (I_m)^{\alpha\beta} &+ \frac{S_\ell^{\alpha\beta} - \overline{S_\ell^\beta}}{ms_0\sqrt{n_k}}\sqrt{\Delta\tau} \\ &+ \frac{\Delta\tau}{2ms_0^2 n_k}\left[(S_\ell^{\alpha\beta} - \overline{S_\ell^\beta})^2 - \left(\overline{(S_\ell^\beta)^2} - (\overline{S_\ell^\beta})^2\right)\right] + \mathcal{O}_{^*\mathbb{P}}(\Delta\tau^{3/2}).\end{aligned} \tag{6.13}$$

The first correction is a centered fluctuation of scale $\sqrt{\Delta\tau}$, while the second is of scale $\Delta\tau$ and contributes through its internal conditional mean.

Insert this expansion into the covariance update $\Phi_{\ell+1} = \frac{1}{N}h_{\ell+1}^\top h_{\ell+1}$. For this no-shortcut update, the internal covariance recursion and its conditional mean over the value weights are exactly

$$\Phi_{\ell+1} = \frac{1}{N^2}(A_\ell^s)^\top h_\ell^\top (W_\ell^v)^\top W_\ell^v h_\ell A_\ell^s, \qquad {}^*\mathbb{E}\big[\Phi_{\ell+1} \mid \mathcal{F}_\ell, A_\ell^s\big] = (A_\ell^s)^\top \Phi_\ell A_\ell^s. \tag{6.20}$$

Indeed, conditioning on $\mathcal{F}_\ell$ and A_ℓ^s leaves W_ℓ^v internally independent with i.i.d. ${}^*\mathcal{N}(0,1)$ entries, and ${}^*\mathbb{E}[(W_\ell^v)^\top W_\ell^v] = NI_N$. Subtracting Φ_ℓ and expanding the second line of (6.20) around the identity gives

$${}^*\mathbb{E}\big[\Phi_{\ell+1} - \Phi_\ell \mid \mathcal{F}_\ell, A_\ell^s\big] = \Phi_\ell(A_\ell^s - I_m) + (A_\ell^s - I_m)^\top \Phi_\ell + (A_\ell^s - I_m)^\top \Phi_\ell (A_\ell^s - I_m). \tag{6.21}$$

After the value-weight average in (6.20), the remaining conditional mean is the fresh query–key average of $(A_\ell^s)^\top \Phi_\ell A_\ell^s$. In the column convention here, the first index of A_ℓ^s is the key and the second is the query, so multiplying the expansion above gives

$$\begin{aligned}{}^*\mathbb{E}\big[(A_\ell^s)^{\nu\alpha}(A_\ell^s)^{\kappa\beta} \mid \mathcal{F}_\ell\big] = \delta_{\nu\alpha}\delta_{\kappa\beta} &+ \frac{\Delta\tau}{s_0^2}\left[\frac{1}{m^2}R_1^{\alpha\nu,\beta\kappa}(\Phi_\ell) + \frac{1}{2m}\Big(\delta_{\kappa\beta}R_2^{\alpha\nu}(\Phi_\ell) + \delta_{\nu\alpha}R_2^{\beta\kappa}(\Phi_\ell)\Big)\right] \\ &+ o(\Delta\tau),\end{aligned} \tag{6.22}$$

where $o(\Delta\tau)$ denotes a remainder that is $o(1)\,\Delta\tau$ uniformly for $\Phi_\ell \in {}^*K$. The linear means vanish; the linear–linear product is the centered score moment in (6.28) and gives R_1, while the two identity–quadratic products use the exact internal conditional expectation

$$\frac{1}{n_k}\,{}^*\mathbb{E}\Big[\big(S_\ell^{\nu\alpha} - \overline{S_\ell^\alpha}\big)^2 - \Big(\overline{(S_\ell^\alpha)^2} - \big(\overline{S_\ell^\alpha}\big)^2\Big) \;\Big|\; \mathcal{F}_\ell\Big] = R_2^{\alpha\nu}(\Phi_\ell). \tag{6.23}$$

Products containing at least three $\sqrt{\Delta\tau}$-scale factors, together with the internal Softmax remainder, are $o(\Delta\tau)$ on ${}^*K$; this is the transferred column-index transpose of the pair calculation in [52, Lemmas C.2–C.3]. Thus the internal tower property and contraction with

Φ_ℓ give

$$
\begin{aligned}
{}^*\mathbb{E}[\Phi_{\ell+1}^{\alpha\beta} \mid \mathcal{F}_\ell] &= \sum_{\nu,\kappa=1}^{m} \Phi_\ell^{\nu\kappa}\, {}^*\mathbb{E}[(A_\ell^s)^{\nu\alpha}(A_\ell^s)^{\kappa\beta} \mid \mathcal{F}_\ell] \\
&= \Phi_\ell^{\alpha\beta} + \frac{\Delta\tau}{s_0^2}\Big[\frac{1}{m^2}\sum_{\nu,\kappa=1}^{m} \Phi_\ell^{\nu\kappa} R_1^{\alpha\nu,\beta\kappa}(\Phi_\ell) \\
&\qquad + \frac{1}{2m}\sum_{\nu=1}^{m}\Big(\Phi_\ell^{\beta\nu} R_2^{\alpha\nu}(\Phi_\ell) + \Phi_\ell^{\alpha\nu} R_2^{\beta\nu}(\Phi_\ell)\Big)\Big] + o(\Delta\tau) \\
&= \Phi_\ell^{\alpha\beta} + b_{\alpha\beta}(\Phi_\ell)\,\Delta\tau + o(\Delta\tau),
\end{aligned}
\tag{6.24}
$$

where symmetry of Φ_ℓ is used in the R_2 contractions, and the final line matches (6.17).

After flattening the upper-triangular entries, the two sources of covariance are separated exactly by the internal conditional covariance formula:

$$
{}^*\mathrm{Cov}(\Phi_{\ell+1} - \Phi_\ell \mid \mathcal{F}_\ell) = {}^*\mathbb{E}[{}^*\mathrm{Cov}(\Phi_{\ell+1} \mid \mathcal{F}_\ell, A_\ell^s) \mid \mathcal{F}_\ell] + {}^*\mathrm{Cov}((A_\ell^s)^\top \Phi_\ell A_\ell^s \mid \mathcal{F}_\ell). \tag{6.25}
$$

The first term is the internal Gaussian sample-covariance fluctuation from the value weights, and the second is the fluctuation of the conditional mean caused by shaped attention. The value-weight contribution can be computed exactly. Conditional on $\mathcal{F}_\ell$ and A_ℓ^s, the N output rows are internally i.i.d. centered Gaussian with covariance $(A_\ell^s)^\top \Phi_\ell A_\ell^s$. Their empirical covariance is $\Phi_{\ell+1}$, so transferred Isserlis' identity gives

$$
{}^*\mathrm{Cov}(\Phi_{\ell+1}^{\alpha\beta}, \Phi_{\ell+1}^{\gamma\delta} \mid \mathcal{F}_\ell, A_\ell^s) = \Delta\tau\, \Sigma_{\mathrm{lin}}^{\alpha\beta,\gamma\delta}((A_\ell^s)^\top \Phi_\ell A_\ell^s). \tag{6.26}
$$

Since $A_\ell^s - I_m = \mathcal{O}_{*\mathbb{P}}(\sqrt{\Delta\tau})$ with bounded conditional moments on ${}^*K$, averaging this identity over shaped attention gives $\Delta\tau\, \Sigma_{\mathrm{lin}}(\Phi_\ell) + o(\Delta\tau)$.

For the attention branch, the score entries are conditionally centered, and internal independence of the key and query row families gives

$$
\frac{1}{n_k}\, {}^*\mathbb{E}[S_\ell^{\alpha\beta} S_\ell^{\gamma\delta} \mid \mathcal{F}_\ell] = \Phi_\ell^{\alpha\gamma}\Phi_\ell^{\beta\delta}, \tag{6.27}
$$

$$
\frac{1}{n_k}\, {}^*\mathbb{E}\Big[(S_\ell^{\kappa\beta} - \overline{S_\ell^{\beta}})(S_\ell^{\nu\delta} - \overline{S_\ell^{\delta}}) \;\Big|\; \mathcal{F}_\ell\Big] = R_1^{\beta\kappa,\delta\nu}(\Phi_\ell). \tag{6.28}
$$

Indeed, when the scores are expanded as sums over the n_k rows, only equal row indices survive in the first line; the second follows by expanding the two column means over the key index. This also explains the index order $R_1^{\beta\kappa,\delta\nu}$: the outside pair records the query covariance, while the centered pair records the key covariance.

At scale $\Delta\tau$ in the conditional covariance, only the two linear summands $\Phi_\ell(A_\ell^s - I_m)$ and $(A_\ell^s - I_m)^\top \Phi_\ell$ contribute to the attention fluctuation. By (6.13), the leading (κ,β) entry of $A_\ell^s - I_m$ is $(S_\ell^{\kappa\beta} - \overline{S_\ell^{\beta}})\sqrt{\Delta\tau}/(ms_0\sqrt{n_k})$. Substituting this term and using the conditional score moments, the rescaled internal covariance of the first linear summand with itself satisfies

$$
N\, {}^*\mathrm{Cov}\Big([\Phi_\ell(A_\ell^s - I_m)]_{\alpha\beta},\, [\Phi_\ell(A_\ell^s - I_m)]_{\gamma\delta} \;\Big|\; \mathcal{F}_\ell\Big) = \frac{1}{m^2 s_0^2}\sum_{\kappa,\nu=1}^{m} \Phi_\ell^{\alpha\kappa}\Phi_\ell^{\gamma\nu} R_1^{\beta\kappa,\delta\nu}(\Phi_\ell) + o(1), \tag{6.29}
$$

with $o(1) \approx 0$ uniformly on ${}^*K$. After removing the common factor s_0^{-2}, this is the first contraction in Σ_{attn}. The other three pairings among the two linear summands give the remaining

three contractions. Although the first-order score fluctuation is conditionally centered, the full $A_\ell^s - I_m$ has a conditional mean of scale $\Delta\tau$, which is why the display uses conditional covariance. Combining these four pairings with (6.25) and (6.26), and using that Φ_ℓ is $\mathcal{F}_\ell$-measurable, gives

$$N\,{}^*\mathrm{Cov}(\Phi_{\ell+1}^{\alpha\beta} - \Phi_\ell^{\alpha\beta},\, \Phi_{\ell+1}^{\gamma\delta} - \Phi_\ell^{\gamma\delta} \mid \mathcal{F}_\ell) = \Sigma_{\mathrm{lin}}^{\alpha\beta,\gamma\delta}(\Phi_\ell) + \frac{1}{s_0^2}\Sigma_{\mathrm{attn}}^{\alpha\beta,\gamma\delta}(\Phi_\ell) + o(1) \approx \Sigma^{\alpha\beta,\gamma\delta}(\Phi_\ell). \tag{6.30}$$

These leading internal conditional moments are summarized by the Keisler–Itô hyperfinite difference form from (5.53):

$$\Phi_{\ell+1} - \Phi_\ell = b(\Phi_\ell)\,\Delta\tau + \Sigma(\Phi_\ell)^{1/2}\,\Delta B_\ell + \tilde{o}_\ell(\Delta\tau). \tag{6.31}$$

Conditional on $\mathcal{F}_\ell$, the hyperfinite Brownian increment $\Delta B_\ell := \sqrt{\Delta\tau}\,\xi_\ell$ is centered with covariance $\Delta\tau\, I_{m(m+1)/2}$ in the upper-triangular coordinates, and the conditional remainder $\tilde{o}_\ell(\Delta\tau)$ satisfies $\|{}^*\mathbb{E}[\tilde{o}_\ell(\Delta\tau) \mid \mathcal{F}_\ell]\|_F = o(\Delta\tau)$ and ${}^*\mathbb{E}[\|\tilde{o}_\ell(\Delta\tau) - {}^*\mathbb{E}[\tilde{o}_\ell(\Delta\tau) \mid \mathcal{F}_\ell]\|_F^2 \mid \mathcal{F}_\ell] = o(\Delta\tau)$ as in (5.52) (together with $N\,{}^*\mathbb{E}[\|\Phi_{\ell+1} - \Phi_\ell\|_F^3 \mid \mathcal{F}_\ell] \approx 0$ uniformly on ${}^*K$). The preceding one-step infinitesimal estimates verify the hypotheses of the hyperfinite Markov-chain diffusion theorem in Theorem A.4, which completes the proof. □

Remark 6.5 (Residual Shortcuts). The theorem isolates the shaped-attention mechanism through an attention-only recursion. Noci et al. also allow a weighted identity shortcut around the shaped-attention branch, with complementary branch strengths chosen to preserve variance; their theorem shows how these strengths rescale the limiting drift and covariance [52, Theorem 4.2]. We suppress that extension here.

Remark 6.6 (Full Transformer Blocks). A full Transformer block composes shaped attention with a feedforward sublayer. In the same no-shortcut specialization as Theorem 6.4, its internal column-token recursion at width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ can be written as

$$\begin{aligned} \widetilde{h}_\ell &= \frac{1}{\sqrt{N}} W_\ell^v h_\ell A_\ell^s, & h_{\ell+1} &= \sqrt{\frac{c}{N}}\, W_\ell^2\, \phi_s\!\left(\frac{1}{\sqrt{N}} W_\ell^1 \widetilde{h}_\ell\right), \\ \phi_s(x) &:= s_+ \max\{x, 0\} + s_- \min\{x, 0\}, & s_\pm &= 1 + \frac{c_\pm}{\sqrt{N}}. \end{aligned} \tag{6.32}$$

Here $\widetilde{h}_\ell$ is the intermediate state after shaped attention, while h_ℓ and $h_{\ell+1}$ are the states at consecutive full-block boundaries. Thus $\Phi_\ell = N^{-1} h_\ell^\top h_\ell$ records the covariance at those boundaries. The activation is applied entrywise, and the width-dependent normalization $c = c(N) \approx 1$ is chosen so that $c^{-1} = {}^*\mathbb{E}[\phi_s(g)^2]$ for $g \sim {}^*\mathcal{N}(0,1)$. Thus ϕ_s is precisely the critically shaped ReLU from (5.50). Here the subscript s refers to the slopes $s_\pm$, not to the attention temperature used in A_ℓ^s. The matrices $W_\ell^1, W_\ell^2 \in {}^*\mathbb{R}^{N\times N}$ have internally i.i.d. ${}^*\mathcal{N}(0,1)$ entries; across the two weight types and all layers, they are mutually independent and independent of all query, key, and value weights.

Let b_{ReLU} denote the shaped-ReLU drift in (5.55). Then [52, Theorem 3.2 and Corollary 4.3] gives the corresponding combined shadow SDE

$$d\Phi_\tau = \left[b(\Phi_\tau) + b_{\mathrm{ReLU}}(\Phi_\tau)\right] d\tau + \left[\Sigma(\Phi_\tau) + 2\Sigma_{\mathrm{lin}}(\Phi_\tau)\right]^{1/2} dB_\tau. \tag{6.33}$$

Here b and Σ are the shaped-attention coefficients in Theorem 6.4, while the feedforward sublayer contributes b_{ReLU} and $2\Sigma_{\mathrm{lin}}$ because it contains two fresh internal Gaussian matrices.

Conditional on the completed attention sublayer, and hence on $\widetilde{h}_\ell$, the fluctuation generated by the fresh feedforward weights is centered. The internal tower property therefore eliminates its cross-covariance with the preceding attention fluctuation at scale $\Delta\tau$. The instantaneous covariance matrices add inside the square root; the two square-root diffusion coefficients are not added directly.

6.2 Spectrum Results for the Covariance SDE

We now return to the deep linear network studied in Chapter 5 and ask what its covariance SDE implies for the spectrum. The affine invariance of the linear model leads to explicit eigenvalue dynamics via infinitesimal Hadamard perturbation and a tractable mean-field spectral limit. The results are adapted from [40].

6.2.1 Recall: The Linear Covariance SDE

Stack the standard inputs $x^1, \dots, x^m \in \mathbb{R}^{n_0}$ as the columns of h_0. At unlimited width $N \in {}^*\mathbb{N}\backslash\mathbb{N}$, let $W_0 \in {}^*\mathbb{R}^{N\times n_0}$ and $W_\ell \in {}^*\mathbb{R}^{N\times N}$ for $\ell \geq 1$ have internally i.i.d. ${}^*\mathcal{N}(0,1)$ entries. As in Chapter 5, set

$$h_1 = \frac{1}{\sqrt{n_0}} W_0 h_0, \quad h_{\ell+1} = \frac{1}{\sqrt{N}} W_\ell h_\ell, \ \ \ell \geq 1, \quad \Phi_0 = \frac{1}{n_0} h_0^\top h_0, \quad \Phi_\ell = \frac{1}{N} h_\ell^\top h_\ell, \ \ \ell \geq 1. \tag{6.34}$$

Fix standard $m \in \mathbb{N}$, let $\bar{m} := m(m+1)/2$, and flatten Φ into its distinct upper-triangular entries, indexed by $\alpha \leq \beta$. On every bounded time interval, the hyperfinite covariance trajectory $\Phi_\tau^{(N)} := \Phi_{\lfloor \tau N \rfloor}$ is S-continuous $L({}^*\mathbb{P})$-a.s., and its standard shadow $\Phi_\tau := \mathrm{st}(\Phi_{\lfloor \tau N \rfloor})$ solves the linear covariance SDE

$$d\Phi_\tau = \Sigma(\Phi_\tau)^{1/2}\, dB_\tau, \qquad \Sigma_{\alpha\beta,\gamma\delta}(\Phi) = \Phi_{\alpha\gamma}\Phi_{\beta\delta} + \Phi_{\alpha\delta}\Phi_{\beta\gamma}, \quad \alpha \leq \beta, \ \gamma \leq \delta. \tag{6.35}$$

Here B is standard Brownian motion in $\mathbb{R}^{\bar{m}}$; at terminal depths $D \in {}^*\mathbb{N} \setminus \mathbb{N}$ with $\mathrm{st}(D/N) = \bar{\tau}$, the terminal covariance shadow therefore satisfies $\mathrm{st}(\Phi_D) \overset{d}{=} \Phi_{\bar{\tau}}$. This is precisely the linear covariance shadow in Theorem 5.6. In particular, Proposition 5.5 and (5.35) identify the diffusion tensor Σ, while (5.40) gives the SDE in the notation of Chapter 5.

6.2.2 Affine Invariance

A key structural feature of (6.35) is its invariance under congruence transformations $\Phi \mapsto A\Phi A^\top$.

Lemma 6.7 (Affine Invariance of the Covariance Markov Chain)**.** *Let P denote the internal one-step covariance update $\Phi_\ell \mapsto \Phi_{\ell+1}$ at width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ induced by the linear recursion (6.34). Then for every $A \in {}^*\mathrm{GL}(m,\mathbb{R})$ (and in particular every standard $A \in \mathrm{GL}(m,\mathbb{R})$),*

$$AP(\Phi)A^\top \overset{d}{=} P(A\Phi A^\top). \tag{6.36}$$

Consequently, the stochastic flow of the shadow SDE (6.35) is affine-invariant: if Φ_τ solves (6.35) with initial condition Φ_0, then $A\Phi_\tau A^\top$ has the same law as the solution started from $A\Phi_0 A^\top$.

Proof. The argument is immediate at the internal Markov-chain level by the Transfer Principle. Condition on Φ and sample N internally i.i.d. Gaussian vectors $(g_i)_{i=1}^N \subset {}^*\mathbb{R}^m$ with $g_i \sim {}^*\mathcal{N}(0,\Phi)$. Then the internal update can be written in distribution as

$$P(\Phi) = \frac{1}{N}\sum_{i=1}^{N} g_i g_i^\top. \tag{6.37}$$

Equivalently, stacking g_i as rows gives $G \in {}^*\mathbb{R}^{N\times m}$ with $\mathrm{vec}(G) \sim {}^*\mathcal{N}(0, \Phi \otimes I_N)$ and $P(\Phi) = \frac{1}{N}G^\top G$. Now replace Φ by $A\Phi A^\top$. By Transfer of the Kronecker identity $(A\Phi A^\top)\otimes I_N = (A\otimes I_N)(\Phi\otimes I_N)(A^\top \otimes I_N)$, we can couple the internal Gaussian matrices so that

$$G' = GA^\top \implies \mathrm{vec}(G') \sim {}^*\mathcal{N}\big(0, (A\Phi A^\top)\otimes I_N\big). \tag{6.38}$$

Then in internal distribution,

$$P(A\Phi A^\top) \overset{d}{=} \frac{1}{N}(G')^\top G' = \frac{1}{N}AG^\top GA^\top = AP(\Phi)A^\top, \tag{6.39}$$

which is the desired internal invariance. For standard $A \in \mathrm{GL}(m,\mathbb{R})$, continuity of $\Phi \mapsto A\Phi A^\top$ commutes with $\mathrm{st}(\cdot)$ along the hyperfinite trajectory $\ell = \lfloor \tau N \rfloor$, transferring congruence invariance directly to the standard shadow SDE (6.35). □

The set $\mathrm{SPD}(m)$ of $m\times m$ symmetric positive-definite matrices is a smooth manifold with tangent space $T_\Phi \mathrm{SPD}(m) \cong \mathrm{Sym}(m)$. This geometry is not needed to use the covariance SDE, but it explains the special structure of its noise.

Proposition 6.8 (Σ^{-1}: Affine-Invariant Metric)**.** *For $\Phi \in \mathrm{SPD}(m)$ and $A, B \in T_\Phi \mathrm{SPD}(m) \cong \mathrm{Sym}(m)$, the affine-invariant Riemannian metric is*

$$g_\Phi(A,B) := \frac{1}{2}\mathrm{Tr}(\Phi^{-1}A\Phi^{-1}B). \tag{6.40}$$

It is invariant under congruences: for every $P \in \mathrm{GL}(m,\mathbb{R})$,

$$g_{P\Phi P^\top}(PAP^\top, PBP^\top) = g_\Phi(A,B). \tag{6.41}$$

In the distinct upper-triangular coordinates on $\mathrm{Sym}(m)$ used in Chapter 5, the matrix representing this metric is $\Sigma(\Phi)^{-1}$. A proof is given in Appendix A.4; see also [40, Proposition 2.5].

Thus Σ is the cometric associated with (6.40); its multiplicative structure leads to the Dyson-type eigenvalue dynamics studied next.

6.2.3 Geometric Dyson Brownian Motion

For the spectral calculation, it is useful to realize the same diffusion in matrix form. Let S_τ be symmetric Brownian motion on $\mathrm{Sym}(m)$, normalized so that

$$d\langle S_{\alpha\beta}, S_{\gamma\delta}\rangle_\tau = (\delta_{\alpha\gamma}\delta_{\beta\delta} + \delta_{\alpha\delta}\delta_{\beta\gamma})\,d\tau, \qquad d\Phi_\tau = \Phi_\tau^{1/2}\,dS_\tau\,\Phi_\tau^{1/2}. \tag{6.42}$$

On the hyperfinite grid $\tau_\ell = \ell\,\Delta\tau$ ($\Delta\tau = 1/N \approx 0$), Anderson's hyperfinite symmetric Brownian increment $\Delta S_\ell \in {}^*\mathrm{Sym}(m)$ satisfies ${}^*\mathrm{Cov}(\Delta S_{\ell,\alpha\beta}, \Delta S_{\ell,\gamma\delta} \mid \mathcal{F}_\ell) = (\delta_{\alpha\gamma}\delta_{\beta\delta} + \delta_{\alpha\delta}\delta_{\beta\gamma})\,\Delta\tau$, and the multiplicative Keisler step $\Delta\Phi_\ell = \Phi_\ell^{1/2}\,\Delta S_\ell\,\Phi_\ell^{1/2}$ has exact conditional covariance $\Sigma(\Phi_\ell)\,\Delta\tau$ matching (6.35), so the two SDEs have the same shadow law.

Let $\lambda_1(\tau) \le \cdots \le \lambda_m(\tau)$ denote the eigenvalues of Φ_τ.

Theorem 6.9 (Geometric Dyson Brownian Motion)**.** *Let Φ_τ solve (6.35) with $\Phi_0 \in \mathrm{SPD}(m)$. Then the spectrum is almost surely simple for every $\tau > 0$, and the ordered eigenvalues satisfy, locally on $(0,\infty)$,*

$$d\lambda_i = \sqrt{2}\,\lambda_i\, dB_\tau^{(i)} + \sum_{j\neq i} \frac{\lambda_i\lambda_j}{\lambda_i - \lambda_j}\, d\tau, \qquad i = 1,\dots,m, \tag{6.43}$$

where $(B_\tau^{(i)})_{i=1}^m$ are independent standard Brownian motions. If Φ_0 has simple spectrum, the system holds from $\tau = 0$ and the eigenvalues never collide.

Proof. We derive the eigenvalue evolution via infinitesimal Hadamard perturbation on the hyperfinite grid (cf. [37] and eigenvalue computations for Wishart-type processes). The calculation is local on ${}^*U$, where $U \subset \mathrm{SPD}(m)$ is the standard open set of matrices with simple spectrum; on any compact $K \subset U$, every eigenvalue gap $\lambda_i - \lambda_j$ $(i \neq j)$ of $\Phi_\ell \in {}^*K$ is appreciable ($\in \mathbb{A}$). At step ℓ, choose an internal orthonormal eigenbasis of Φ_ℓ; by the transferred orthogonal invariance in Lemma 6.7, we may compute in coordinates where $\Phi_\ell = \mathrm{diag}(\lambda_1,\dots,\lambda_m)$. For an internal symmetric perturbation $\Psi \in {}^*\mathrm{Sym}(m)$, transferring the first and second Hadamard variation formulae gives

$$\langle \nabla\lambda_i(\Phi_\ell), \Psi\rangle = \Psi_{ii}, \qquad \nabla^2\lambda_i(\Phi_\ell)[\Psi,\Psi] = 2\sum_{j\neq i} \frac{\Psi_{ij}^2}{\lambda_i - \lambda_j}. \tag{6.44}$$

In this eigenbasis, the hyperfinite realization $\Delta\Phi_\ell = \Phi_\ell^{1/2}\,\Delta S_\ell\,\Phi_\ell^{1/2}$ of (6.42) gives $\Delta\Phi_{ij} = \sqrt{\lambda_i\lambda_j}\,\Delta S_{\ell,ij}$, and hence the conditional quadratic moments

$${}^*\mathbb{E}[\Delta\Phi_{ii}\,\Delta\Phi_{jj} \mid \mathcal{F}_\ell] = 2\lambda_i^2\delta_{ij}\,\Delta\tau, \qquad {}^*\mathbb{E}[(\Delta\Phi_{ij})^2 \mid \mathcal{F}_\ell] = \lambda_i\lambda_j\,\Delta\tau \quad (i\neq j). \tag{6.45}$$

Since $\lambda_i - \lambda_j \in \mathbb{A}$ for $j \neq i$ on ${}^*K$ and $\|\Delta\Phi_\ell\|_F = \mathcal{O}_{*\mathbb{P}}(\sqrt{\Delta\tau}) \approx 0$, internal second-order Taylor expansion of $\lambda_i(\Phi_\ell + \Delta\Phi_\ell)$ yields the infinitesimal Hadamard expansion

$$\Delta\lambda_i = \Delta\Phi_{ii} + \sum_{j\neq i} \frac{(\Delta\Phi_{ij})^2}{\lambda_i - \lambda_j} + o(\Delta\tau), \tag{6.46}$$

where the cubic Taylor remainder $\mathcal{O}(\|\Delta\Phi_\ell\|_F^3)$ has conditional expectation and conditional L^2 norm of order $\Delta\tau^{3/2} = o(\Delta\tau)$ on ${}^*K$. Moreover, for $j \neq i$, the centered square $(\Delta\Phi_{ij})^2 - \lambda_i\lambda_j\,\Delta\tau$ has conditional mean zero and conditional variance $\mathcal{O}(\Delta\tau^2) = o(\Delta\tau)$, so each off-diagonal square concentrates onto $\lambda_i\lambda_j\,\Delta\tau$ up to $\tilde{o}_\ell(\Delta\tau)$. Meanwhile, the diagonal moments in (6.45) show that $\Delta\Phi_{ii} = \sqrt{2}\,\lambda_i\,\Delta B_\ell^{(i)}$ for uncorrelated hyperfinite Brownian increments $(\Delta B_\ell^{(i)})_{i=1}^m$. Taking standard parts via Theorem A.4 gives (6.43) up to the first collision. (Geometrically in $\mathcal{C}\ell_m$, transporting the orthonormal eigenframe via a spinor rotor $u_k(\tau) = R_\tau e_k R_\tau^\sim$ with $R_\tau \in \mathrm{Spin}(m) \subset \mathcal{C}\ell_m^+$ and bivector angular-velocity generator $d\Omega_\tau := (dR_\tau)R_\tau^\sim = \frac{1}{2}\sum_{i<j} d\omega_{ij}\,(e_i \wedge e_j) \in \bigwedge^2(\mathbb{R}^m)$, the unit bivector identity $(e_i \wedge e_j)^2 = -1$ expands the two-sided sandwich $\exp(\frac{1}{2}d\omega_{ij}e_i\wedge e_j)\,e_i\,\exp(-\frac{1}{2}d\omega_{ij}e_i\wedge e_j) = (1 - \frac{1}{2}(d\omega_{ij})^2)e_i - d\omega_{ij}e_j + o(d\tau)$. Substituting into $\Phi_\tau = \sum_k \lambda_k u_k u_k^\top$ gives $d\Phi_{ij} = (\lambda_j - \lambda_i)\,d\omega_{ij}$ and $d\Phi_{ii} = d\lambda_i - (\lambda_i - \lambda_j)(d\omega_{ij})^2 + o(d\tau)$; matching ${}^*\mathbb{E}[(d\Phi_{ij})^2] = \lambda_i\lambda_j\,d\tau$ and ${}^*\mathbb{E}[d\Phi_{ii}] = 0$ immediately forces the repulsive drift $+(\lambda_i - \lambda_j)(d\omega_{ij})^2 = \frac{\lambda_i\lambda_j}{\lambda_i - \lambda_j}\,d\tau$ as a direct consequence of $(e_i\wedge e_j)^2 = -1$ in $\mathrm{Spin}(m)$ [29, 18].) The absence of collisions from a simple initial spectrum, and instantaneous simplicity from an arbitrary positive-definite initial spectrum, are proved in [40, Theorem 1.1 and Appendix B]. □

6.2.4 Mean Field Limit, a Burgers PDE, and a Fixed Point Equation

Having obtained the standard eigenvalue diffusion (6.43) for every standard token dimension $m \in \mathbb{N}$, we now transfer that diffusion family to an unlimited spectral dimension $M \in {}^*\mathbb{N} \setminus \mathbb{N}$ on the accelerated spectral clock τ/M. Write $\mu_i(\tau) := {}^*\lambda_i(\tau/M)$ for $i = 1, \dots, M$. Then the hyperfinite empirical spectral measure of the time-changed covariance is

$$\rho_\tau^{(M)} := \frac{1}{M} \sum_{i=1}^{M} \delta_{\mu_i(\tau)}. \tag{6.47}$$

The time change $\tau \mapsto \tau/M$ normalizes the pairwise interaction drift in (6.43) to $\mathbb{L}$ (order one) at unlimited $M \in {}^*\mathbb{N}\setminus\mathbb{N}$, and simultaneously shrinks each eigenvalue's diffusion scale to $M^{-1/2} \approx 0$. The scales here are sequential (corresponding to $1 \ll M \ll N$): first the proportional depth–width shadow at fixed dimension gives the covariance SDE and its eigenvalue diffusion (6.43), and only then do we apply the time change $\tau \mapsto \tau/M$ at an unlimited spectral dimension $M \in {}^*\mathbb{N} \setminus \mathbb{N}$. No simultaneous three-parameter limit without scale separation is asserted.

The T-transform is useful here because the multiplicative eigenvalue interaction becomes a closed equation for a single analytic function. For a standard probability measure ρ_τ on $\mathbb{R}_{\geq 0}$, define the T-transform

$$G_\tau(z) := \int_{\mathbb{R}} \frac{x}{z - x}\, \rho_\tau(dx), \qquad z \in \mathbb{C} \setminus \mathbb{R}_{\geq 0}. \tag{6.48}$$

(Equivalently, $G_\tau(z) = z\, g_\tau(z) - 1$ where g_τ is the Stieltjes transform.)

Remark 6.10 (T-Transform Inversion and $\mathcal{C}\ell_2^+$). Consider the Stieltjes transform $g_\tau(z) := \int_{\mathbb{R}} (z-x)^{-1}\, \rho_\tau(dx)$. Identifying the spectral plane $\mathbb{C} \cong \mathcal{C}\ell_2^+$ via the unit bivector $\boldsymbol{i} := e_1 e_2 \in \mathcal{C}\ell_2^+$ ($\boldsymbol{i}^2 = -1$, $\boldsymbol{i}^\sim = -\boldsymbol{i}$) realizes $(z-x)^{-1} = (z-x)^\sim / |z-x|^2$ ($z = x_0 + \varepsilon \boldsymbol{i}$) as the two-dimensional monogenic Cauchy–Dirac kernel [9]. Whenever ρ_τ admits a continuous density (which we denote by the same symbol) at a point $x \in \mathbb{R}$, the standard Stieltjes inversion formula extracts the bivector grade at an infinitesimal vertical offset $\varepsilon \in {}^*\mathbb{R}_{>0} \cap \mathbb{I}$:

$$\rho_\tau(x) = -\frac{1}{\pi} \lim_{\varepsilon \downarrow 0} \Im g_\tau(x + i\varepsilon) = -\frac{1}{\pi}\, \mathrm{st}\big(\langle (-\boldsymbol{i})\, {}^*g_\tau(x + \varepsilon \boldsymbol{i}) \rangle_0\big) \quad (\varepsilon \in {}^*\mathbb{R}_{>0} \cap \mathbb{I}). \tag{6.49}$$

Since $g_\tau(z) = \frac{G_\tau(z)+1}{z}$, we can equivalently write this in terms of the T-transform:

$$\rho_\tau(x) = -\frac{1}{\pi} \lim_{\varepsilon \downarrow 0} \Im\left(\frac{G_\tau(x + i\varepsilon) + 1}{x + i\varepsilon} \right). \tag{6.50}$$

The path-level limit and Burgers equation below are proved in [40, Theorem 1.2]; the identity-start fixed point is [40, Corollary 1.3].

Theorem 6.11 (Burgers PDE and Fixed Point)**.** *Assume that at unlimited $M \in {}^*\mathbb{N} \setminus \mathbb{N}$ the initial hyperfinite empirical measure $\rho_0^{(M)}$ has limited first moment $\int_{{}^*\mathbb{R}_{\geq 0}} x\, \rho_0^{(M)}(dx) \in \mathbb{L}$ and deterministic Loeb weak shadow $\rho_0 = L(\rho_0^{(M)}) \circ \mathrm{st}^{-1}$ (equivalently, $\rho_0^{(m)} \xrightarrow{w} \rho_0$ with uniformly bounded first moments). Then $\rho_\tau^{(M)}$ concentrates $L({}^*\mathbb{P})$-infinitesimally, uniformly on compact standard time intervals in the weak topology, around a deterministic continuous measure path ρ_τ whose T-transform satisfies the inviscid Burgers-type PDE*

$$\partial_\tau G_\tau(z) = -z\, G_\tau(z)\, \partial_z G_\tau(z). \tag{6.51}$$

If moreover $\rho_0 = \delta_1$, then $G_\tau(z)$ is the physical branch of the fixed point equation

$$G_\tau(z) = \frac{1}{z\, e^{-\tau G_\tau(z)} - 1}. \tag{6.52}$$

Remark 6.12 (Free Log-Normal Fixed Point). The identity-start limit ρ_τ turns out to be the mean-one free log-normal law, the multiplicative free-convolution analogue of the classical log-normal law. This identification is most transparent through the S-transform, which turns multiplicative free convolution into multiplication [6]. For the compactly supported law ρ_τ [40, Corollary 4.1], define $\psi_\tau(u) := G_\tau(u^{-1})$, and let χ_τ be its local compositional inverse near the origin. The S-transform of ρ_τ is then $S_{\rho_\tau}(w) := (1+w)\chi_\tau(w)/w$. For z near infinity, taking $w = G_\tau(z)$ gives $\chi_\tau(w) = z^{-1}$, while (6.52) rearranges to $z^{-1} = e^{-\tau w} w/(1+w)$. Consequently, $S_{\rho_\tau}(w) = e^{-\tau w}$, the S-transform of the mean-one free log-normal law [40, Corollary 1.3 and Section 4]. See also [54, Section 16.3] for a physics-oriented treatment.

Remark 6.13 (Physical Branch). In the identity-start case, the fixed-point equation may admit several analytic branches. The T-transform in Theorem 6.11 is selected by $G_\tau(z) \sim z^{-1}$ as $z \to \infty$; this choice is called the physical branch.

Proof Sketch for Theorem 6.11. We first derive the Burgers equation via hyperfinite empirical concentration. Fix standard $z \in \mathbb{C} \setminus \mathbb{R}_{\geq 0}$ and let $f_z(x) := \frac{x}{z-x}$. Note that $x f_z'(x) = \frac{zx}{(z-x)^2}$ and $x^2 f_z''(x) = \frac{2zx^2}{(z-x)^3}$ are uniformly bounded on $\mathbb{R}_{\geq 0}$. On each positive-time interval, after relabeling the rescaled transferred Brownian motions $\sqrt{M}\, {}^*B^{(i)}_{\tau/M}$ as $B^{(i)}_\tau$, the transferred time-changed form of (6.43) is

$$d\mu_i(\tau) = \sqrt{\frac{2}{M}}\, \mu_i(\tau)\, dB^{(i)}_\tau + \frac{1}{M} \sum_{j \neq i} \frac{\mu_i(\tau)\mu_j(\tau)}{\mu_i(\tau) - \mu_j(\tau)}\, d\tau. \tag{6.53}$$

Applying transferred Itô's formula to $\left\langle {}^*f_z, \rho^{(M)}_\tau \right\rangle = \frac{1}{M} \sum_{i=1}^M {}^*f_z(\mu_i(\tau))$, the martingale part has quadratic variation $\frac{2}{M^3} \sum_{i=1}^M \int_0^\tau \left| \mu_i(s)\, {}^*f_z'(\mu_i(s)) \right|^2 ds = \mathcal{O}(M^{-2}) \approx 0$ (hence vanishes uniformly in $L({}^*\mathbb{P})$-probability by Doob's inequality), while the second-order Itô correction $\frac{1}{M^2} \sum_{i=1}^M \mu_i^2\, {}^*f_z''(\mu_i)$ is bounded by $\mathcal{O}(M^{-1}) \approx 0$. Suppressing the common time argument, the S-derivative of the empirical T-transform therefore satisfies the infinitesimal relation

$$\frac{d}{d\tau} \left\langle {}^*f_z, \rho^{(M)}_\tau \right\rangle \approx \frac{1}{M^2} \sum_{i \neq j} {}^*f_z'(\mu_i)\, \frac{\mu_i \mu_j}{\mu_i - \mu_j}. \tag{6.54}$$

Now symmetrize in (i, j):

$$\sum_{i \neq j} {}^*f_z'(\mu_i)\, \frac{\mu_i \mu_j}{\mu_i - \mu_j} = \frac{1}{2} \sum_{i \neq j} \mu_i \mu_j\, \frac{{}^*f_z'(\mu_i) - {}^*f_z'(\mu_j)}{\mu_i - \mu_j}. \tag{6.55}$$

Because the diagonal limit $x^2 f_z''(x)$ of the bounded continuous kernel $(x, y) \mapsto xy \frac{f_z'(x) - f_z'(y)}{x - y}$ contributes at most $\mathcal{O}(M^{-1}) \approx 0$ to $\frac{1}{2M^2} \sum_{i=1}^M$, taking the standard part replaces $\frac{1}{M^2} \sum_{i \neq j}$ by Loeb product integration against the shadow $\rho_\tau \otimes \rho_\tau$, giving

$$\partial_\tau G_\tau(z) = \frac{1}{2} \iint xy\, \frac{f_z'(x) - f_z'(y)}{x - y}\, \rho_\tau(dx)\, \rho_\tau(dy). \tag{6.56}$$

A direct algebraic calculation gives the symmetric identity

$$xy\,\frac{f_z'(x)-f_z'(y)}{x-y} = -z\big(f_z(x)\,\partial_z f_z(y) + f_z(y)\,\partial_z f_z(x)\big). \tag{6.57}$$

The two terms agree after integration against $\rho_\tau \otimes \rho_\tau$, so the factor $1/2$ gives precisely $\partial_\tau G_\tau(z) = -z\,G_\tau(z)\,\partial_z G_\tau(z)$.

We next derive the fixed-point equation by characteristics. The idea is to follow a curve along which the two terms in the PDE cancel. Starting from z_0, let $z(\tau)$ solve $\partial_\tau z(\tau) = z(\tau)\,G(\tau, z(\tau))$ with $z(0) = z_0$. Then

$$\frac{d}{d\tau} G(\tau, z(\tau)) = \partial_\tau G + \partial_\tau z\,\partial_z G = \partial_\tau G + zG\,\partial_z G = 0, \tag{6.58}$$

so the transform is constant along the curve: $G(\tau, z(\tau)) = G_0(z_0)$. The characteristic equation therefore reduces to the linear ODE $\partial_\tau z(\tau) = G_0(z_0)\,z(\tau)$, whose solution is $z(\tau) = z_0 \exp(\tau G_0(z_0))$. Now fix the terminal time τ and write $z = z(\tau)$. Since $G_\tau(z) = G_0(z_0)$, the last relation becomes $z = z_0 \exp(\tau G_\tau(z))$, or $z_0 = z\,e^{-\tau G_\tau(z)}$. Substituting this starting point into $G_\tau(z) = G_0(z_0)$ gives the general characteristic identity

$$G_\tau(z) = G_0\big(z\,e^{-\tau G_\tau(z)}\big). \tag{6.59}$$

For $\rho_0 = \delta_1$, we have $G_0(z) = (z-1)^{-1}$, and (6.59) becomes (6.52). This complex-characteristic calculation is formal. □

6.2.5 A Small-τ Approximation in Closed Form

Making the first-order replacement $e^{-\tau G_\tau(z)} \approx 1 - \tau G_\tau(z)$ for small (or positive infinitesimal) $\tau > 0$ in (6.52) leads to the quadratic approximation [40, Section 4.2]

$$z\tau\,G_\tau(z)^2 - (z-1)G_\tau(z) + 1 = 0. \tag{6.60}$$

For $\tau > 0$, the physical root is selected by $G_\tau(z) \sim z^{-1}$ as $z \to \infty$; equivalently, the square root is chosen to behave like z at infinity. This gives

$$G_\tau(z) \approx \frac{(z-1) - \sqrt{(z-1)^2 - 4z\tau}}{2z\tau}. \tag{6.61}$$

Let $\lambda_\pm := (\sqrt{1+\tau} \pm \sqrt{\tau})^2$. Since $(z-1)^2 - 4z\tau = (z-\lambda_-)(z-\lambda_+)$, substituting the selected root into (6.50) gives, for $x > 0$,

$$\begin{aligned}
\rho_\tau(x) &\approx -\frac{1}{\pi x} \lim_{\varepsilon \downarrow 0} \Im G_\tau(x + i\varepsilon) \\
&\approx -\frac{1}{\pi x}\left(-\frac{\sqrt{(x-\lambda_-)(\lambda_+ - x)}}{2x\tau}\right) \mathbb{1}_{\{x \in [\lambda_-, \lambda_+]\}} \\
&= \frac{\sqrt{(x-\lambda_-)(\lambda_+ - x)}}{2\pi x^2 \tau}\, \mathbb{1}_{\{x \in [\lambda_-, \lambda_+]\}}.
\end{aligned} \tag{6.62}$$

Within this approximation, the displayed density has total mass one. More precisely, it is the x^{-1}-reweighting of the Marchenko–Pastur law with variance parameter $1+\tau$ and aspect ratio $\tau/(1+\tau)$.

6.3 One Step of Feature Learning in the Proportional Limit

The preceding two sections study covariance dynamics at initialization. We now turn to training and ask how a single gradient step changes features in the proportional galaxy $D, N \in {}^*\mathbb{N} \setminus \mathbb{N}$ with $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$ and $\Delta\tau = 1/N \approx 0$. A one-sample deep linear network is the simplest setting in which simultaneous updates across proportionally many hyperfinite layers interact nontrivially. The broader multi-sample nonlinear theory is the subject of ongoing work [41]. Since that manuscript is still in preparation, the internal hyperfinite identities below are derived directly by Transfer, while the continuum shadow equations serve as a formal guide. Even in this scalar setting, the proportional regime produces nontrivial stochastic dynamics in depth.

We begin with a fixed standard input–output pair (x, y), with nonzero $x \in \mathbb{R}^{n_0}$ and $y \in \mathbb{R}$. We write $h_0 = x$ for the input representation and use the same input boundary as in (5.14):

$$h_0 := x, \qquad h_1 := \frac{1}{\sqrt{n_0}} W_0 x, \qquad h_{\ell+1} := \frac{1}{\sqrt{N}} W_\ell h_\ell, \quad 1 \le \ell \le D-1. \tag{6.63}$$

The scalar readout is

$$f := \frac{1}{\sqrt{N}} W_D h_D. \tag{6.64}$$

Here $W_0 \in {}^*\mathbb{R}^{N \times n_0}$, $W_\ell \in {}^*\mathbb{R}^{N \times N}$ for $1 \le \ell \le D-1$, and $W_D \in {}^*\mathbb{R}^{1 \times N}$ have mutually independent internal ${}^*\mathcal{N}(0,1)$ entries at initialization. To focus on the bulk depth mechanism, we hold W_0 and W_D fixed and train only $W_1, \dots, W_{D-1}$.

For the backward pass, define the scaled internal output sensitivity

$$g_\ell := \sqrt{N} \frac{\partial f}{\partial h_\ell} \in {}^*\mathbb{R}^N, \qquad 1 \le \ell \le D. \tag{6.65}$$

Then $g_D = W_D^\top$, and the transferred chain rule gives

$$g_\ell = \frac{1}{\sqrt{N}} W_\ell^\top g_{\ell+1}, \qquad \ell = D-1, \dots, 1. \tag{6.66}$$

6.3.1 Scalar Kernels and Depth-Time SDEs at Initialization

All Gram objects reduce to internal scalars in ${}^*\mathbb{R}$. Define

$$\Phi_0 := \frac{1}{n_0} \|x\|^2, \qquad \Phi_\ell := \frac{1}{N} \|h_\ell\|^2, \qquad G_\ell := \frac{1}{N} \|g_\ell\|^2, \qquad Q_\ell := \frac{1}{\sqrt{N}} \langle h_\ell, g_\ell \rangle. \tag{6.67}$$

The last three quantities are defined for $1 \le \ell \le D$.

Set the hyperfinite depth-time grid $\tau_\ell = \ell/N = \ell\,\Delta\tau$ with $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$. We use $\mathcal{F}^h$ for the internal forward filtration, $\mathcal{F}^g$ for the reverse filtration, and $\mathcal{F}^{\Delta h}$ below for the feature change. In the one-sample linear setting, the forward kernel has a geometric Brownian motion shadow.

Remark 6.14 (Recall: Geometric Brownian Motion for the Forward Kernel Φ). In the deep linear case, let $\mathcal{F}_\ell^h := \sigma(h_0, \dots, h_\ell)$ be the internal forward filtration. For $0 \le \ell \le D-1$, conditioning on $\mathcal{F}_\ell^h$ gives

$$\Phi_{\ell+1} \mid \mathcal{F}_\ell^h \overset{d}{=} \Phi_\ell \frac{\chi_N^2}{N}, \tag{6.68}$$

where $\chi^2_N \sim {}^*\chi^2(N)$ is internally independent of $\mathcal{F}^h_\ell$. As recalled in (5.23), when $D, N \in {}^*\mathbb{N} \setminus \mathbb{N}$ satisfy $\mathrm{st}(D/N) = \bar{\tau}$, the hyperfinite trajectory $\Phi^{(N)}_\tau := \Phi_{\lfloor \tau N \rfloor}$ is S-continuous $L({}^*\mathbb{P})$-a.s., and its standard shadow $\Phi_\tau := \mathrm{st}\big(\Phi_{\lfloor \tau N \rfloor}\big)$ solves the SDE

$$d\Phi_\tau = \sqrt{2}\,\Phi_\tau\, dB^\Phi_\tau, \qquad \Phi_0 = \frac{1}{n_0}\,\|x\|^2 . \tag{6.69}$$

The linear forward and backward recursions also imply that Q_ℓ is exactly invariant across hyperfinite depth. Indeed, using $h_{\ell+1} = \frac{1}{\sqrt{N}} W_\ell h_\ell$ and $g_\ell = \frac{1}{\sqrt{N}} W_\ell^\top g_{\ell+1}$,

$$Q_{\ell+1} = \frac{1}{\sqrt{N}}\,\langle h_{\ell+1}, g_{\ell+1}\rangle = \frac{1}{N}\,\langle W_\ell h_\ell, g_{\ell+1}\rangle = \frac{1}{N}\,\Big\langle h_\ell, W_\ell^\top g_{\ell+1}\Big\rangle = \frac{1}{\sqrt{N}}\,\langle h_\ell, g_\ell\rangle = Q_\ell. \tag{6.70}$$

Thus $Q_\ell \equiv Q$ for all $1 \le \ell \le D$. At the readout boundary $\ell = D$ this common value is the network output itself, while the backward kernel has the terminal behavior

$$Q = Q_D = f = \sqrt{\Phi_D}\, Z_D, \qquad G_D = \frac{1}{N}\,\|W_D\|^2 \overset{L({}^*\mathbb{P})}{\approx} 1, \tag{6.71}$$

where $Z_D := \Big\langle h_D, W_D^\top \Big\rangle / \|h_D\| \sim {}^*\mathcal{N}(0,1)$ is internally independent of the forward trajectory $\mathcal{F}^h_D$, and $G_D \approx 1$ $L({}^*\mathbb{P})$-a.s. by the Loeb strong law of large numbers. Writing $Z := \mathrm{st}(Z_D) \sim \mathcal{N}(0,1)$, the shadow constant $\mathrm{st}(Q) = \sqrt{\Phi_{\bar{\tau}}}\, Z$ remains coupled to the terminal forward kernel $\Phi_{\bar{\tau}}$, even though $\mathrm{st}(G_D) = 1$.

To analyze the backward kernel G, complete the forward pass and then expose the backward variables from the readout toward the input. Define the internal reverse filtration

$$\mathcal{F}^g_\ell := \sigma(\mathcal{F}^h_D, \{g_k\}_{k=\ell}^D), \qquad 1 \le \ell \le D. \tag{6.72}$$

Thus $\mathcal{F}^g_{\ell+1} \subset \mathcal{F}^g_\ell$. The nontrivial transition is therefore $g_\ell \mid \mathcal{F}^g_{\ell+1}$ for $1 \le \ell \le D-1$.

Conditional on $\mathcal{F}^h_D$, the layerwise internal Gaussian residual blocks are mutually independent. The variables $g_{\ell+1}, \dots, g_D$ use only higher-layer residual blocks, so adjoining them to $\mathcal{F}^h_D$ to form $\mathcal{F}^g_{\ell+1}$ reveals W_ℓ only through the forward constraint $W_\ell h_\ell = \sqrt{N}\, h_{\ell+1}$. Writing $P^\perp_{h_\ell} := I_N - h_\ell h_\ell^\top / \|h_\ell\|^2$, the transferred one-sided Gaussian conditioning identity in Lemma 2.26 leaves the $P^\perp_{h_\ell}$ block fresh. Multiplying its transpose by $g_{\ell+1}/\sqrt{N}$ gives the internal conditional law

$$g_\ell = \frac{1}{\sqrt{N}} W_\ell^\top g_{\ell+1} \;\Big|\; \mathcal{F}^g_{\ell+1} \overset{d}{=} \frac{Q_{\ell+1}}{\Phi_\ell}\,\frac{h_\ell}{\sqrt{N}} + \sqrt{G_{\ell+1}}\, P^\perp_{h_\ell}\xi_\ell. \tag{6.73}$$

Here $\xi_\ell \sim {}^*\mathcal{N}(0, I_N)$ is internally independent of $\mathcal{F}^g_{\ell+1}$.

The two terms in this representation are orthogonal in ${}^*\mathbb{R}^N$, while $P^\perp_{h_\ell}$ has internal rank $N-1$. Thus, writing $\chi^2_{N-1} := \Big\|P^\perp_{h_\ell}\xi_\ell\Big\|^2$, we have $\chi^2_{N-1} \sim {}^*\chi^2(N-1)$ with an internal conditional law independent of $\mathcal{F}^g_{\ell+1}$, and we obtain the exact one-step hyperfinite formula

$$G_\ell = \frac{1}{N}\,\|g_\ell\|^2 = G_{\ell+1}\,\frac{\chi^2_{N-1}}{N} + \frac{Q^2_{\ell+1}}{N\Phi_\ell}. \tag{6.74}$$

From ${}^*\mathbb{E}[\chi^2_{N-1}] = N-1$ and ${}^*\mathrm{Var}(\chi^2_{N-1}) = 2(N-1)$ we read off the internal conditional drift and conditional variance with reverse step $\Delta s := \Delta\tau = 1/N \approx 0$:

$${}^*\mathbb{E}[G_\ell \mid \mathcal{F}^g_{\ell+1}] = G_{\ell+1} + \Delta s\left(\frac{Q^2_{\ell+1}}{\Phi_\ell} - G_{\ell+1}\right), \tag{6.75}$$

$${}^*\mathrm{Var}(G_\ell \mid \mathcal{F}^g_{\ell+1}) = 2\,\Delta s\,(1-\Delta s)\,G^2_{\ell+1}. \tag{6.76}$$

For this reverse recursion on the reverse grid $s_\ell := (D-\ell)/N$, let $\tilde{o}^g_{\ell+1}(\Delta s)$ denote the scalar analogue of the conditional remainder in (5.52), with conditioning on $\mathcal{F}^g_{\ell+1}$ and interpreted locally after stopping G on limited sets. Define the standardized internal chi-square innovation $\xi^G_{\ell,N} := (\chi^2_{N-1} - (N-1))/\sqrt{2(N-1)}$ and write the one-step recursion in Keisler–Euler form as

$$\begin{aligned} G_\ell - G_{\ell+1} &= \left(\frac{Q^2}{\Phi_\ell} - G_{\ell+1}\right)\Delta s + \sqrt{2\,\Delta s}\sqrt{1-\Delta s}\,G_{\ell+1}\,\xi^G_{\ell,N} \\ &= \left(\frac{Q^2}{\Phi_\ell} - G_{\ell+1}\right)\Delta s + \sqrt{2}\,G_{\ell+1}\,\Delta B^G_\ell + \tilde{o}^g_{\ell+1}(\Delta s), \end{aligned} \tag{6.77}$$

where $\Delta B^G_\ell := \sqrt{\Delta s}\,\xi^G_{\ell,N}$. Conditional on $\mathcal{F}^g_{\ell+1}$, the innovation $\xi^G_{\ell,N}$ has internal mean zero and variance one, its law does not depend on the filtration, and $\mathrm{st}(\xi^G_{\ell,N}) \sim \mathcal{N}(0,1)$ by the hyperfinite central limit theorem. The remainder in the last line is precisely $\sqrt{2\Delta s}\,(\sqrt{1-\Delta s}-1)G_{\ell+1}\xi^G_{\ell,N}$; on stopped sets where $G_{\ell+1} \in \mathbb{L}$, it has conditional mean zero and conditional second moment $\mathcal{O}(\Delta s^3)$, and hence has conditional order $\tilde{o}^g_{\ell+1}(\Delta s)$.

Finally, (6.77) has the form of a hyperfinite Keisler–Itô step when the layers are traversed along the reverse grid $s_\ell = (D-\ell)/N$ from the readout toward the input. At the standard shadow level, set $s = \bar{\tau} - \tau$ and $\widehat{G}_s := G_{\bar{\tau}-s}$, so that $s = 0$ corresponds to the readout. In this reverse clock, the infinitesimal increment suggests an ordinary forward Itô equation in s without requiring a backward stochastic-integral convention.

Proposition 6.15 (Formal Limit for (Q,G) at Initialization (One Sample, Linear))**.** *Assume $D, N \in {}^*\mathbb{N} \setminus \mathbb{N}$ satisfy $\mathrm{st}(D/N) = \bar{\tau} \in (0,\infty)$. Then $Q_\ell \equiv Q$ is constant across hyperfinite depth, with shadow $\mathrm{st}(Q) = \sqrt{\Phi_{\bar{\tau}}}\,Z$. Conditional on the realized forward shadow path $(\Phi_\tau)_{0\le\tau\le\bar{\tau}}$ and the terminal variable $Z = \mathrm{st}(Z_D)$ from (6.71), the reverse Keisler–Euler expansion (6.77) suggests the formal reverse-depth shadow SDE*

$$d\widehat{G}_s = \left(\frac{\mathrm{st}(Q)^2}{\Phi_{\bar{\tau}-s}} - \widehat{G}_s\right)ds + \sqrt{2}\,\widehat{G}_s\,dB^G_s, \qquad \widehat{G}_0 = 1, \tag{6.78}$$

where we identify Q with its standard part $\mathrm{st}(Q)$ in the continuum system. The Brownian motion B^G is independent of the forward path and Z. The exact hyperfinite content is (6.71), (6.74), and (6.77); the stochastic differential equation is its formal shadow summary.

6.3.2 One Training Step: Overlap Processes R^h, R^g, and $\dot{\Phi}$

We now take one simultaneous gradient-descent step, with all gradients evaluated at the initialized network, while keeping the proportional depth–width galaxy $\mathrm{st}(D/N) = \bar{\tau}$. Let $L(f,y)$ be a standard continuously differentiable loss and write the internal scalar loss derivative

$$r := \partial_f{}^* L(f,y). \tag{6.79}$$

(For the mean-squared error $L(f,y) = \frac{1}{2}(f-y)^2$ this is the usual residual $r = f - y$.) We use the infinitesimal learning-rate scaling $\eta = \eta_0/D \approx (\eta_0/\bar{\tau})\,\Delta\tau$ for fixed standard $\eta_0 > 0$.

For $\ell = 1, \dots, D-1$ the internal gradient of the scalar output f with respect to W_ℓ is

$$\nabla_{W_\ell} f = \frac{1}{N} g_{\ell+1} h_\ell^\top, \tag{6.80}$$

so the simultaneous hidden-layer update is

$$W_\ell^+ := W_\ell - \eta\, r\, \nabla_{W_\ell} f = W_\ell - \frac{\eta\, r}{N} g_{\ell+1} h_\ell^\top \qquad \text{(linear, one sample)}. \tag{6.81}$$

Let h_ℓ^+ denote the resulting internal forward features and let Δh_ℓ denote their change:

$$h_1^+ = h_1, \qquad h_{\ell+1}^+ = \frac{1}{\sqrt{N}} W_\ell^+ h_\ell^+, \qquad \Delta h_\ell := h_\ell^+ - h_\ell. \tag{6.82}$$

To track this feature change across hyperfinite depth, define the internal scalar overlaps

$$R_\ell^h := \frac{1}{N} \langle \Delta h_\ell, h_\ell \rangle, \qquad R_\ell^g := \frac{1}{\sqrt{N}} \langle \Delta h_\ell, g_\ell \rangle, \qquad \dot{\Phi}_\ell := \frac{1}{N} \|\Delta h_\ell\|^2. \tag{6.83}$$

Thus $\dot{\Phi}_\ell$ is not a time derivative of Φ_ℓ; it is the normalized squared norm of the feature change. If $\Phi_\ell^+ := N^{-1} \left\| h_\ell^+ \right\|^2$, then the actual kernel change is exactly

$$\Phi_\ell^+ - \Phi_\ell = 2R_\ell^h + \dot{\Phi}_\ell. \tag{6.84}$$

Expanding the updated forward recursion gives the exact internal difference equation

$$\Delta h_{\ell+1} = \frac{1}{\sqrt{N}} W_\ell \Delta h_\ell - \eta\, r(\Phi_\ell + R_\ell^h) \frac{1}{\sqrt{N}} g_{\ell+1}, \qquad \Delta h_1 = 0. \tag{6.85}$$

The extra R_ℓ^h comes from the nonlinear interaction between the weight change and the accumulated feature change within the same simultaneous step.

The overlap R^g satisfies an exact deterministic-given-kernels recursion. Pair (6.85) with $g_{\ell+1}$ and use $g_\ell = \frac{1}{\sqrt{N}} W_\ell^\top g_{\ell+1}$:

$$R_{\ell+1}^g = \frac{1}{\sqrt{N}} \langle \Delta h_{\ell+1}, g_{\ell+1} \rangle = \frac{1}{\sqrt{N}} \langle \Delta h_\ell, g_\ell \rangle - \eta\, r(\Phi_\ell + R_\ell^h) \frac{1}{N} \|g_{\ell+1}\|^2. \tag{6.86}$$

Since $G_{\ell+1} = \frac{1}{N} \|g_{\ell+1}\|^2$, we obtain the exact layerwise increment

$$R_{\ell+1}^g - R_\ell^g = -\eta\, r(\Phi_\ell + R_\ell^h) G_{\ell+1}. \tag{6.87}$$

Since $\eta = \eta_0/D = (\eta_0/\bar{\tau} + o(1))\Delta\tau$, passing to standard depth-time $\tau = \mathrm{st}(\ell/N)$ formally yields the continuum shadow ODE

$$dR_\tau^g = -\frac{\eta_0}{\bar{\tau}}\, r(\Phi_\tau + R_\tau^h) G_\tau\, d\tau, \qquad R_0^g = 0. \tag{6.88}$$

Equivalently, integrating (6.88) gives the integral form

$$R_\tau^g = -\frac{\eta_0}{\bar{\tau}}\, r \int_0^\tau (\Phi_s + R_s^h) G_s\, ds. \tag{6.89}$$

The remaining overlaps R^h and $\dot{\Phi}$ pick up $\mathcal{O}(\sqrt{\Delta\tau})$ fluctuations from the internal random matrix product. Their one-step conditional laws require conditioning on both the initialized

forward and backward passes. The active matrix $W_\ell \in {}^*\mathbb{R}^{N\times N}$ is then constrained simultaneously by $W_\ell h_\ell = \sqrt{N}\, h_{\ell+1}$ and $W_\ell^\top g_{\ell+1} = \sqrt{N}\, g_\ell$. For this analysis, use the increasing internal filtration

$$\mathcal{F}_0^{\Delta h} := \mathcal{F}_1^g, \qquad \mathcal{F}_\ell^{\Delta h} := \sigma(\mathcal{F}_1^g, \{\Delta h_k\}_{1\le k\le \ell}), \quad 1 \le \ell \le D. \tag{6.90}$$

All initialized gradients and weight changes are measurable with respect to $\mathcal{F}_1^g$. Given this σ-field, layerwise internal conditional independence and the change recursion (6.85) imply that each Δh_k with $k \le \ell$ uses only residual blocks from matrices W_j with $j < \ell$. Thus adjoining the earlier feature changes reveals no additional action of the active matrix W_ℓ.

The transferred two-sided projection decomposition in Lemma A.5 shows that the internal conditional law of $\Delta h_{\ell+1}$ given $\mathcal{F}_\ell^{\Delta h}$ is a (possibly degenerate) internal Gaussian on ${}^*\mathbb{R}^N$ with

$$\begin{aligned} {}^*\mathbb{E}[\Delta h_{\ell+1} \mid \mathcal{F}_\ell^{\Delta h}] &= \frac{R_\ell^h}{\Phi_\ell}\, h_{\ell+1} + \frac{1}{\sqrt{N}\, G_{\ell+1}} \left(R_{\ell+1}^g - \frac{Q}{\Phi_\ell} R_\ell^h \right) g_{\ell+1}, \\ {}^*\mathrm{Cov}(\Delta h_{\ell+1} \mid \mathcal{F}_\ell^{\Delta h}) &= \left(\dot\Phi_\ell - \frac{(R_\ell^h)^2}{\Phi_\ell} \right) P_{g_{\ell+1}}^\perp, \qquad 1 \le \ell \le D-1. \end{aligned} \tag{6.91}$$

Here $P_{g_{\ell+1}}^\perp := I_N - g_{\ell+1} g_{\ell+1}^\top / \left\|g_{\ell+1}\right\|^2$ is the internal orthogonal projector defined analogously to $P_{h_\ell}^\perp$ above. The exact recursion (6.87) makes $R_{\ell+1}^g$ measurable with respect to $\mathcal{F}_\ell^{\Delta h}$, so its appearance in the conditional mean is not circular.

The whole forward trajectory belongs to $\mathcal{F}_\ell^{\Delta h}$, so the forward increment $\Phi_{\ell+1} - \Phi_\ell$ is fixed under this conditioning. Taking internal scalar moments of (6.91) gives the exact conditional mean increments (with $\Delta\tau = 1/N$)

$$\begin{aligned} {}^*\mathbb{E}[R_{\ell+1}^h - R_\ell^h \mid \mathcal{F}_\ell^{\Delta h}] &= \frac{R_\ell^h}{\Phi_\ell}(\Phi_{\ell+1} - \Phi_\ell) + \frac{Q\,\Delta\tau}{G_{\ell+1}} \left(R_{\ell+1}^g - \frac{Q}{\Phi_\ell} R_\ell^h \right), \\ {}^*\mathbb{E}[\dot\Phi_{\ell+1} - \dot\Phi_\ell \mid \mathcal{F}_\ell^{\Delta h}] &= \frac{(R_\ell^h)^2}{\Phi_\ell^2}(\Phi_{\ell+1} - \Phi_\ell) + \Delta\tau \left(\frac{(R_{\ell+1}^g)^2}{G_{\ell+1}} - \frac{Q^2 (R_\ell^h)^2}{G_{\ell+1}\Phi_\ell^2} - \dot\Phi_\ell + \frac{(R_\ell^h)^2}{\Phi_\ell} \right). \end{aligned} \tag{6.92}$$

The corresponding exact internal conditional covariances are

$$\begin{aligned} N\, {}^*\mathrm{Var}(R_{\ell+1}^h \mid \mathcal{F}_\ell^{\Delta h}) &= \left(\dot\Phi_\ell - \frac{(R_\ell^h)^2}{\Phi_\ell} \right) \left(\Phi_{\ell+1} - \frac{Q^2}{N G_{\ell+1}} \right), \\ N\, {}^*\mathrm{Cov}(R_{\ell+1}^h, \dot\Phi_{\ell+1} \mid \mathcal{F}_\ell^{\Delta h}) &= \frac{2R_\ell^h}{\Phi_\ell} \left(\dot\Phi_\ell - \frac{(R_\ell^h)^2}{\Phi_\ell} \right) \left(\Phi_{\ell+1} - \frac{Q^2}{N G_{\ell+1}} \right), \\ N\, {}^*\mathrm{Var}(\dot\Phi_{\ell+1} \mid \mathcal{F}_\ell^{\Delta h}) &= 2(1-\Delta\tau) \left(\dot\Phi_\ell - \frac{(R_\ell^h)^2}{\Phi_\ell} \right)^2 \\ &\quad + \frac{4(R_\ell^h)^2}{\Phi_\ell^2} \left(\dot\Phi_\ell - \frac{(R_\ell^h)^2}{\Phi_\ell} \right) \left(\Phi_{\ell+1} - \frac{Q^2}{N G_{\ell+1}} \right). \end{aligned} \tag{6.93}$$

Here $\Phi_{\ell+1} - Q^2/(NG_{\ell+1}) = N^{-1} \left\| P_{g_{\ell+1}}^\perp h_{\ell+1} \right\|^2 \ge 0$. The part of the conditional mean parallel to $g_{\ell+1}$ contributes to the drifts but not to these covariances, because the fresh internal Gaussian block lies in $g_{\ell+1}^\perp$.

On limited state sets where $\Phi_\ell, G_{\ell+1} \in \mathbb{A}_{>0}$, substituting (6.87) into (6.92) shows that replacing $R_{\ell+1}^g$ by R_ℓ^g changes both conditional mean increments by $\mathcal{O}(\Delta\tau^2)$; in the second

identity, the pure square of the $\mathcal{O}(\Delta\tau)$ overlap difference contributes only $\mathcal{O}(\Delta\tau^3)$. We absorb these terms into $\tilde{o}_\ell^{\Delta h}(\Delta\tau)$, used componentwise in the sense of the conditional remainder in (5.52), with conditioning on $\mathcal{F}_\ell^{\Delta h}$ and locally where Φ_ℓ and $G_{\ell+1}$ remain appreciable. Since S-continuity of the forward path gives $\Phi_{\ell+1} - \Phi_\ell \approx 0$, while the alignment and finite-rank corrections $Q^2/(NG_{\ell+1})$ and $\Delta\tau$ in (6.93) are $\mathcal{O}(\Delta\tau) \approx 0$ locally, the exact identities reduce to the leading internal conditional covariance at Keisler–Euler scale:

$$N\,{}^*\mathrm{Cov}\left(\begin{pmatrix} R_{\ell+1}^h \\ \dot{\Phi}_{\ell+1} \end{pmatrix} \,\middle|\, \mathcal{F}_\ell^{\Delta h}\right) = \begin{pmatrix} \Phi_\ell\dot{\Phi}_\ell - (R_\ell^h)^2 & 2R_\ell^h\left(\dot{\Phi}_\ell - \dfrac{(R_\ell^h)^2}{\Phi_\ell}\right) \\ 2R_\ell^h\left(\dot{\Phi}_\ell - \dfrac{(R_\ell^h)^2}{\Phi_\ell}\right) & 2\left(\dot{\Phi}_\ell^2 - \dfrac{(R_\ell^h)^4}{\Phi_\ell^2}\right) \end{pmatrix} + o(1), \tag{6.94}$$

where $o(1) \approx 0$ on limited appreciable state sets. By transferred Cauchy–Schwarz, $(R_\ell^h)^2 \le \Phi_\ell\dot{\Phi}_\ell$, so the radicands below are nonnegative.

Combining the exact conditional means in (6.92) with a Cholesky factorization of the leading covariance in (6.94) gives the hyperfinite Keisler–Euler expansions

$$R_{\ell+1}^h - R_\ell^h = \frac{R_\ell^h}{\Phi_\ell}(\Phi_{\ell+1} - \Phi_\ell) + \sqrt{\Delta\tau}\,\sqrt{\Phi_\ell\dot{\Phi}_\ell - (R_\ell^h)^2}\,\xi_\ell^h + \frac{Q\,\Delta\tau}{G_{\ell+1}}\left(R_\ell^g - \frac{Q}{\Phi_\ell}R_\ell^h\right) + \tilde{o}_\ell^{\Delta h}(\Delta\tau), \tag{6.95}$$

$$\begin{aligned}\dot{\Phi}_{\ell+1} - \dot{\Phi}_\ell = {} & \frac{(R_\ell^h)^2}{\Phi_\ell^2}(\Phi_{\ell+1} - \Phi_\ell) + \frac{2R_\ell^h}{\Phi_\ell}\sqrt{\Delta\tau}\,\sqrt{\Phi_\ell\dot{\Phi}_\ell - (R_\ell^h)^2}\,\xi_\ell^h + \sqrt{2\,\Delta\tau}\left(\dot{\Phi}_\ell - \frac{(R_\ell^h)^2}{\Phi_\ell}\right)\xi_\ell^{\dot{\Phi}} \\ & + \Delta\tau\left(\frac{(R_\ell^g)^2}{G_{\ell+1}} - \frac{Q^2}{G_{\ell+1}}\frac{(R_\ell^h)^2}{\Phi_\ell^2} - \dot{\Phi}_\ell + \frac{(R_\ell^h)^2}{\Phi_\ell}\right) + \tilde{o}_\ell^{\Delta h}(\Delta\tau).\end{aligned} \tag{6.96}$$

In (6.95) and (6.96), ξ_ℓ^h is the internal Gaussian fluctuation shared by both increments, whereas $\xi_\ell^{\dot{\Phi}}$ records the remaining chi-square fluctuation in the squared feature change and appears only in the second. Once the quantities in $\mathcal{F}_\ell^{\Delta h}$ are held fixed, these two internal random variables are uncorrelated (and independent in the standard shadow): $\xi_\ell^h \sim {}^*\mathcal{N}(0,1)$, while $\mathrm{st}(\xi_\ell^{\dot{\Phi}}) \sim \mathcal{N}(0,1)$. The exact $\dot{\Phi}$ increment also contains a smaller term proportional to $((\xi_\ell^h)^2 - 1)\,\Delta\tau$; its conditional second moment is $\mathcal{O}(\Delta\tau^2)$, so it is absorbed into $\tilde{o}_\ell^{\Delta h}(\Delta\tau)$. Thus (6.92) and (6.93) are exact internal hyperfinite identities, whereas (6.95) and (6.96) are local infinitesimal conditional expansions.

At a formal continuum shadow level (identifying all variables with their standard parts), the Keisler–Euler expansions suggest the following coupled equations:

$$dR_\tau^h = \sqrt{2}\,R_\tau^h\,dB_\tau^\Phi + \sqrt{\Phi_\tau\dot{\Phi}_\tau - (R_\tau^h)^2}\,dB_\tau^h + \left(\frac{Q}{G_\tau}R_\tau^g - \frac{Q^2}{\Phi_\tau G_\tau}R_\tau^h\right)d\tau, \tag{6.97}$$

$$\begin{aligned}d\dot{\Phi}_\tau = {} & \sqrt{2}\,\frac{(R_\tau^h)^2}{\Phi_\tau}\,dB_\tau^\Phi + \frac{2R_\tau^h}{\Phi_\tau}\sqrt{\Phi_\tau\dot{\Phi}_\tau - (R_\tau^h)^2}\,dB_\tau^h + \sqrt{2}\left(\dot{\Phi}_\tau - \frac{(R_\tau^h)^2}{\Phi_\tau}\right)dB_\tau^{\dot{\Phi}} \\ & + \left(\frac{(R_\tau^g)^2}{G_\tau} - \frac{Q^2}{G_\tau}\frac{(R_\tau^h)^2}{\Phi_\tau^2} - \dot{\Phi}_\tau + \frac{(R_\tau^h)^2}{\Phi_\tau}\right)d\tau.\end{aligned} \tag{6.98}$$

In this formal covariance factorization, B^h and $B^{\dot{\Phi}}$ may be taken independent of each other and of the forward driver B^Φ in (6.69). In particular, this factorization gives $d\left\langle R^h, \dot{\Phi}\right\rangle_\tau = 2R_\tau^h\dot{\Phi}_\tau\,d\tau$.

Together with (6.88) and the formal backward equation (6.78), these relations motivate the consolidated shadow summary below.

Theorem 6.16 (Scalar Proportional-Limit Kernel Dynamics (Formal))**.** *In the proportional galaxy $D, N \in {}^*\mathbb{N} \setminus \mathbb{N}$ with $\mathrm{st}(D/N) = \bar{\tau} \in (0, \infty)$, the preceding hyperfinite calculations formally suggest the following six-equation continuum shadow system for the standard parts $(\Phi_\tau, Q, G_\tau, R^g_\tau, R^h_\tau, \dot{\Phi}_\tau)$. For the backward kernel, write $s = \bar{\tau} - \tau$ and $\widehat{G}_s = G_{\bar{\tau}-s}$:*

$$d\Phi_\tau = \sqrt{2}\,\Phi_\tau\, dB^\Phi_\tau, \qquad \Phi_0 = \frac{1}{n_0}\|x\|^2, \qquad \textit{[forward in } \tau\textit{]}, \tag{6.99}$$

$$Q_\tau = Q \quad \textit{(constant in depth)}, \qquad Q = \sqrt{\Phi_{\bar{\tau}}}\, Z, \qquad \textit{[terminal coupling]}, \tag{6.100}$$

$$d\widehat{G}_s = \left(\frac{Q^2}{\Phi_{\bar{\tau}-s}} - \widehat{G}_s\right) ds + \sqrt{2}\,\widehat{G}_s\, dB^G_s, \qquad \widehat{G}_0 = 1, \qquad \textit{[reverse depth } s\textit{]}, \tag{6.101}$$

$$dR^g_\tau = -\frac{\eta_0}{\bar{\tau}}\, r(\Phi_\tau + R^h_\tau) G_\tau\, d\tau, \qquad R^g_0 = 0, \qquad \textit{[forward in } \tau\textit{]}. \tag{6.102}$$

The two feature-change overlaps also use the forward clock τ:

$$dR^h_\tau = \sqrt{2}\, R^h_\tau\, dB^\Phi_\tau + \sqrt{\Phi_\tau \dot{\Phi}_\tau - (R^h_\tau)^2}\, dB^h_\tau + \left(\frac{Q}{G_\tau} R^g_\tau - \frac{Q^2}{\Phi_\tau G_\tau} R^h_\tau\right) d\tau, \qquad R^h_0 = 0, \tag{6.103}$$

$$\begin{aligned} d\dot{\Phi}_\tau = {} & \sqrt{2}\,\frac{(R^h_\tau)^2}{\Phi_\tau}\, dB^\Phi_\tau + \frac{2R^h_\tau}{\Phi_\tau}\sqrt{\Phi_\tau \dot{\Phi}_\tau - (R^h_\tau)^2}\, dB^h_\tau + \sqrt{2}\left(\dot{\Phi}_\tau - \frac{(R^h_\tau)^2}{\Phi_\tau}\right) dB^{\dot{\Phi}}_\tau \\ & + \left(\frac{(R^g_\tau)^2}{G_\tau} - \frac{Q^2 (R^h_\tau)^2}{G_\tau \Phi^2_\tau} - \dot{\Phi}_\tau + \frac{(R^h_\tau)^2}{\Phi_\tau}\right) d\tau, \qquad \dot{\Phi}_0 = 0. \end{aligned} \tag{6.104}$$

Here $Z \sim \mathcal{N}(0,1)$ is independent of the forward path, and $r = \partial_f L(Q, y)$ denotes the shadow loss derivative. The Brownian motions B^h and $B^{\dot{\Phi}}$ may be taken independent of each other and of B^Φ. The backward driver B^G is independent of the forward path and Z; no independence from B^h or $B^{\dot{\Phi}}$ is asserted.

To read off the exact effect of the training step on the prediction, let $f^+ := \frac{1}{\sqrt{N}} W_D h^+_D$. The readout is fixed and $g_D = W_D^\top$, so telescoping (6.87) across hyperfinite layers from $R^g_1 = 0$ gives

$$f^+ - f = \frac{1}{\sqrt{N}} \langle \Delta h_D, g_D \rangle = R^g_D = -\eta\, r \sum_{\ell=1}^{D-1} (\Phi_\ell + R^h_\ell) G_{\ell+1}. \tag{6.105}$$

Thus the terminal overlap R^g_D gives the exact prediction change (and $\mathrm{st}(f^+ - f) = R^g_{\bar{\tau}}$ in the continuum shadow), while (6.84) recovers the feature-kernel change from R^h_ℓ and $\dot{\Phi}_\ell$.

The factor $\Phi_\ell + R^h_\ell = N^{-1} \left\langle h_\ell, h^+_\ell \right\rangle$ is the internal overlap between the original and updated features. Replacing it by Φ_ℓ gives the linearized prediction update $-\eta\, r \sum_{\ell=1}^{D-1} \Phi_\ell G_{\ell+1}$. Indeed, $\|\nabla_{W_\ell} f\|^2_F = \Phi_\ell G_{\ell+1}$, so summing these squared gradient norms gives the initialized scalar NTK restricted to the trained hidden matrices; compare Proposition 2.25. The remaining $R^h_\ell G_{\ell+1}$ contribution records interactions among the simultaneous layer updates across proportional hyperfinite depth: an earlier feature change is passed through later matrices that have also been updated. All gradients are still evaluated at initialization. The network is linear in its input, but not jointly linear in all its weights.

Extensions to multiple inputs and nonlinear activations remain part of the ongoing work [41].

Appendix A

Technical Complements

A.1 Formal Derivation of the Transport PDE

Here is the nonstandard weak-form derivation of (1.11) from the hyperfinite mean-field system in (1.8), with the velocity field b defined in (1.10).

At an unlimited particle count $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, let $\rho_t^{(N)} = \frac{1}{N}\sum_{i=1}^N \delta_{X_i(t)}$ be the internal hyperfinite empirical measure from (1.9), let $\rho_t := L(\rho_t^{(N)}) \circ \mathrm{st}^{-1}$ denote its Loeb pushforward shadow on $\mathbb{R}$ [44, 15], and let $q : \mathbb{R} \to \mathbb{R}$ be a standard smooth compactly supported test function. Then the internal pairing is

$$\int_{{}^*\mathbb{R}} {}^*q(x)\, \rho_t^{(N)}(dx) = \frac{1}{N}\sum_{i=1}^N {}^*q(X_i(t)). \tag{A.1}$$

Differentiate internally in time by Transfer of the chain rule and substitute (1.8):

$$\begin{aligned} \frac{d}{dt}\int_{{}^*\mathbb{R}} {}^*q(x)\, \rho_t^{(N)}(dx) &= \frac{1}{N}\sum_{i=1}^N {}^*q'(X_i(t))\, \partial_t X_i(t) \\ &= \int_{{}^*\mathbb{R}} {}^*q'(x)\, {}^*b(x, \rho_t^{(N)})\, \rho_t^{(N)}(dx). \end{aligned} \tag{A.2}$$

Because q' and the interaction kernel K in (1.10) are standard bounded Lipschitz functions, every particle velocity $\partial_t X_i(t)$ is uniformly bounded, so the right-hand side of (A.2) is bounded and S-continuous in t on bounded time intervals. Equivalently, the internal pairing $t \mapsto \int_{{}^*\mathbb{R}} {}^*q(x)\, \rho_t^{(N)}(dx)$ is S-differentiable. Integrating (A.2) over $[0, t]$, passing to standard parts via Loeb integration of bounded S-continuous functions, and differentiating in standard time t gives

$$\frac{d}{dt}\int_{\mathbb{R}} q(x)\, \rho_t(dx) = \int_{\mathbb{R}} q'(x)\, b(x, \rho_t)\, \rho_t(dx). \tag{A.3}$$

This is precisely the distributional formulation of (1.11); in the Clifford calculus on $\mathbb{R}^k \subset \mathcal{C}\ell_k$ with vector Dirac operator $\partial_x = \sum_{j=1}^k e_j \partial_{x_j}$, the pairing $\int \langle (\partial_x q)\, b \rangle_0\, \rho_t(dx) = -\int q\, \langle \partial_x (b\rho_t) \rangle_0\, dx$ expresses the distributional adjointness of the vector gradient $\partial_x q$ and the scalar Dirac divergence $\langle \partial_x (b\rho_t) \rangle_0 = \partial_x \lrcorner (b\rho_t)$ [29, 9], without requiring ρ_t to have a smooth density. The equation is a prototype of a McKean–Vlasov PDE because the velocity field depends on the shadow measure ρ_t itself.

A.2 A Sufficient Condition for Stochastic Equicontinuity

This technical lemma supplies the S-continuity (equivalently, tightness) condition used in the function-space strengthening following Theorem 2.3. Let f_N denote the internal width-N network in (2.1) at an unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ (and f_n its standard width-n counterpart).

Lemma A.1 (Hölder Activation Implies S-Continuity and Tightness)**.** *Assume that for some standard $\gamma \in (0,1]$ and $C_\phi > 0$,*

$$|\phi(u) - \phi(v)| \le C_\phi \, |u - v|^\gamma \qquad \forall\, u, v \in \mathbb{R}. \tag{A.4}$$

Fix a standard compact set $K \subset \mathbb{R}^{n_0}$. For every standard $p \ge 2$, there is a standard constant $C_{p,K} < \infty$ such that, for every $N \in {}^*\mathbb{N}$ and all $x, x' \in {}^*K$,

$${}^*\mathbb{E}\big[|f_N(x) - f_N(x')|^p\big] \le C_{p,K} \, \|x - x'\|^{\gamma p}. \tag{A.5}$$

In particular, choose standard $p \ge 2$ so that $\gamma p > n_0$ and set $\beta := \gamma p - n_0 > 0$, making the right-hand side of (A.5) equal to $C_{p,K} \, \|x - x'\|^{n_0+\beta}$. Then at any unlimited width $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, $L({}^*\mathbb{P})$-almost surely $f_N(x) \in \mathbb{L}$ for all $x \in {}^*K$ and f_N is S-continuous on ${}^*K$:

$$x \approx x' \text{ in } {}^*K \implies f_N(x) \approx f_N(x') \qquad L({}^*\mathbb{P})\text{-a.s.}$$

(equivalently, $\{f_n|_K\}_{n \ge 1}$ is tight in $C(K)$).

Proof. Fix $N \in {}^*\mathbb{N}$ and $x, x' \in {}^*K$, and let $\mathcal{F} := \sigma(h_i(x), h_i(x') : i \in [N])$ be the internal hidden-layer σ-field. By Transfer of the conditional Gaussian output-layer identity (2.3), the increment $f_N(x) - f_N(x')$, conditional on $\mathcal{F}$, is centered internal Gaussian with variance $N^{-1} \sum_{i=1}^N |{}^*\phi(h_i(x)) - {}^*\phi(h_i(x'))|^2$. Hence the transferred Gaussian moment formula, internal Jensen's inequality (for $p/2 \ge 1$), and Transfer of (A.4), with C_p denoting the standard pth absolute moment of $\mathcal{N}(0,1)$, give

$$\begin{aligned}
{}^*\mathbb{E}\big[|f_N(x) - f_N(x')|^p\big] &= C_p \, {}^*\mathbb{E}\left[\left(\frac{1}{N}\sum_{i=1}^N |{}^*\phi(h_i(x)) - {}^*\phi(h_i(x'))|^2\right)^{p/2}\right] \\
&\le C_p \, {}^*\mathbb{E}\left[\frac{1}{N}\sum_{i=1}^N |{}^*\phi(h_i(x)) - {}^*\phi(h_i(x'))|^p\right] \\
&\le C_p C_\phi^p \, {}^*\mathbb{E}\left[\left|\frac{\langle w, x - x'\rangle}{\sqrt{n_0}}\right|^{\gamma p}\right] \le C_{p,K} \, \|x - x'\|^{\gamma p}.
\end{aligned} \tag{A.6}$$

The same conditional Gaussian identity at one fixed standard $x_0 \in K$, together with $|{}^*\phi(u)| \le |\phi(0)| + C_\phi \, |u|^\gamma$, gives ${}^*\mathbb{E}[|f_N(x_0)|^p] \le C'_{p,K} < \infty$ for a standard constant $C'_{p,K}$. Choosing standard p with $\gamma p > n_0$ and $\alpha \in (0, \min\{\gamma, \beta/p\})$, each internal sample path of f_N is internally continuous because ${}^*\phi$ is Hölder continuous, and Transfer of the Kolmogorov–Chentsov maximal moment bound on K yields a standard constant $\widetilde{C}_{p,\alpha,K} < \infty$ such that

$${}^*\mathbb{E}\left[\sup_{\substack{x,x' \in {}^*K \\ x \ne x'}} \frac{|f_N(x) - f_N(x')|^p}{\|x - x'\|^{\alpha p}}\right] \le \widetilde{C}_{p,\alpha,K}.$$

Consequently, at unlimited $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, both $|f_N(x_0)|$ and the internal α-Hölder constant $M_N := \sup_{x \neq x' \in {}^*K} |f_N(x) - f_N(x')| / \|x - x'\|^\alpha$ belong to $\mathbb{L}\, L({}^*\mathbb{P})$-almost surely [44, 15]. On this $L({}^*\mathbb{P})$-full-measure event, $x \approx x'$ in ${}^*K$ implies $|f_N(x) - f_N(x')| \leq M_N \|x - x'\|^\alpha \approx 0$, establishing $L({}^*\mathbb{P})$-a.s. S-continuity on ${}^*K$ (and, by standard equivalence, tightness of $\{f_n|_K\}_{n\geq 1}$ in $C(K)$). □

Remark A.2. On the $L({}^*\mathbb{P})$-full-measure event where f_N is limited and S-continuous on ${}^*K$, the standard shadow $f^\circ(\mathrm{st}(x)) := \mathrm{st}(f_N(x))$ $(x \in {}^*K)$ defines a continuous random function in $C(K)$ satisfying $\sup_{x \in {}^*K} |f_N(x) - {}^*(f^\circ)(x)| \approx 0$. Combined with the finite-dimensional Gaussian shadows in Theorem 2.3, f° is a continuous version of the limiting Gaussian process on K (equivalently, the laws of $f_n|_K$ converge weakly in $C(K)$ to that of f°).

A.3 Convergence in the Skorohod Topology

Many processes in these notes arise by piecewise-constant interpolation of a layer-indexed Markov chain on a hyperfinite grid $\mathbb{T}_N := \{0, \Delta t, 2\Delta t, \ldots, \lfloor NT \rfloor \Delta t\}$ with $\Delta t = 1/N \approx 0$ (e.g. $\Phi_\tau^{(N)} := \Phi_{\lfloor \tau N \rfloor}$). At finite n, such interpolations are càdlàg—right-continuous with left limits—and therefore live naturally in a Skorohod path space. This appendix records the classical Skorohod definitions needed to interpret statements like

$$X^{(n)} \xrightarrow{d} X \qquad \text{in } D([0,T], S), \tag{A.7}$$

where S is a metric state space, and then states their nonstandard characterization via *S-continuity*, following [42, Appendix A], [20, 35], and the hyperfinite stochastic-process theory of Anderson, Keisler, and Lindstrøm [3, 37, 43].

Let (S, d_S) be a complete separable metric space, and write $D([0,T], S)$ for the set of càdlàg functions $x : [0,T] \to S$. When $S = \mathbb{R}^k$, we use the Euclidean distance $d_S(x,y) = \|x - y\|$. To compare two càdlàg paths up to small shifts of their jump times, let Λ_T be the set of increasing homeomorphisms $\lambda : [0,T] \to [0,T]$ with $\lambda(0) = 0$ and $\lambda(T) = T$.

Skorohod J_1 metric. For $x, y \in D([0,T], S)$ define the (equivalent) J_1 metric

$$d_{J_1}(x,y) := \inf_{\lambda \in \Lambda_T} \left\{ \sup_{t \in [0,T]} d_S\big(x(\lambda(t)), y(t)\big) \vee \sup_{t \in [0,T]} |\lambda(t) - t| \right\}. \tag{A.8}$$

Then $x_n \to x$ in the Skorohod J_1 topology if and only if there exist $\lambda_n \in \Lambda_T$ such that

$$\sup_{t \in [0,T]} |\lambda_n(t) - t| \to 0 \qquad \text{and} \qquad \sup_{t \in [0,T]} d_S\big(x_n(\lambda_n(t)), x(t)\big) \to 0, \tag{A.9}$$

which is the same criterion stated as [42, (A.1)].

Nonstandard characterization via S-continuity. The Skorohod space accommodates piecewise-constant approximations by allowing small shifts in jump times. However, when the limiting path $x \in C([0,T], S)$ is *continuous*—as for every diffusion studied in these notes—convergence in the J_1 topology reduces to uniform convergence, and in Nonstandard Analysis the jump-time homeomorphisms $\lambda \in \Lambda_T$ disappear completely. Specifically, at an unlimited

step scale $N \in {}^*\mathbb{N} \setminus \mathbb{N}$, an internal path $X^{(N)} \in {}^*D([0,T],S)$ satisfies ${}^*d_{J_1}(X^{(N)}, {}^*x) \approx 0$ for a standard continuous path $x \in C([0,T],S)$ if and only if $\sup_{t\in{}^*[0,T]} {}^*d_S\big(X_t^{(N)}, {}^*x(t)\big) \approx 0$, which holds if and only if $X^{(N)}$ is nearstandard-valued and *S-continuous* on ${}^*[0,T]$:

$$s \approx t \text{ in } {}^*[0,T] \implies {}^*d_S\big(X^{(N)}_{\lfloor Ns\rfloor}, X^{(N)}_{\lfloor Nt\rfloor}\big) \approx 0,$$

with $x(\mathrm{st}(t)) = \mathrm{st}\big(X^{(N)}_{\lfloor Nt\rfloor}\big)$ [3, 37, 43].

Equipped with the J_1 topology, $D([0,T],S)$ is a Polish space (separable and completely metrizable), so whenever an internal piecewise-constant process $X^{(N)}_{\lfloor Nt\rfloor}$ is $L({}^*\mathbb{P})$-almost surely nearstandard and S-continuous on ${}^*[0,T]$, its pointwise standard part $X_t^\circ(\omega) := \mathrm{st}\big(X^{(N)}_{\lfloor Nt\rfloor}(\omega)\big)$ defines a $L({}^*\mathbb{P})$-measurable continuous random path in $C([0,T],S) \subset D([0,T],S)$ whose Loeb pushforward law is the J_1 weak limit of $X^{(n)}$.

Definition A.3 (Local S-Continuous Shadow and Local Skorohod Convergence)**.** Let $U \subseteq \mathbb{R}^k$ be standard open, and write $D \Subset U$ when D is open and bounded and $\overline{D}$ is a compact subset of U. Let $X^{(N)}$ ($N \in {}^*\mathbb{N} \setminus \mathbb{N}$) be an internal càdlàg ${}^*U$-valued process (arising from a standard sequence $X^{(n)}$), and let X be a continuous standard process defined up to its lifetime in U. For $D \Subset U$, define the standard exit time

$$\tau_D(Y) := \inf\{t \in [0,T] : Y_t \notin D\}, \tag{A.10}$$

and the corresponding internal exit time $\tau_{{}^*D}(X^{(N)}) := \inf\{t \in {}^*[0,T] : X_t^{(N)} \notin {}^*D\}$, with the convention $\inf\varnothing := T$. The stopped paths are frozen at their exit values and are therefore defined on all of $[0,T]$ (respectively ${}^*[0,T]$), even if X later leaves U or explodes. We say $X^{(N)}$ has *local S-continuous shadow* X in distribution (equivalently, $X^{(n)}$ converges *locally* to X in distribution) if, for every such $D \Subset U$ at which the stopping map is almost surely continuous under X, the stopped hyperfinite path $X^{(N)}_{\cdot\wedge\tau_{{}^*D}(X^{(N)})}$ is $L({}^*\mathbb{P})$-a.s. S-continuous on ${}^*[0,T]$ and its standard shadow satisfies

$$\mathrm{st}\Big(X^{(N)}_{\cdot\wedge\tau_{{}^*D}(X^{(N)})}\Big) \overset{d}{=} X_{\cdot\wedge\tau_D(X)} \qquad \text{in } C([0,T],\mathbb{R}^k) \subset D([0,T],\mathbb{R}^k) \tag{A.11}$$

(equivalently, $X^{(n)}_{\cdot\wedge\tau_D(X^{(n)})} \xrightarrow{d} X_{\cdot\wedge\tau_D(X)}$ in $D([0,T],\mathbb{R}^k)$). A standard sufficient condition for continuity of the stopping map is $\tau_D(X) = \tau_{\overline{D}}(X)$ almost surely.

This formulation is useful when the limiting diffusion can leave compact sets or explode. The stopping times in Theorem 5.10 localize the process against norm growth. A compact localization inside $\mathrm{SPD}(m)$ must in addition stay an appreciable distance from the singular boundary.

A standard generator route combines a Feller construction—see [20, Chapter 8, Theorem 2.5] under smoother coefficient assumptions—with the discrete-time criterion [35, Theorem 17.28]; [42, Appendix A] develops this route for the Neural Covariance SDE. In Nonstandard Analysis, Keisler's hyperfinite martingale-lifting theory [37, 43] (combined with the localized Stroock–Varadhan martingale characterization [59]) yields the following convenient infinitesimal moment criterion.

Theorem A.4 (Hyperfinite Diffusion Approximation for Markov Chains)**.** *Let $U \subseteq \mathbb{R}^k$ be standard open, fix an unlimited step scale $N \in {}^*\mathbb{N} \setminus \mathbb{N}$ with $\Delta t := 1/N \approx 0$, and let $(X_j^{(N)})_{j\geq 0}$ be an internal time-homogeneous Markov chain taking values in ${}^*U$ (transferred from a standard sequence $(X_j^{(n)})_{j\geq 0}$). Define the hyperfinite piecewise-constant interpolation $X_t^{(N)} := X_{\lfloor Nt \rfloor}^{(N)}$. Assume $X_0^{(N)}$ is nearstandard in ${}^*U$ $L({}^*\mathbb{P})$-a.s. with $\mathrm{st}(X_0^{(N)}) \overset{d}{=} X_0$, where X_0 takes values in U. Assume there exist standard functions $b : U \to \mathbb{R}^k$ and $\Sigma : U \to \mathbb{R}^{k\times k}$ such that for every standard compact $K \subset U$ and every $x \in {}^*K$,*

$$N\, {}^*\mathbb{E}\big[X_1^{(N)} - X_0^{(N)} \mid X_0^{(N)} = x\big] \approx {}^*b(x), \tag{A.12}$$

$$N\, {}^*\mathrm{Cov}\big(X_1^{(N)} - X_0^{(N)} \mid X_0^{(N)} = x\big) \approx {}^*\Sigma(x), \tag{A.13}$$

$$N\, {}^*\mathbb{E}\big[\big\|X_1^{(N)} - X_0^{(N)}\big\|^3 \mid X_0^{(N)} = x\big] \approx 0. \tag{A.14}$$

Here the internal conditional moments refer to the one-step transition kernel started from x (by Robinson's compactness characterization [55], (A.12)–(A.14) are equivalent to uniform convergence of the rescaled moments on every standard compact $K \subset U$). Let $\sigma : U \to \mathbb{R}^{k\times k}$ satisfy $\sigma(x)\sigma(x)^\top = \Sigma(x)$, and assume that b and σ are locally Lipschitz on U. Let X be the maximal solution, started from X_0, of

$$dX_t = b(X_t)\, dt + \sigma(X_t)\, dB_t. \tag{A.15}$$

Then, for each standard $T > 0$, $X^{(N)}$ has local S-continuous shadow X in the sense of Definition A.3: for every admissible localization domain $D \Subset U$, the stopped hyperfinite chain $X^{(N)}_{\cdot\wedge\tau_{{}^*D}(X^{(N)})}$ is $L({}^*\mathbb{P})$-a.s. S-continuous on ${}^*[0,T]$ and its Loeb shadow solves (A.15) up to $\tau_D(X)$.

Proof. Fix an admissible localization domain $D \Subset U$ from Definition A.3 and choose a standard compact $K \subset U$ with $\overline{D} \subset K$. Let P_N be the internal transition kernel of $X^{(N)}$, and let $\Delta_N(x)$ have the internal one-step increment law started from $x \in {}^*K$. Decompose each pre-exit step into its predictable drift and martingale increment: $X_{j+1}^{(N)} - X_j^{(N)} = \mu_N(X_j^{(N)}) + M_{j+1}^{(N)}$, where $\mu_N(x) := {}^*\mathbb{E}[\Delta_N(x)]$.

By (A.12), $\mu_N(x) = \Delta t\,({}^*b(x) + \eta_N(x))$ with $\sup_{x\in {}^*K} \|\eta_N(x)\| \approx 0$, so the stopped cumulative drift sum has bounded difference quotients on $\mathbb{T}_N$ and is deterministically S-continuous. For the stopped martingale sum, (A.13) bounds the predictable quadratic variation increment by $\mathcal{O}(\Delta t)$ on ${}^*K$, while (A.14) and internal Markov's inequality give $N\, {}^*\mathbb{P}(\|\Delta_N(x)\| > \varepsilon) \leq \varepsilon^{-3} N\, {}^*\mathbb{E}[\|\Delta_N(x)\|^3] \approx 0$ uniformly on ${}^*K$ for every standard $\varepsilon > 0$. By Lindstrøm's and Keisler's S-continuity theorem for hyperfinite martingales with bounded quadratic variation and infinitesimal rare-jump rate [43, Section III.3], [37], the stopped martingale sum—and hence $X^{(N)}_{\cdot\wedge\tau_{{}^*D}(X^{(N)})}$—is $L({}^*\mathbb{P})$-almost surely S-continuous on ${}^*[0,T]$.

To identify the shadow law, note that if $\mu_N(x) = {}^*\mathbb{E}[\Delta_N(x)]$, then the raw internal second moment is $N\, {}^*\mathrm{Cov}(\Delta_N(x)) + N\,\mu_N(x)\mu_N(x)^\top$. Because $N\,\mu_N(x)$ is bounded on ${}^*K$, the second term is $\mathcal{O}(N^{-1}) \approx 0$, so $N\, {}^*\mathbb{E}[\Delta_N(x)\Delta_N(x)^\top] \approx {}^*\Sigma(x)$ uniformly on ${}^*K$; the third-moment bound (A.14) likewise shows that truncating jumps at any standard $\varepsilon > 0$ alters the first two rescaled moments only infinitesimally (compare [59, Lemma 11.2.1] and [42, Appendix A, Proposition A.6]). Expanding ${}^*f(y) - {}^*f(x)$ to second order for any standard $f \in C_c^\infty(U)$

therefore gives, uniformly for $x \in {}^*K$,

$$\begin{aligned} \mathcal{A}_N {}^*f(x) &:= N \int_{{}^*U} ({}^*f(y) - {}^*f(x))\, P_N(x, dy) \\ &\approx {}^*b(x)^\top \nabla {}^*f(x) + \frac{1}{2} \operatorname{Tr}({}^*\Sigma(x) D^{2*} f(x)) =: {}^*(\mathcal{A}f)(x). \end{aligned} \tag{A.16}$$

Taking standard parts of the stopped internal Dynkin martingale $M_t^f := {}^*f(X^{(N)}_{\lfloor Nt \rfloor}) - {}^*f(X_0^{(N)}) - \Delta t \sum_{j=0}^{\lfloor Nt \rfloor - 1} \mathcal{A}_N {}^*f(X_j^{(N)})$ on $[0,T]$ shows that the continuous Loeb shadow $X_t^\circ = \operatorname{st}(X^{(N)}_{t \wedge \tau_{*D}})$ solves the stopped martingale problem for $\mathcal{A}$ started from $\operatorname{st}(X_0^{(N)}) \overset{d}{=} X_0$ [37, 43] (equivalently, by [59, Theorem 11.2.3]). Local Lipschitz continuity of b and σ gives uniqueness of the solution to (A.15) up to exit from D, and the boundary continuity condition on D ensures $\operatorname{st}(\tau_{*D}(X^{(N)})) = \tau_D(X^\circ)$ $L({}^*\mathbb{P})$-a.s., yielding (A.11) for every admissible $D \Subset U$. □

A.4 Affine-Invariant Metric for the Covariance SDE

Proof of Proposition 6.8. Congruence invariance follows from $(P\Phi P^\top)^{-1} = P^{-\top}\Phi^{-1}P^{-1}$ and cyclicity of the trace. It remains to identify the metric in the coordinates of (6.35). For $A \in \operatorname{Sym}(m)$, let $F_A(\Phi) := \operatorname{Tr}(A\Phi)$, so that $dF_A(\Phi)[V] = \operatorname{Tr}(AV)$. On the hyperfinite grid $\tau_\ell = \ell\,\Delta\tau$ with $\Delta\Phi_\ell = \Phi_\ell^{1/2}\,\Delta S_\ell\,\Phi_\ell^{1/2}$ realizing (6.42), the normalization ${}^*\mathrm{Cov}(\Delta S_{\ell,\alpha\beta}, \Delta S_{\ell,\gamma\delta} \mid \mathcal{F}_\ell) = (\delta_{\alpha\gamma}\delta_{\beta\delta} + \delta_{\alpha\delta}\delta_{\beta\gamma})\,\Delta\tau$ gives ${}^*\mathbb{E}[\operatorname{Tr}(A\,\Delta\Phi_\ell)\operatorname{Tr}(B\,\Delta\Phi_\ell) \mid \mathcal{F}_\ell] = 2\operatorname{Tr}(A\Phi_\ell B\Phi_\ell)\,\Delta\tau$, whose standard shadow quadratic variation is

$$d\,\langle F_A(\Phi_\tau), F_B(\Phi_\tau)\rangle = 2\operatorname{Tr}(A\Phi_\tau B\Phi_\tau)\,d\tau. \tag{A.17}$$

The metric-dual tangent vector of $dF_A(\Phi)$ is $2\Phi A\Phi$, since

$$g_\Phi(2\Phi A\Phi, V) = \operatorname{Tr}(AV). \tag{A.18}$$

Writing g_Φ^{-1} for the cometric, we therefore have

$$g_\Phi^{-1}(dF_A, dF_B) = g_\Phi(2\Phi A\Phi, 2\Phi B\Phi) = 2\operatorname{Tr}(A\Phi B\Phi), \tag{A.19}$$

so the covariance tensor of the SDE is the cometric associated with g_Φ.

We now pass explicitly to the distinct upper-triangular coordinates. For $\alpha \le \beta$, let $x^{\alpha\beta}(\Phi) := \Phi_{\alpha\beta}$ and $C_{\alpha\beta} := (E_{\alpha\beta} + E_{\beta\alpha})/2$, where $E_{\alpha\beta}$ is the standard matrix unit. Then $dx^{\alpha\beta}(V) = V_{\alpha\beta} = \operatorname{Tr}(C_{\alpha\beta}V)$, and hence

$$\begin{aligned} g_\Phi^{-1}(dx^{\alpha\beta}, dx^{\gamma\delta}) &= 2\operatorname{Tr}(C_{\alpha\beta}\Phi C_{\gamma\delta}\Phi) \\ &= \Phi_{\alpha\gamma}\Phi_{\beta\delta} + \Phi_{\alpha\delta}\Phi_{\beta\gamma} = \Sigma_{\alpha\beta,\gamma\delta}(\Phi). \end{aligned} \tag{A.20}$$

Thus $\Sigma(\Phi)$ is exactly the matrix of the cometric in the coordinate coframe $(dx^{\alpha\beta})_{\alpha\le\beta}$. The metric matrix in the dual coordinate tangent basis is therefore $\Sigma(\Phi)^{-1}$. □

A.5 Gaussian Conditioning

This section collects two internal Gaussian conditioning identities valid at arbitrary hyperfinite dimensions $N, p, q, m \in {}^*\mathbb{N}$ by the Transfer Principle [55, 22]. The first is a one-sided conditioning identity used when we condition only on the forward information $W\phi$ (see Lemma 2.26). The second is a two-sided projection decomposition used when we condition simultaneously on forward and backward information $(W\phi, W^\top g)$ (see Lemma A.5). Formally, the two-sided statement reduces to the one-sided one when the backward information is absent (e.g. by setting $g = 0$ in Lemma A.5, which makes $P_g = 0$).

A.5.1 One-Sided Gaussian Conditioning

This subsection provides a proof of Lemma 2.26 for internal matrices $W \in {}^*\mathbb{R}^{N\times p}$ with i.i.d. ${}^*\mathcal{N}(0,1)$ entries and internally deterministic $\phi \in {}^*\mathbb{R}^{p\times q}$ at arbitrary hyperfinite dimensions $N, p, q \in {}^*\mathbb{N}$.

Proof of Lemma 2.26. By the Transfer Principle, every first-order finite-dimensional linear-algebraic and Gaussian conditioning statement over $\mathbb{R}$ holds verbatim for internal hyperfinite-dimensional matrices over ${}^*\mathbb{R}$. Write the N rows of $W \in {}^*\mathbb{R}^{N\times p}$ as $w_1, \dots, w_N \in {}^*\mathbb{R}^{1\times p}$. Since the rows are internally independent and identically distributed, it suffices to prove the statement for a single internal row $w \sim {}^*\mathcal{N}(0, I_p)$.

Let $\phi \in {}^*\mathbb{R}^{p\times q}$ be internally deterministic and define the internal orthogonal projector $P_\phi = \phi(\phi^\top\phi)^\dagger\phi^\top$ onto $\mathrm{col}(\phi) \subseteq {}^*\mathbb{R}^p$ via the transferred Moore–Penrose pseudoinverse $(\cdot)^\dagger$, with $P_\phi^\perp = I_p - P_\phi$. (Equivalently, choosing any nonzero simple internal blade $\mathbf{\Phi} = u_1 \wedge \cdots \wedge u_{r_\phi} \in \bigwedge^{r_\phi}({}^*\mathbb{R}^p) \subset {}^*\mathcal{C}\ell_p$ spanning $\mathrm{col}(\phi)$, Hestenes' contraction–wedge decomposition $w\mathbf{\Phi} = w \lrcorner \mathbf{\Phi} + w \wedge \mathbf{\Phi}$ expresses the row projection and rejection without choosing a basis as $wP_\phi = (w \lrcorner \mathbf{\Phi})\mathbf{\Phi}^{-1}$ and $wP_\phi^\perp = (w \wedge \mathbf{\Phi})\mathbf{\Phi}^{-1}$ [29]; when $r_\phi = 0$, $P_\phi = 0$ and $P_\phi^\perp = I_p$.) Decompose

$$w = wP_\phi + wP_\phi^\perp. \tag{A.21}$$

The internal random vector $wP_\phi = (w\phi)(\phi^\top\phi)^\dagger\phi^\top$ is internally measurable with respect to $w\phi$ because P_ϕ depends on ϕ only and wP_ϕ is the unique minimum-norm internal linear function of $w\phi$ matching the constraint in least squares. Moreover, $wP_\phi^\perp$ is a centered internal Gaussian with covariance $P_\phi^\perp$, and it is internally independent of $w\phi$ because

$${}^*\mathrm{Cov}(wP_\phi^\perp, w\phi) = P_\phi^\perp\phi = 0, \tag{A.22}$$

so transferred joint Gaussianity implies internal independence.

Let $\widetilde{w} \sim {}^*\mathcal{N}(0, I_p)$ be an internally independent copy of w. Then $\widetilde{w}P_\phi^\perp \sim {}^*\mathcal{N}(0, P_\phi^\perp)$ and is internally independent of $w\phi$. Therefore, in internal conditional distribution,

$$w \mid \sigma(w\phi) \overset{d}{=} wP_\phi + \widetilde{w}P_\phi^\perp. \tag{A.23}$$

Stacking the N hyperfinite rows back together yields $W \mid \sigma(W\phi) \overset{d}{=} WP_\phi + \widetilde{W}P_\phi^\perp$ with $\widetilde{W}$ an internally independent copy of W, as claimed. □

A.5.2 Two-Sided Gaussian Conditioning

Lemma A.5 (Gaussian Conditioning (Two-Sided Projection Decomposition))**.** *For internal dimensions $N, m \in {}^*\mathbb{N}$, let $W \in {}^*\mathbb{R}^{N\times N}$ have internally i.i.d. ${}^*\mathcal{N}(0,1)$ entries and let $\phi, g \in {}^*\mathbb{R}^{N\times m}$ be internally deterministic. Define the internal orthogonal projections onto the column spans*

$$P_\phi = \phi(\phi^\top\phi)^\dagger\phi^\top, \qquad P_g = g(g^\top g)^\dagger g^\top, \tag{A.24}$$

where $(\cdot)^\dagger$ is the transferred Moore–Penrose pseudoinverse. Then in internal conditional distribution,

$$W \mid (W\phi, W^\top g) \overset{d}{=} P_g W + W P_\phi - P_g W P_\phi + P_g^\perp \widetilde{W} P_\phi^\perp, \tag{A.25}$$

where $P^\perp = I_N - P$ and $\widetilde{W} \in {}^*\mathbb{R}^{N\times N}$ is an internally independent copy of W.

This subsection proves Lemma A.5. The statement is a convenient hyperfinite matrix form of linear-Gaussian regression obtained by Transfer: conditioning an internal Gaussian matrix W on the internal linear observations $W\phi$ and $W^\top g$ amounts to removing the components of W that are "seen" along the internal spans of ϕ and g, and leaving an internally independent Gaussian on the orthogonal complement.

Proof of Lemma A.5. By the Transfer Principle, write the internal Frobenius inner product on ${}^*\mathbb{R}^{N\times N}$ as $\langle A, B\rangle_F := \mathrm{Tr}(A^\top B)$. Let $\mathcal{U} := \mathrm{col}(\phi) \subseteq {}^*\mathbb{R}^N$ and $\mathcal{V} := \mathrm{col}(g) \subseteq {}^*\mathbb{R}^N$ and denote the corresponding internal orthogonal projections by P_ϕ and P_g (which, for nonzero simple blades $\boldsymbol{\Phi} \in \bigwedge^{r_\phi}({}^*\mathbb{R}^N)$ and $\boldsymbol{G} \in \bigwedge^{r_g}({}^*\mathbb{R}^N)$ representing $\mathcal{U}$ and $\mathcal{V}$ in ${}^*\mathcal{C}\ell_N$, act on vectors by $P_\phi(u) = (u \lrcorner \boldsymbol{\Phi})\boldsymbol{\Phi}^{-1}$, $P_\phi^\perp(u) = (u \wedge \boldsymbol{\Phi})\boldsymbol{\Phi}^{-1}$ and $P_g(v) = (v \lrcorner \boldsymbol{G})\boldsymbol{G}^{-1}$, $P_g^\perp(v) = (v \wedge \boldsymbol{G})\boldsymbol{G}^{-1}$).

Decompose W into four internal blocks with respect to the orthogonal splittings ${}^*\mathbb{R}^N = \mathcal{V} \oplus \mathcal{V}^\perp$ (rows) and ${}^*\mathbb{R}^N = \mathcal{U} \oplus \mathcal{U}^\perp$ (columns):

$$W = \underbrace{P_g W P_\phi}_{\text{seen by both}} + \underbrace{P_g W P_\phi^\perp}_{\text{seen by } W^\top g} + \underbrace{P_g^\perp W P_\phi}_{\text{seen by } W\phi} + \underbrace{P_g^\perp W P_\phi^\perp}_{\text{unseen}}. \tag{A.26}$$

The key internal observations are:

- $W\phi$ depends only on the columns of W in $\mathcal{U}$, i.e. on WP_ϕ (equivalently the sum of the first and third blocks);
- $W^\top g$ depends only on the rows of W in $\mathcal{V}$, i.e. on $P_g W$ (equivalently the sum of the first and second blocks);
- the "unseen" block $P_g^\perp W P_\phi^\perp$ is internally independent of $(WP_\phi, P_g W)$ because W has internally i.i.d. Gaussian entries and these are Frobenius-orthogonal internal linear functionals of $\mathrm{vec}(W)$.

More formally, $\mathrm{vec}(W) \sim {}^*\mathcal{N}(0, I_{N^2})$ and the internal maps

$$L_1 : W \mapsto WP_\phi, \qquad L_2 : W \mapsto P_g W, \qquad L_3 : W \mapsto P_g^\perp W P_\phi^\perp \tag{A.27}$$

are internally linear. One checks that L_3 is orthogonal (as an internal linear map on ${}^*\mathbb{R}^{N\times N}$ with the internal Frobenius inner product) to both L_1 and L_2: for any internal matrices $A, B \in {}^*\mathbb{R}^{N\times N}$,

$$\left\langle P_g^\perp A P_\phi^\perp, BP_\phi \right\rangle_F = \mathrm{Tr}((P_g^\perp A P_\phi^\perp)^\top B P_\phi) = 0, \qquad \left\langle P_g^\perp A P_\phi^\perp, P_g B \right\rangle_F = 0, \tag{A.28}$$

since $P_\phi^\perp P_\phi = 0$ and $P_g P_g^\perp = 0$. By Transfer of Gaussian orthogonality-implies-independence, $L_3(W)$ is internally independent of $(L_1(W), L_2(W))$, and its internal conditional law given $(W\phi, W^\top g)$ is unchanged.

It remains to identify the conditional mean. Given $W\phi$ and $W^\top g$, the blocks $P_g^\perp W P_\phi$ and $P_g W P_\phi^\perp$ are fixed by the constraints, and the intersection block $P_g W P_\phi$ is fixed twice but consistently. A short algebraic rearrangement gives

$$P_g W + W P_\phi - P_g W P_\phi = P_g W P_\phi + P_g W P_\phi^\perp + P_g^\perp W P_\phi, \tag{A.29}$$

which is exactly the sum of the internal blocks determined by $(W^\top g, W\phi)$. Thus the internal conditional law is "deterministic part + independent unseen Gaussian block", i.e.

$$W \mid (W\phi, W^\top g) \stackrel{d}{=} (P_g W + W P_\phi - P_g W P_\phi) + P_g^\perp \widetilde{W} P_\phi^\perp, \tag{A.30}$$

where $\widetilde{W}$ is an internally independent copy of W. □

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